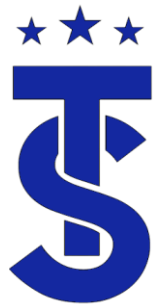




STS Training Series

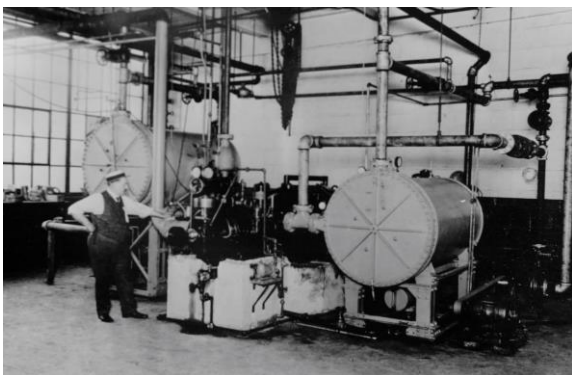
Ductless Heat Pump Basic Refrigeration
2025





Air Conditioning through the ages

- As much as things change, they stay the same?



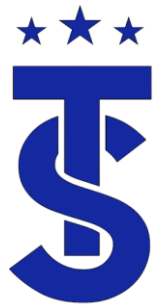
1902



1950

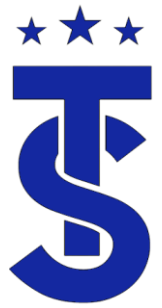


2024



Why Attend Training?





Are there differences in manufacturers? Are There Similarities?



Are there differences in operation, performance, service tools and techniques between manufacturers?

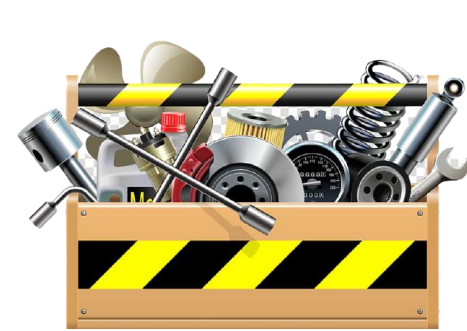
☆☆☆ T Do you have the right tools and knowledge to work on the equipment?



What is your
expectation for
service?



OR



What is your
customers
expectation for
service?



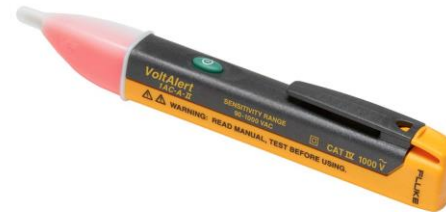
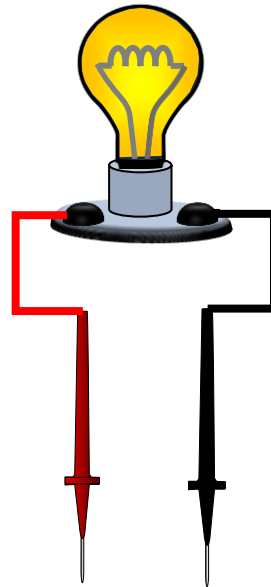
OR

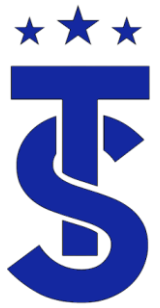
And



Why

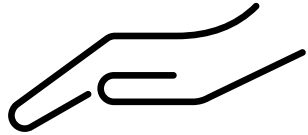
- Why do have a meter?
- You can do the same with an induction tester or a light bulb?





Do the Tools Make the Technician?

- Do you have the right tools for the job?
- Why does it matter?
- What information can you gather with these tools?



Are you limited to this?





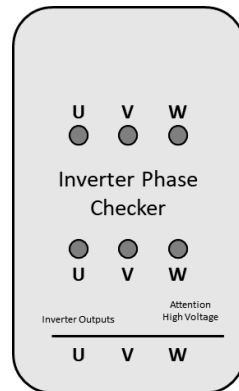
- Are these specialty tools?
- What value do these tool bring?
- How proficient can you be with out them?



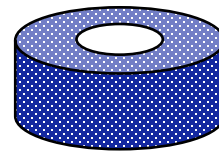
HAVC App.
Phone with internet



S-Net Pro
Converter



Inverter Tester



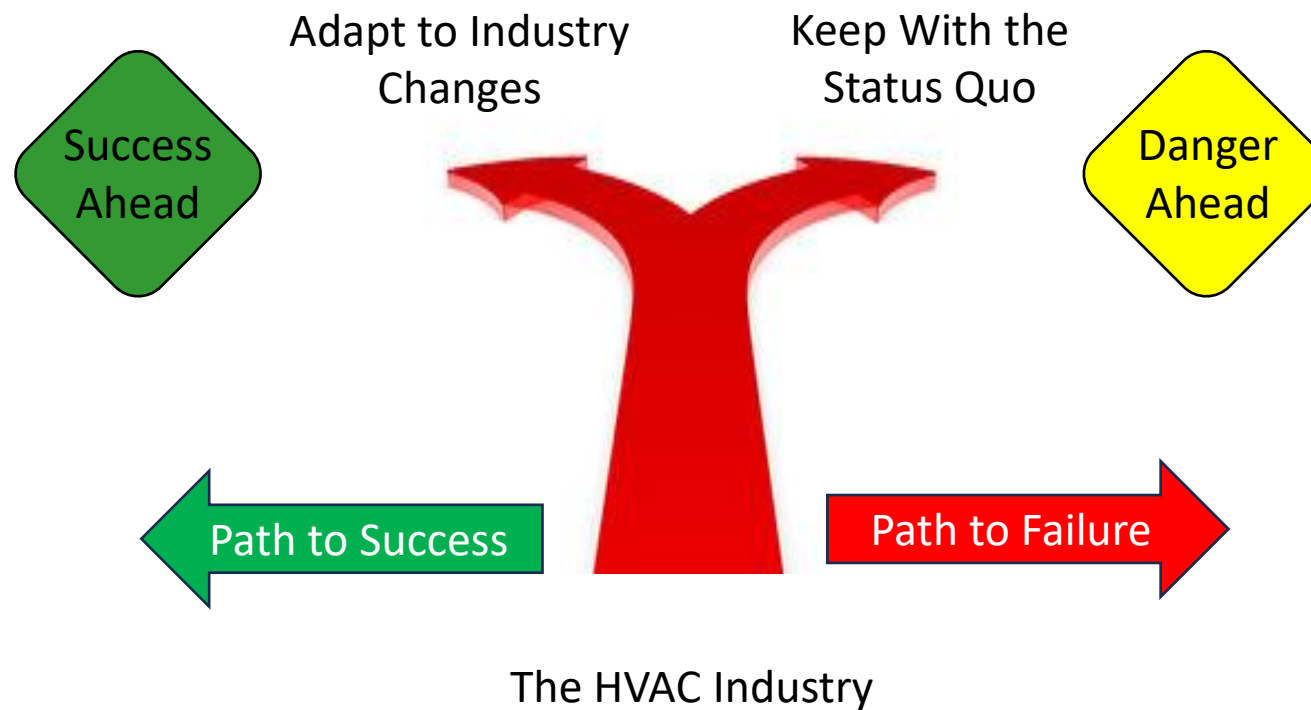
EEV & Solenoid
Magnet

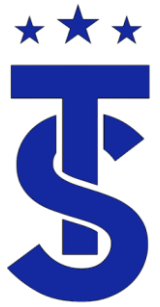


Laptop with
manufacturers service
and design software



Choose your path the choice is yours

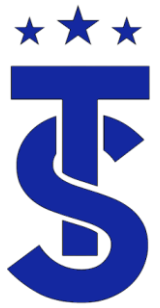




Review

Accessing Prior Knowledge

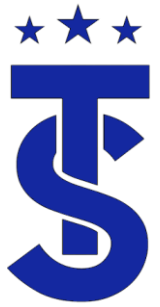
- Given the following system determine the fan speed settings for heat and cooling .
 - 70,000 BTU Input furnace?
 - 2 tons air 20 SEER air conditioning?
- Heat Speed.
- Cooling Speed
- More information
 - 95% 70,000 BTU Input furnace with a design TD of 55 degrees
 - 2 tons air high efficiency air conditioning requiring 350 cfm/ton?
- The CFM requirement for A/C is 700 CFM. The heating CFM can be determined on the following page



Calculating CFM

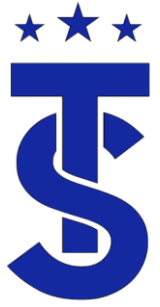
Gas Furnace

- When installing a gas burning piece of equipment you must calculate the required CFM needed for proper operation.
- To calculate the required CFM you need to know the systems Btu Output and the design TD.
- The formula is:
 - **CFM = BTU Output / TD x 1.08** (the specific heat of a cu' of air).
- Calculate the CFM requirement for a 95% 70,000 BTU Input furnace with a design TD of 55 degrees?
- $CFM = 66,500 / 55 \times 1.08$
- $CFM = 1,103$



Conclusion

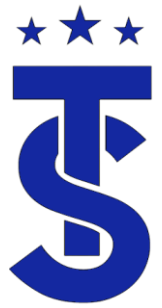
- The heating system requires more air than the air conditioning system.
- To get the correct performance out of the air conditioning system it requires 700 CFM not 800.
- Why is this important?
- Does it really make a difference?
- Do your contractors know this?



Have you ever wondered?

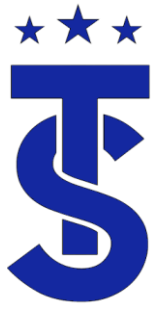
- How a variable speed system operates?
- What is the system controlling to?
- What causes the compressor to ramp up and slow down.
- How is the easiest way to test the operation.
- How come there are no pressure transducers?

What else are you wondering?

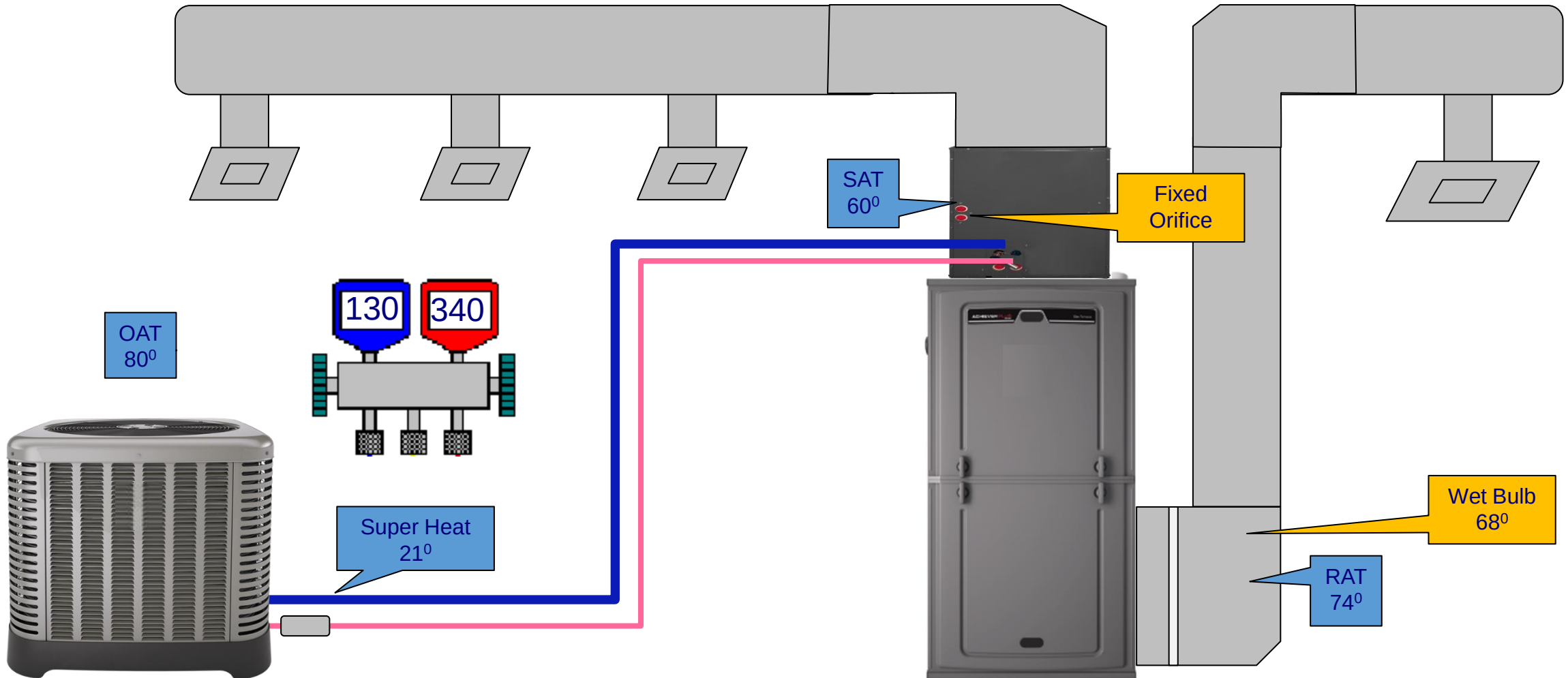


Why this class?

- To effectively troubleshoot any system, it is important to have a basic understanding of the refrigeration cycle, its components and how they interact with one another.
- Understanding superheat and sub-cooling are vital to understanding how a system is operating.
- Knowing how each component operates makes knowing the system as a whole easier.
- In order to effectively troubleshoot a system, you need to know its sequence of operation or how it is supposed to operate.
- Let's try something



Evaluate the System Operation





Fixed Orifice Charging Charts

Condenser Entering Air Temperature

Indoor Wet Bulb Temperature		55	60	65	70	75	80	85	90	95
	52	12	10	6	-	-	-	-	-	-
	54	14	12	10	7	-	-	-	-	-
	56	17	15	13	10	6	-	-	-	-
	58	20	18	16	13	9	5	-	-	-
	60	23	21	19	16	12	8	-	-	-
	62	26	24	21	19	15	12	8	5	-
	64	29	27	24	21	18	15	11	9	6
	66	32	30	27	24	21	18	15	13	10
	68	35	33	30	27	24	21	19	16	14
	70	37	35	33	30	28	25	22	18	15
	72	40	38	36	33	31	28	26	24	22

Required Super Heat at Service Valve $\pm 5^{\circ}\text{F}$

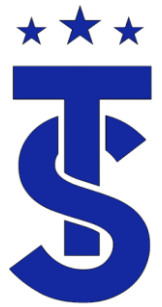
Super Heat Calculator

Indoor Entering Wet Bulb Temperature

Indoor Dry Bulb Temperature		57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
	70	51	51	52	52	53	53	54	55	55	56	57	59	60	-	-	-
	72	52	52	53	53	54	55	55	56	57	57	58	59	60	61	62	63
	74	53	53	53	54	55	55	56	57	58	58	59	60	61	62	63	64
	76	54	54	54	55	55	56	57	57	58	59	60	61	62	63	64	65
	78	55	55	55	56	56	57	57	58	59	60	61	62	63	64	65	66
	80	56	56	56	56	57	58	58	59	60	61	62	63	64	65	66	67
	82	57	57	57	57	58	59	60	60	61	62	63	64	65	66	67	68
	84	-	58	59	59	60	60	61	61	62	63	63	64	65	66	67	68

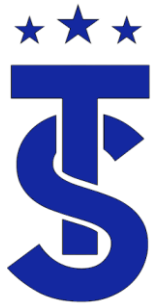
Proper Evaporator Leaving Dry Bulb Temperature $\pm 3^{\circ}\text{F}$

Air Flow Calculator



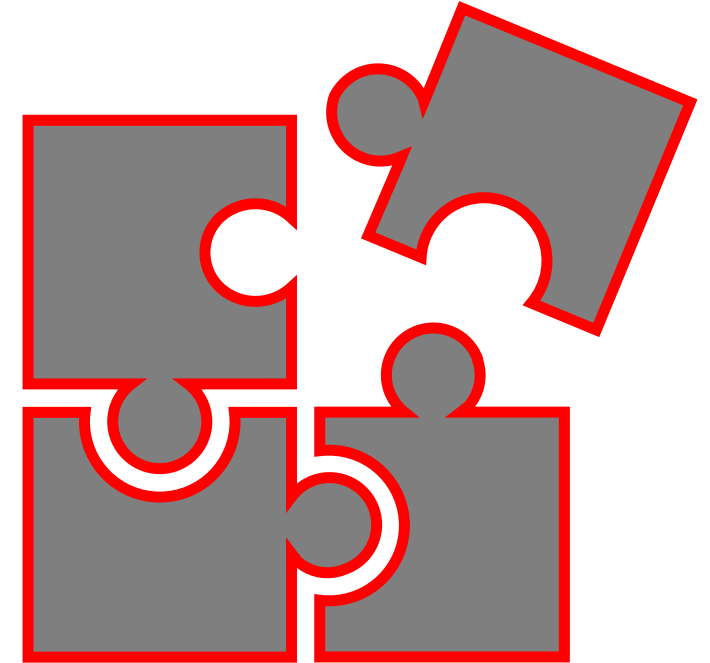
Take Aways

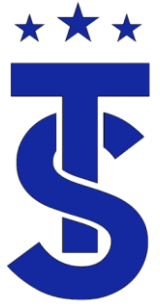
- Our industry is changing we need to adapt.
- Are you asking?
- Are you using the correct information for troubleshooting.



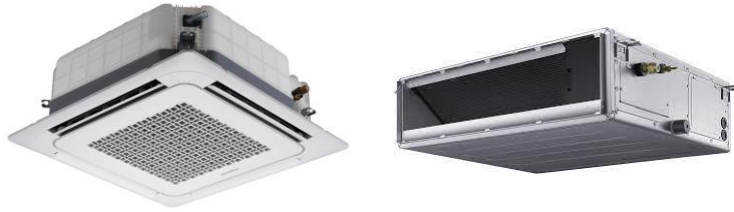
Overview

- This class will cover the basics of refrigeration and apply those concepts to troubleshooting ductless refrigeration circuits.
- We will review the components found in the ductless heat pump system individually and then will discuss how they interact with the system as a whole.
- We will discuss the sequence of operation, review the systems target operation parameters and discuss the control parameter for system components.

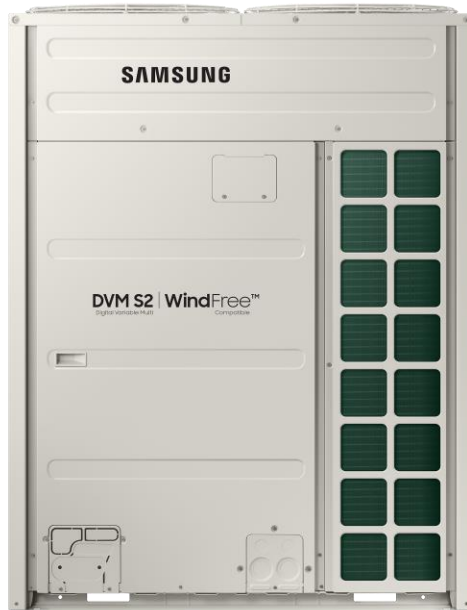




What do these have in common?



Ductless



VRF



Unitary A/C

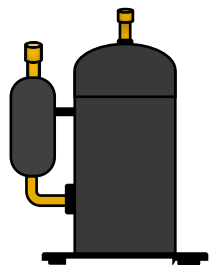


Refrigeration

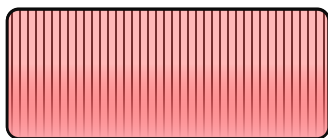


Basic Refrigeration

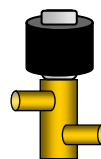
- All manufacturers are bound by the same Laws of thermodynamics and refrigeration theories when designing their equipment.
- A compressor, condenser, evaporator and metering devices functions do not change once applied to a VRF system.
- Understanding the basics of a refrigeration cycle is key to understanding a VRF system.
- Understanding the function of each component individually makes understanding the system as a whole easier.



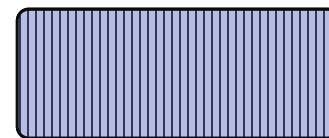
Compressor



Condenser



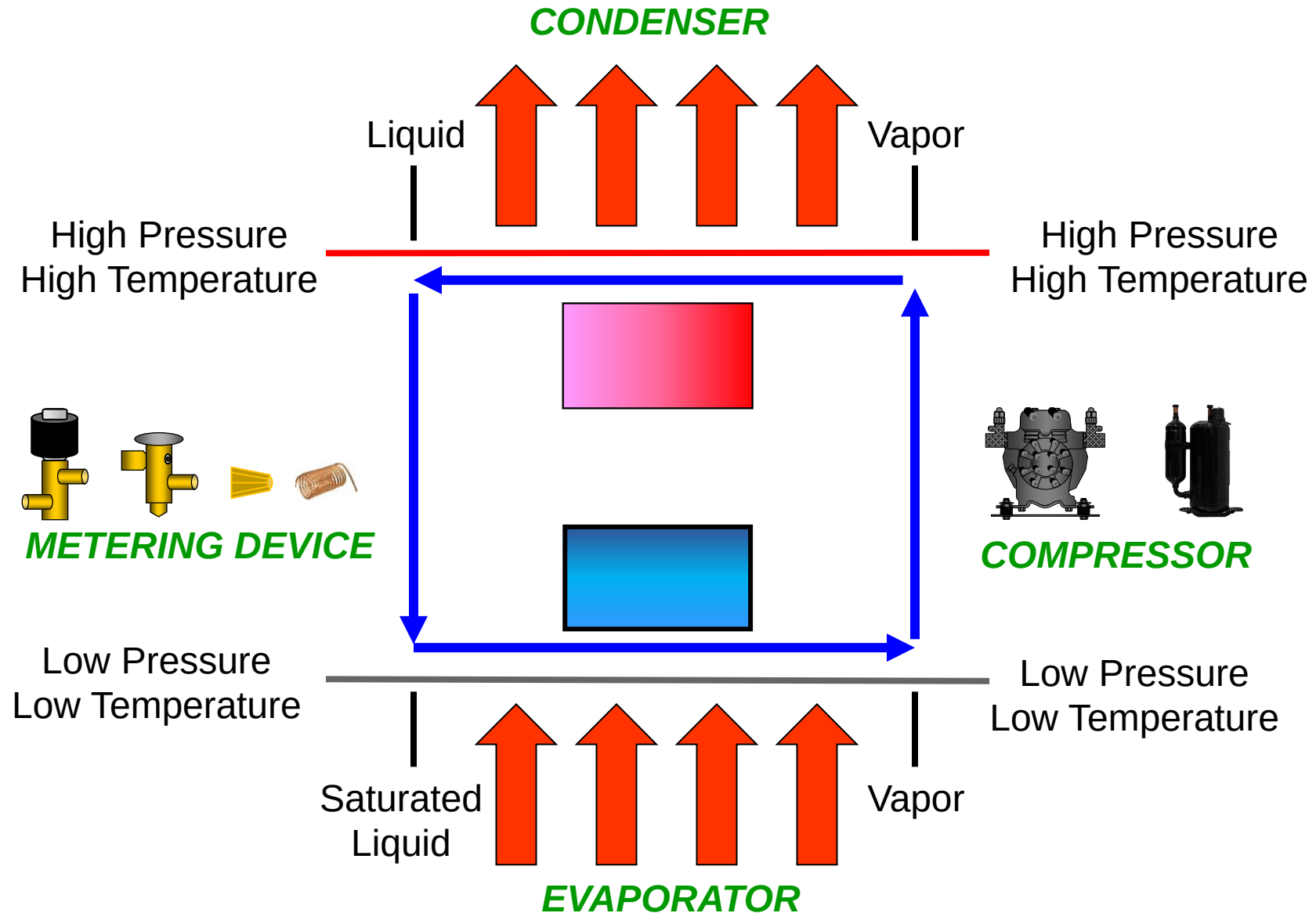
Metering Device



Evaporator

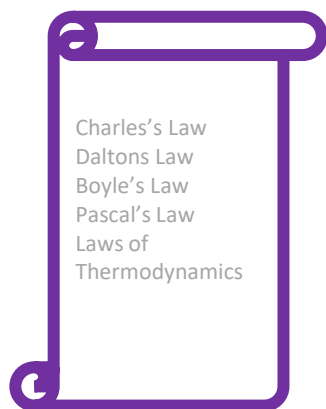


Refrigeration Cycle Review





The Science of Refrigeration



Theory



Chemicals



Mechanical

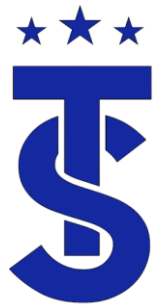


Science



Troubleshooting

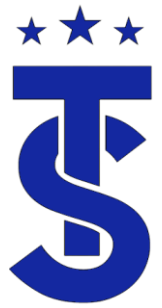
$$A + B = C$$



Review of The Basics

Gas Laws

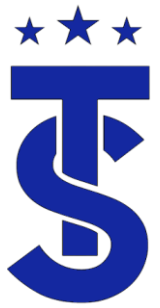
- When a gas is heated it expands. When it is cooled its volume is decreased.
 - The Relationship between Temperature and Volume
- As a gas is compressed its pressure increases
 - The Relationship between Volume and Pressure
- Total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of the gases in the mixture.
 - Characteristics of blended refrigerants
- The pressure of a fixed sample of gas is proportional to the temperature.
 - The relationship between temperature and pressure



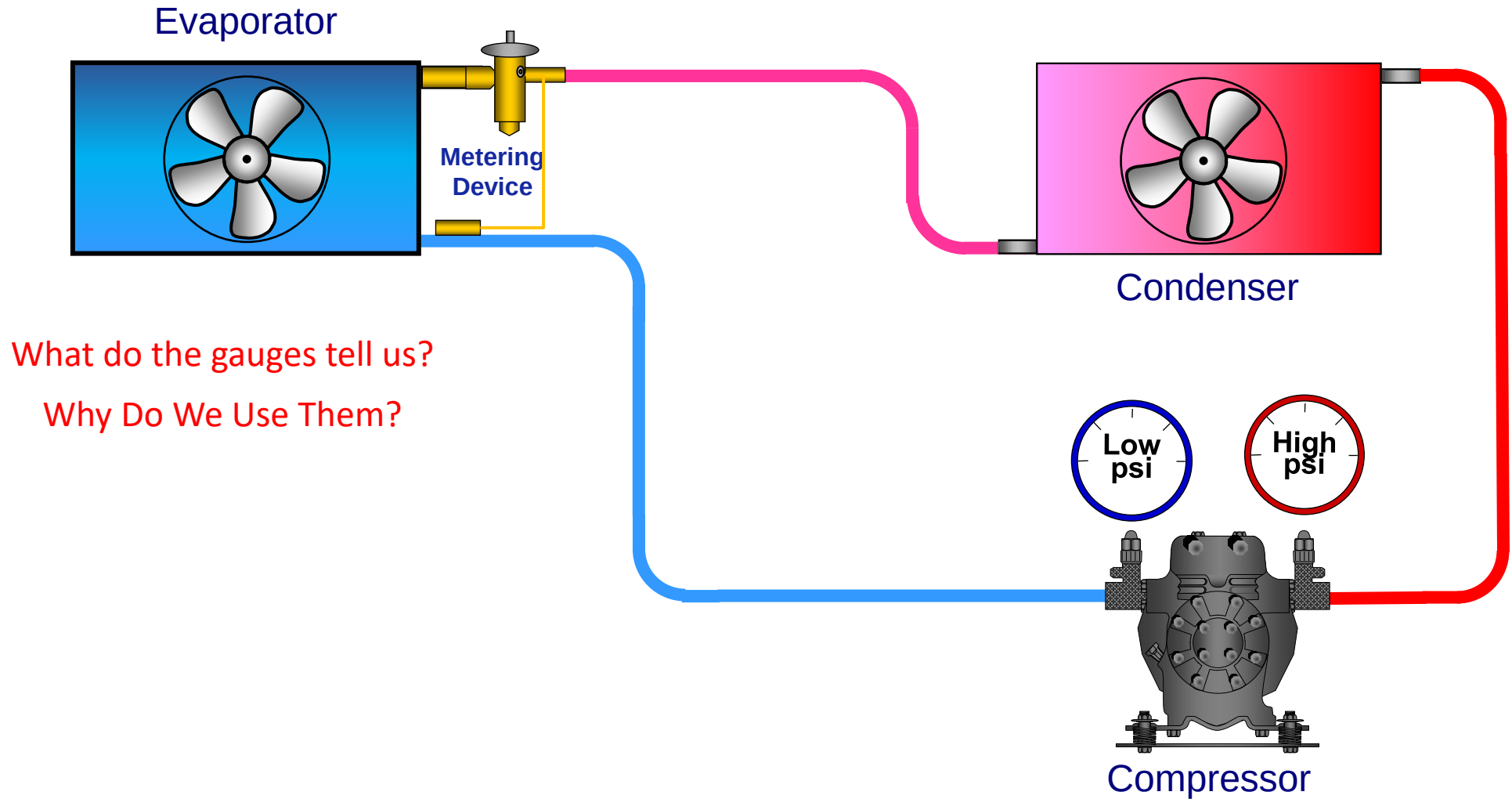
Review of The Basics

Who's Gas Laws? Who Cares?

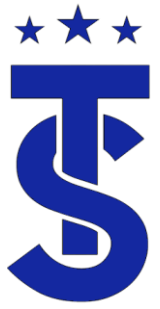
- When a gas is heated it expands. When it is cooled its volume is decreased.
 - The Relationship between temperature and Volume
 - Charles Law
- As a gas is compressed its pressure increases
 - The Relationship between Volume and Pressure
 - Boyles Law
- Total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of the gases in the mixture.
 - Characteristics of blended refrigerants
 - Dalton Law
- The pressure of a fixed sample of gas is proportional to the temperature.
 - The relationship between temperature and pressure
 - Gay-Lussac's Law



Gas Law Applications



What do the gauges tell us?
Why Do We Use Them?



Thermodynamics

- Refrigeration is based on the principle of thermodynamics.
- The laws of thermodynamics state:
 - Heat travels from hot to cold.
 - In order of heat to be transferred that there must be a difference in temperature.
 - The rate at which heat can be transferred depends on surface area, contact time and temperature difference.

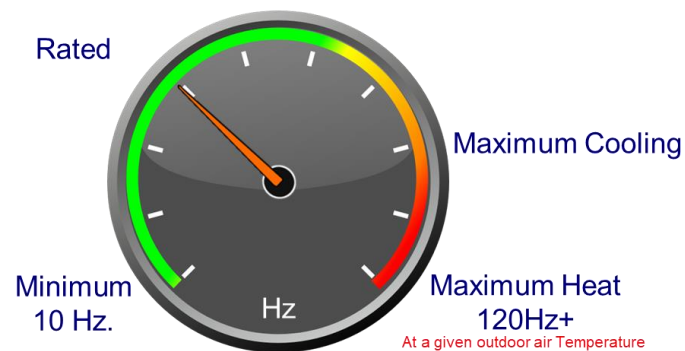
Always remember that refrigeration is based on temperatures NOT pressures

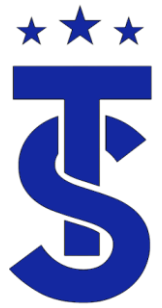


Contact Time

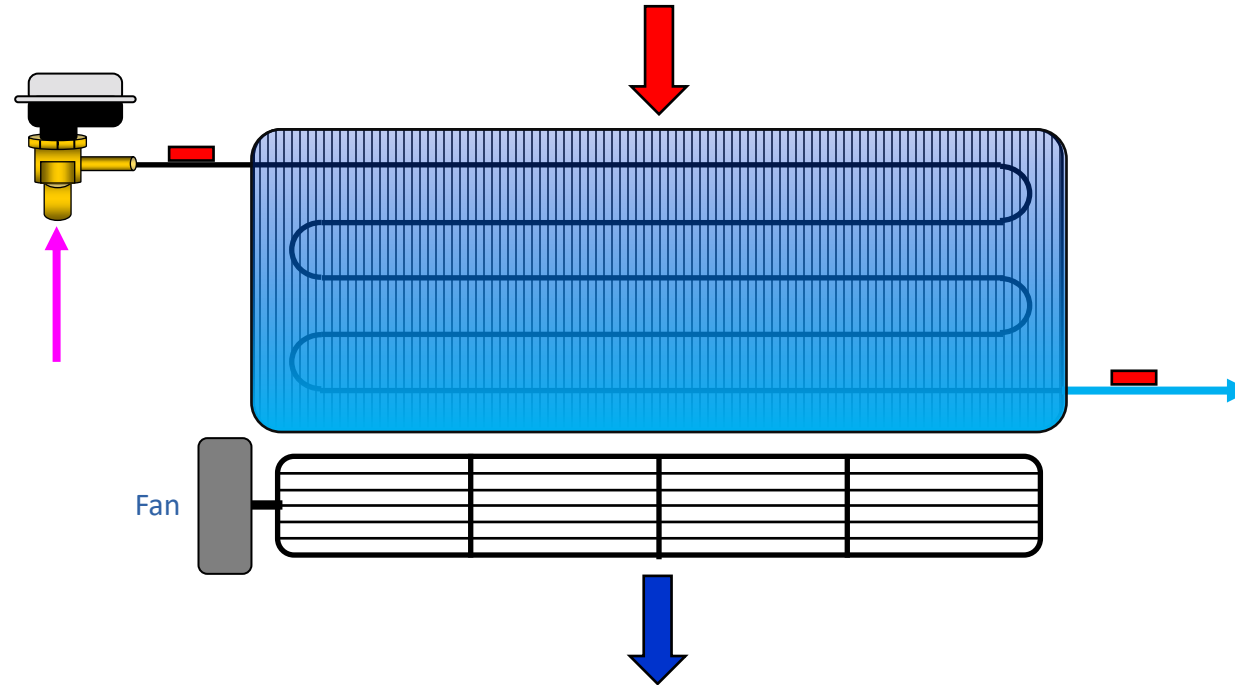
Variable Speed Compressor

- The speed of the refrigerant is directly related to the operating frequency.
- The lower the Hz the slower the refrigerant moves through the system. (more Contact Time with the coil)
- The higher the Hz the faster the refrigerant moves through the system. (less Contact Time with the coil)





Surface Area & Contact Time Air Flow



High efficiency coils are usually larger in size.

The CFM determines the amount of time the air is in contact with the coil.

There is no “standard” CFM when it comes to ductless heat pumps.

To Achieve proper heat transfer the CFM must be correct for the application.



Air Flow Example

- In order to achieve the proper operation of the system the air flow must be correct.
- Indoor unit capacities are given at high fan speed.
- Always check the specification for the proper air flow of each system.
- What effect does too much air flow have on the system?
- What effect does too little air have on the system?

How many CFM are required for these indoor units?

AJ018BNHDCH/AA

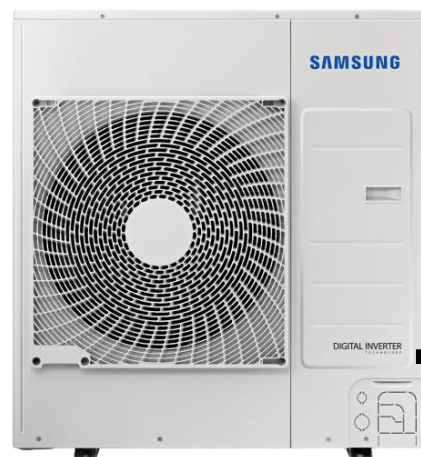
Airflow CFM:
L-346 M-417 H- 502



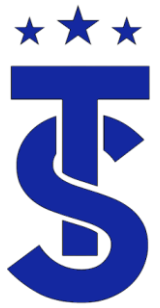
AR18BSFCMWKNCV



Heating (CFM):
L-465 M- 525 H-584 T-623
Cooling CFM:
L-401 M-435 H-469 T-486



AJ036BXJ4CH/AA



How does air flow effect efficiency?

- 2 ton systems



Air Volume	CFM (L/M/H)	466 / 537 / 629
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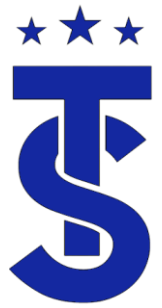
19.5 Seer



16.9 Seer

Air Volume	CFM (L/M/H)	547 / 636 / 759
------------	-------------	-----------------

The required air flow depends on the equipment and its efficiency!



Caution:

How much cooling will this system provided?

Is this a bad design?

What must be considered when servicing this system?



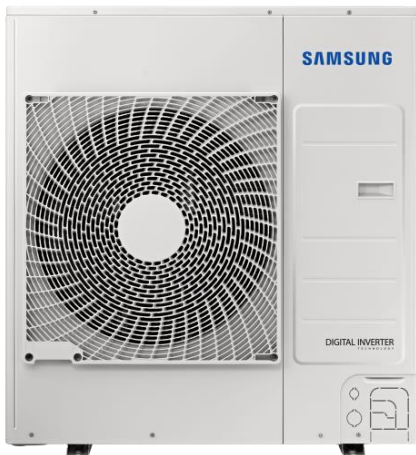
15K Indoor unit



15K Indoor unit

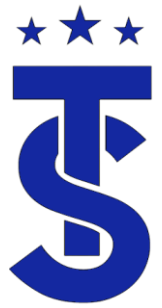


15K Indoor unit

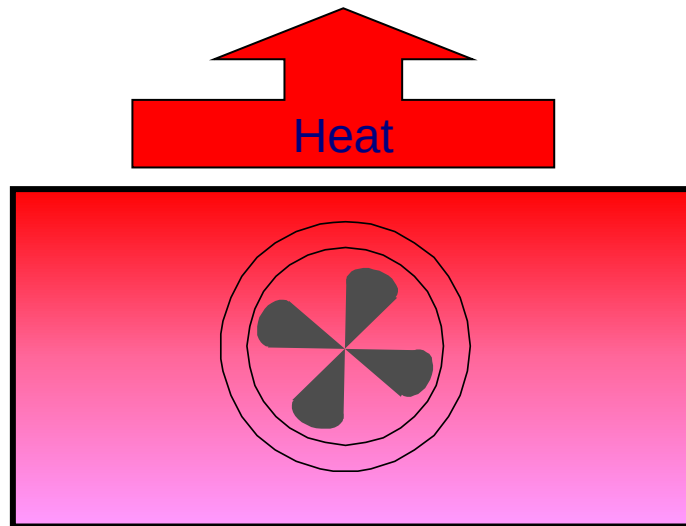


36 K outdoor unit

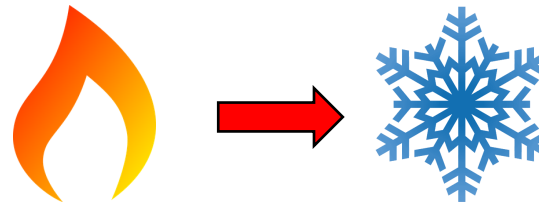
Do not test system with more air flow than the outdoor unit's capacity
What effect does too much airflow have on the system?



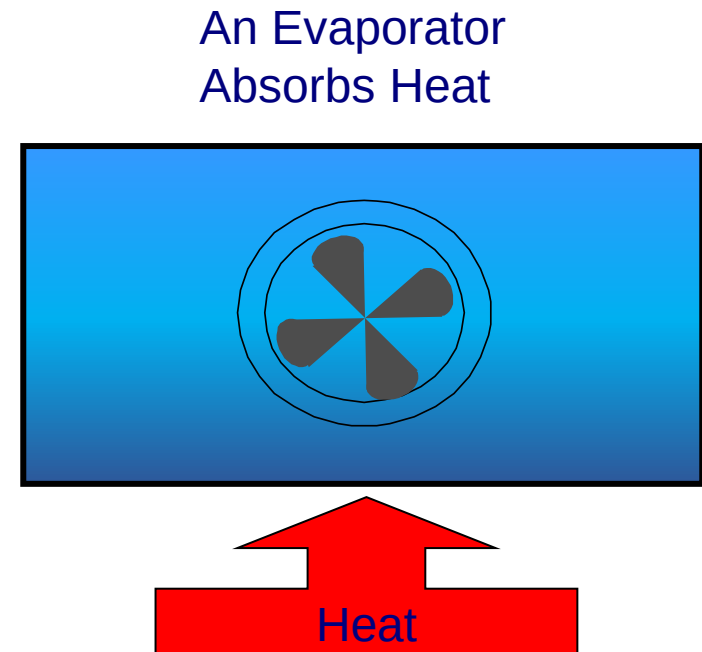
Heat Transfer



A Condenser
Rejects Heat



Heat Travels for hot to cold



An Evaporator
Absorbs Heat

What is a BTU?

What is Sensible Heat?

What is Latent Heat?

What is Specific Heat?



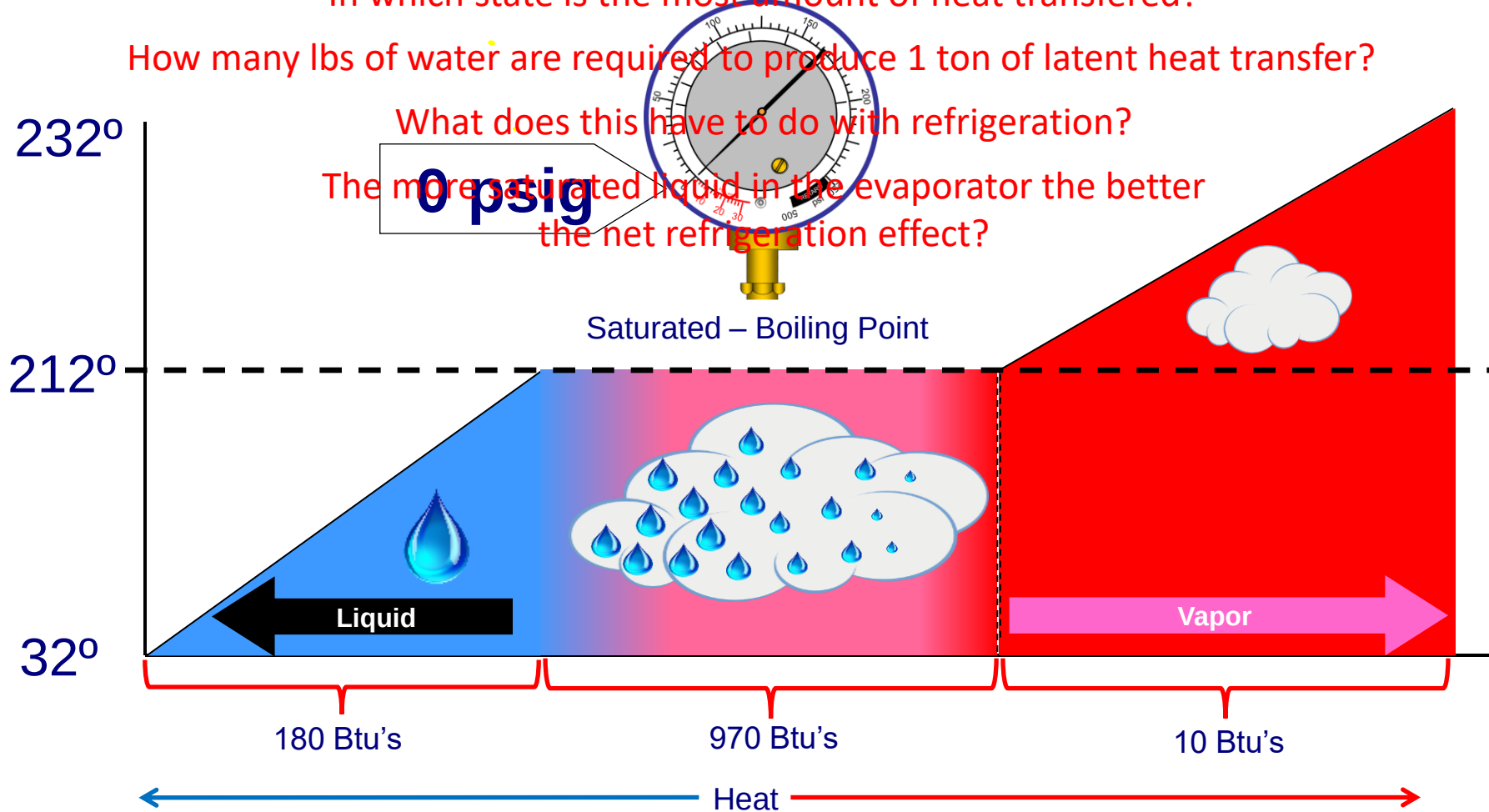
Heat Transfer Rates for Different States of Water

In which state is the most amount of heat transferred?

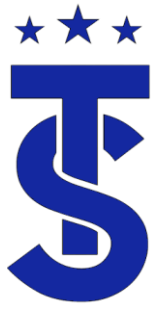
How many lbs of water are required to produce 1 ton of latent heat transfer?

What does this have to do with refrigeration?

The more saturated liquid in the evaporator the better the net refrigeration effect?



12.37 Lb.
198 Oz.
3.1 Gal.



Heat Transfer

Surface Area & Temperature Differences



10 SEER
30° TD

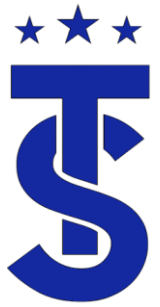


14 SEER
20° TD



20 SEER
10-15° TD

A larger Volume of the outdoor unit reduces the high side pressure
Larger coils allow more heat transfer with lower splits



Temperature Differences & Head Pressures

(A+B=C)

90°-95° Saturation Temp

110° Saturation Temp

100° Saturation Temp



20 SEER
10-15° TD



14 SEER
20° TD

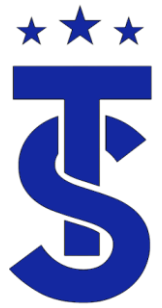


10 SEER
30° TD



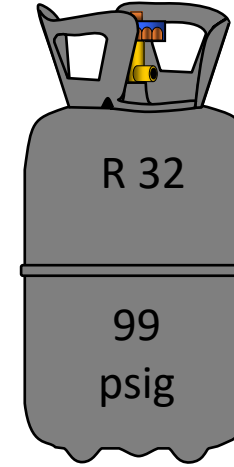
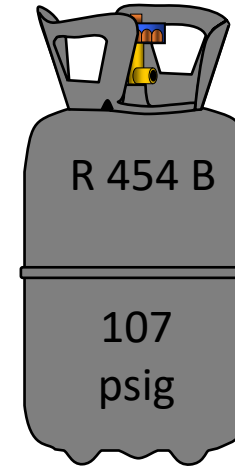
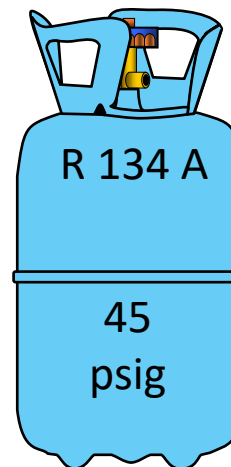
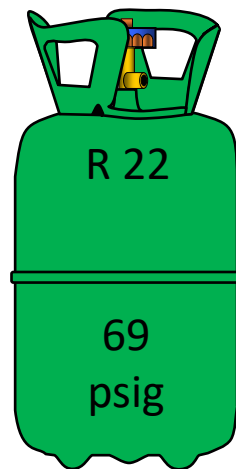
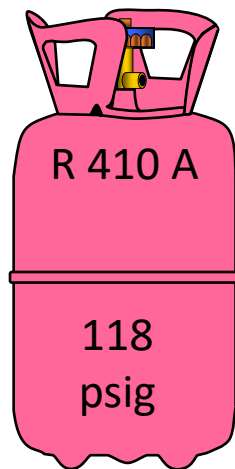
80° Outdoor Air
Temperature

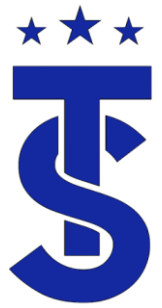
Head Pressure



What is a normal low side pressure for air conditioning?

- The better question is what is a typical evaporator saturation temperature?
- A “typical” evaporator temperature for unitary air condition is 40 degrees
- At a temperature of 40 degrees the low side pressure is dependent upon the refrigerant being used





What Effect Does Lower Head Pressures Have on a System?

A typical evaporator – low side saturation temperature is 40° which has a corresponding pressure of 118 psig.



10 SEER
30° TD



20+ SEER
10° TD

The difference between high side and low side pressures is the compression ratio. High compression ratios cause higher heat of compression. High compression ratios require more energy to operate the compressor.

Which Compressor Has to work harder ?
Which Uses More Energy and why?



Refrigerants





Properties of Water

- Water has the same characteristics as refrigerants.
- The boiling temperature of water changes with the pressure it is under.
- The higher the elevation the lower the boiling point.
 - The boiling point of water in Denver is 202°F
- It can be evaporated or condensed.
- Because of these similarities we will use water as an example refrigerant and discuss its properties.

Pressure	Boiling Point
30 psig	271°
12 psig	242°
0 psig	212°
25,000 Microns	80°
500 Microns	0°



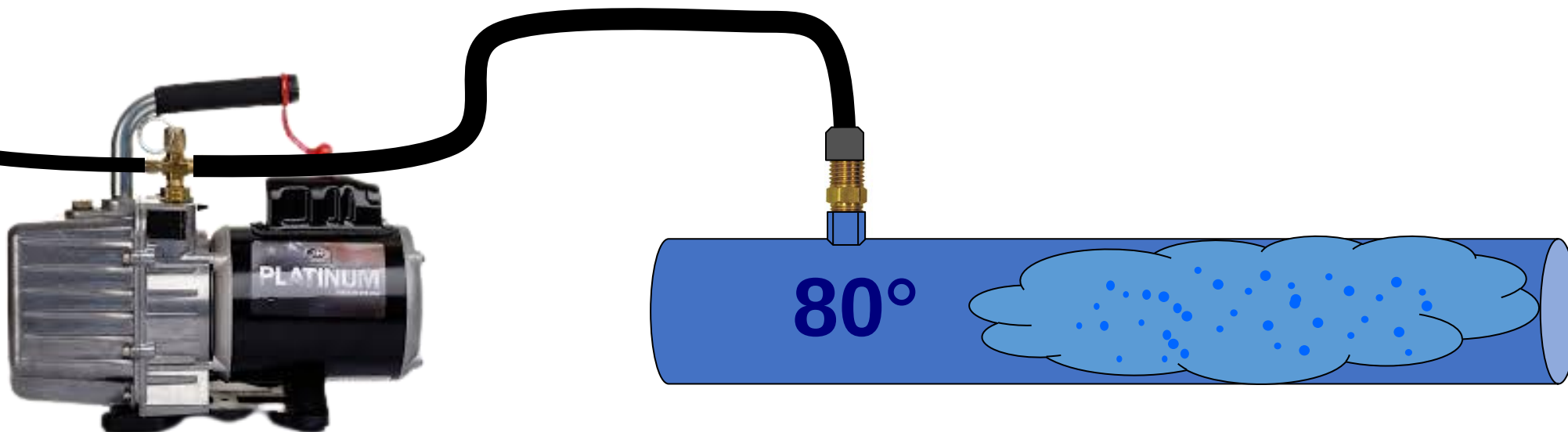


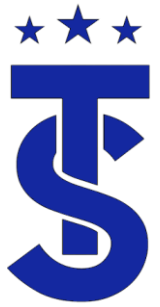
Why do we evacuate a system?



<u>Microns</u>	<u>Boil Point</u>
760,000	212°F
25,000	80°F
500	0°F

Water has a boiling point of 0 degrees at 500 microns. What this means is if there is moisture in the pipe and the pressure within the pipe is at 500 microns the water within pipe boils to vapor allowing the pump to remove it from the system.



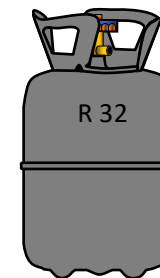
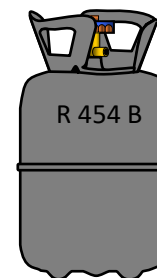
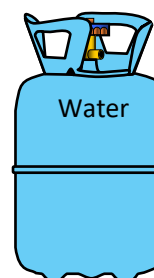
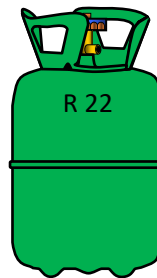


Refrigerant

Are R-454B and / or R-32 The Future? What Changes?

- A fluid that picks up heat by evaporating at a low temperature and pressure and gives up heat by condensing at higher temperature and pressure.
- Refrigerants have a specific temperature / pressure relationship.
- Knowing one you can find the other using a P/T chart.
- Refrigerants can exist in one of three states :
 - Saturated
 - Superheated
 - Subcooled

Refrigeration is based on
TEMPERATURE not pressures



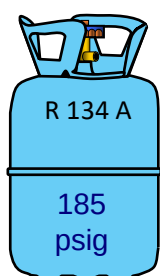
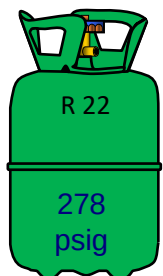
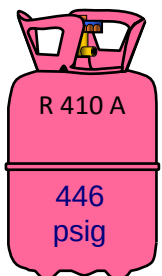


Refrigerants

Pressure Temperature Relation

Refrigerants have a specific pressure-temperature relationship

Pressure Temperature (PT) Charts show the relationship between the pressure and saturation temperature (boiling point) of a refrigerant.



125°

PSIG	134a	22	410A
180	123	94	63
185	125	96	65
190	127	98	66
195	128	100	68
200	130	101	70
210	134	105	73
220	137	108	76
230	140	111	79
240	143	114	82
250	146	117	84
260	149	120	87
270	152	123	89
280	155	126	92
290		128	94
300		131	96
325		137	102
350		143	107
375		149	112
400		155	117
425			121
450			126
475			130

Saturation Temperatures

Example:

The condenser coil is designed to transfer the proper amount of heat if the refrigerant temperature is 125 degrees.

What would the high side pressure be if the following refrigerants were used:

R 410A

R 22

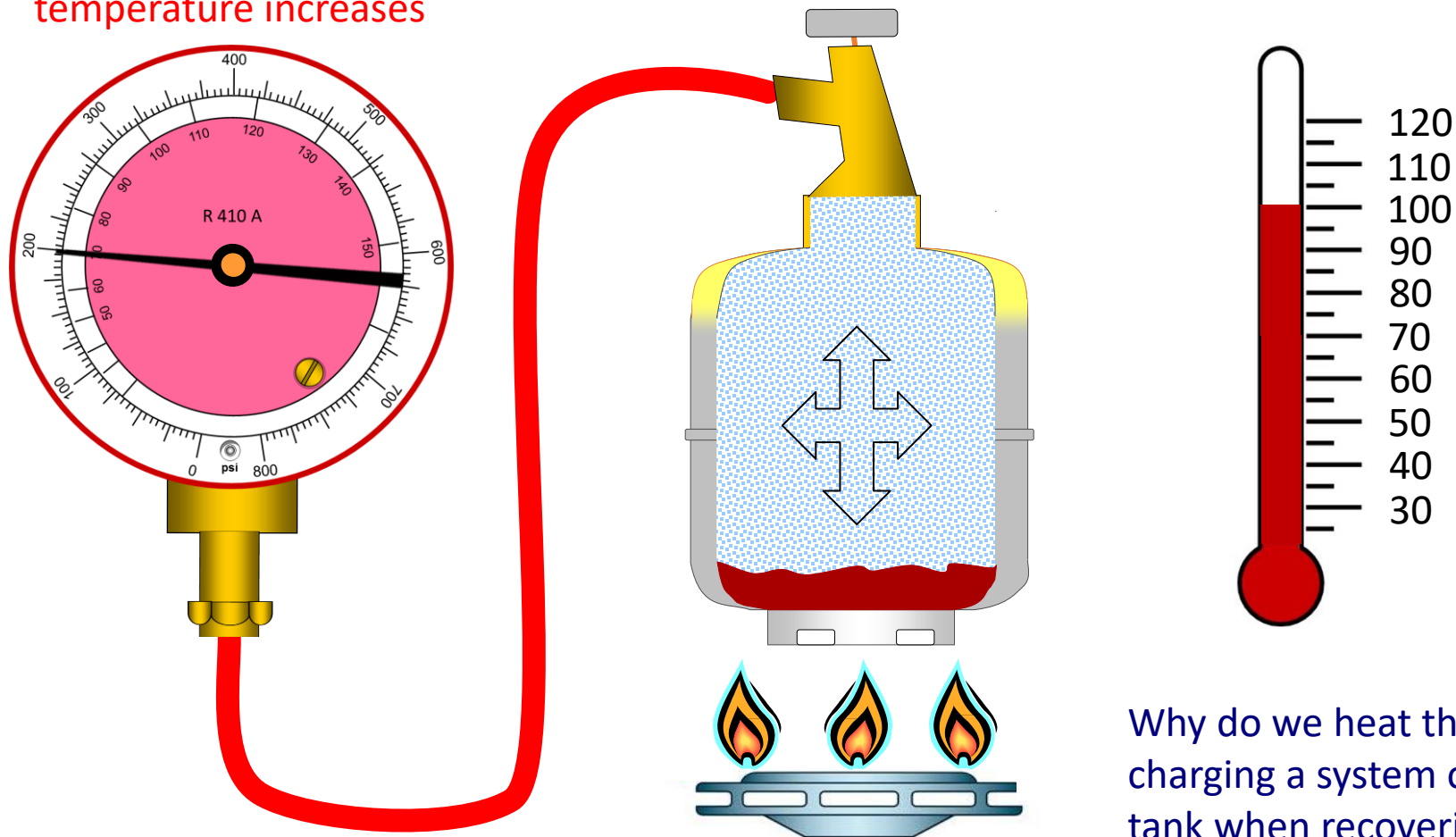
R 134A

Remember that refrigeration is based on Temperature NOT pressure.

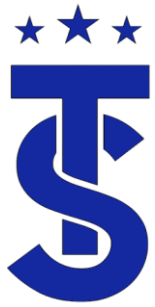


Pressure Temperature Relationship Example

Tank Psi increases as
temperature increases

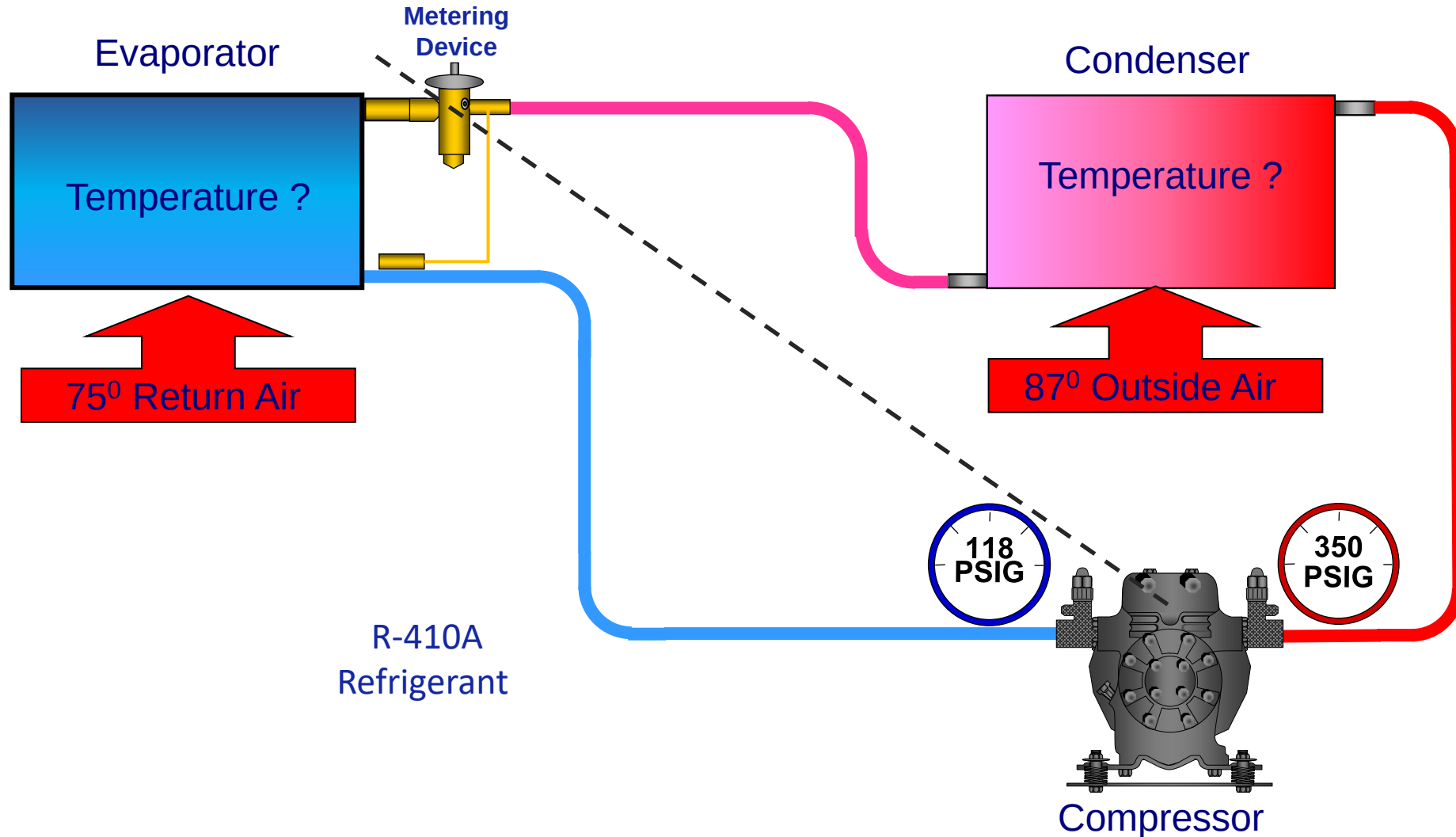


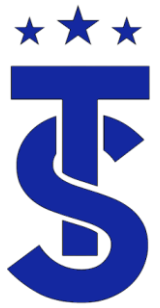
Why do we heat the tank when we are
charging a system or why do we cool a
tank when recovering refrigerant?



What Do Gauges Tell Us?

System Pressures and Temperatures





States of a refrigerant

(Water as an example at 0 psig / sea level)



Liquid
Sub-Cooled

$< 212^{\circ}$



Mixture of Liquid and Vapor
Saturated

212°



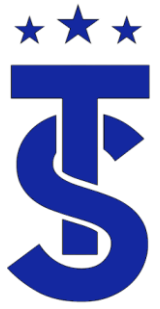
Vapor/Gas
Super Heated

$> 212^{\circ}$

Saturation

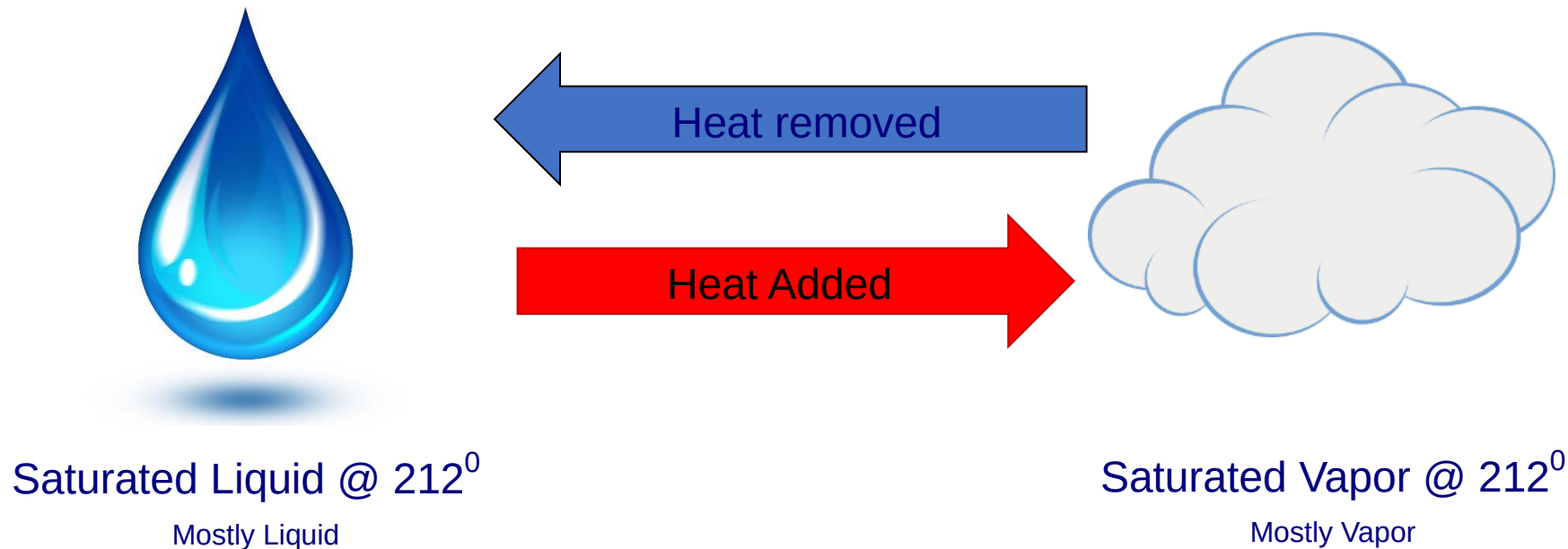
- Saturation Temperature is the temperature where liquid and vapor can exist at a given pressure.
- Also referred to as the boiling point of a refrigerant.
- Saturated refrigerant normally exist in the condenser and evaporator.
- May be called the condensing temperature or evaporator temperature.
- The saturation point is directly related to the pressure.
 - The lower the pressure the lower the boiling point.

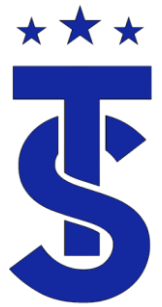




Saturation

- The point where liquid and vapor exist and where heat added or removed will cause a change of state



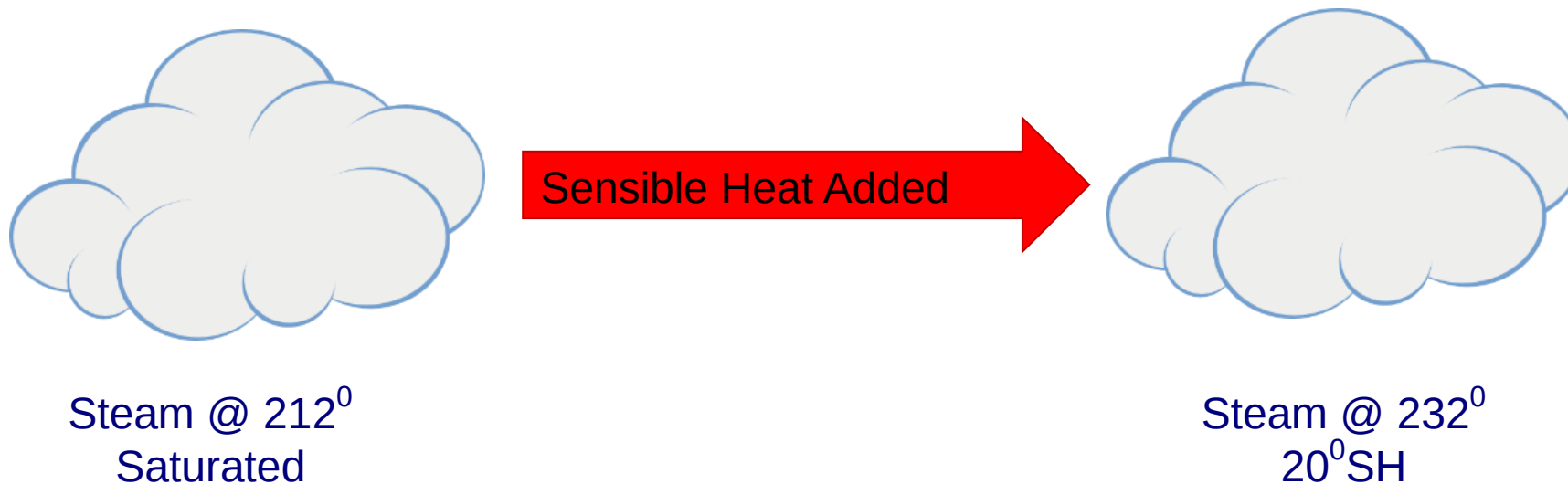


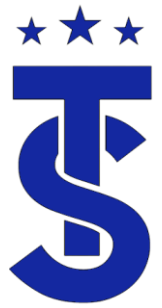
Superheat

- Superheat refers to the temperature of the refrigerants above its saturation temperature for the pressure it is under.
- Is an indicator of how well the evaporator is working.
- If a refrigerant is superheated no liquid is present.
 - Only a gas can be superheated
- Superheated gas normally exist in the suction and discharge lines.

Super Heat

- Sensible Heat added to a vapor above its boiling point



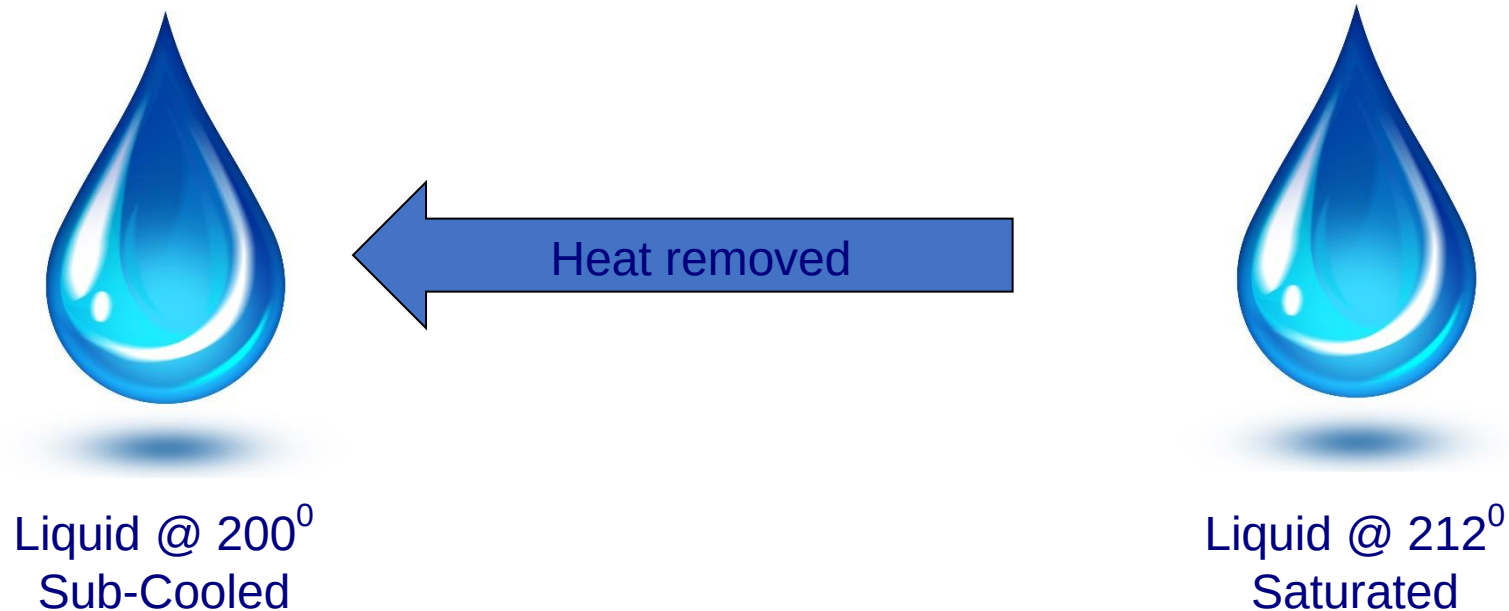


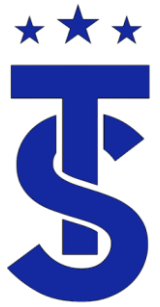
Subcooling

- Sub-cooled refers to the temperature of the refrigerant below its saturation temperature for the pressure it is under.
- If a refrigerant is sub-cooled no vapor is present.
 - Only a liquid can be sub-cooled.
- Is an indicator of the amount of charge in the system.
- Sub-cooled liquid normally exist in the liquid line.
- Why do ductless units require greater subcooling?

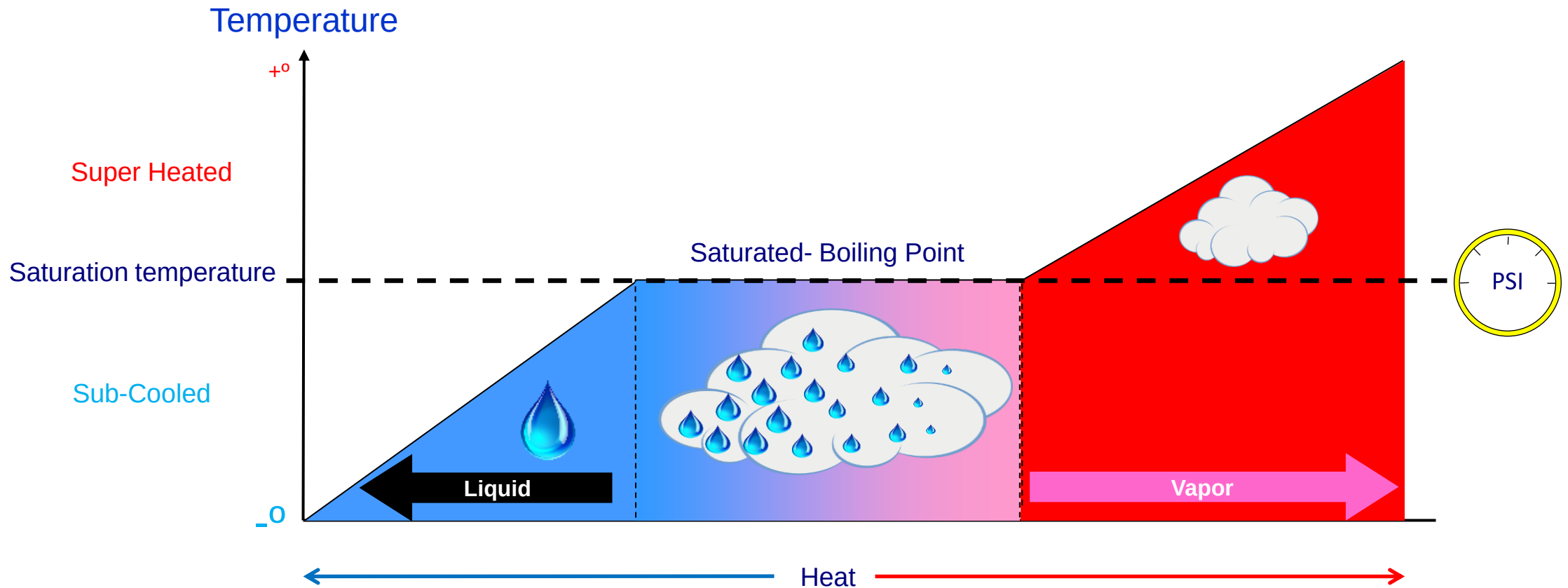
Sub-Cooling

- Sensible Heat removed to a lower a liquid below its boiling point





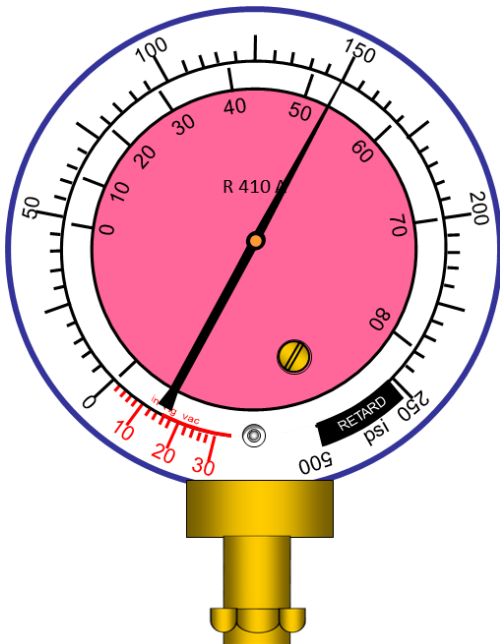
States of Refrigerants

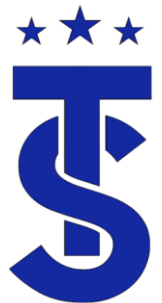


Application

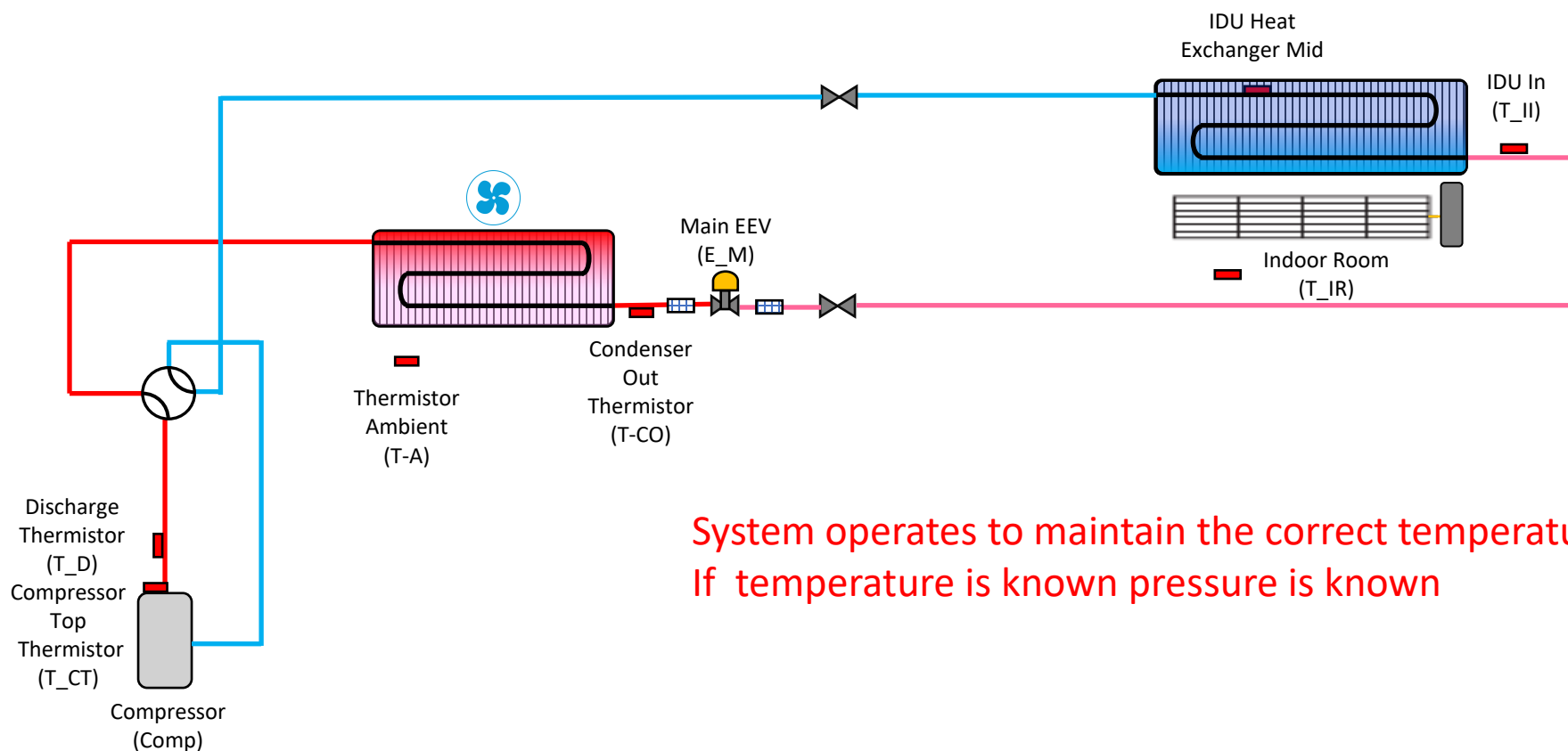
- The IDU Pipe In temperature is 53 degrees.
- The IDU Pipe Out temperature is 55 degrees

What is the pressure at the IDU?





Why are there no pressure gauges in a ductless system?

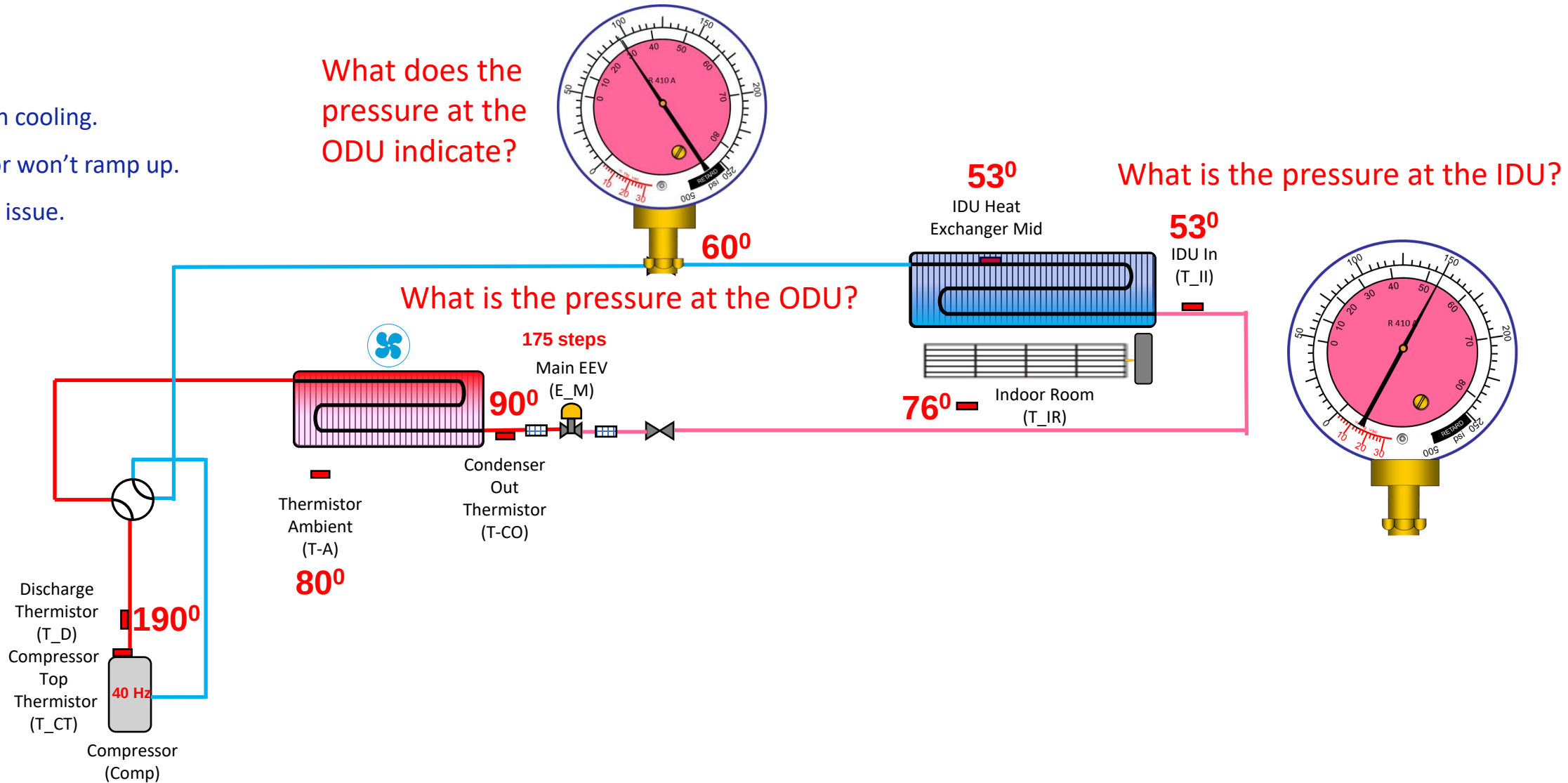


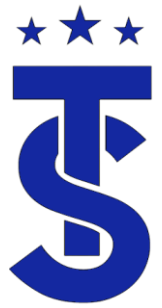
System operates to maintain the correct temperatures.
If temperature is known pressure is known



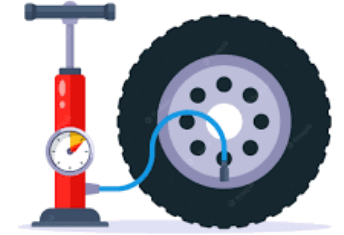
Application

- Not enough cooling.
- Compressor won't ramp up.
- What's the issue.

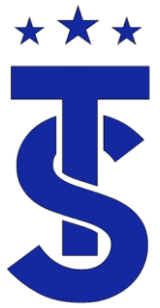




Why is charge so critical?

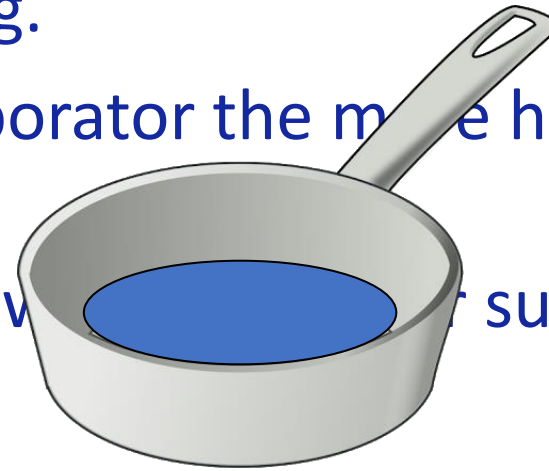


- Systems are designed with a specific refrigerant charge.
- Having the correct charge ensures you transfer the correct amount of heat.
- Changing the charge changes the pressure in the system in effect changing the temperatures.
- Too much charge causes the pressures to rise.
- Too little caused the pressures to drop.



What's the use of all of this??

- Having a good understanding of refrigerant properties is essential to understanding what the system is doing.
- The more liquid refrigerant in the evaporator the more heat that can be removed from the space.
- This is why efficiency units have low superheats and high subcools?



Which pot requires more heat to boil all the water?

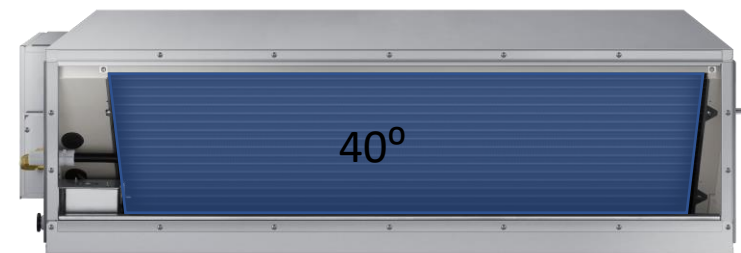
Why?



R 410A to R32

What Changes?

- Refrigeration is based on the transfer of heat.
- A quantity of heat is temperature.
- The design temperatures for air conditioning have remained the same since the beginning.
- A 40° evaporator is a common for air condition
- The only thing that changes is the operating pressure because that is dictated by the refrigerant in the system



R 410 A Pressure Temperature Chart			
°F	°C	PSI	KPA
13.8	-10.1	68	468.8
17.5	-8.1	74	510.2
21	-6.1	80	551.6
24.3	-4.3	86	592.9
28.5	-1.9	94	648.1
32.2	0.1	102	703.3
35.5	1.9	110	758.4
39.5	4.2	118	813.6
40.5	4.7	120	827.4
43	6.1	126	868.7
46.3	7.9	134	923.9
50.3	10.2	144	992.8

R-32 Pressure Temperature Chart			
°F	°C	PSI	KPA
14	-10	69.79	481.2
17.6	-8	75.81	522.7
21.2	-6	82.15	566.4
24.8	-4	88.82	612.4
28.4	-2	95.84	660.8
32	0	103.21	711.6
35.6	2	110.95	765
39.2	4	119.07	821
42.8	6	127.58	879.6
46.4	8	136.49	941.1
50	10	145.81	1005.3



R 22



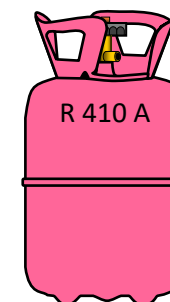
R 410 A

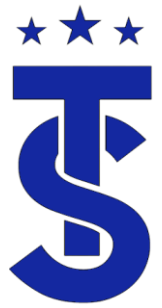


R 32



R 454 B





What Changes are coming R – 32

- Is an A2L refrigerant
- Requires service tools rated for A2L refrigerants.
- System service valves will require Left-Handed Thread adapters or Hoses.
- Samsung will continue to use 5/16 connectors.
- The troubleshooting process for system with R-32 does not change!
- It is the responsibility of the contractor to get educated on the new refrigerants

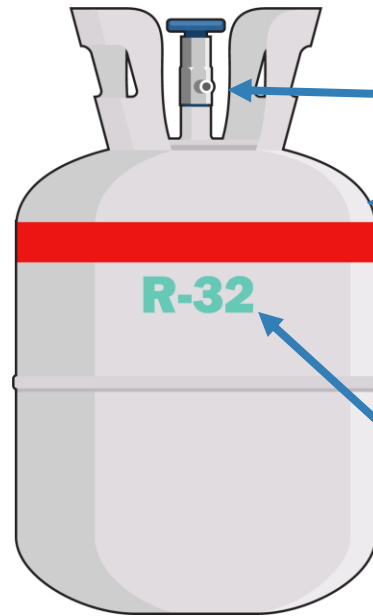


A2L connectors will have a double knurled rings on the fitting

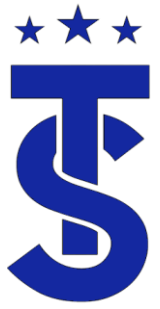




A New Look

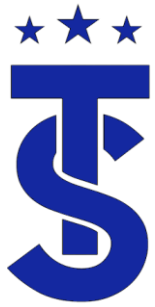


- Reverse threads on port
- No tank color designation
- Added red band to identify A2L refrigerant
- Refrigerant type may follow traditional color code



Working A2L Refrigerant

- Vacuum pumps, recovery machines and Leak detectors must be rated for A2L Refrigerants
- Leak detectors should be a ultra sonic or infrared type.
- Existing Gauge sets can be used
- Nitrogen MUST be purged anytime a torch is applied during the installation and service process.
- System must be leak checked after installation
- System must hold a vacuum during installation.

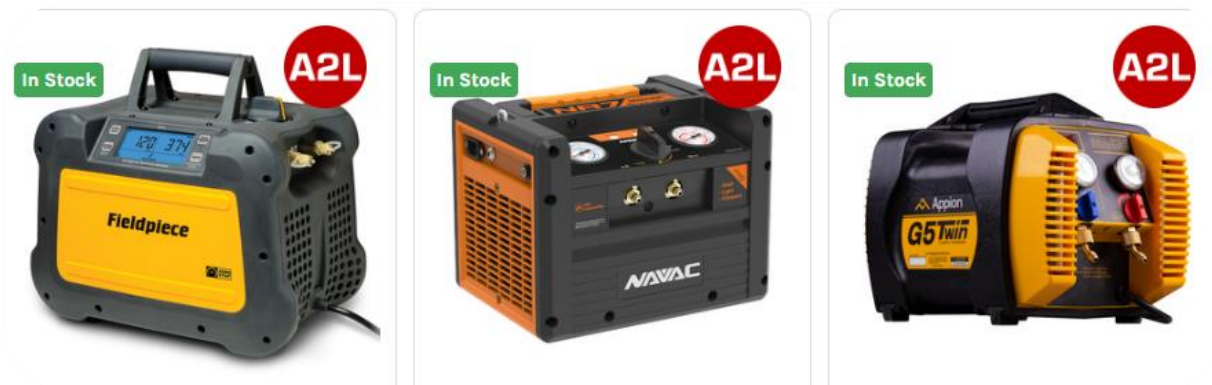


New A2L Service Tools

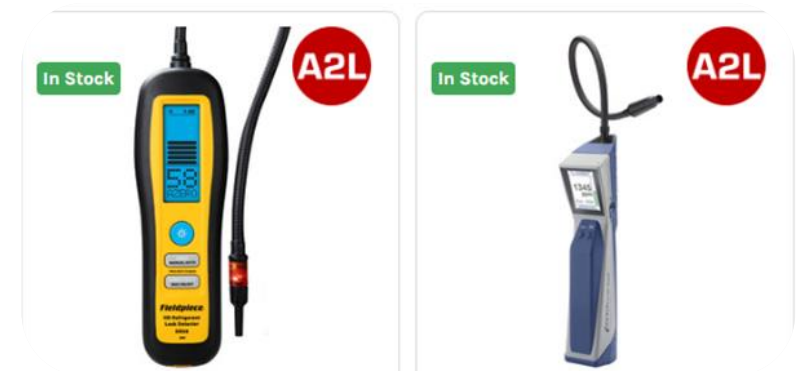
Vacuum Pump



Recover Machine



Leak Detector





UL Service Requirements

- A2L refrigerants must be recovered.
- Tubing may be un-brazed.
- Ventilation required during work. It is assumed that the fan should be UL507 listed.
- Pressure tests must be conducted after repair
- It is also required to perform a leak test on the system after repairs have been made





IDU's With Refrigerant Detection

R-32 FJM Indoor Units

RDS = Installed at factory

RDS = Field installed accessory



FJM - Only

RDS
WindFree™ Standard(w/ RDS)

RDS
Duct S

RDS
MPAH

RAC - Shared

WindFree™ Premium

WindFree™ Standard

WindFree™ IAQ

CAC - Shared

RDS
1-Way

RDS
WindFree™ Mini 4-Way

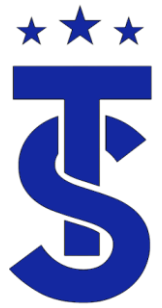
RDS
Slim Duct

RDS
Console



The Refrigeration Cycle and Components





Variable Speed Vs Unitary

What the difference?

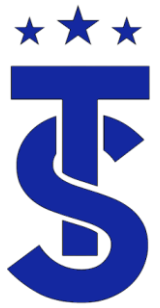


- Same
 - Evaporator
 - Condenser
 - Compressor
 - Metering device

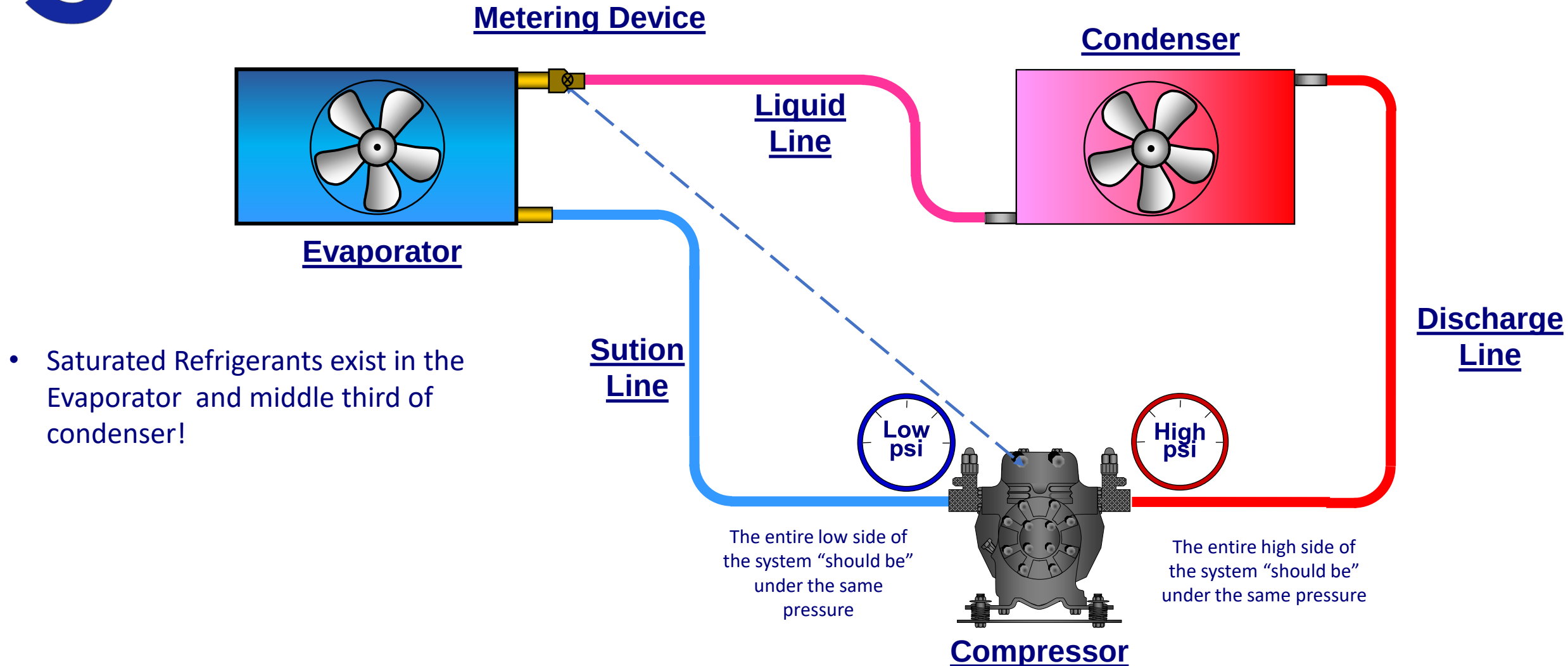


- Differences:
 - Variable Speed Compressors
 - EEV's
 - Sensor Controlled





Conventional Refrigeration Cycle Review



The Compressor

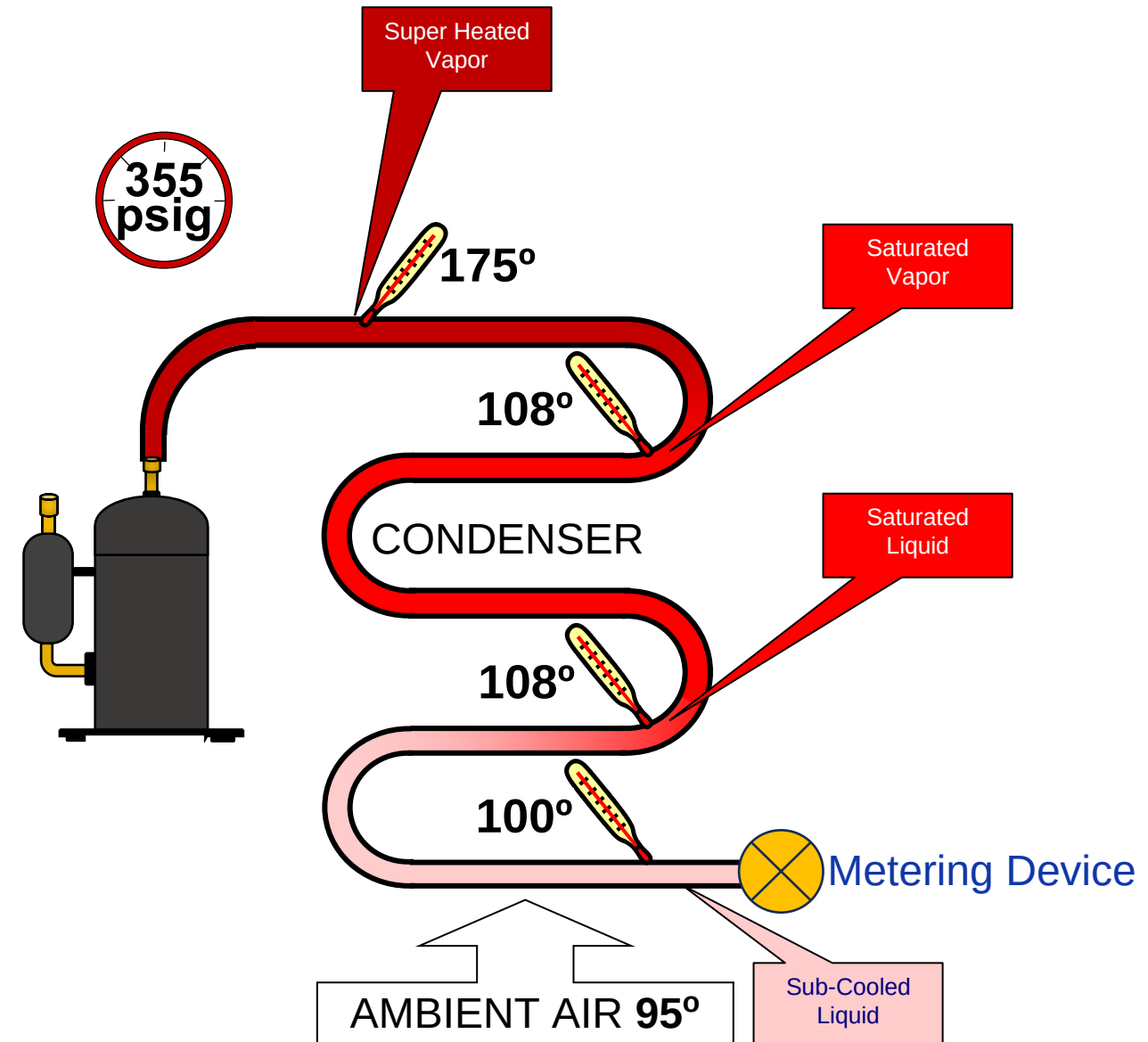
- It circulates refrigerant through the system.
- Gas enters through the suction side of the compressor and leaves through the discharge line.
- It compress low pressure gas to high pressure gas.
- The compression ratio is the difference between the low side and high pressures
- Compressors are more efficient when it operates as low compression Ratios.
- The discharge temperature of the compressor is directly related the temperature of the suction gas.

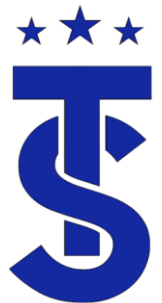




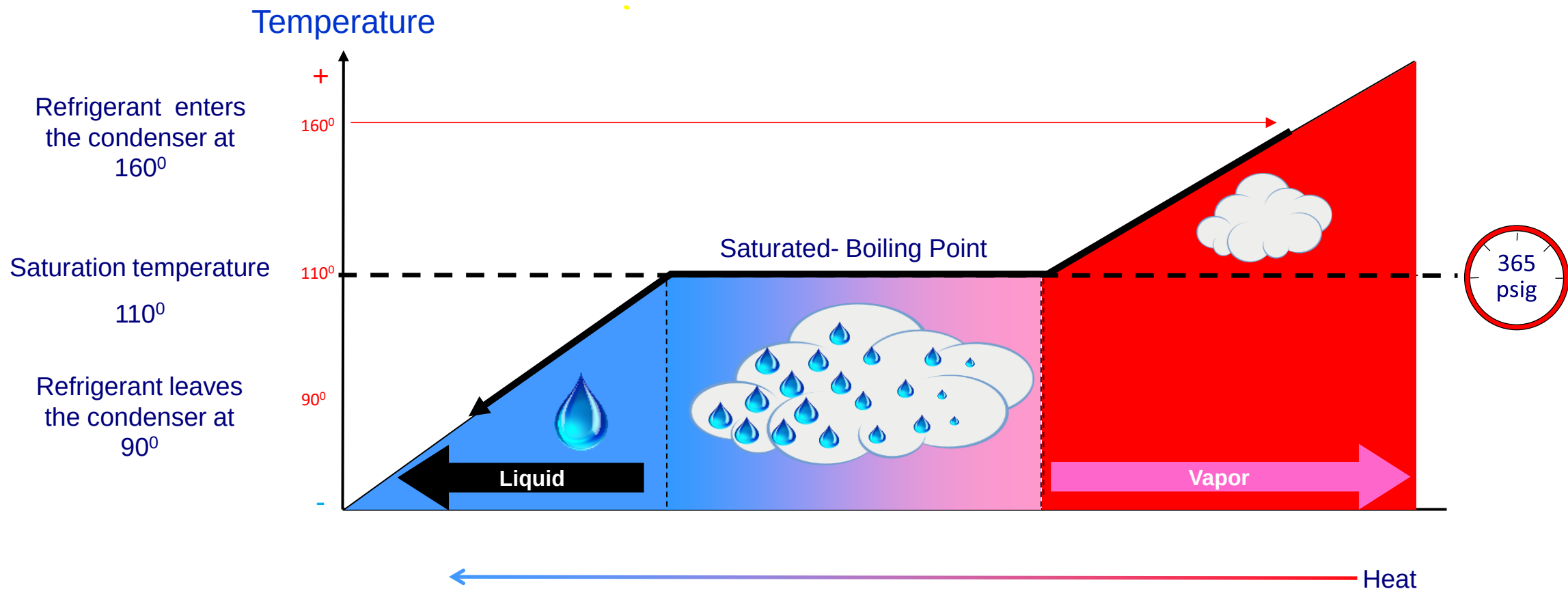
The Condenser

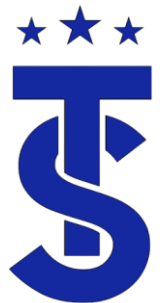
- Its function is to remove the heat from the superheated refrigerant and in doing so changes the vapor refrigerant into a sub-cooled liquid.
- The Condenser HAS TO BE HOTTER than the air going across it for heat to be transferred.
- Does its function change in a heat pumps system?





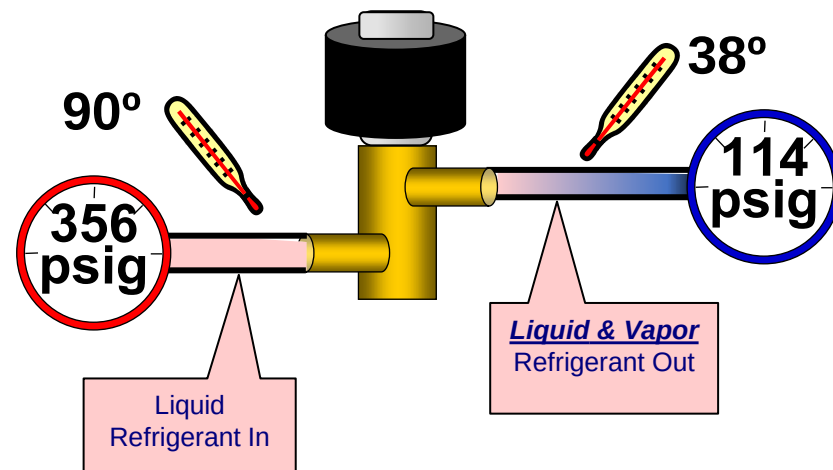
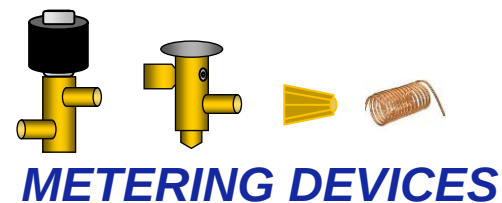
Example- Refrigerant flow through a condenser





Metering Device Operation

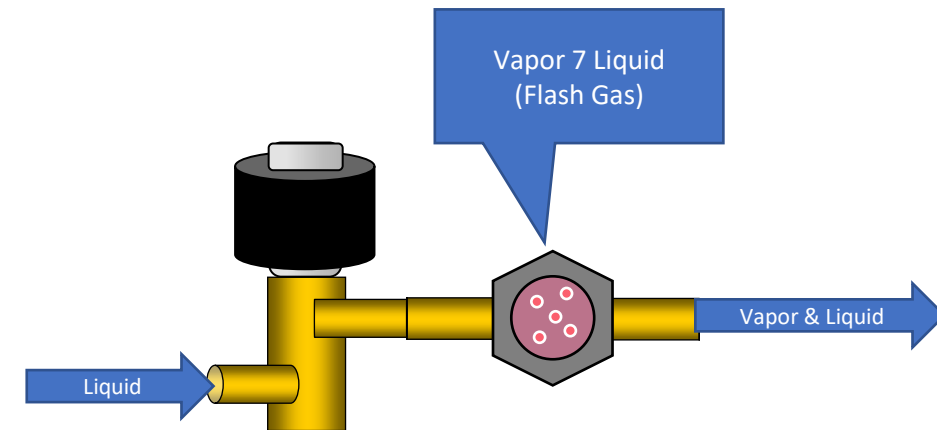
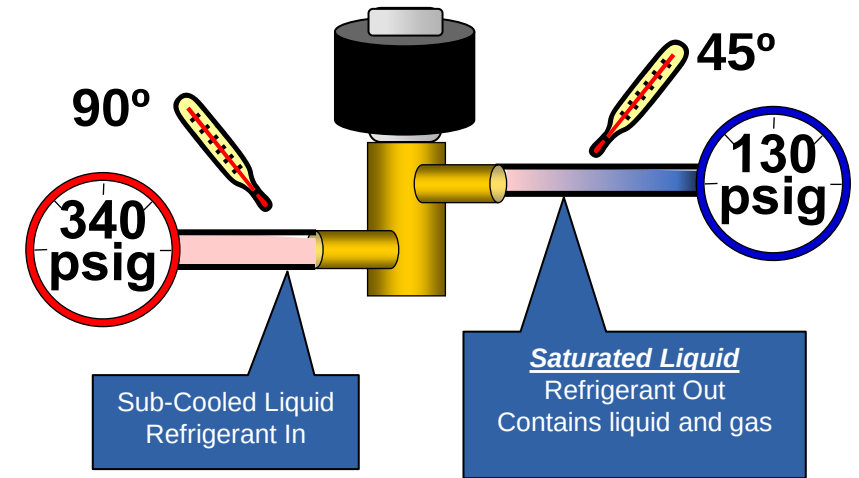
- It creates a pressure drop.
- It is designed to take Condensing pressure (temperature) and lower it to a design evaporator pressure so that its corresponding temperature is below the design space temperature
- It regulates flow to maintain a design superheat through the component.



The refrigerant leaving the metering device is **MOSTLY** liquid.

Flash Gas

- When refrigerants change pressure the temperature at which it boils changes.
- A metering device changes the pressure and boiling point of the refrigerant.
- The refrigerant leaves the metering device as a liquid rich refrigerant that contains some gas (saturated).
- This gas is referred to as flash gas.

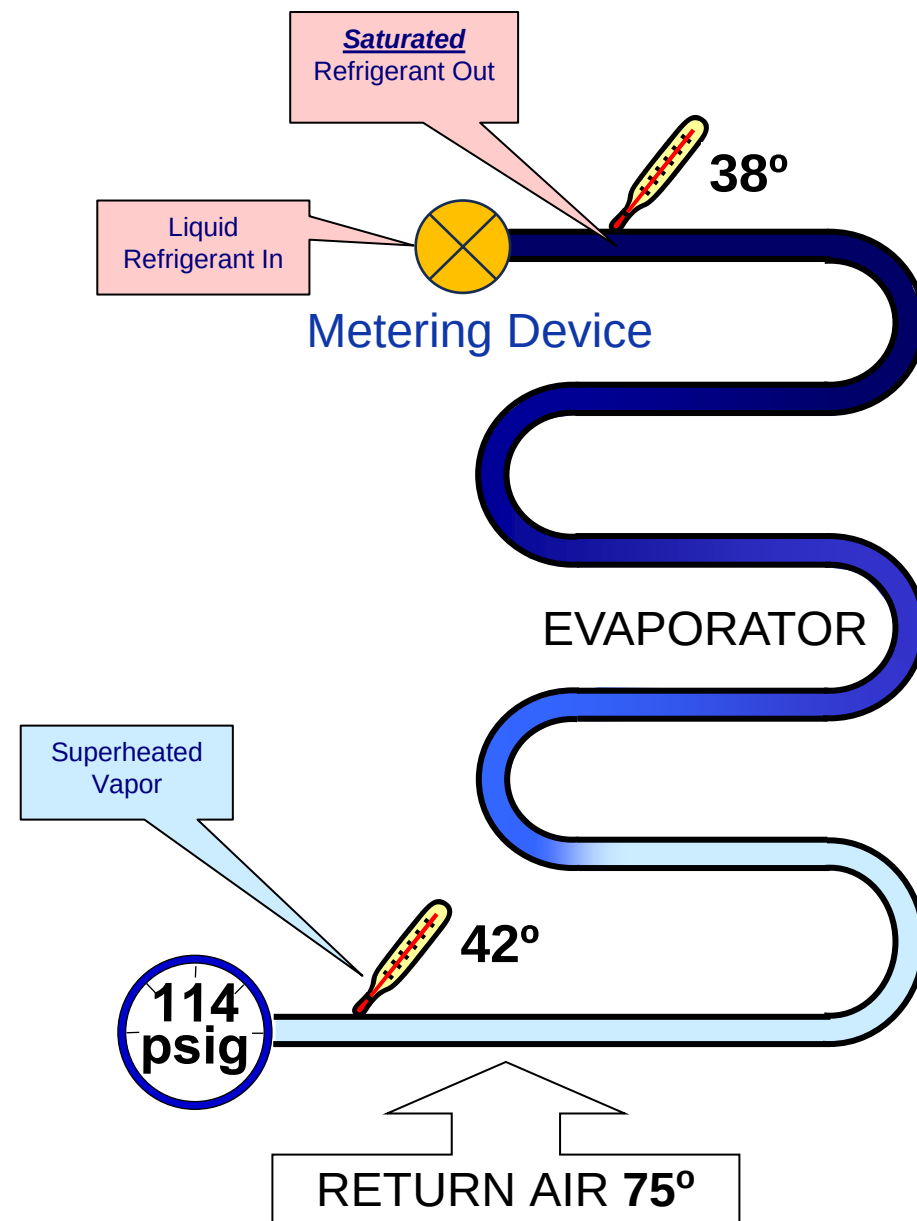


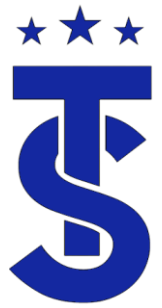
Flash gas reduces the capacity of the evaporator.



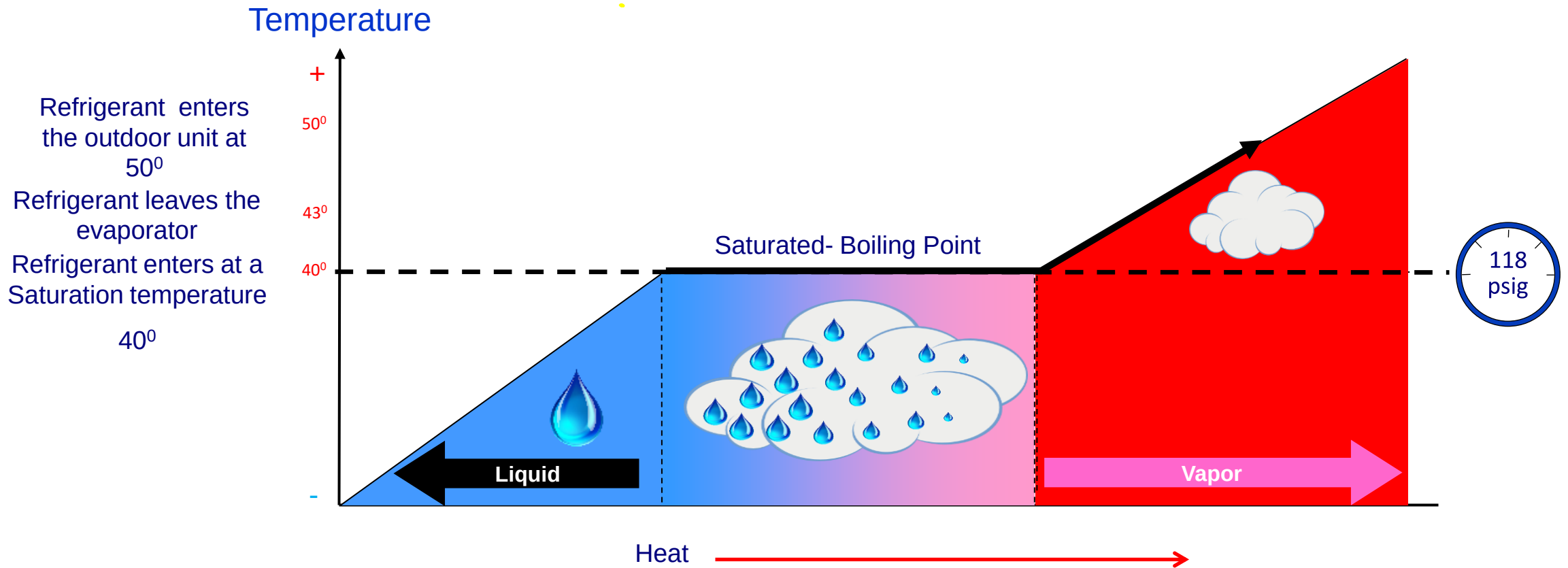
The Evaporator

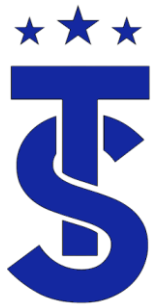
- Its function is to absorb heat into the refrigerant.
- In doing so it changes the saturated liquid refrigerant into a superheated vapor.
- For heat to be absorbed into the evaporator it **MUST BE COLDER** than the air going across it.
- The evaporator capacity is directly related to its airflow, coil temperature and the space humidity and temperature.
- Does the function of the evaporator change in a heat pump?



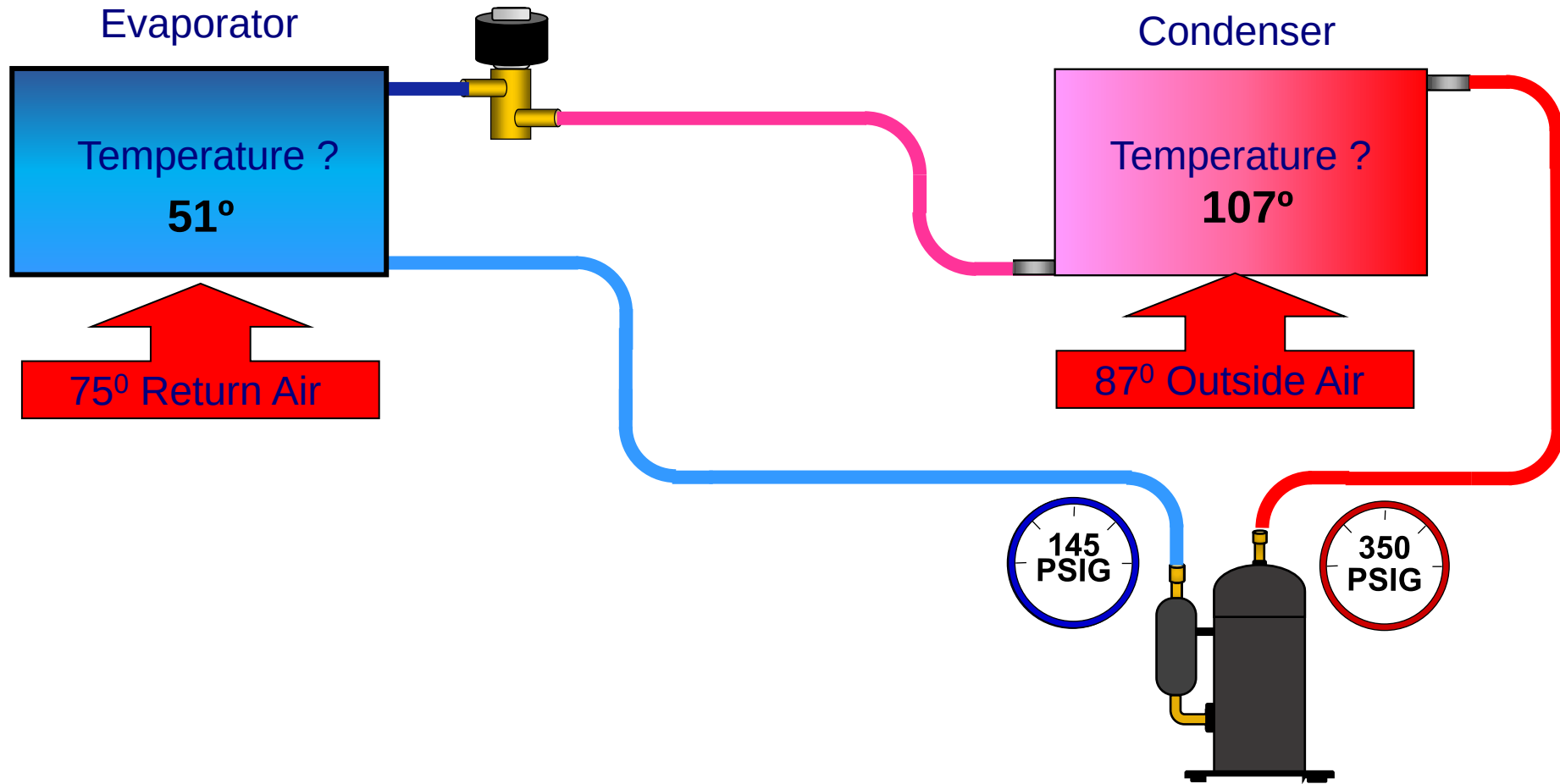


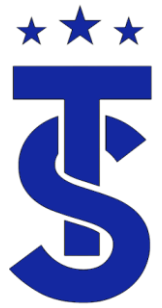
Example- Refrigerant flow through a evaporator





What Can Saturation Temperatures be used for in ductless heat pumps?





Basic Components of Ductless Refrigeration Circuit



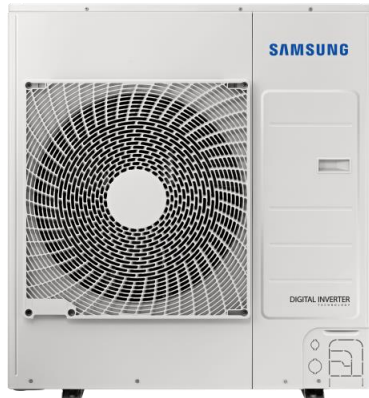
Indoor Coil



Fan



System Piping



outdoor Coil



Compressor



Fan



Reversing Valve



Metering Device

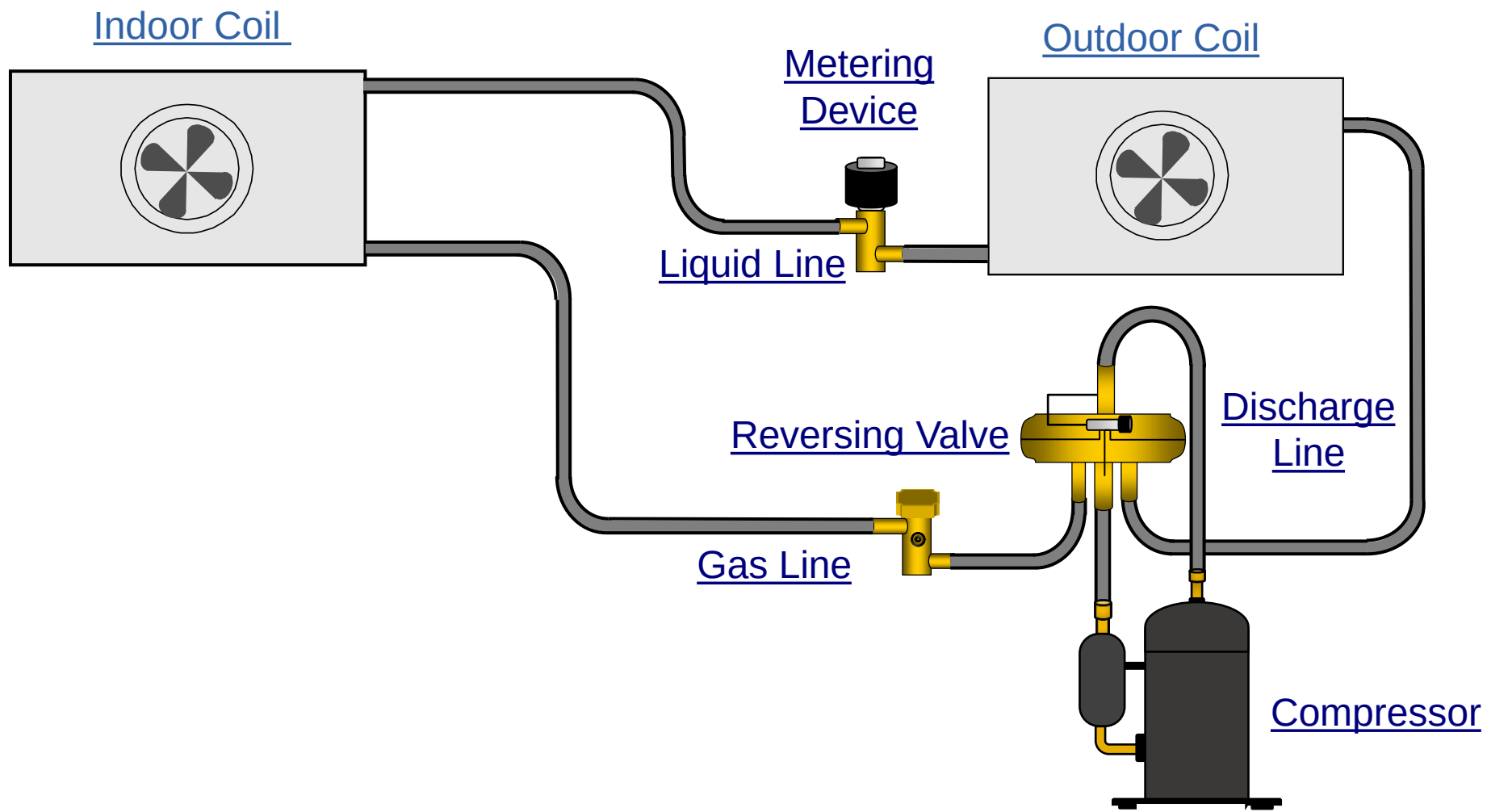


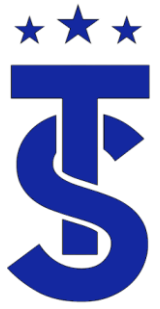
Piping systems

- The piping system is commonly overlooked when it comes to refrigeration system.
- A basic refrigeration system will contain:
 - Discharge line / Hot gas line
 - Connects the discharge of the compressor to the inlet of the condenser.
 - Carries high pressure, high temperature superheated vapor.
 - Liquid line
 - Connects the outlet of the condenser to the inlet of the metering device.
 - Carried high pressure high temperature sub-cooled liquid.
 - Suction Line
 - Connects the outlet of the evaporator to the inlet of the compressor.
 - Carries low pressure low temperature superheated vapor.



Basic Components of a Ductless Heat Pump



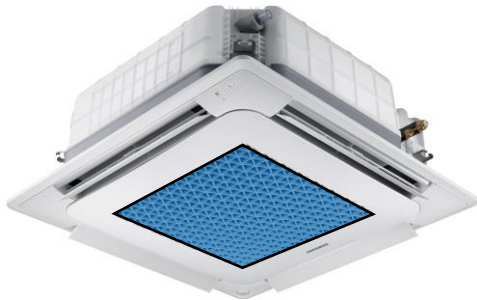


Indoor Coil



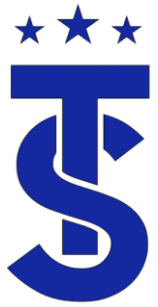
Heating

- In the heating mode the indoor coil in the Condenser



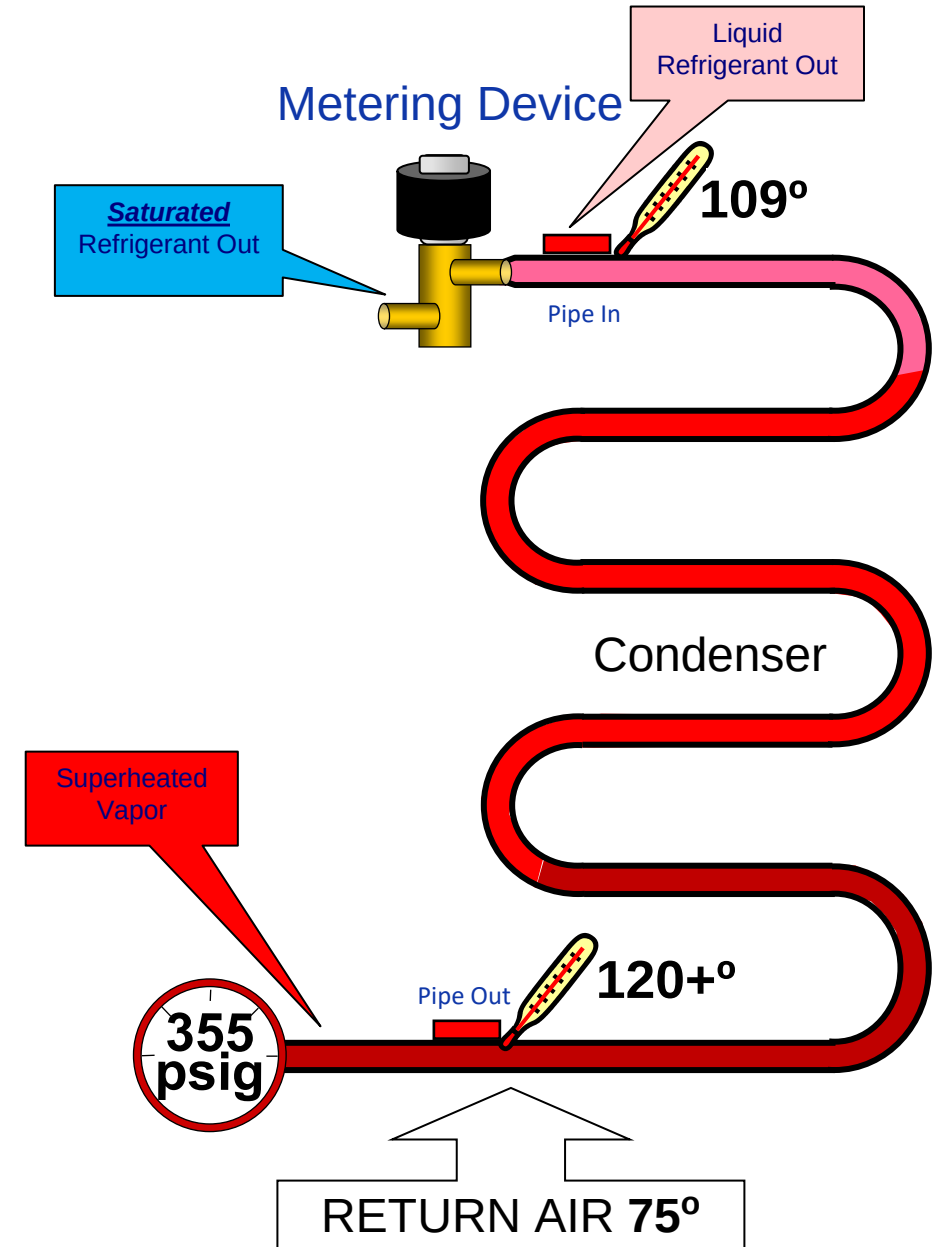
Cooling

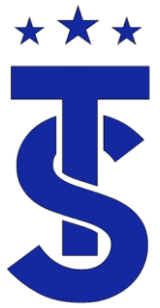
- In the cooling mode the indoor coil is the Evaporator



Indoor Coil Heating Mode

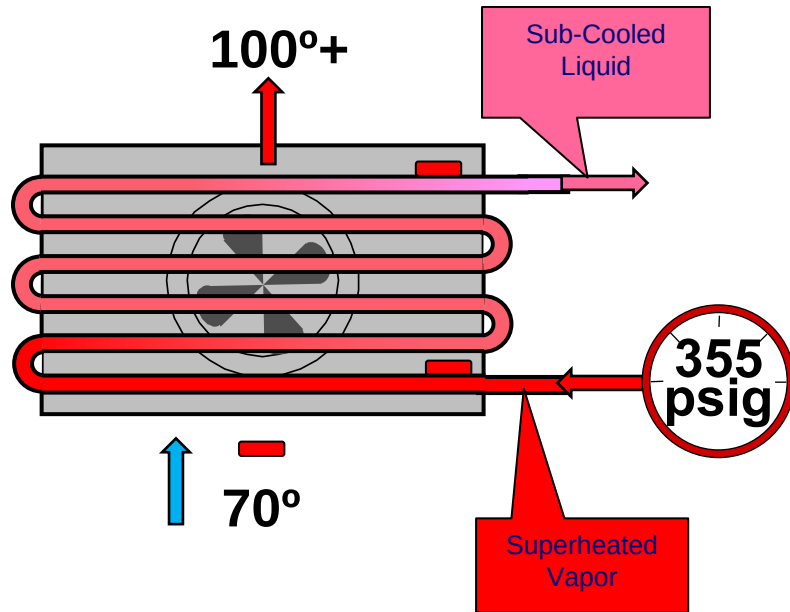
- Its function is to reject heat from the refrigerant into the space
- In doing so it changes the superheated vapor refrigerant into a sub-cooled liquid
- The condenser must be hotter than the air for heat to be transferred
- The condenser capacity is directly related to its airflow, coil temperature and the space temperature.
- Does the function of the condenser change in a heat pump?





States of refrigerant within a system

Heating Mode

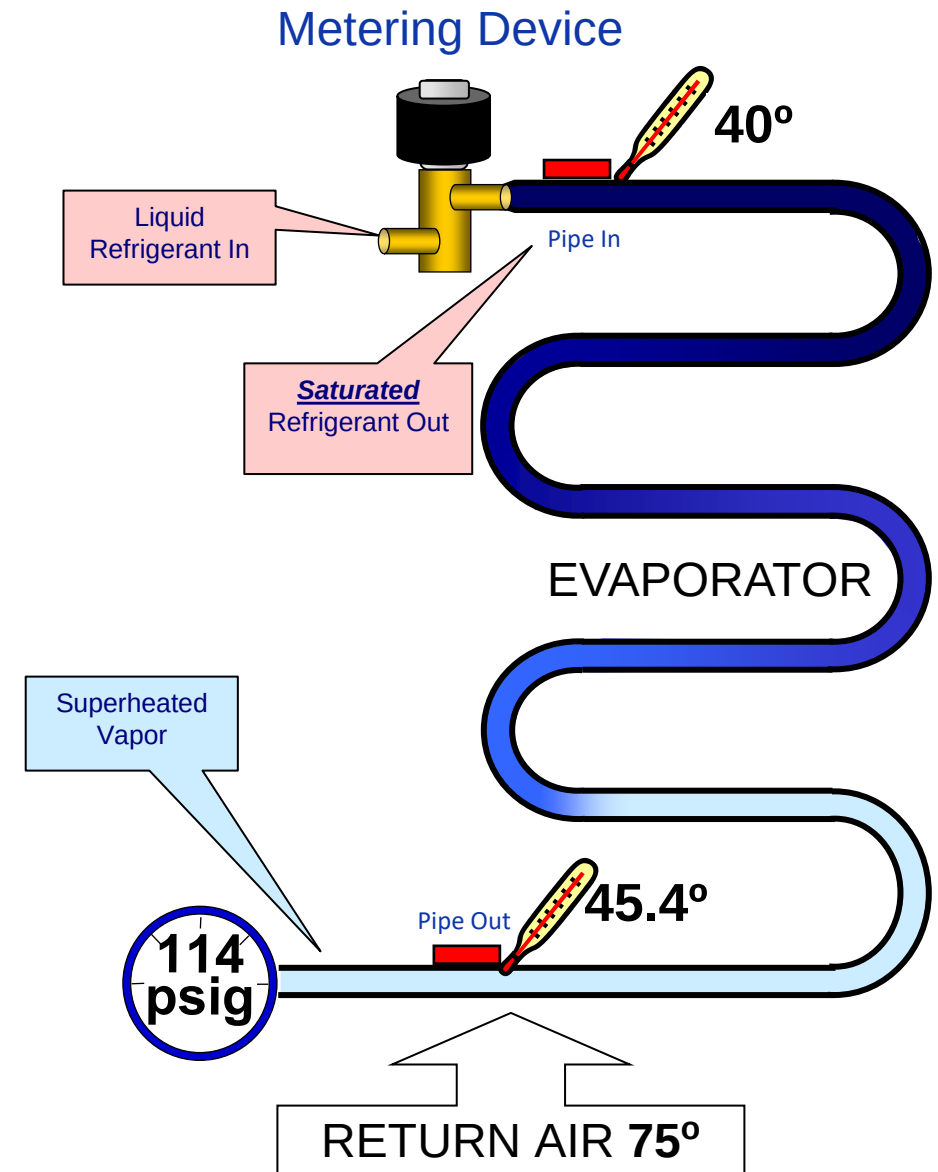


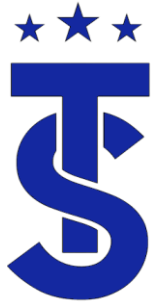
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- The condenser capacity is directly related to its airflow, coil temperature and the space temperature.
- Does the function of the condenser change in a heat pump?



Indoor Coil Cooling Mode

- Its function is to absorb heat into the refrigerant.
- In doing so it changes the saturated liquid refrigerant into a superheated vapor.
- For heat to be absorbed into the evaporator it **MUST BE COLDER** than the air going across it.
- The evaporator capacity is directly related to its airflow, coil temperature and the space humidity and temperature.
- Does the function of the evaporator change in a heat pump?



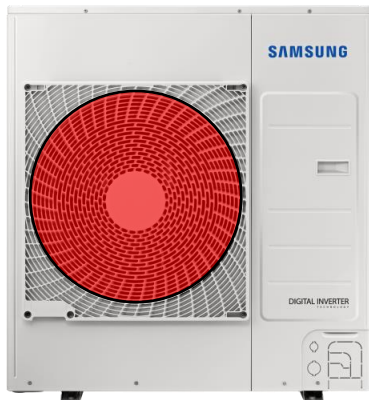


Outdoor Coil



Heating

- In the heating mode the outdoor coil is the Evaporator

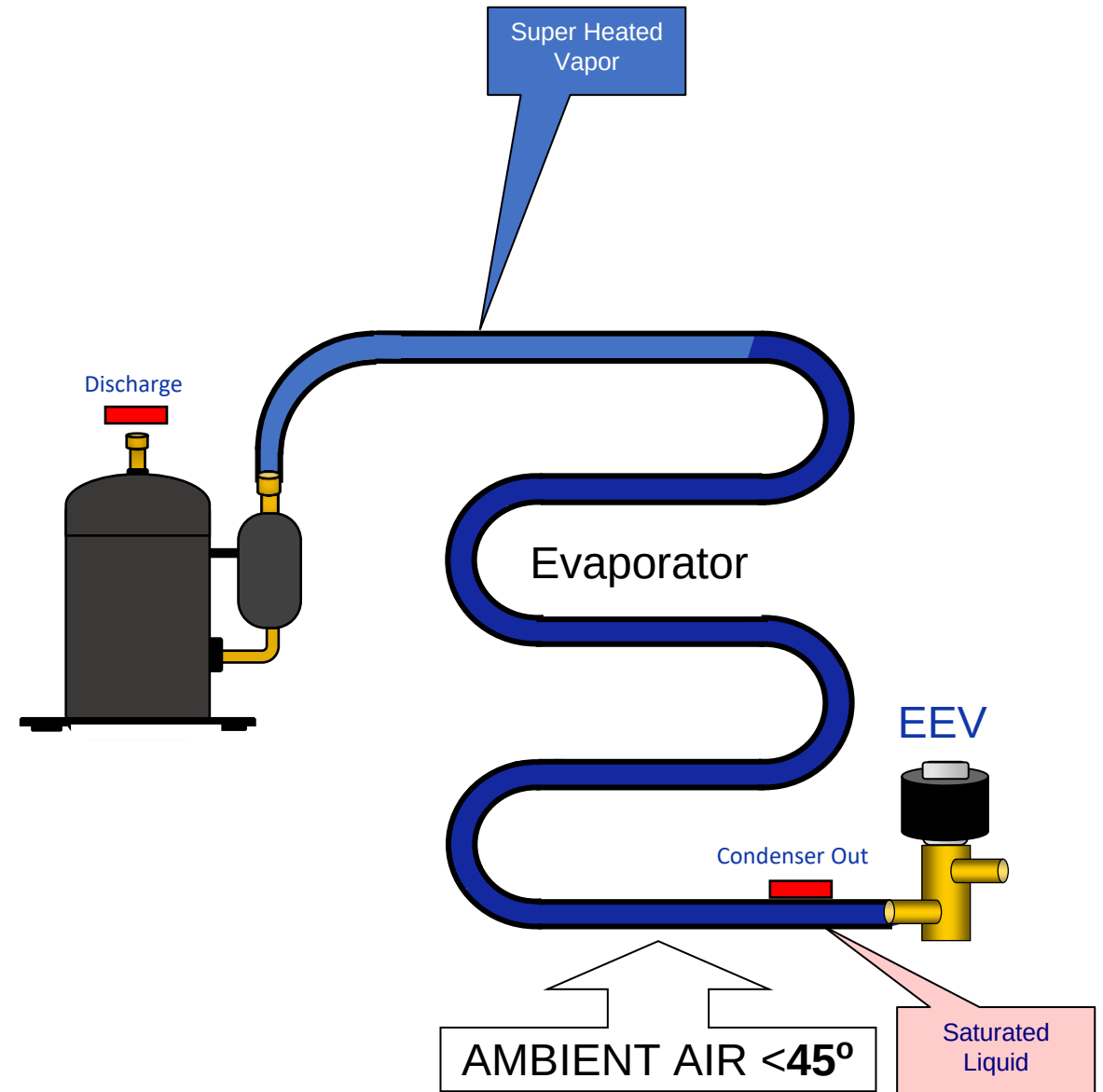


Cooling

- In the cooling mode the outdoor coil is the Condenser

★★★ Outdoor Coil Heating Mode

- Its function is to absorb heat into the refrigerant.
- In doing so it changes the saturated liquid refrigerant into a superheated vapor.
- For heat to be absorbed into the evaporator it **MUST BE COLDER** than the air going across it.
- The evaporator capacity is directly related to its airflow, coil temperature and the space humidity and temperature.
- Does the function of the evaporator change in a heat pump?

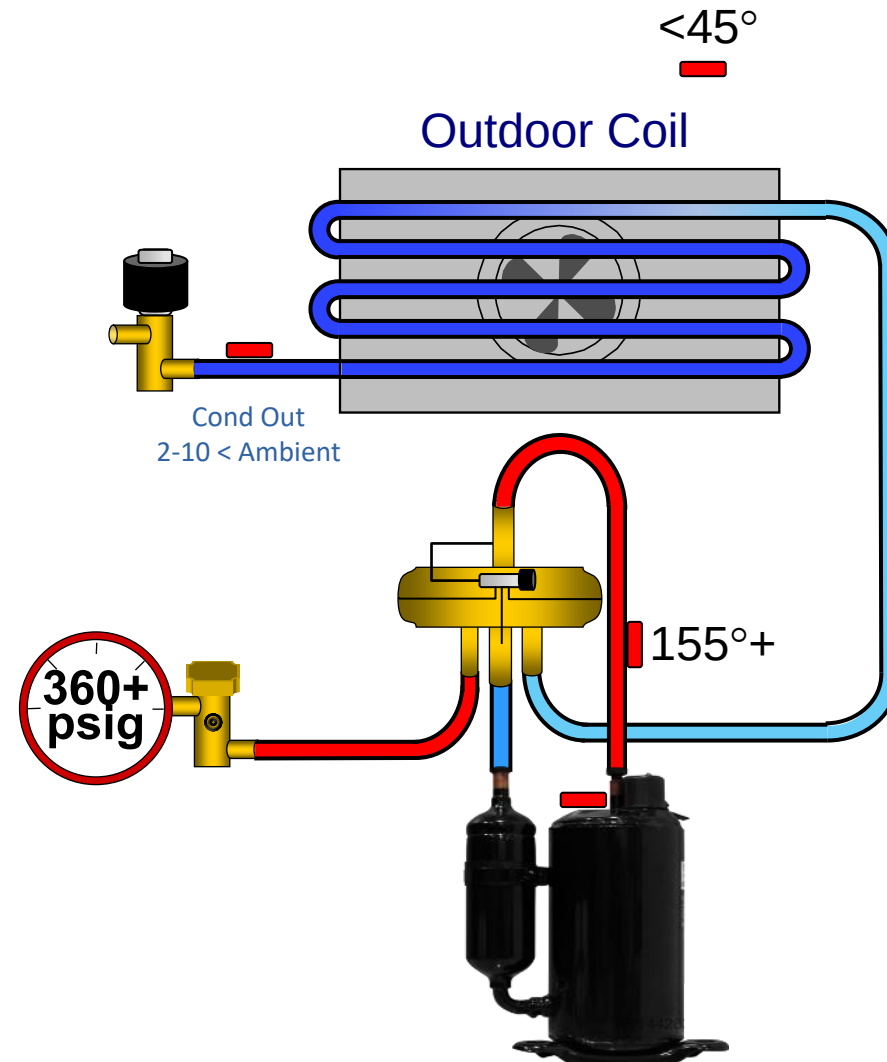




Outdoor Coil

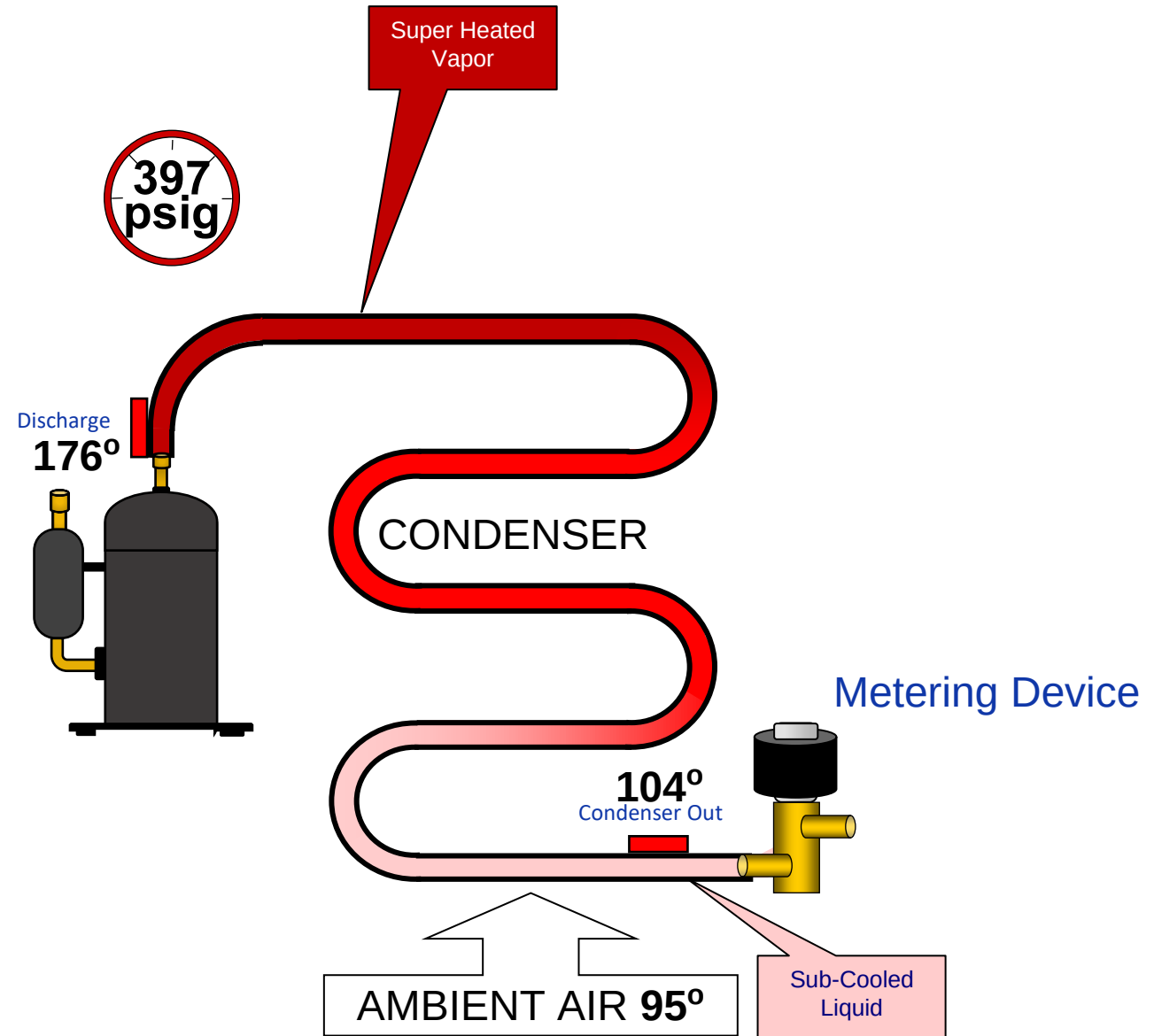
Heating Mode

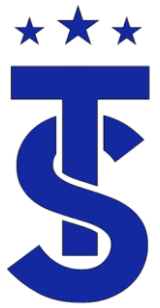
- Its function is to absorb heat into the refrigerant.
- In doing so it changes the saturated liquid refrigerant into a superheated vapor.
- For heat to be absorbed into the evaporator it **MUST BE COLDER** than the air going across it.
- The evaporator capacity is directly related to its airflow, coil temperature and the space humidity and temperature.
- Does the function of the evaporator change in a heat pump?



★★★ The Outdoor Coil Cooling Mode

- Its function is to remove the heat from the superheated refrigerant and in doing so changes the vapor refrigerant into a sub-cooled liquid.
- The Condenser HAS TO BE HOTTER than the air going across it for heat to be transferred.
- Does its function change in a heat pumps system?

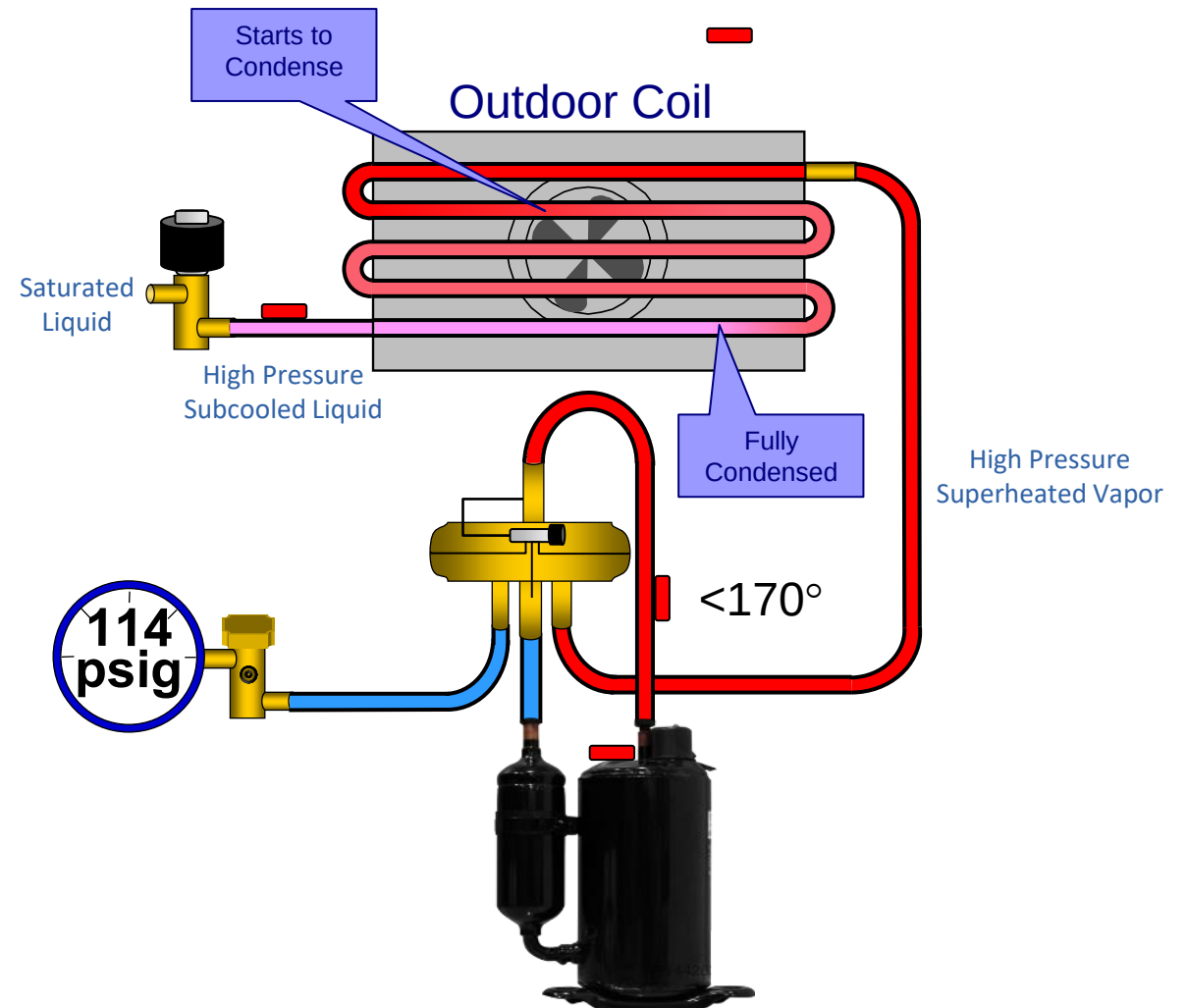




States of refrigerant within a ductless heat pump system

Cooling Mode

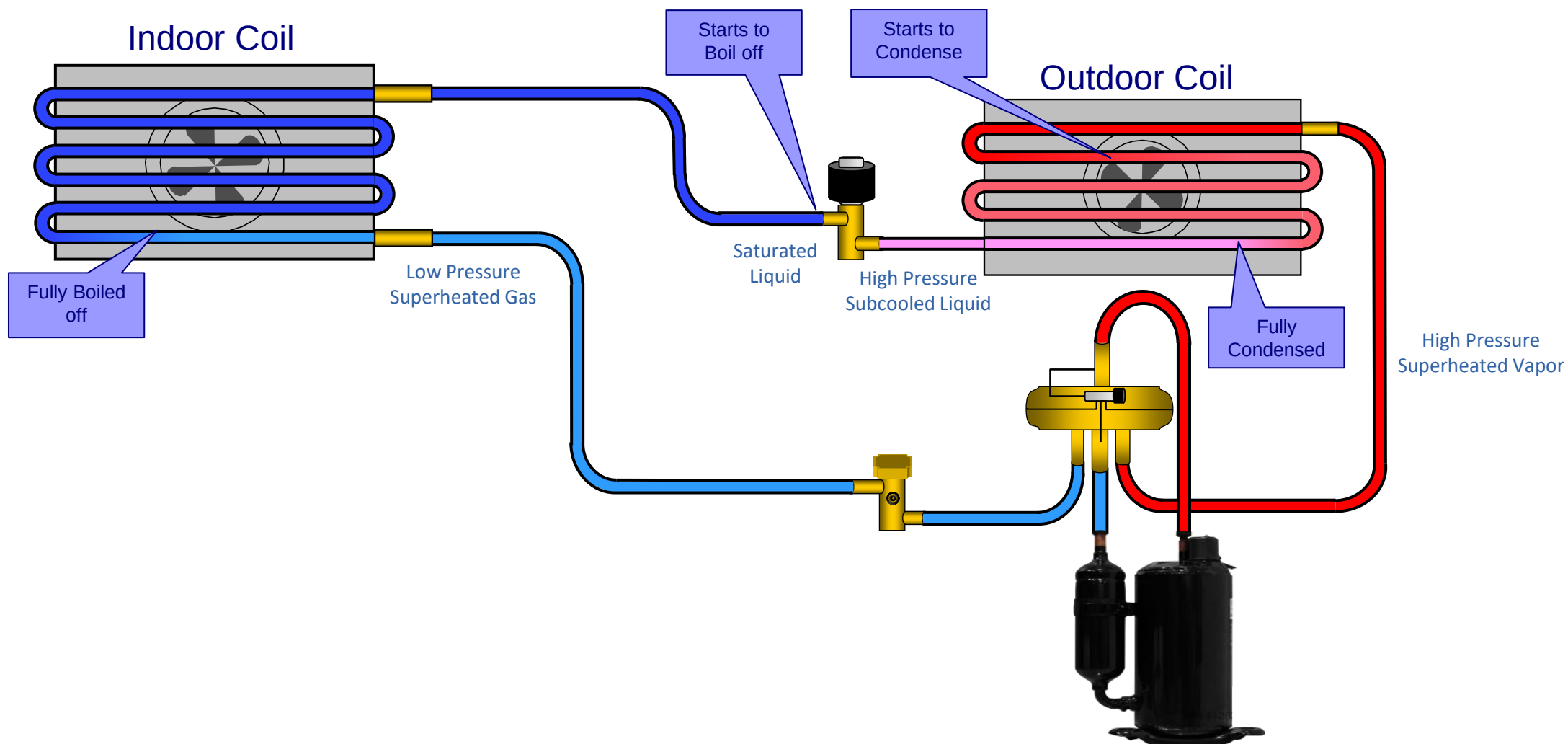
- Its function is to remove the heat from the superheated refrigerant and in doing so changes the vapor refrigerant into a sub-cooled liquid.
- The Condenser HAS TO BE HOTTER than the air going across it for heat to be transferred.
- Does its function change in a heat pumps system?

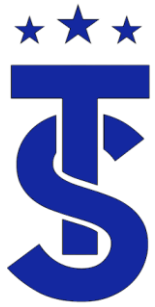




States of refrigerant within a ductless heat pump system

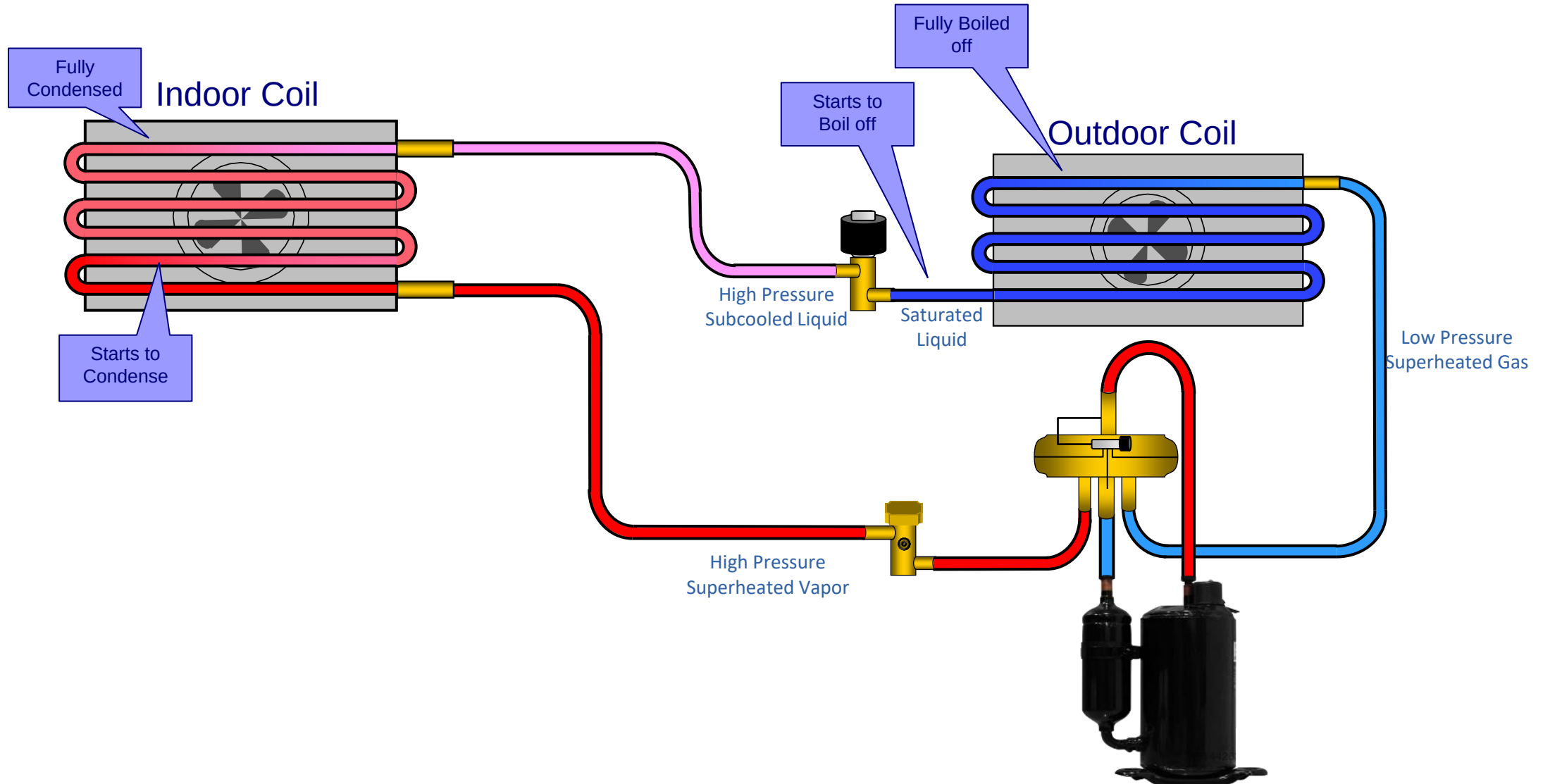
Cooling Mode

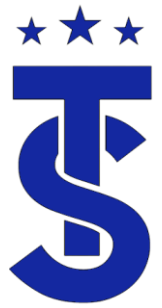




States of refrigerant within a ductless heat pump system

Heating Mode



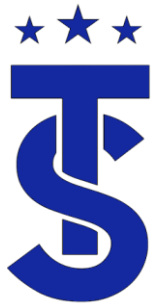


Electronic Expansion Valves

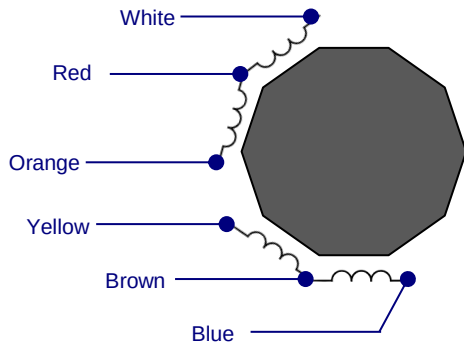
EEV

- EEV's are used as metering devices in commercial and residential VFR systems.
- The valve contains 2 parts the body and driver.
- The drives create a rotating magnetic field to open and close the valve.
- Resistances of the driver motor varies depending on the style.



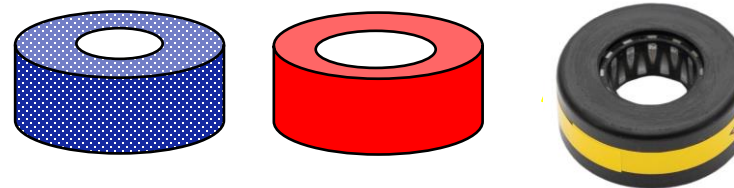
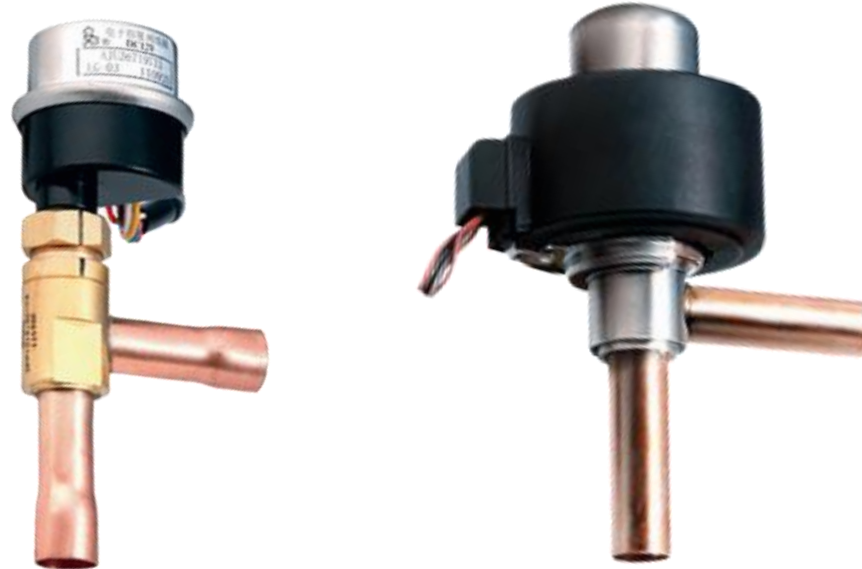


EEV Coil Resistance



RED – WHITE	$150 \pm 15\Omega$
RED – ORANGE	"
BROWN-YELLOW	"
BRWON-BLU	"

COM : RED, BROWN

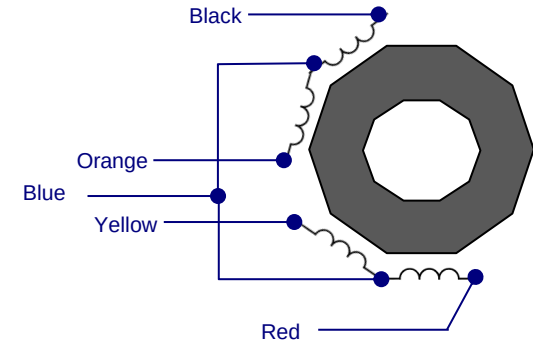


Testing Magnets

BLUE-BLACK	$92\Omega \pm 2\Omega$
BLUE-YELLOW	$92\Omega \pm 2\Omega$
BLUE-RED	$92\Omega \pm 2\Omega$
BLUE-ORANGE	$92\Omega \pm 2\Omega$

COM : Blue

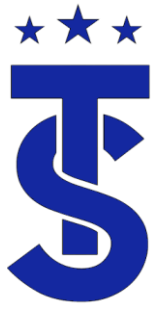
DVMS EVI
EEV



BLUE-BLACK	$46 \pm 2\Omega$
BLUE-YELLOW	"
BLUE-RED	"
BLUE-ORANGE	"

COM : Blue

RAC CAC
FJM



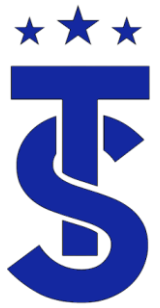
Reversing Valve

- Is energized in the heating mode.
 - Coil Voltage is:
 - 208+ VAC
- It's the difference in refrigerant pressure that causes the valve to shift

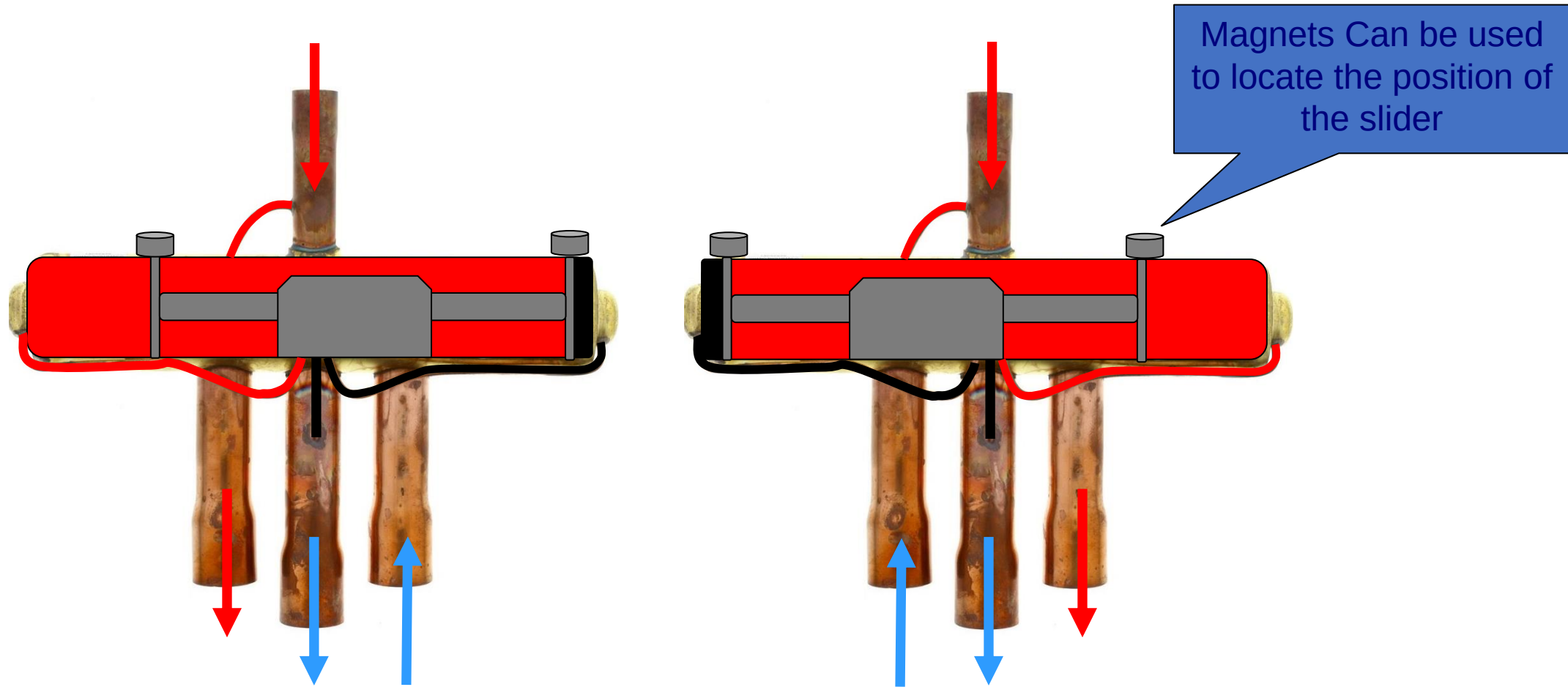


* Varies by model

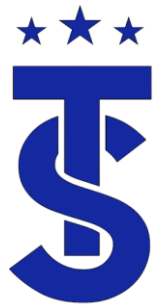
Pressure differential between high side and low side causes the valve to shift!



Refrigerant Flow

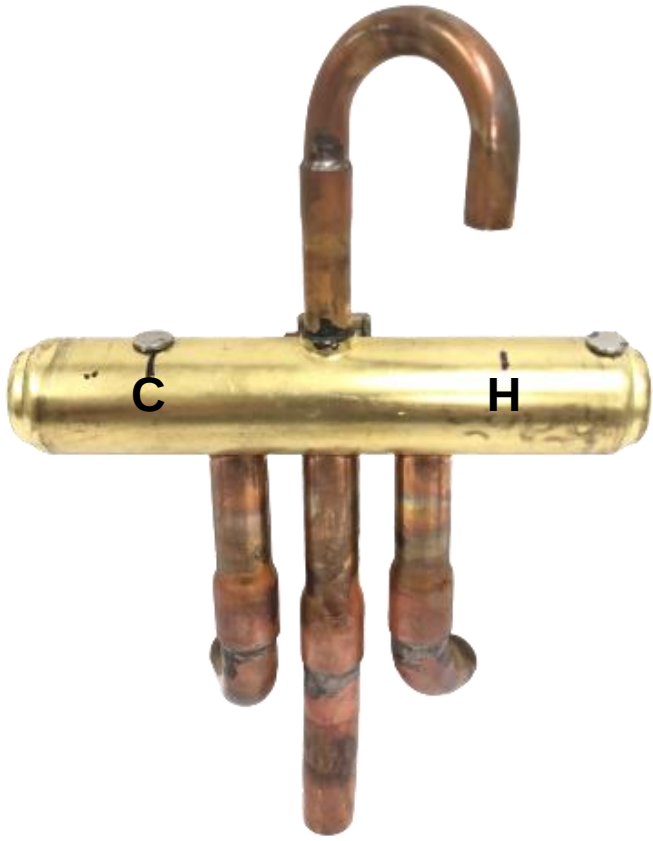


A 27 PSIG differential between high and low side pressure is required to move the slider. It may shift with as low as 18 PSIG differential.

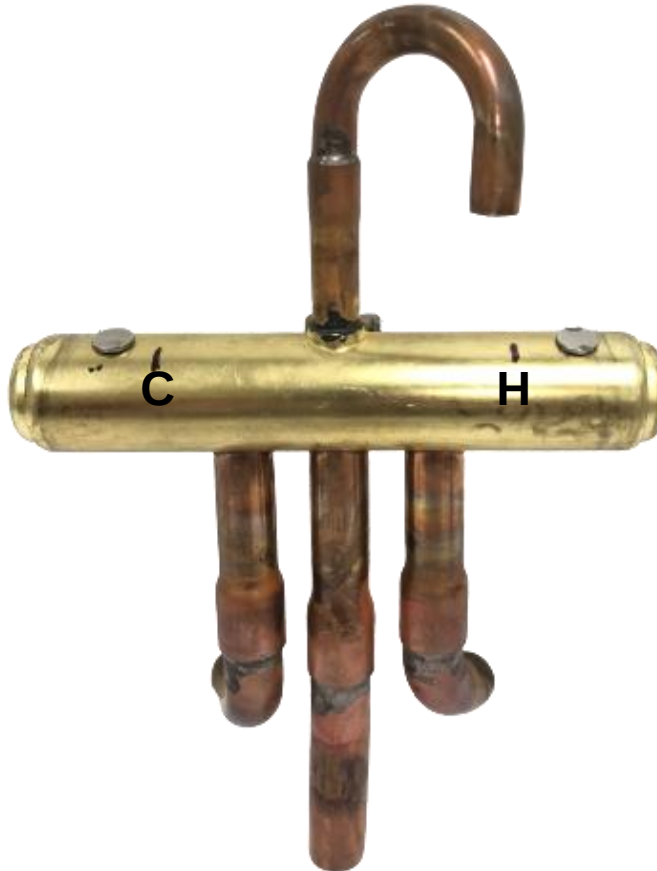


Magnets Used as a Service Tool

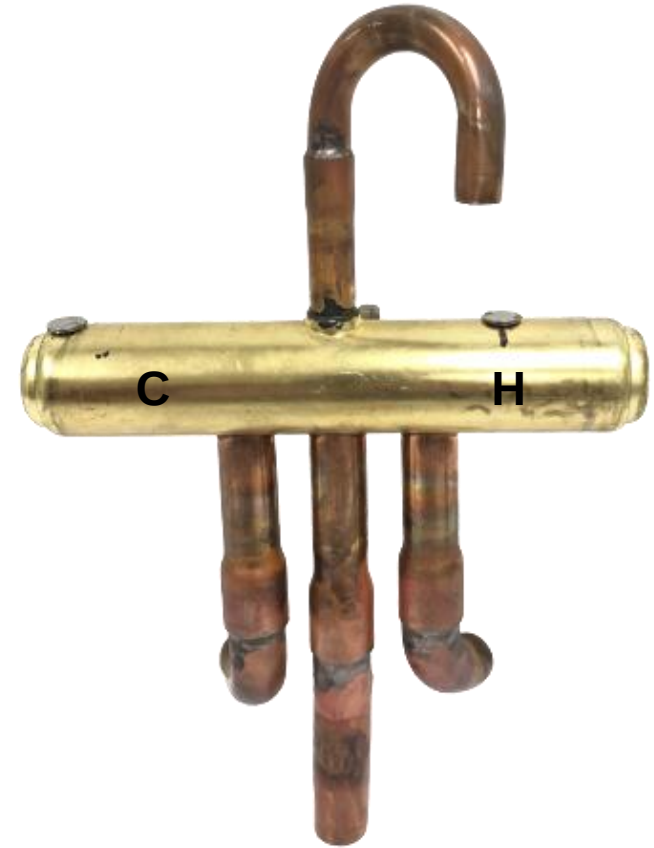
Normal Position Deenergized



Cooling Position



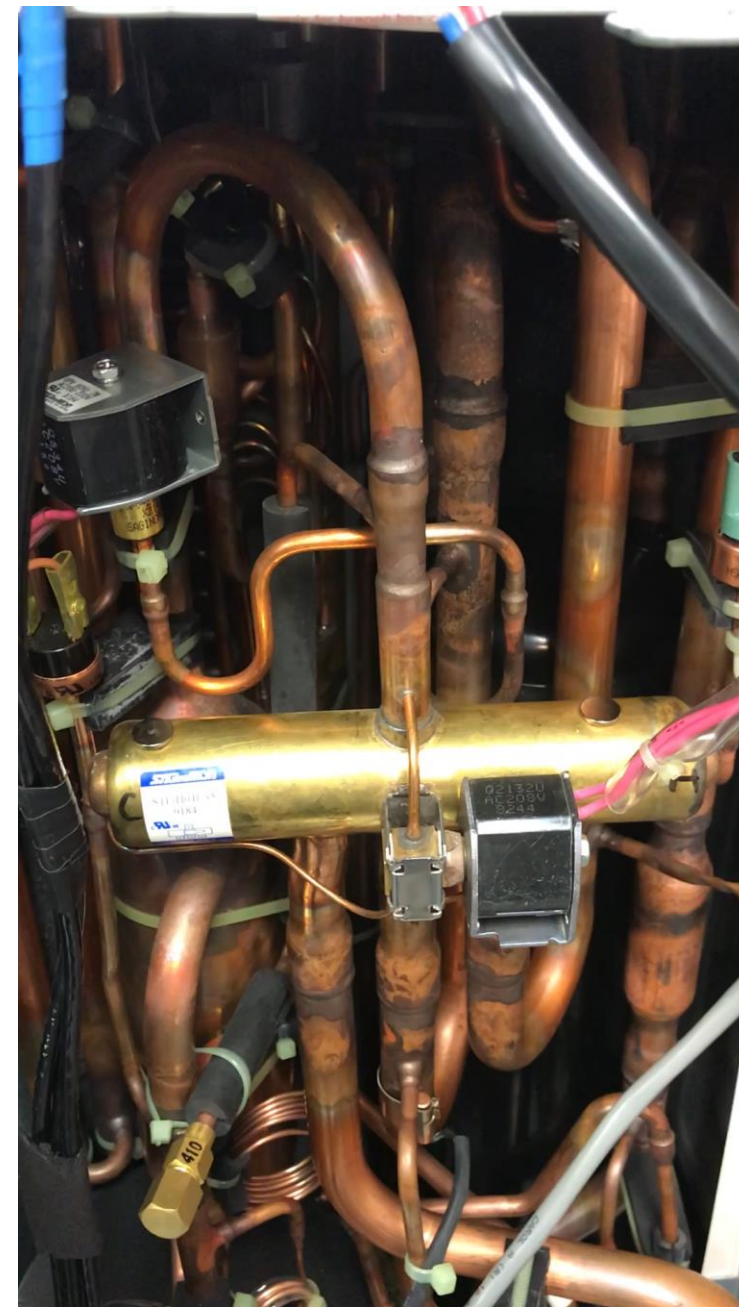
Mid Seated

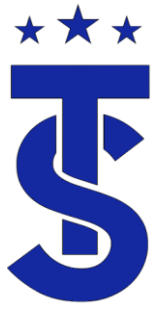


Heating Position



Reversing Valve Operation





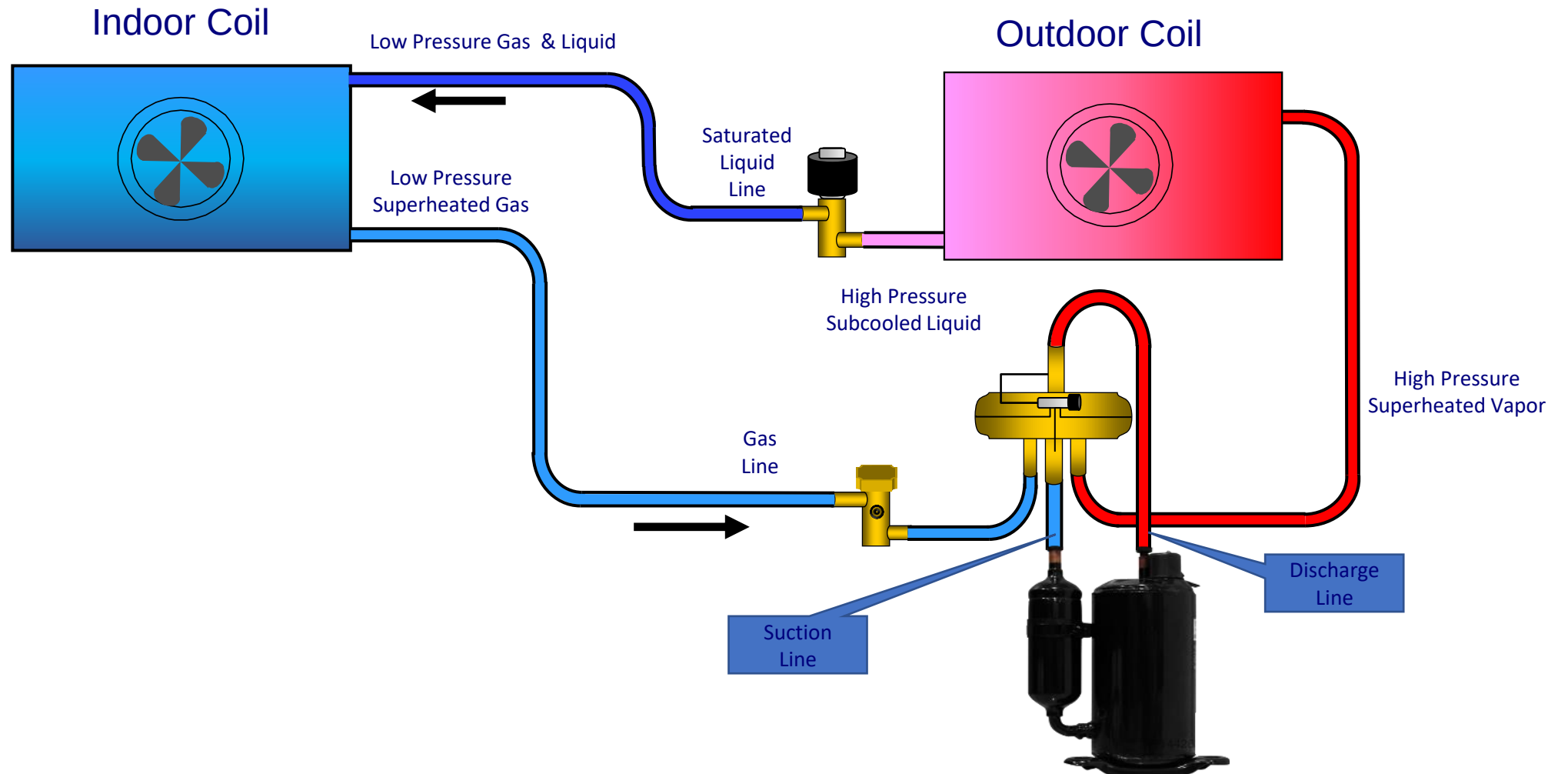
Reversing Valve

- This video show how you can use a magnet to find the position of the reversing valve slider and see its operation in the cooling mode.



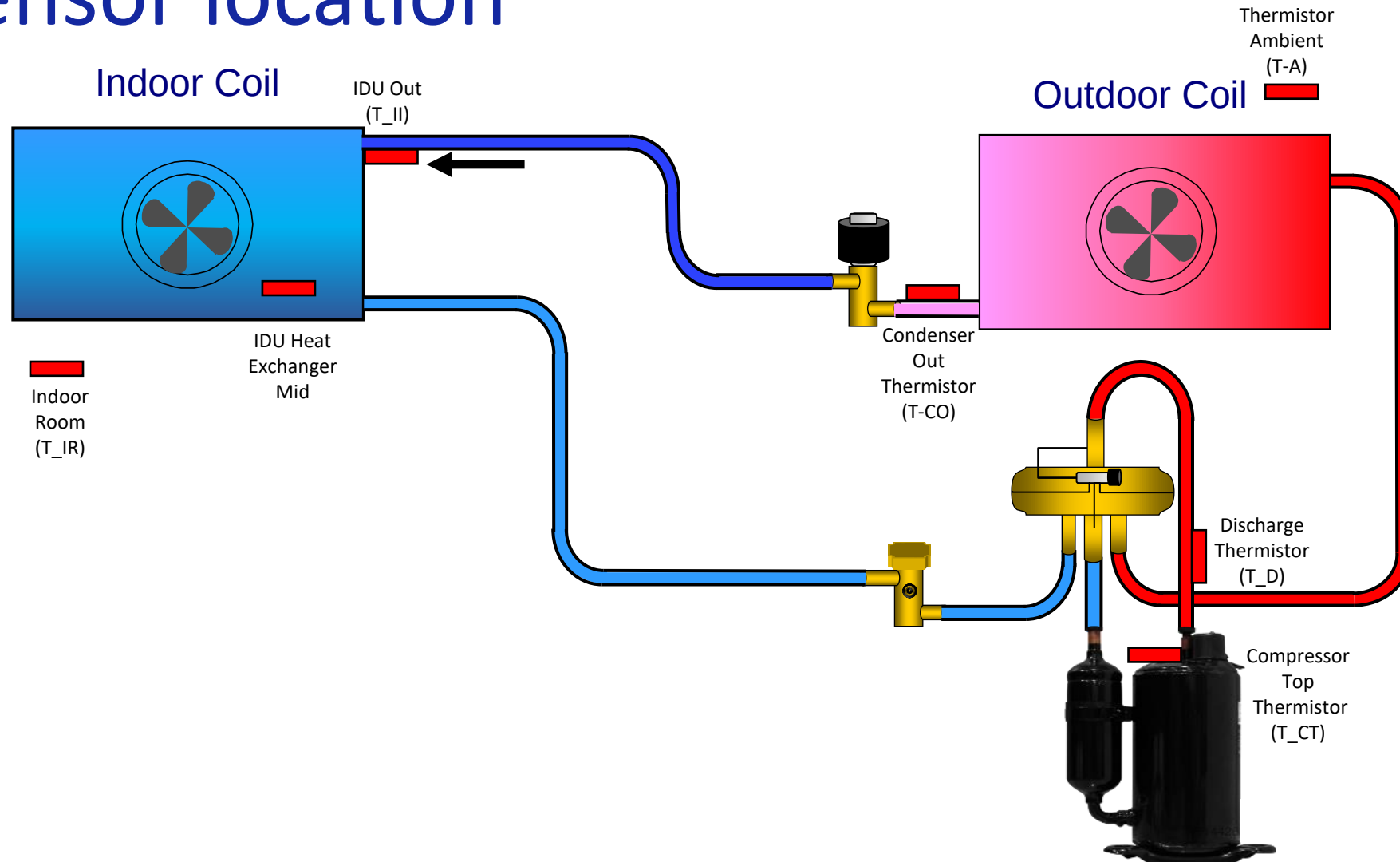


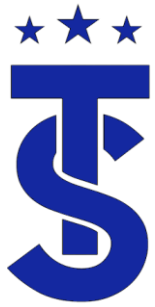
Ductless Heat Pump Cooling Cycle



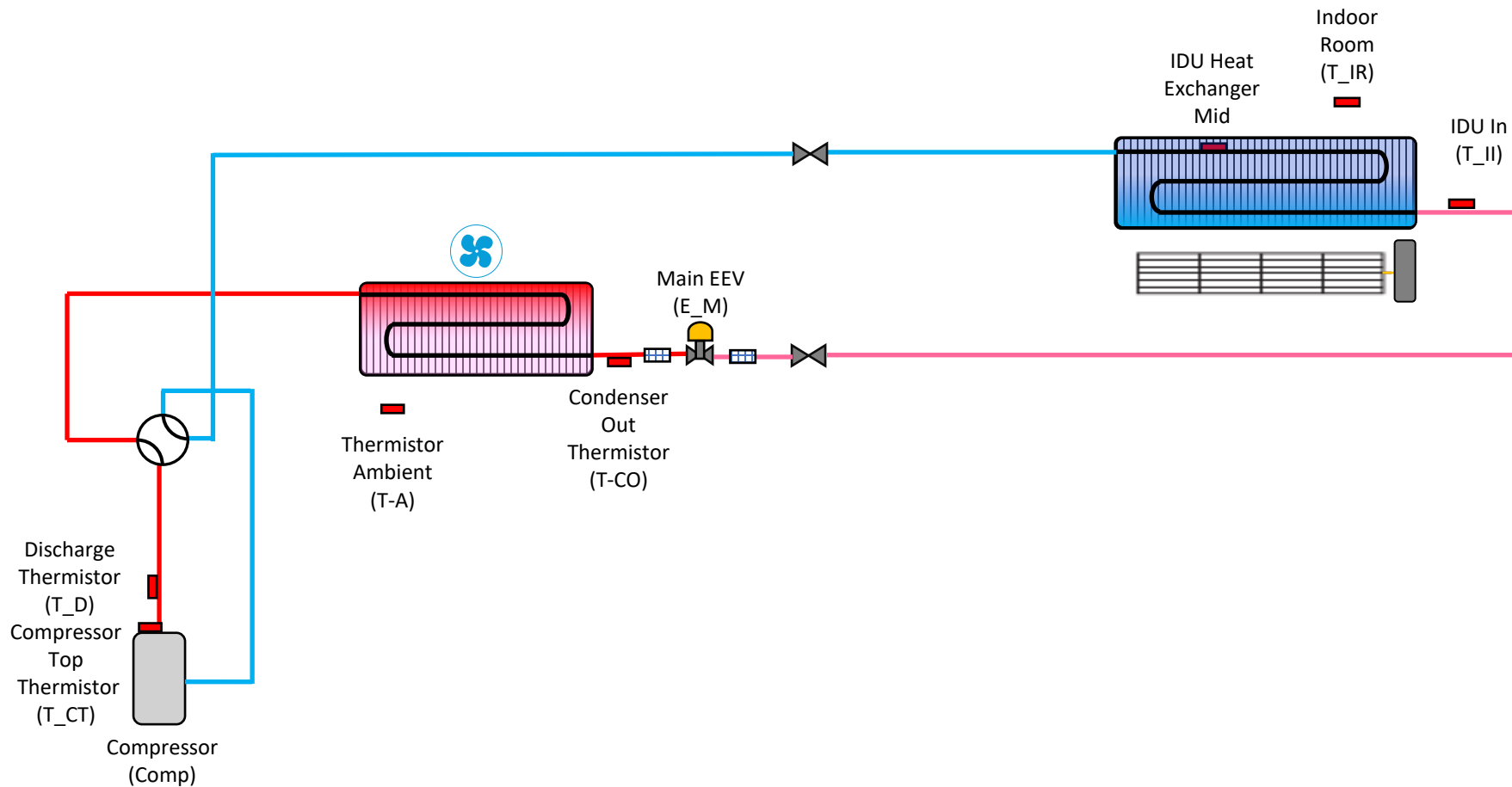


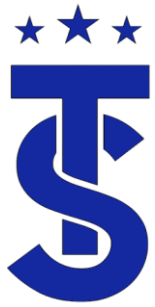
Ductless Heat Pump Sensor location



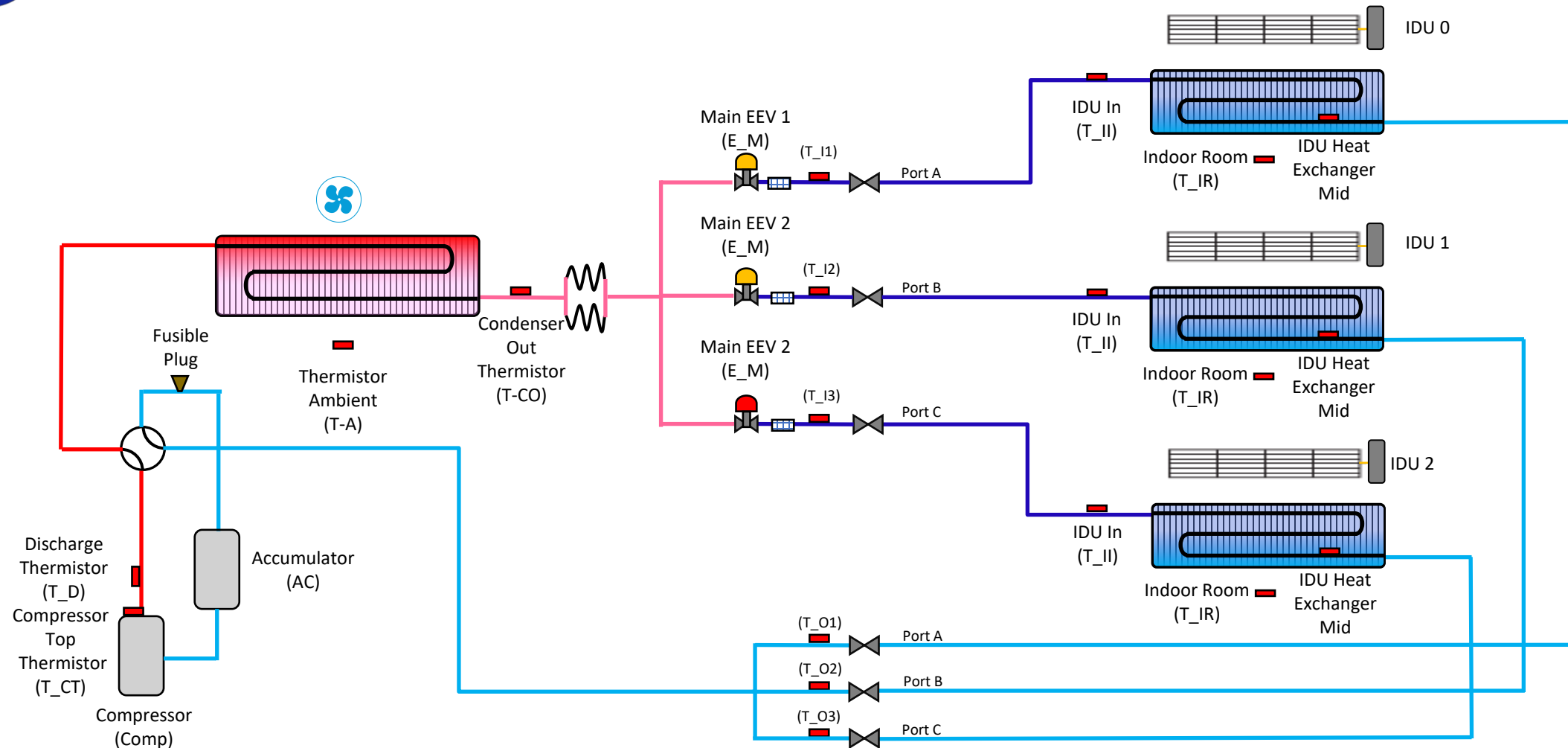


1 to 1 Cooling Mode



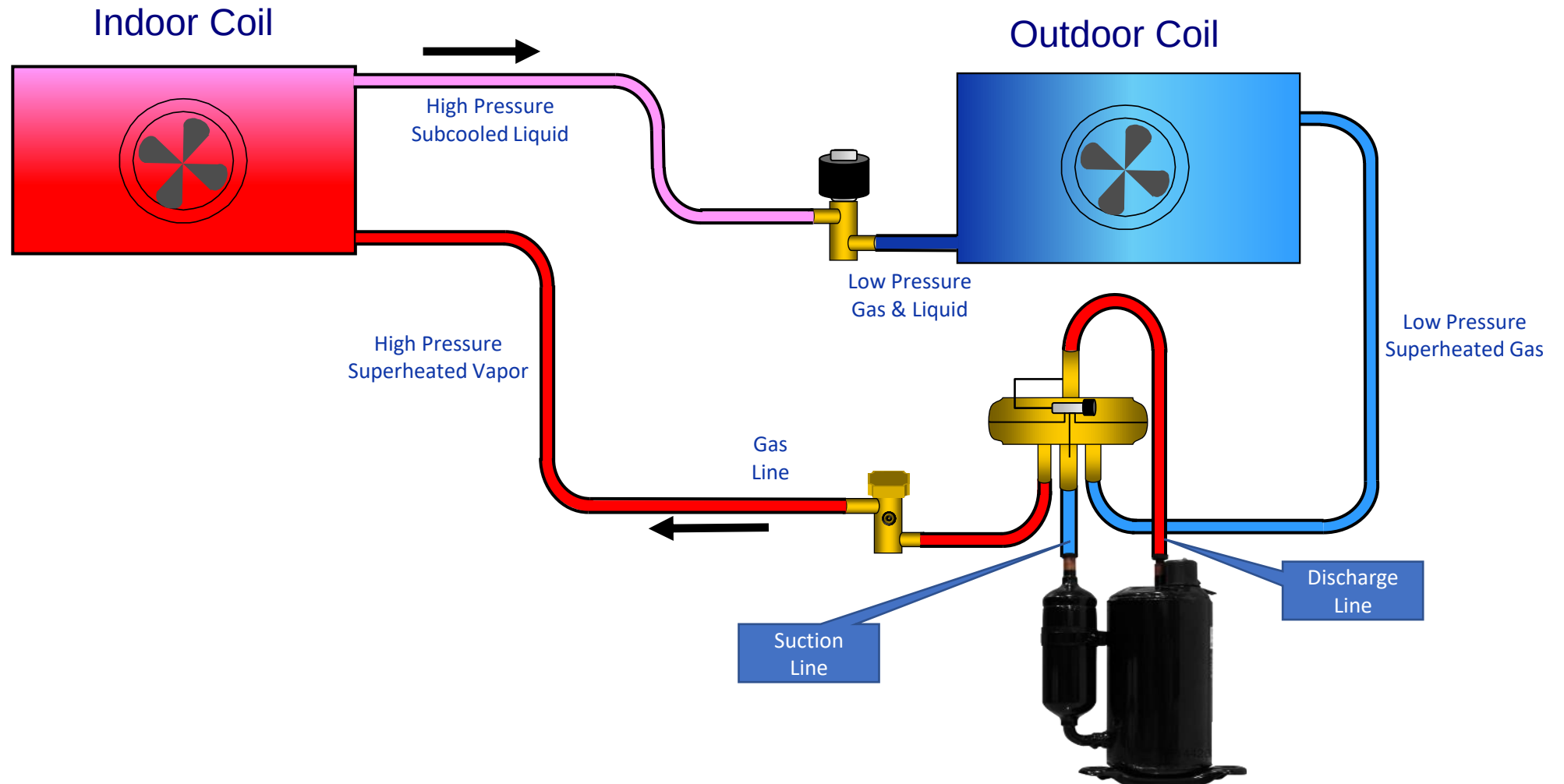


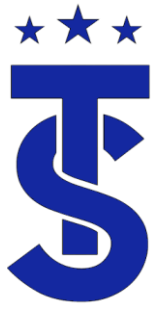
FJM Cooling Mode



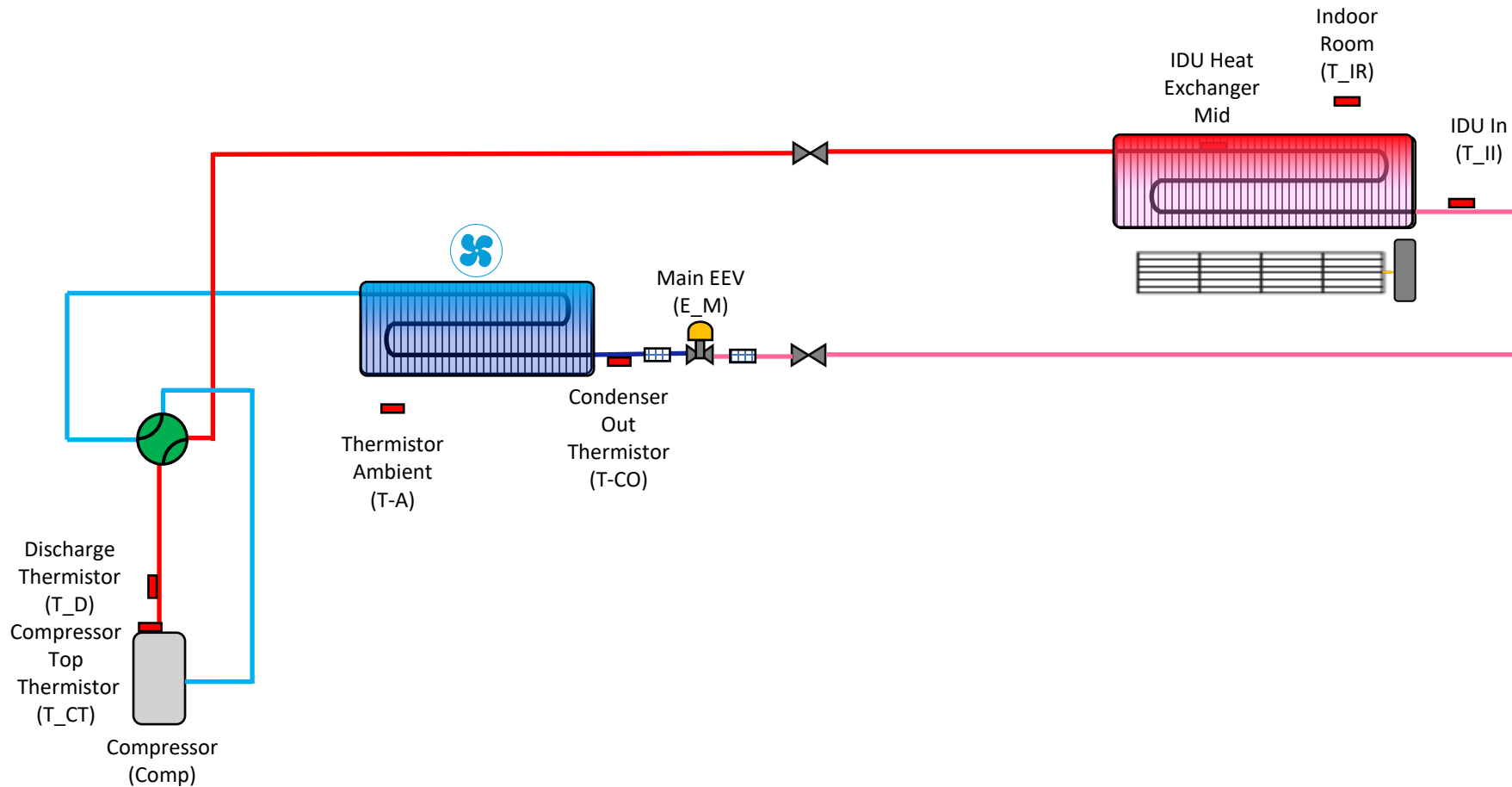


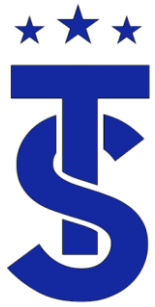
Ductless Heat Pump Heating Cycle



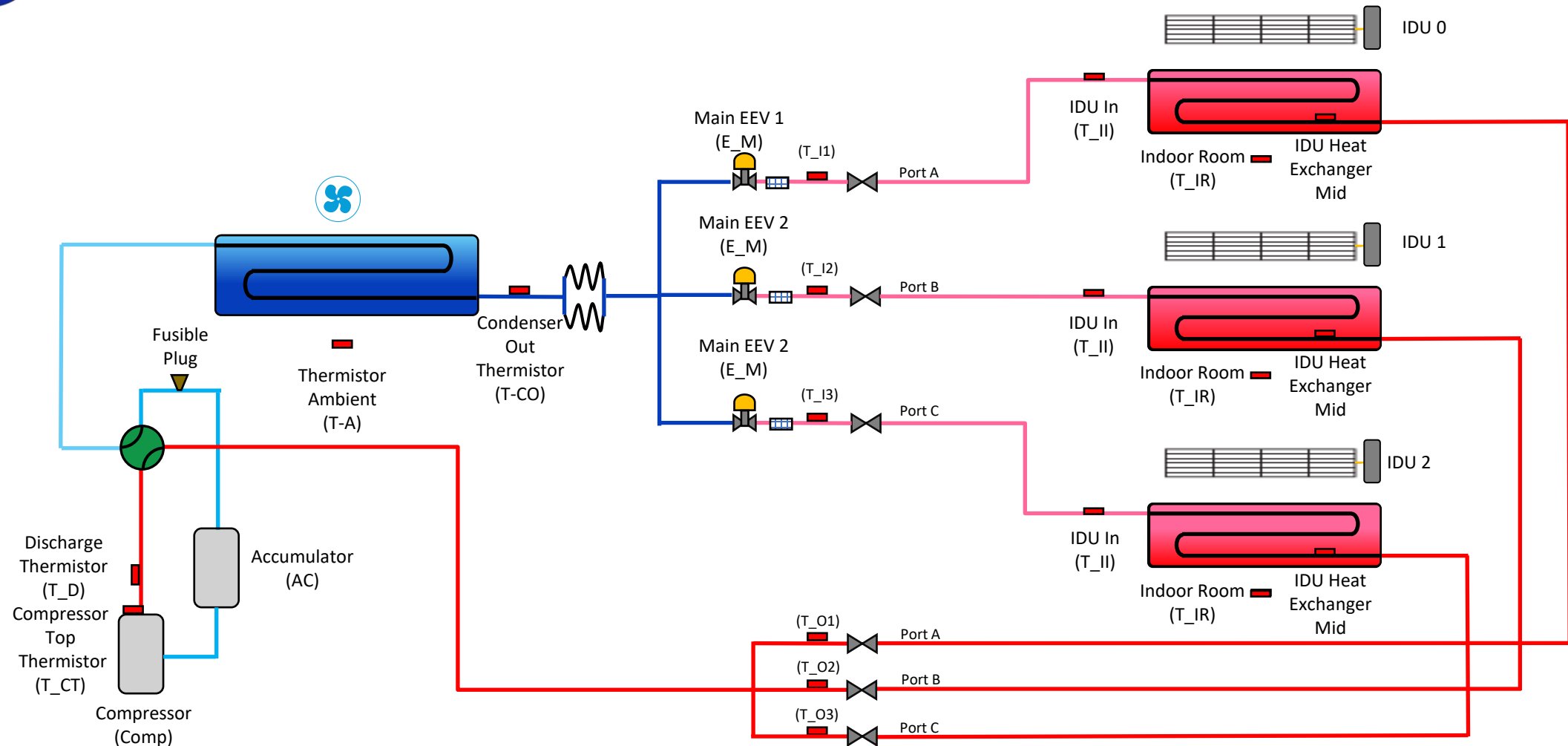


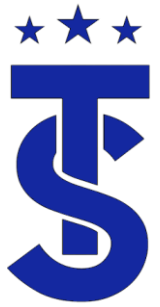
1 to 1 Heating Mode





FJM Heating Mode



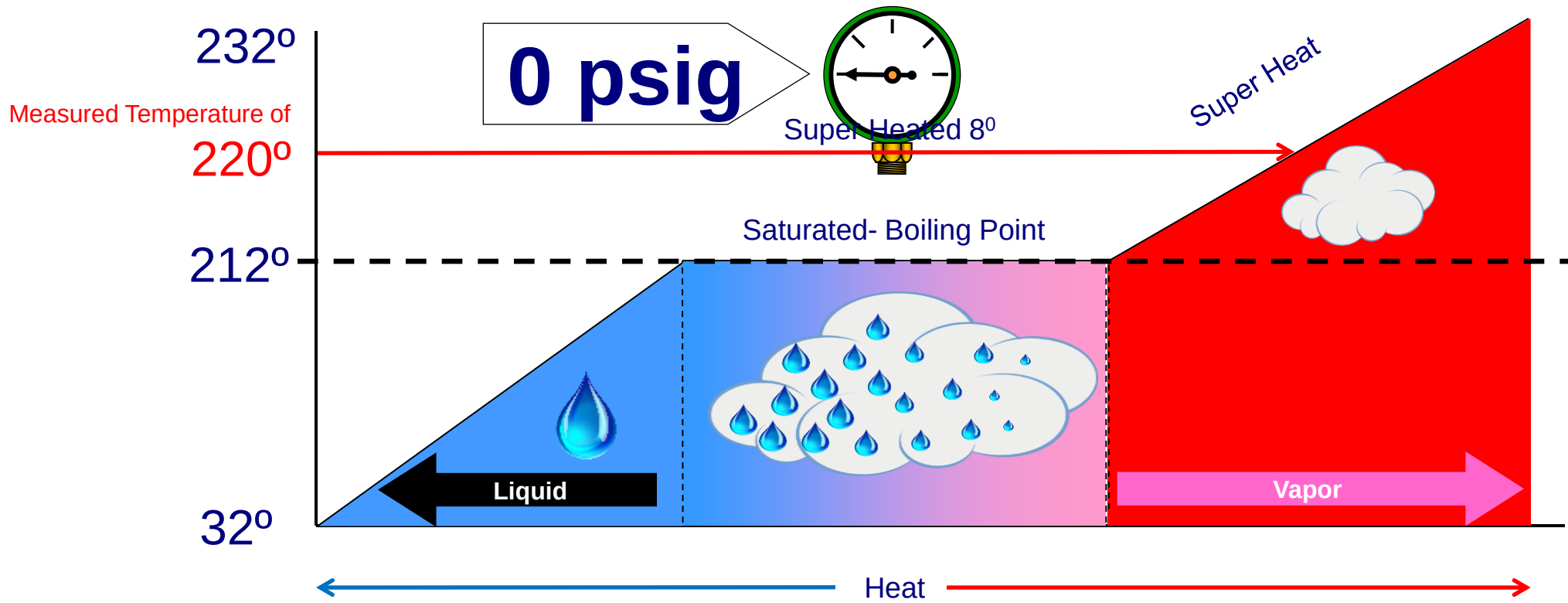


How to Calculate Super Heat

- Measure the refrigerant pressure.
 - Best practice are to measure the pressure at the location where you are calculating superheat. (this is not always possible).
- Use this pressure to determine the saturation temperature from the PT chart.
- Measure the temperature of the refrigerant pipe.
- Superheat = Measured line temperature – Saturation Temperature.

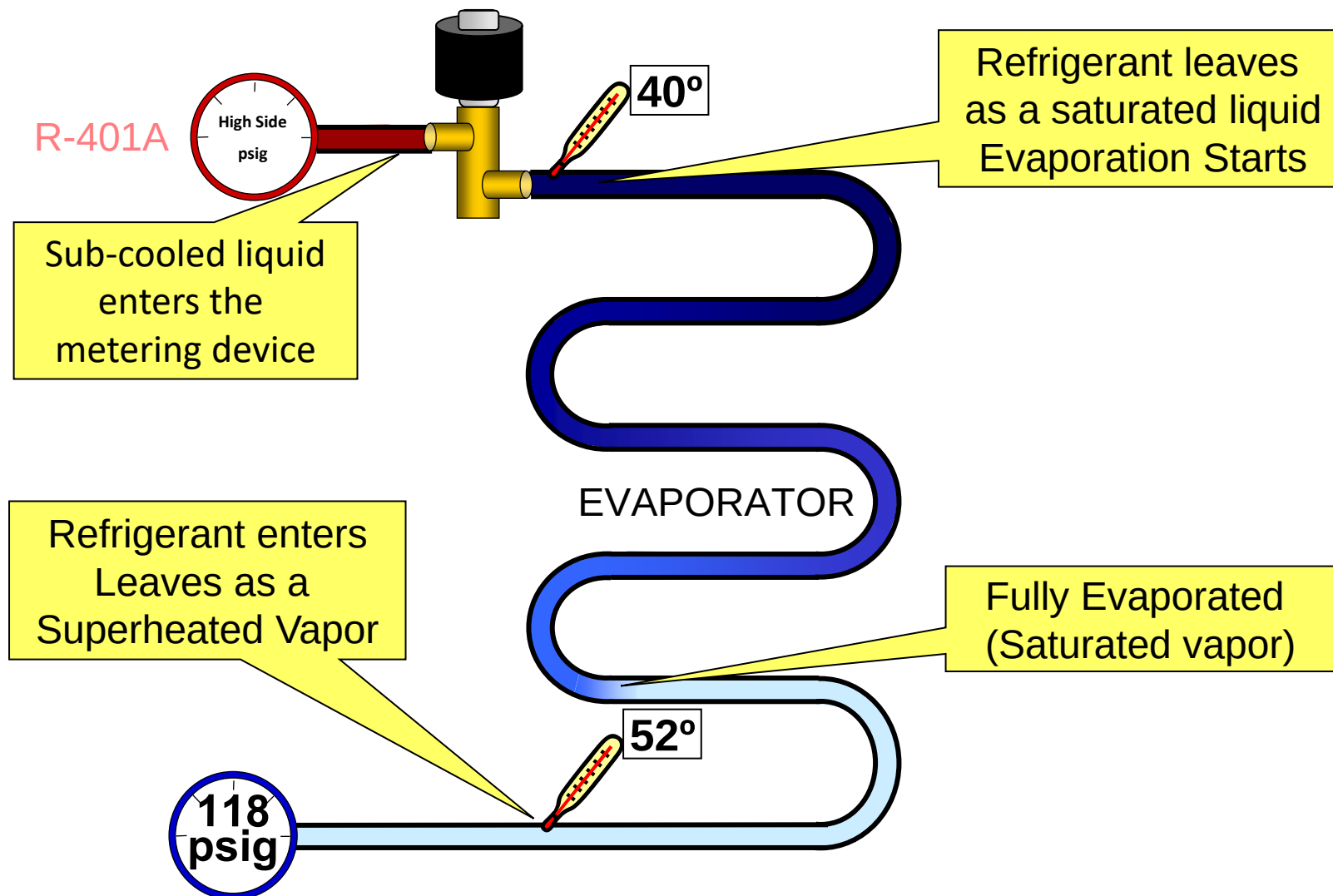


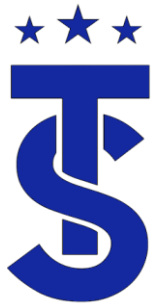
State of Water at 0 psig and a temperature of 220 degrees



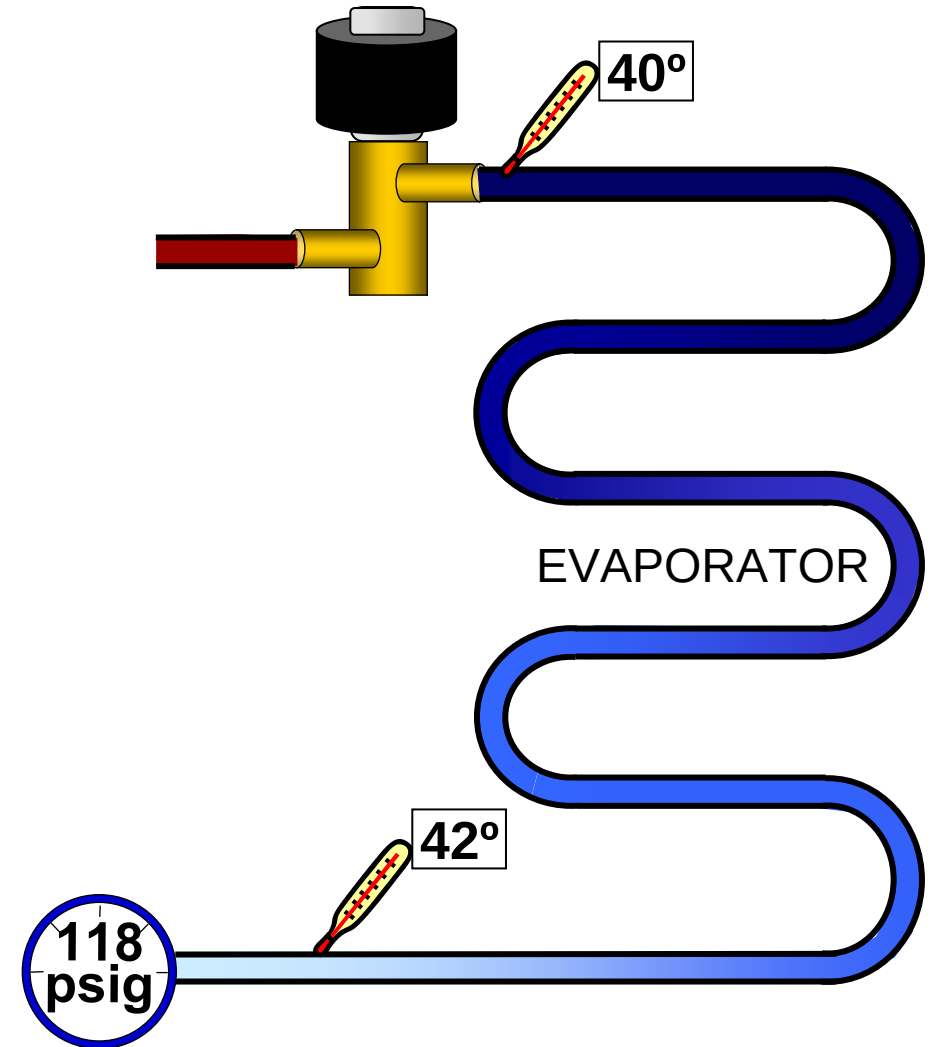
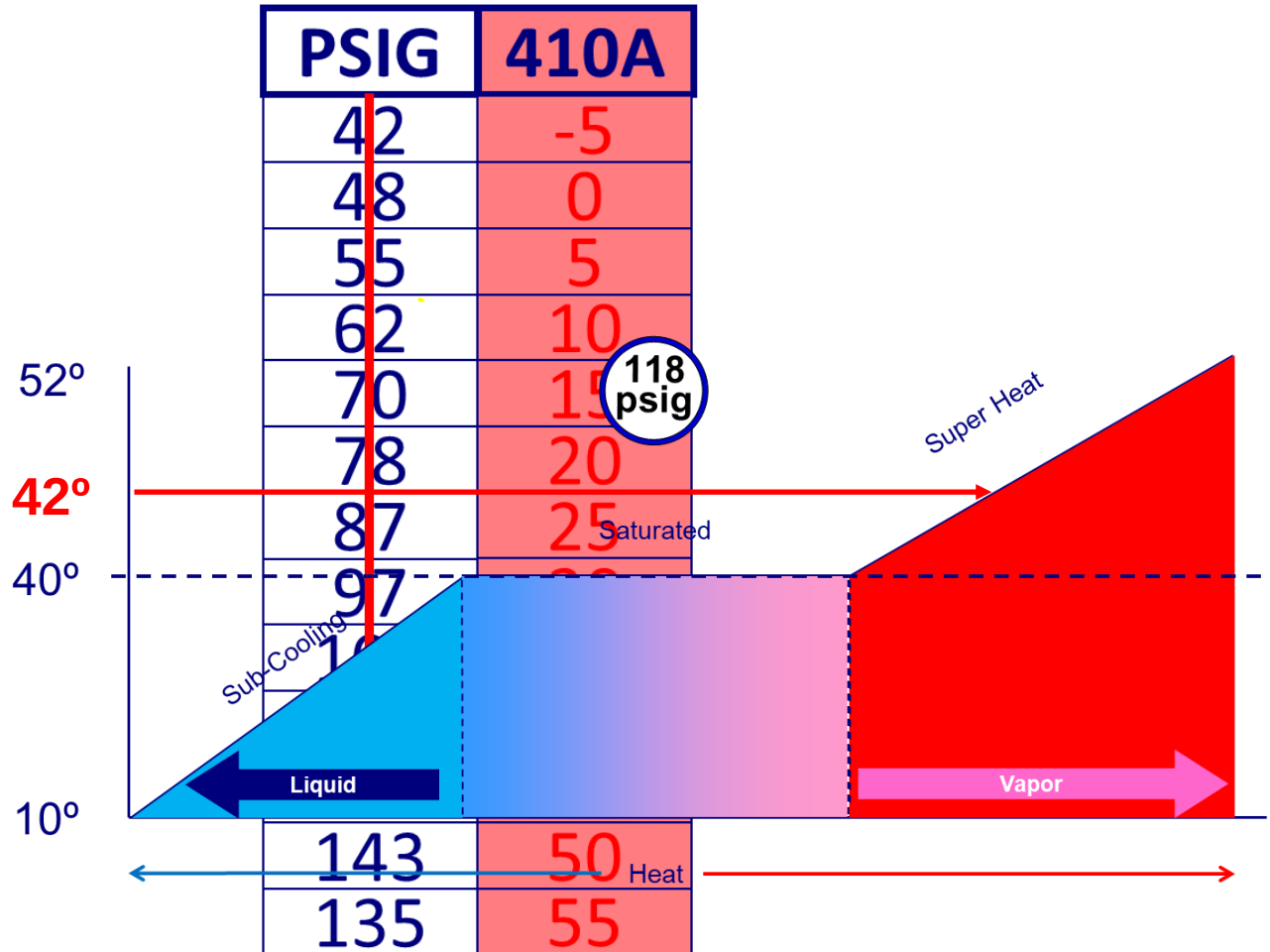


Evaporator Operation





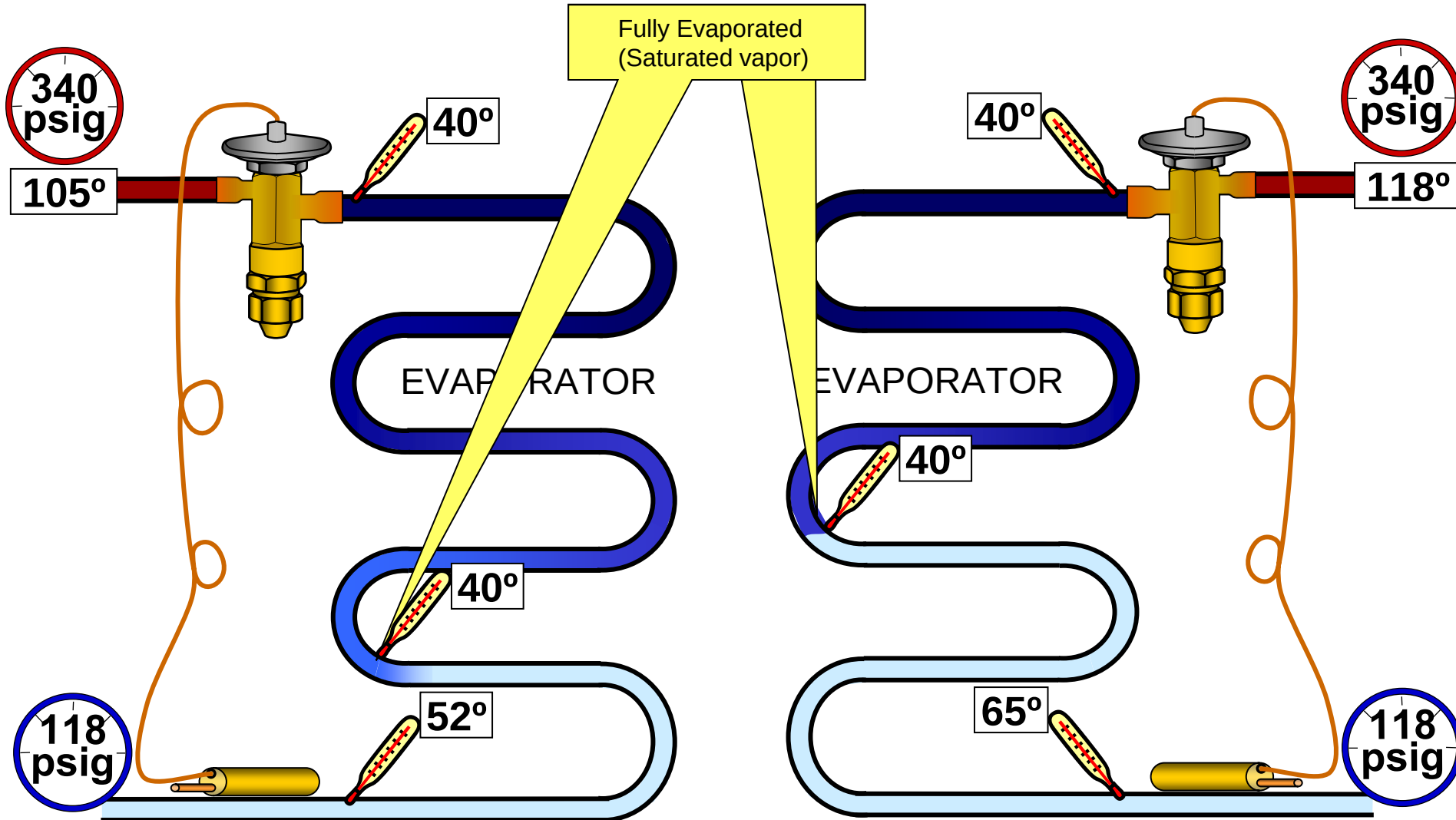
Super Heat Example

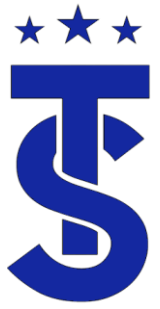


Measured Superheat is 2 degrees



Which evaporator absorbs more heat?



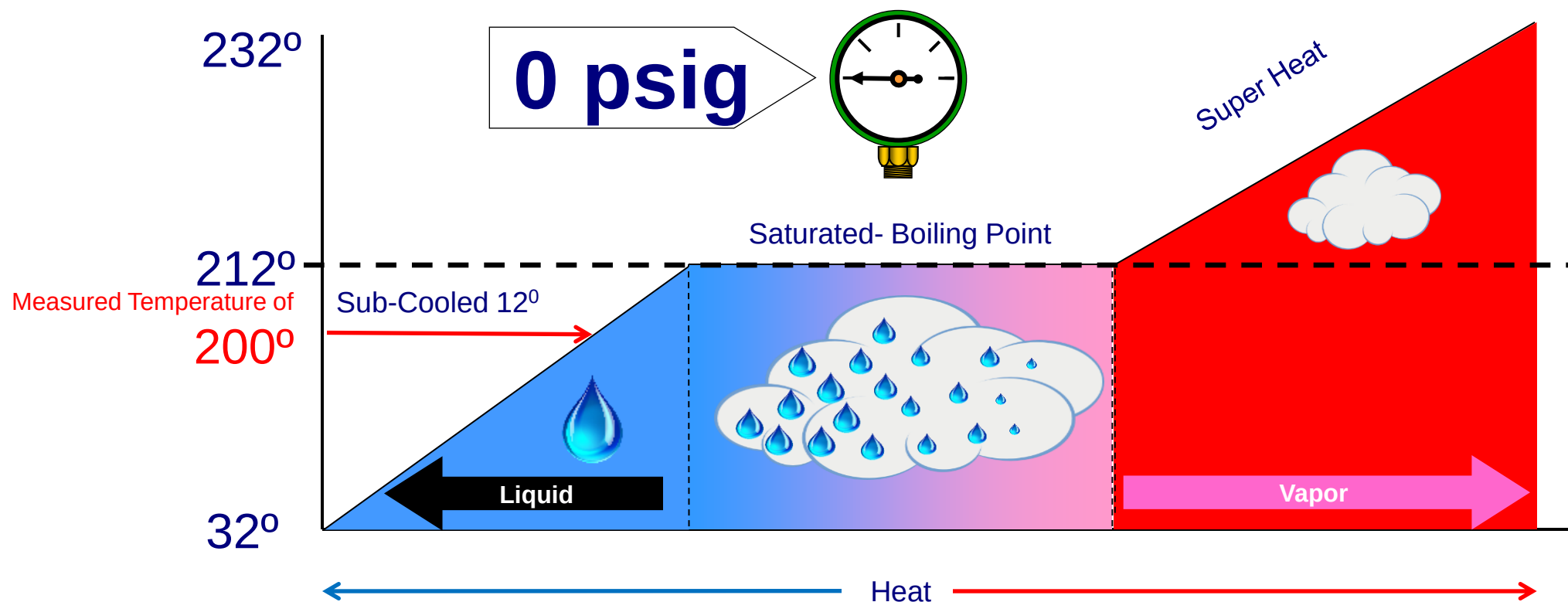


How to Calculate Sub-Cooling

- Measure high side pressure.
 - Best practice are to measure the pressure at the location where you are calculating sub-cooling. (this is not always possible)
- Use this pressure to determine the saturation temperature from the PT chart.
- Measure the Liquid Line temperature.
- Subtract the line temperature from the saturation temperature.
- The difference is the degrees of sub-Cooling.

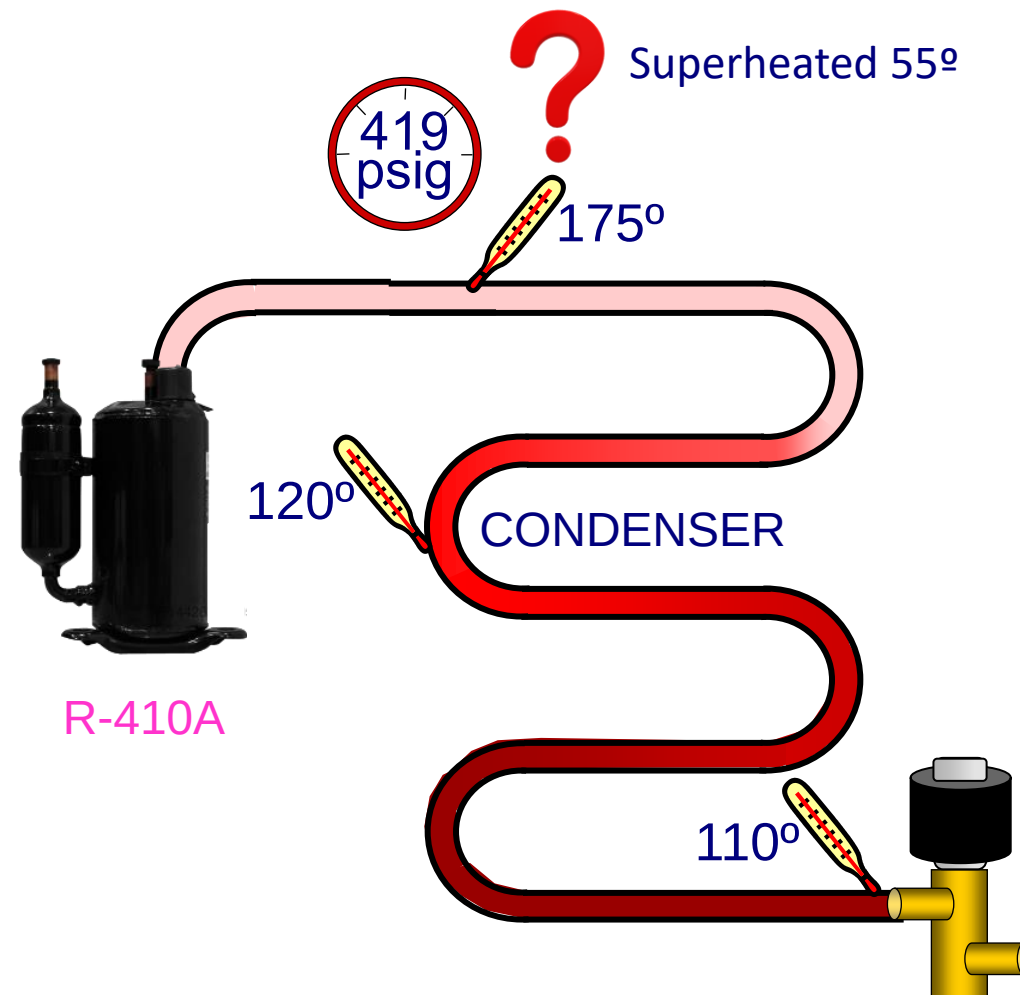
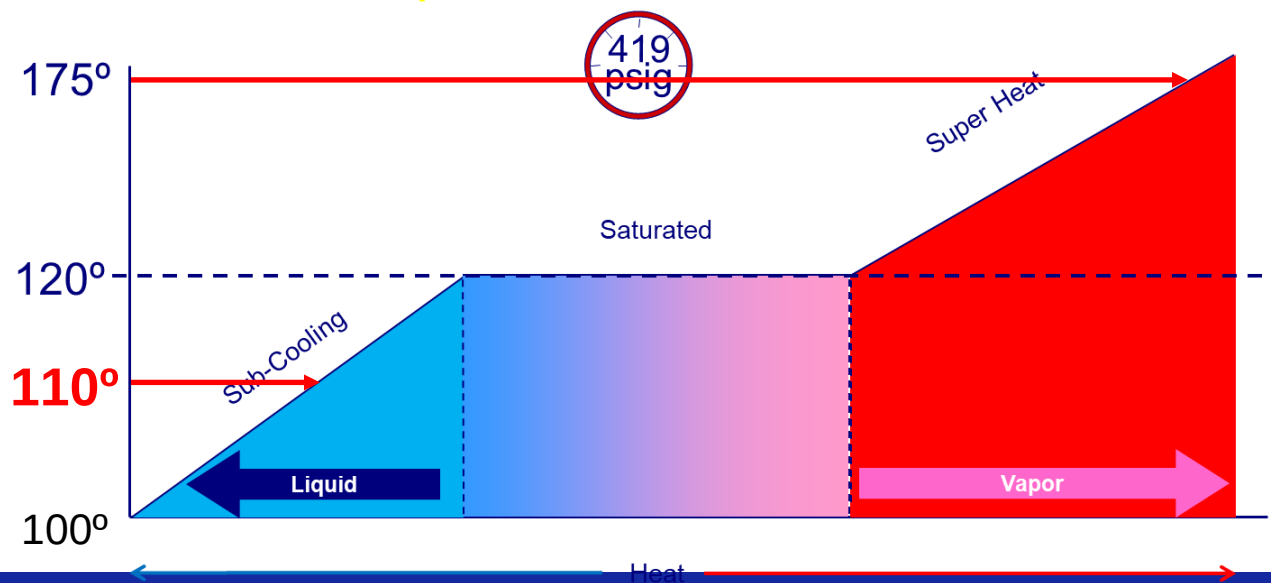


State of Water at 0 psig and a temperature of 200 degrees





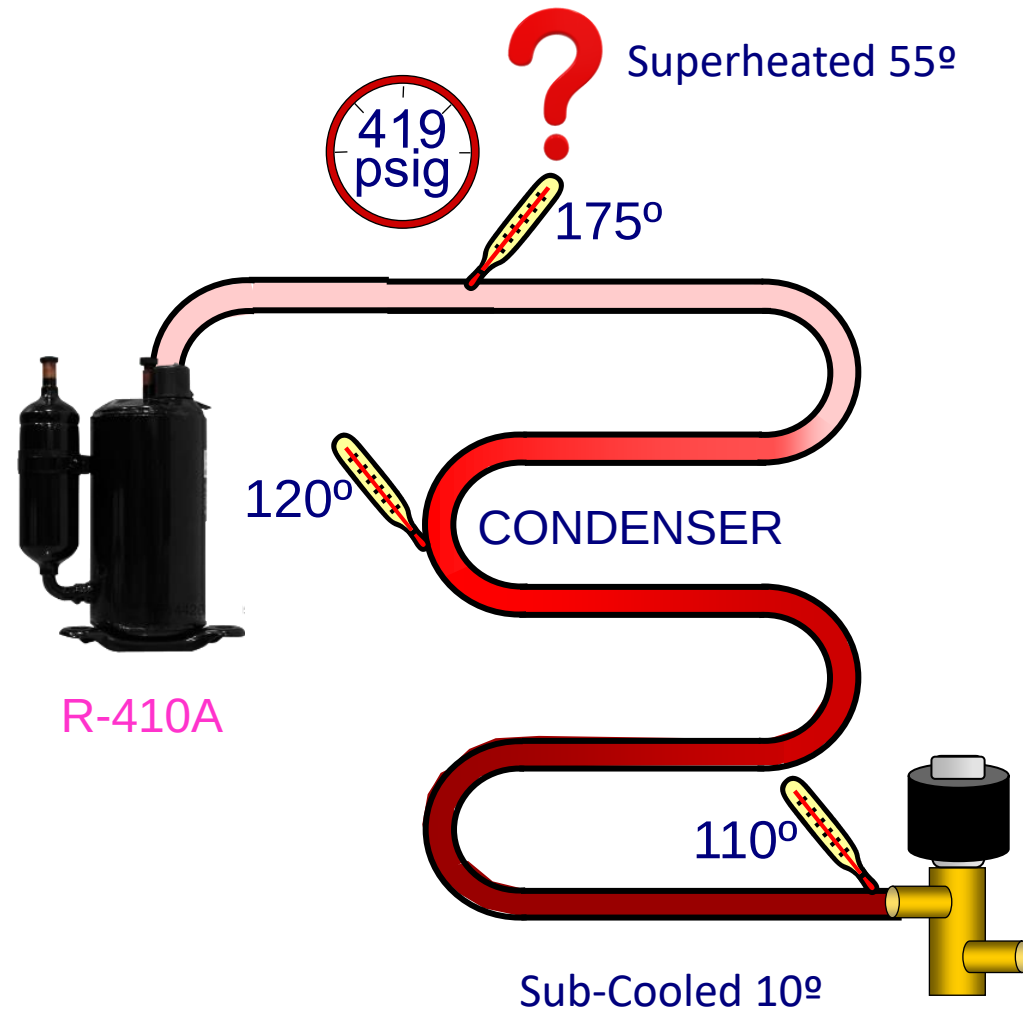
Sub-Cooling Example

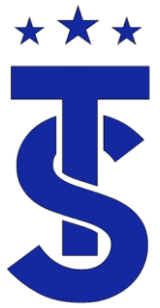




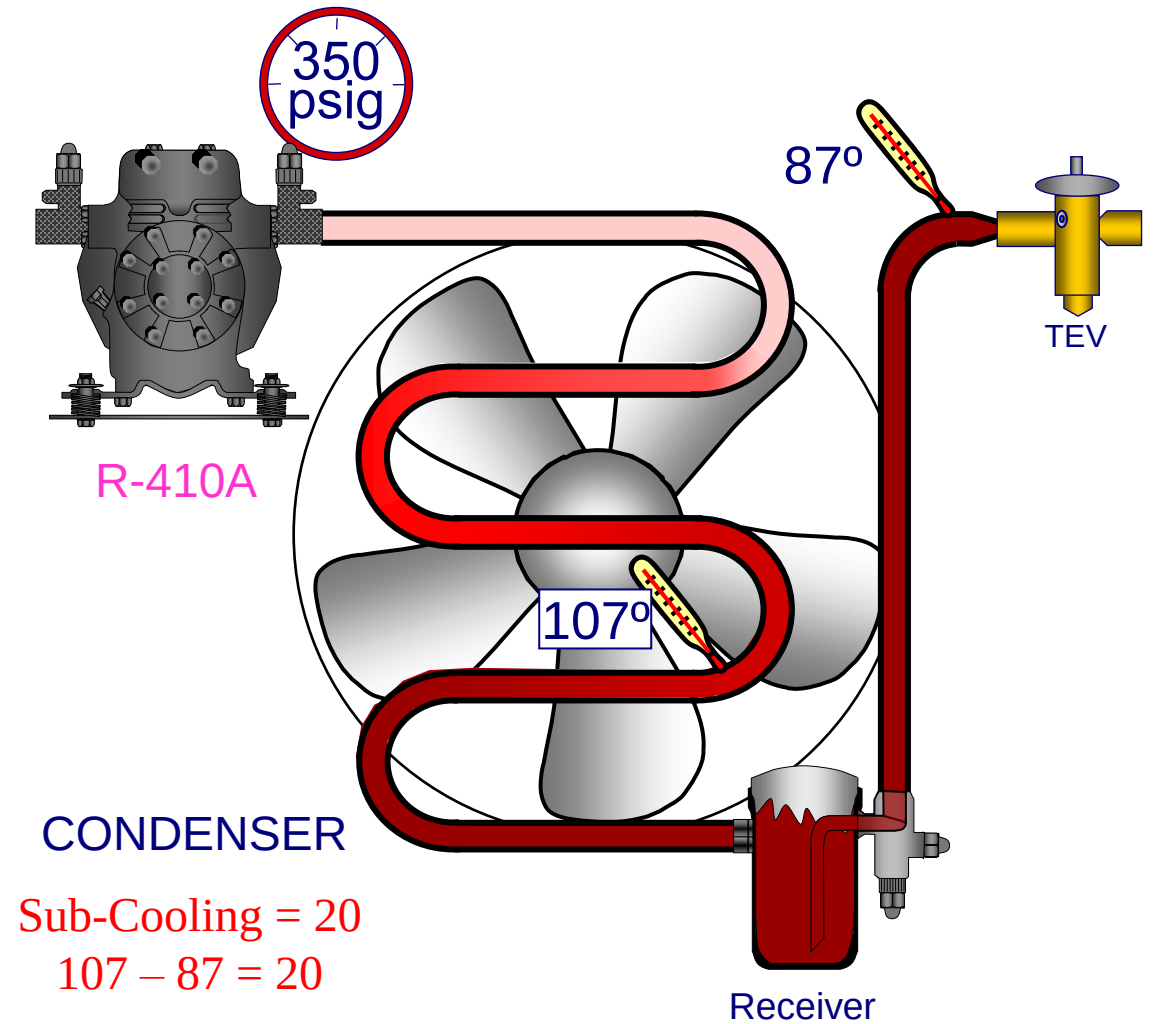
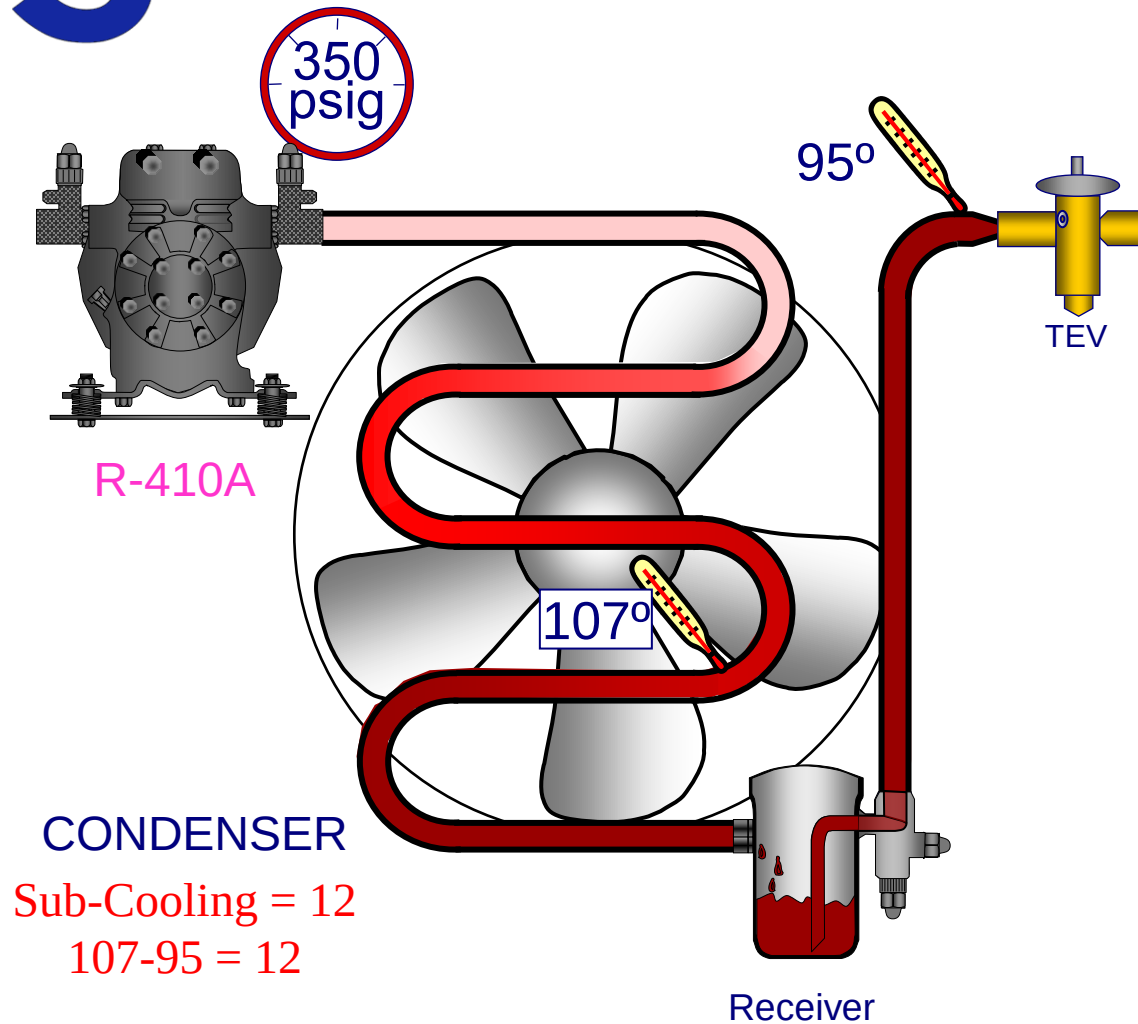
Sub-Cooling Example

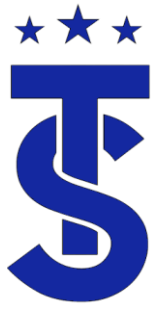
PSIG	134a	22	410A
180	123	94	63
185	125	96	65
190	127	98	66
195	128	100	68
200	130	101	70
210	134	105	73
220	137	108	76
230	140	111	79
240	143	114	82
250	146	117	84
260	149	120	87
270	152	123	89
280	155	126	92
290		128	94
300		131	96
325		137	102
350		143	107
375		149	112
400		155	117
425			121
450			126
475			130





Which system has more refrigerant

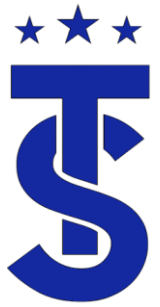




Ductless Refrigerant Charging

- How do you make sure the charge is correct in a ductless system?
- Remove it and weigh it in!
- Do not:
 - “Gas and Go”
 - Do not charge to subcooling or superheat.
 - Do not charge to beer can cold
- Why?





What Does All This Mean ?

- In order to troubleshoot a system you should first know what it should be doing.
- You must be able to determine the condition of the refrigerant through the system
- If you don't have a good refrigeration foundation you will struggle with understanding the system's operation
- This makes troubleshooting extremely difficult!





Engineering

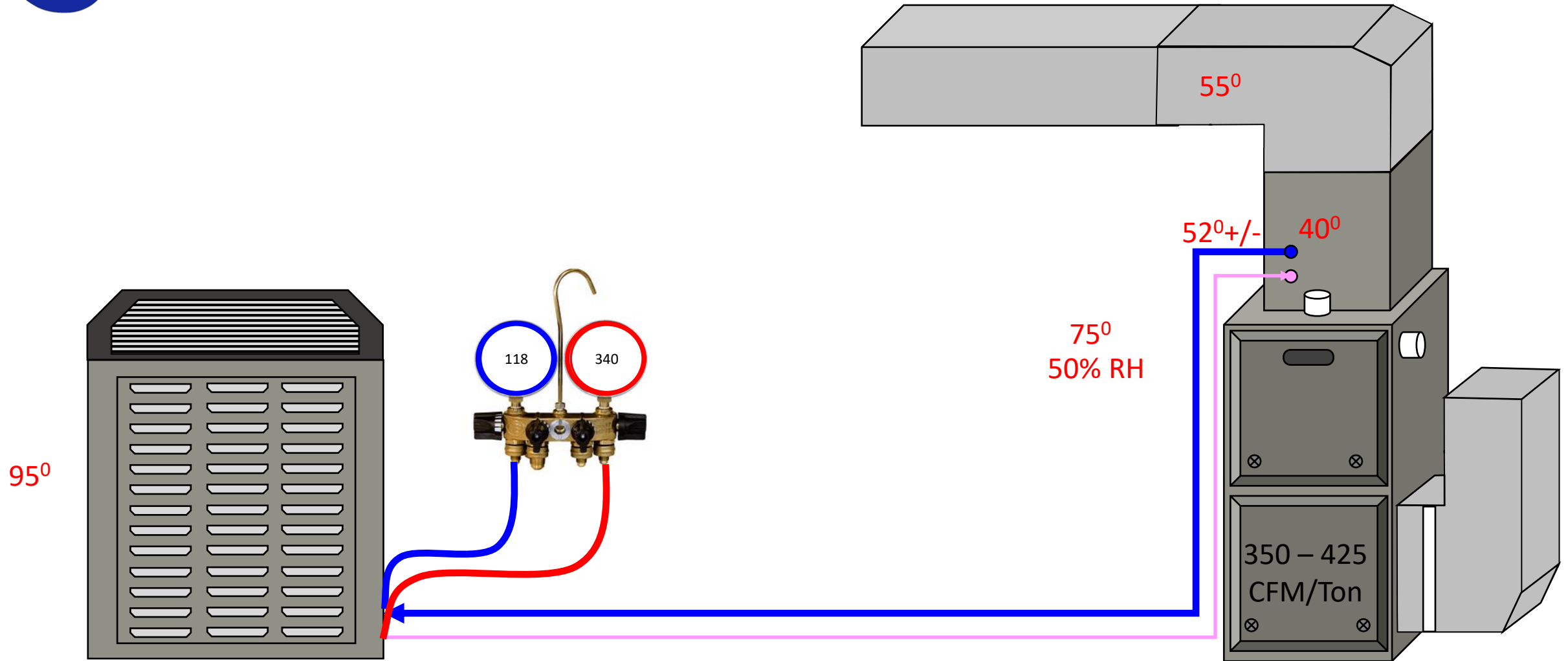
Heat-Pump Design

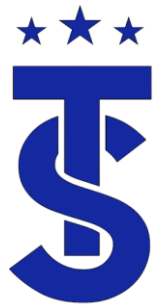




Review

Unitary Air Conditioning Design Conditions





Heat Pump Operation

- What's the function of a heat pump?

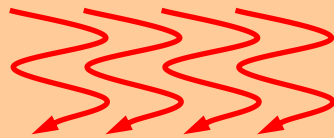


Heating Mode



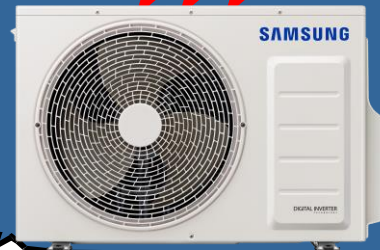
Cooling Mode

Where does the heat come from?



4 KW of heat

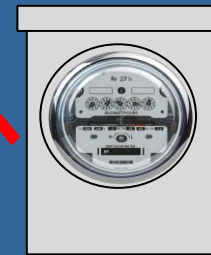
3 KW of heat



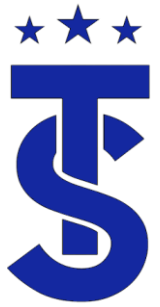
Outdoor Unit

Picks up Heat
From Outside Air and Electricity

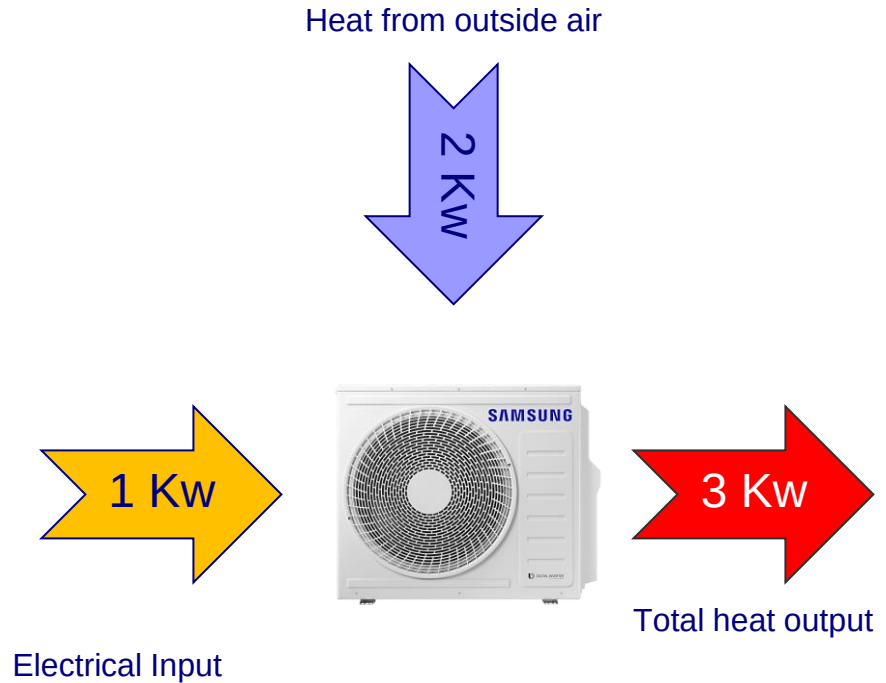
1 KW of heat



 = 1 KW of heat



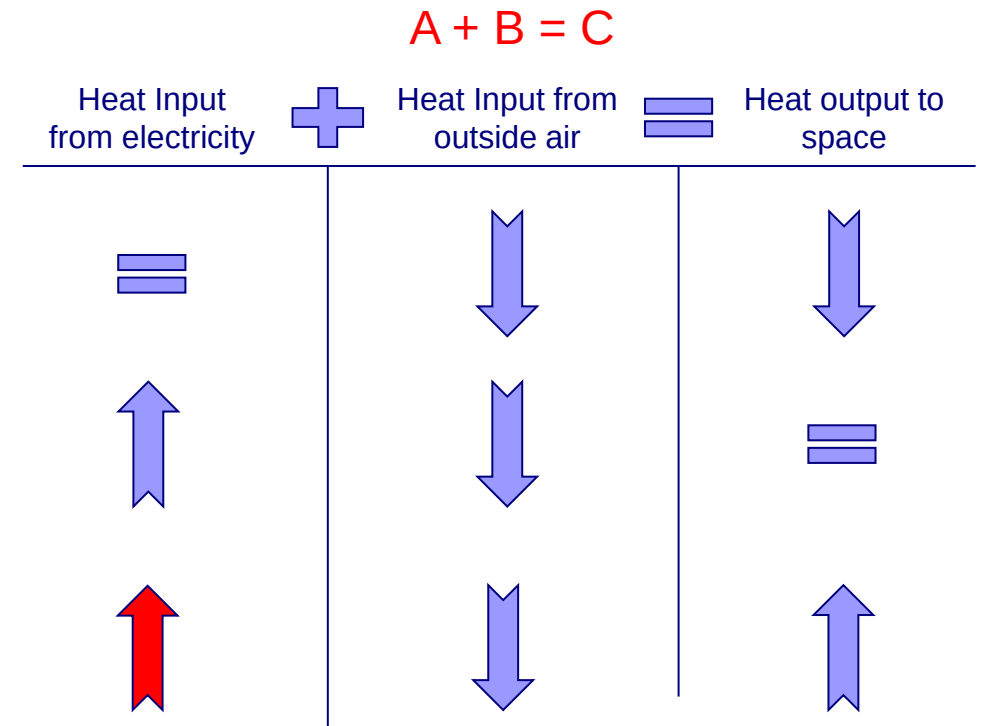
Heat Pump Operation



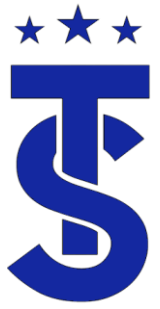
Electrical BTU Input + Outdoor air BTU input = System Output

$$\text{COP} = 3:1$$

COP is the ratio of how much useful heat a heat pump will produce at a given energy input.



Where does the heat come from?



Heat Pump Design

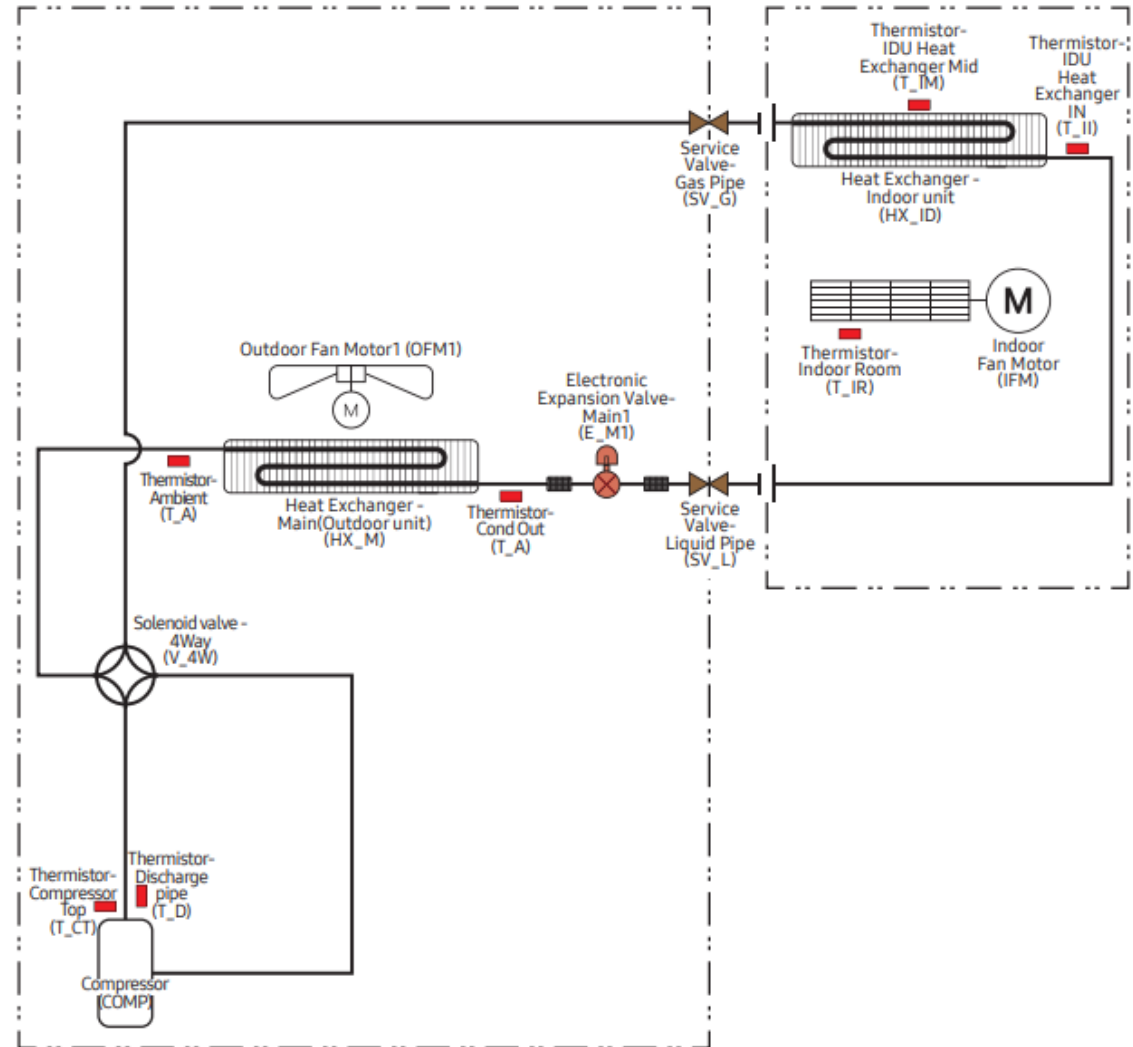
- All manufacturers are bound by the laws of thermodynamics.
- Heat pumps are designed to perform a specific function under specific set of conditions.
- They must produce a specific output at a given condition “Rated Condition”.
- Knowing this information is key to understanding how the system operates and how to trouble shoot system issues.
- Remember that refrigeration is based on temperature!

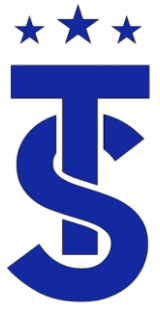


Heat Pump Design

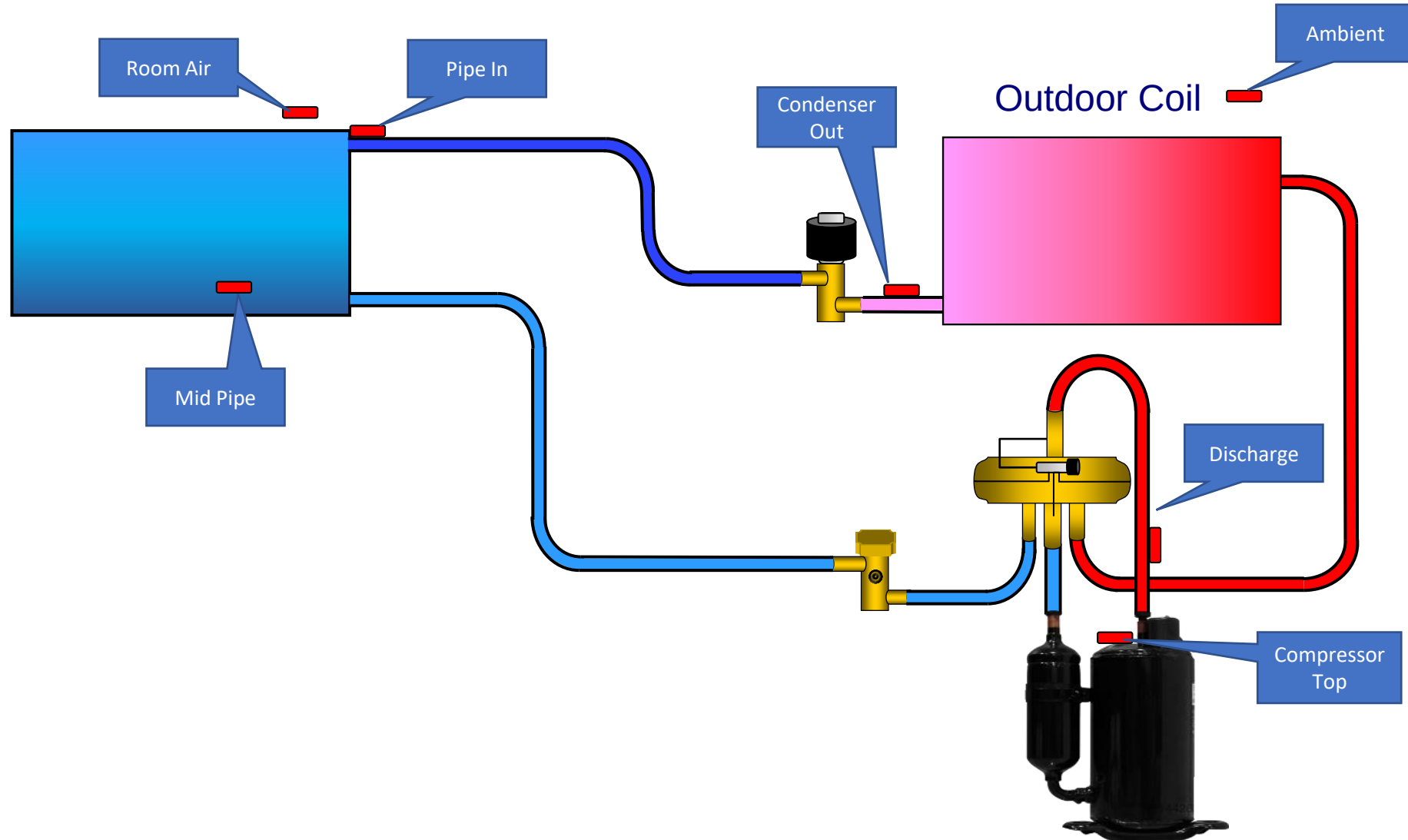
1:1 Piping Diagram

- What do you not see in the piping diagram?
 - Pressure Transducers.
- How does the system operate if it can't determine the system pressure?
 - There is a specific relationship between pressure and temperature of a refrigerant.
- How does the system operate off temperature and how are those temperatures determined.
 - Pressure Enthalpy diagrams



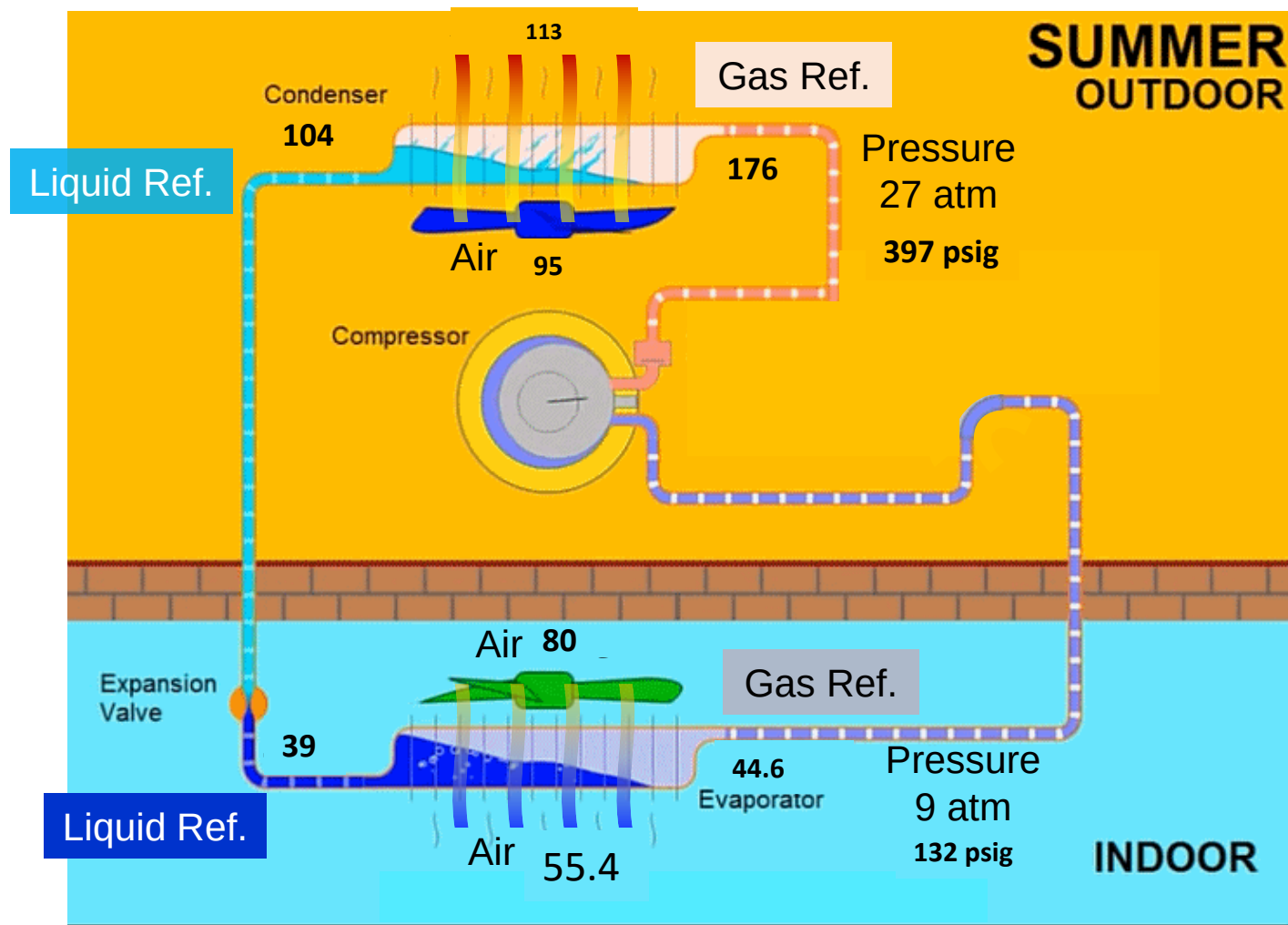


Thermistor Locations





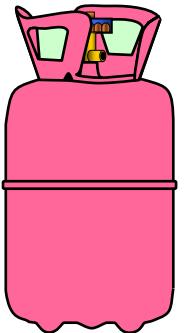
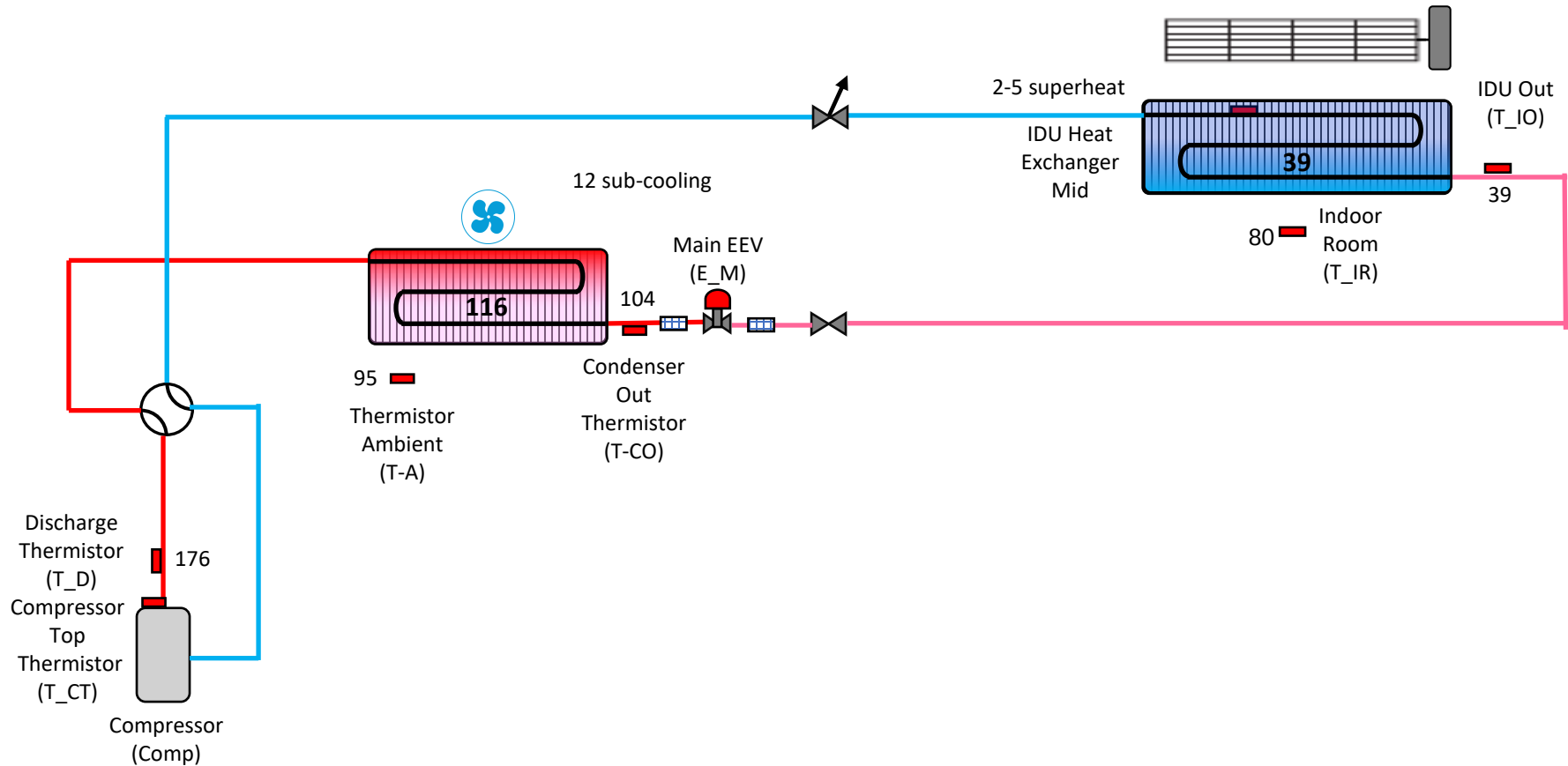
Design Conditions





System Design

12,000 Btu.





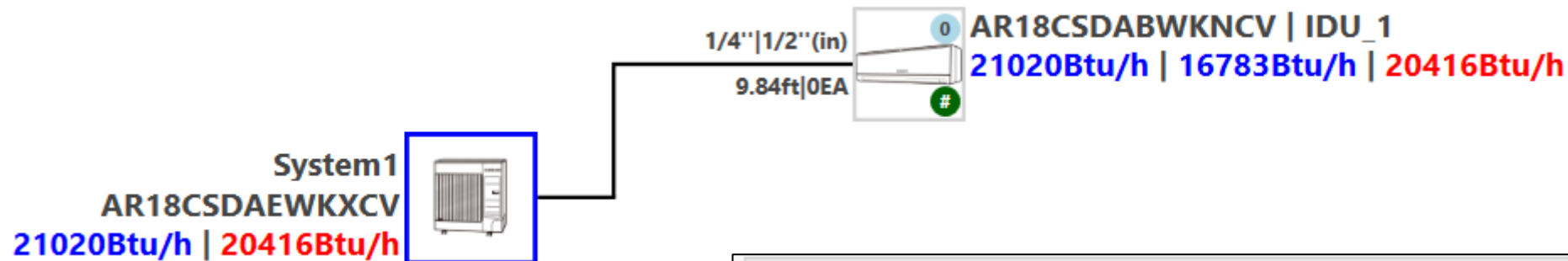
System Operation at Rated Conditions

- In order to know how the system operates and what are considered normal temperatures within the system you need to know how systems are rated.
- The Rated Capacity of a unit is determined in a lab, operating at a set of specific parameters.

Model	Indoor Unit Model Number (US Code)		AR18CSDABWKNCV (RNS18ABC)
	Outdoor Unit Model Number (US Code)		AR18CSDAEWKXCV (RXS18AEC)
Performance	Nominal Capacity	Cooling / Heating (Btu/h)	18,000 / 20,000
	Capacity Range	Cooling (Btu/h)	7,500 - 25,000
		Heating (Btu/h)	7,500 - 41,000
	SEER2 / EER2		21 / 13.15
	HSPF2 (Region IV)		9.7



Are they sized Correctly?



Model	Indoor Unit Model Number (US Code)		AR18CSDABWKNCV (RNS18ABC)
	Outdoor Unit Model Number (US Code)		AR18CSDAEWKXCV (RXS18AEC)
Performance	Nominal Capacity	Cooling / Heating (Btu/h)	18,000 / 20,000
	Capacity Range	Cooling (Btu/h)	7,500 - 25,000
		Heating (Btu/h)	7,500 - 41,000
	SEER2 / EER2		21 / 13.15
	HSPF2 (Region IV)		9.7

Design Condition

Country

USA

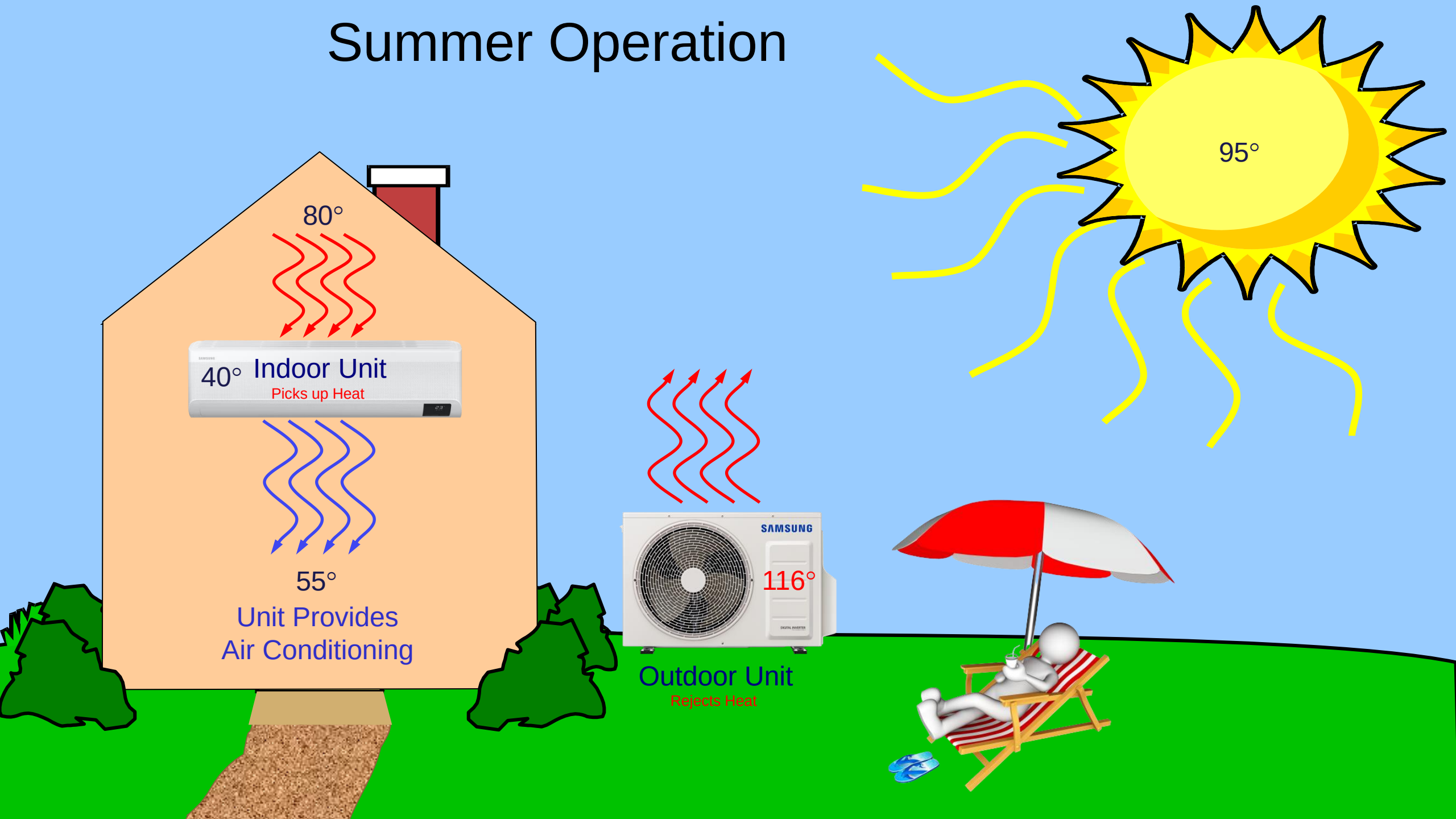
City

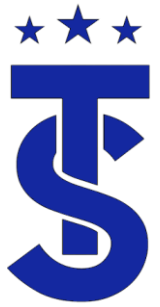
Albany, NY

Design Condition (Air)

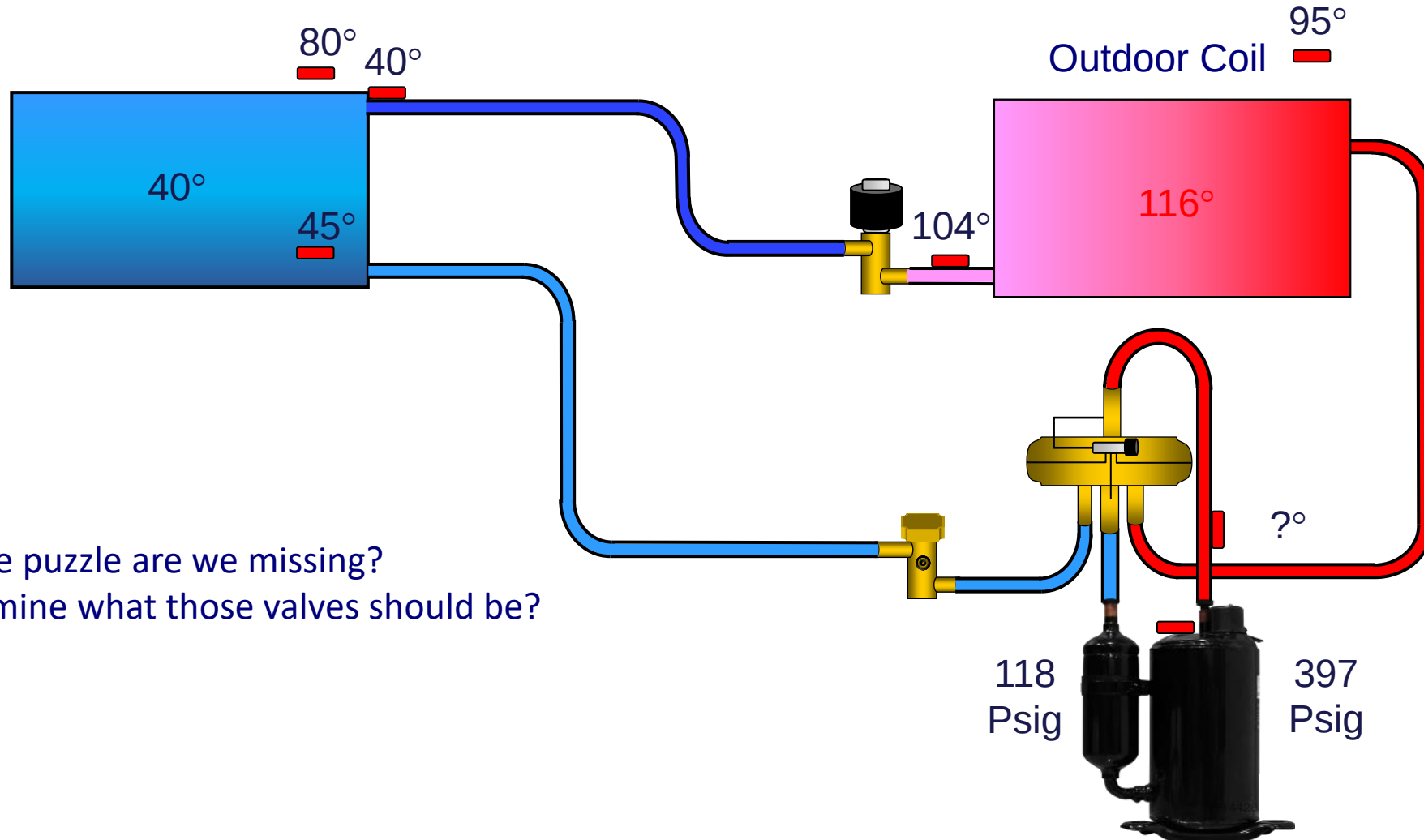
°F	Cooling		Heating	
	Outdoor	Indoor	Outdoor	Indoor
DB Temperature	91.0	80.6	-4.0	68.0
WB Temperature	73.0	66.2	-5.4	59.0
Relative Humidity	42.7	47.0	50.0	59.0

Summer Operation





Cooling Cycle



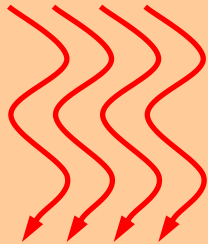
What Pieces of the puzzle are we missing?
How do we determine what those valves should be?

Winter Operation

70°



120° Indoor Unit
Rejects Heat



Unit Provides
Heating

Changes the temperature of the air by
30-35 degrees at RATED Conditions



Outdoor Unit

Picks up Heat

@ Rated Conditions

45°

5°

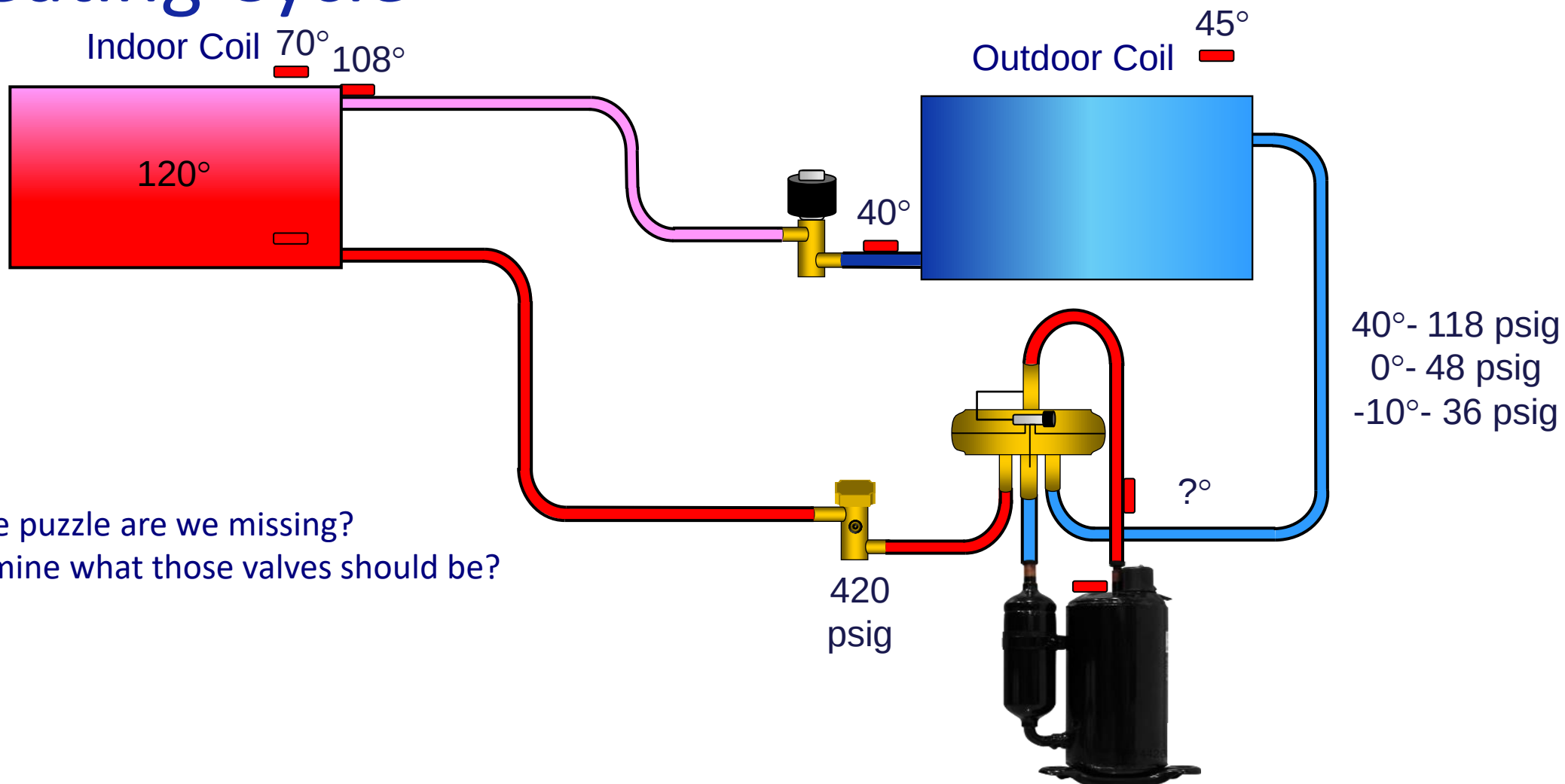
-5°



The outside coil temperature is 2 - 10°C colder than
outside air temperature



Ductless Heat Pump Heating Cycle



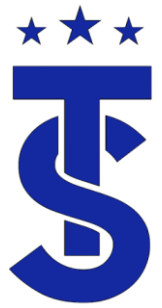
What Pieces of the puzzle are we missing?
How do we determine what those valves should be?

System Operation

- The operation of a refrigeration is very predictable.
- At a given set of condition, you can determine the operating temperatures and pressures of a refrigeration system.
- These know values are used as operating targets.
- The system adjust as the actual measurements deviate for the targets.



There is no mystery!



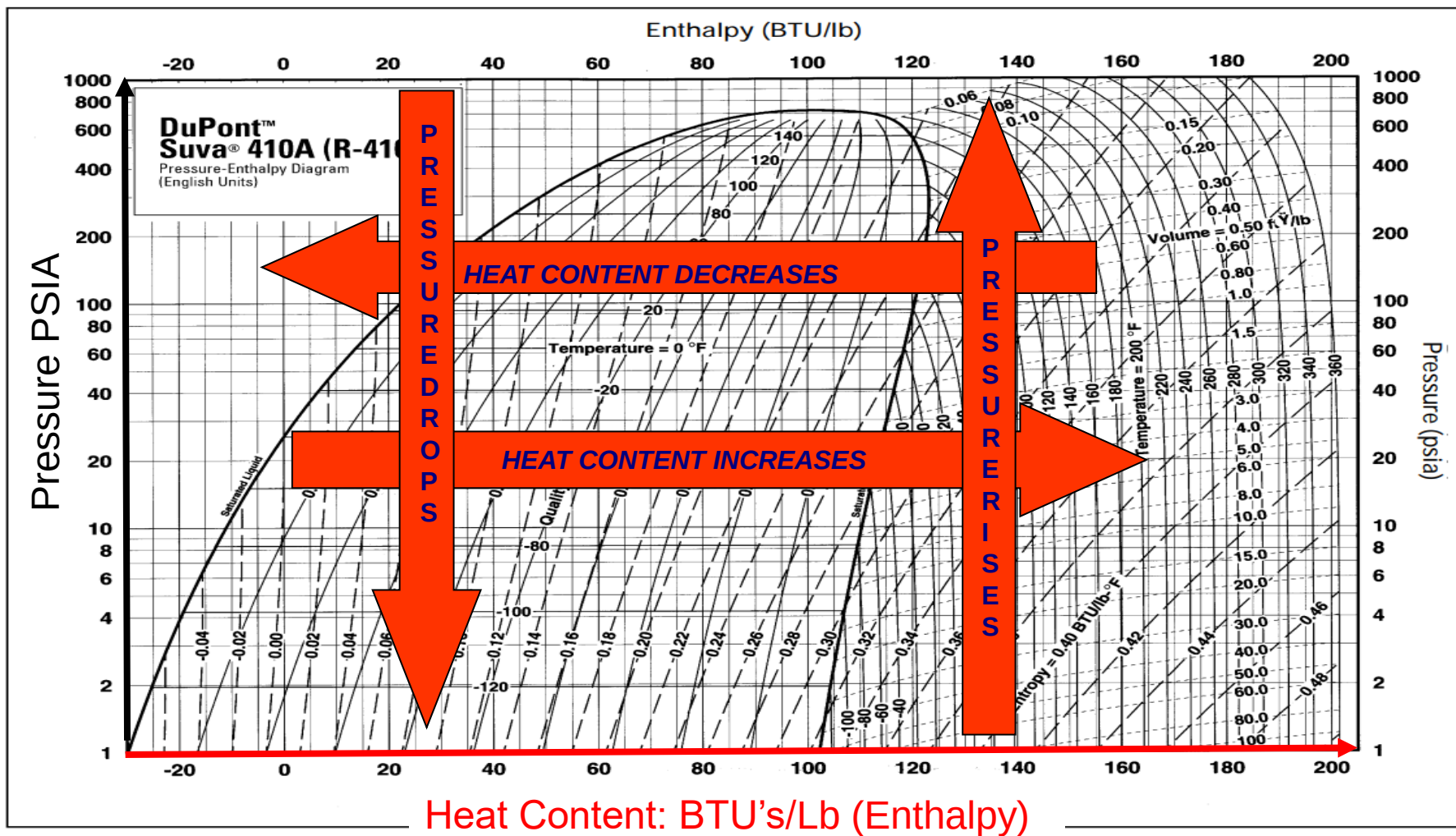
PE Diagrams

- Are charts that describe the relationship between pressure and enthalpy (total heat content)of a select refrigerant.
- They are the engineering charts for a refrigeration system.
- They show the total heat transfer within a system.
- A refrigeration's system operation is extremely predictable.
- PE diagrams show the operating characteristics of a system at various operating conditions.
- They are used to calculate:
 - Net Refrigeration Effect of the Evaporator
 - Heat of Compression of the compressor
 - Total heat of rejection from the condenser
 - Discharge temperature
 - The amount of refrigerant required for system operation



PE Diagrams Basics

Enthalpy and Pressure



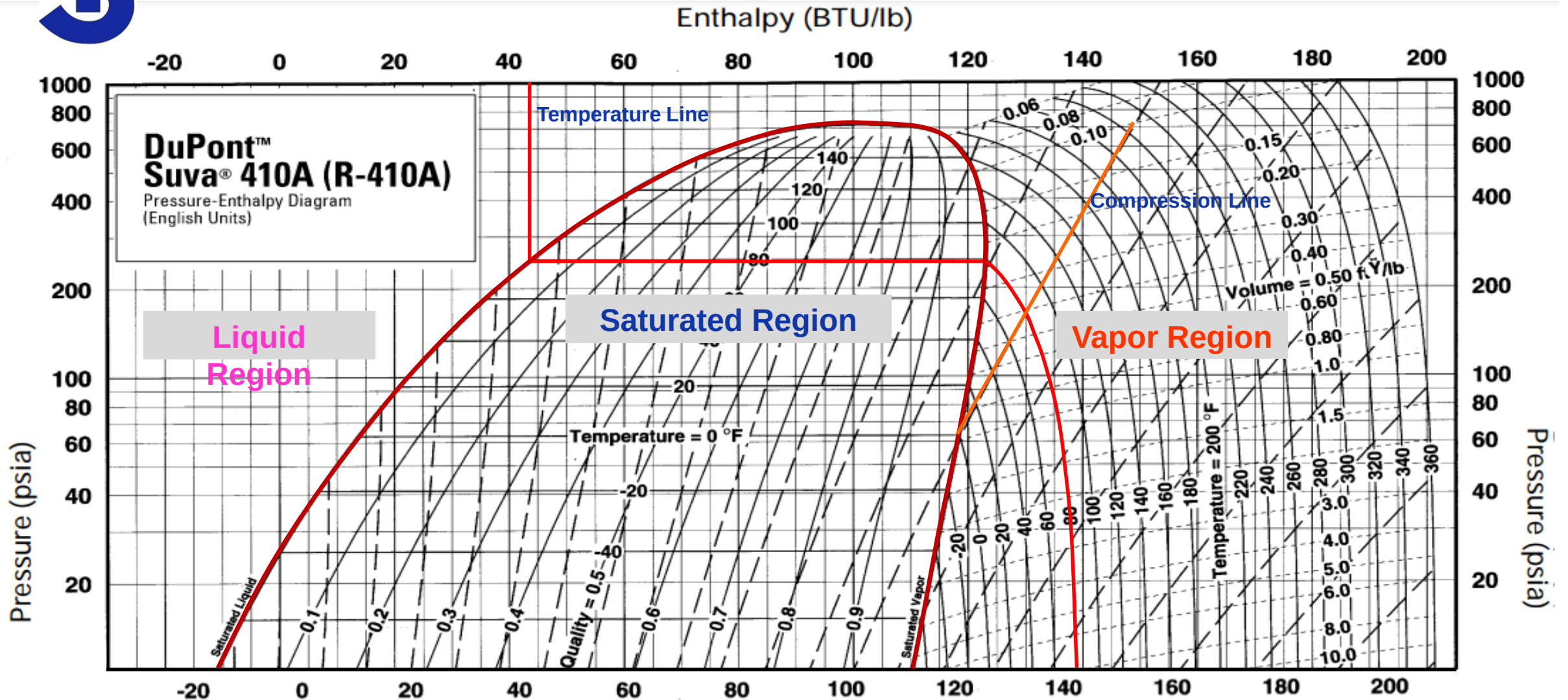
PSIA =

Gauge Pressure + 14.7



PE Diagram Basics

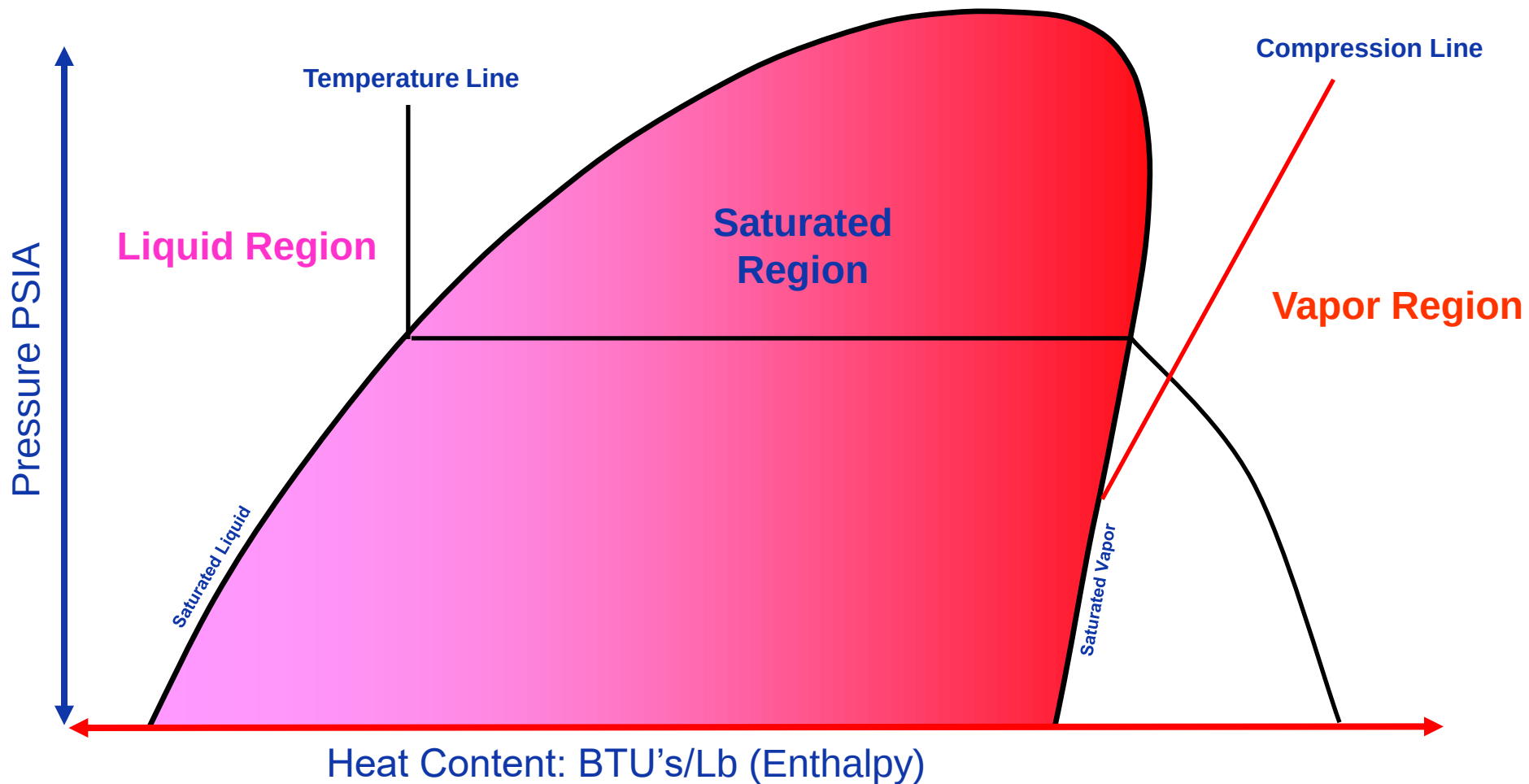
States of Refrigerant, Lines of Temperature and Compression





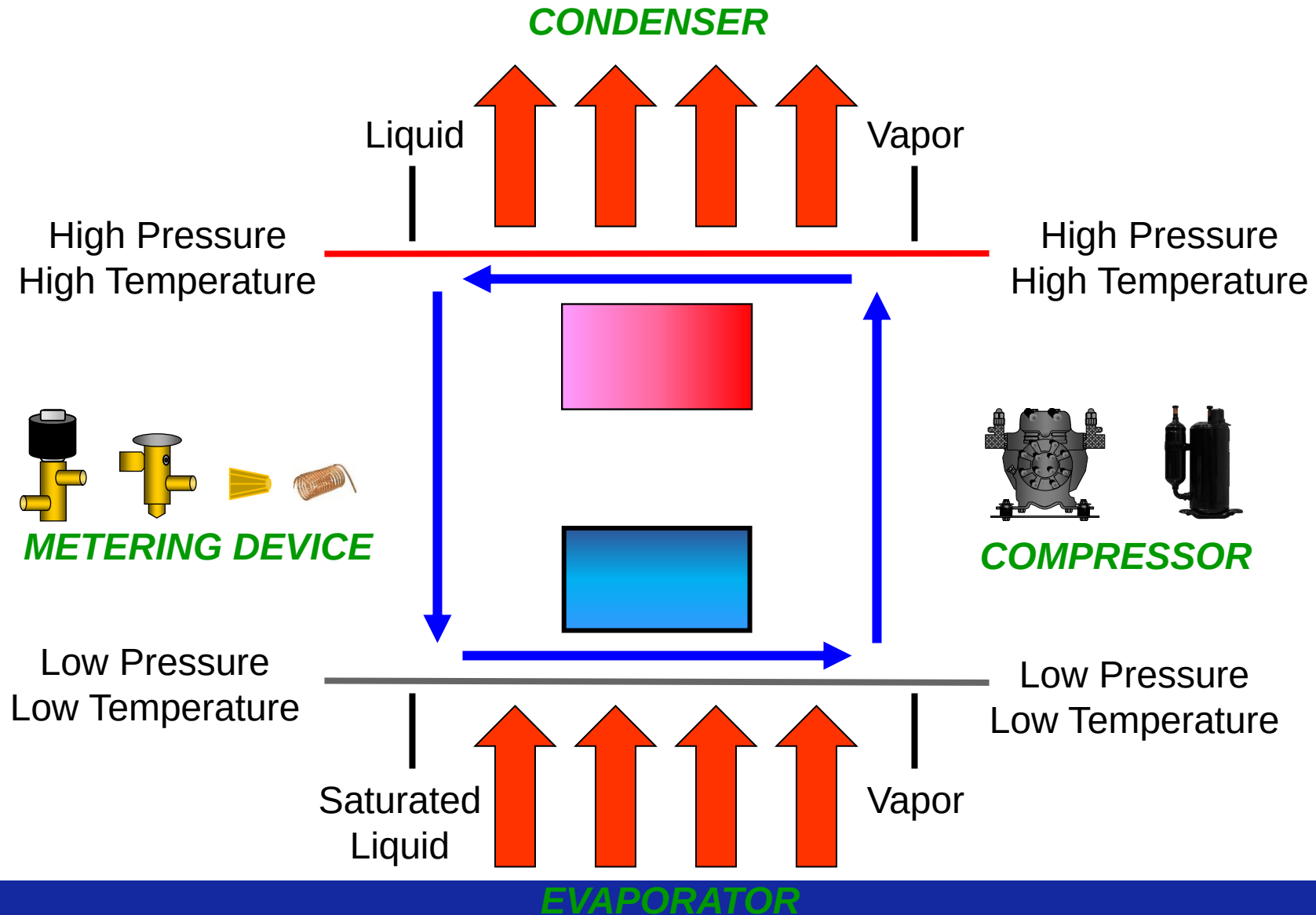
PE Diagrams

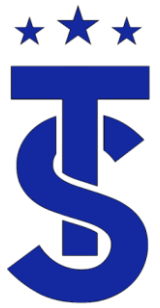
States of Refrigerant, Lines of Temperature and Compression



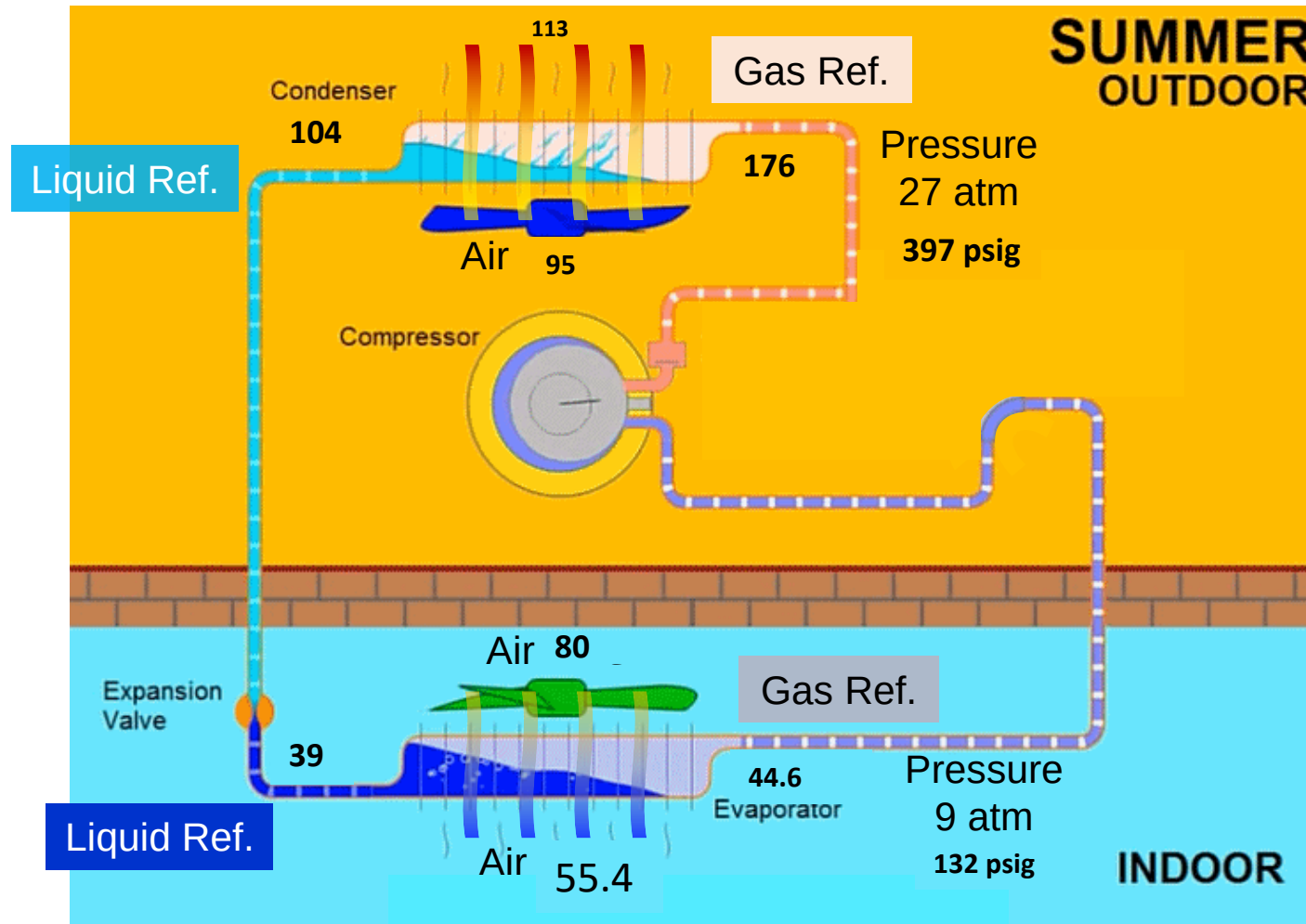


Refrigeration Cycle Review





Design Conditions





PE Diagram Basics

System Design

Saturation
Temperature

116°

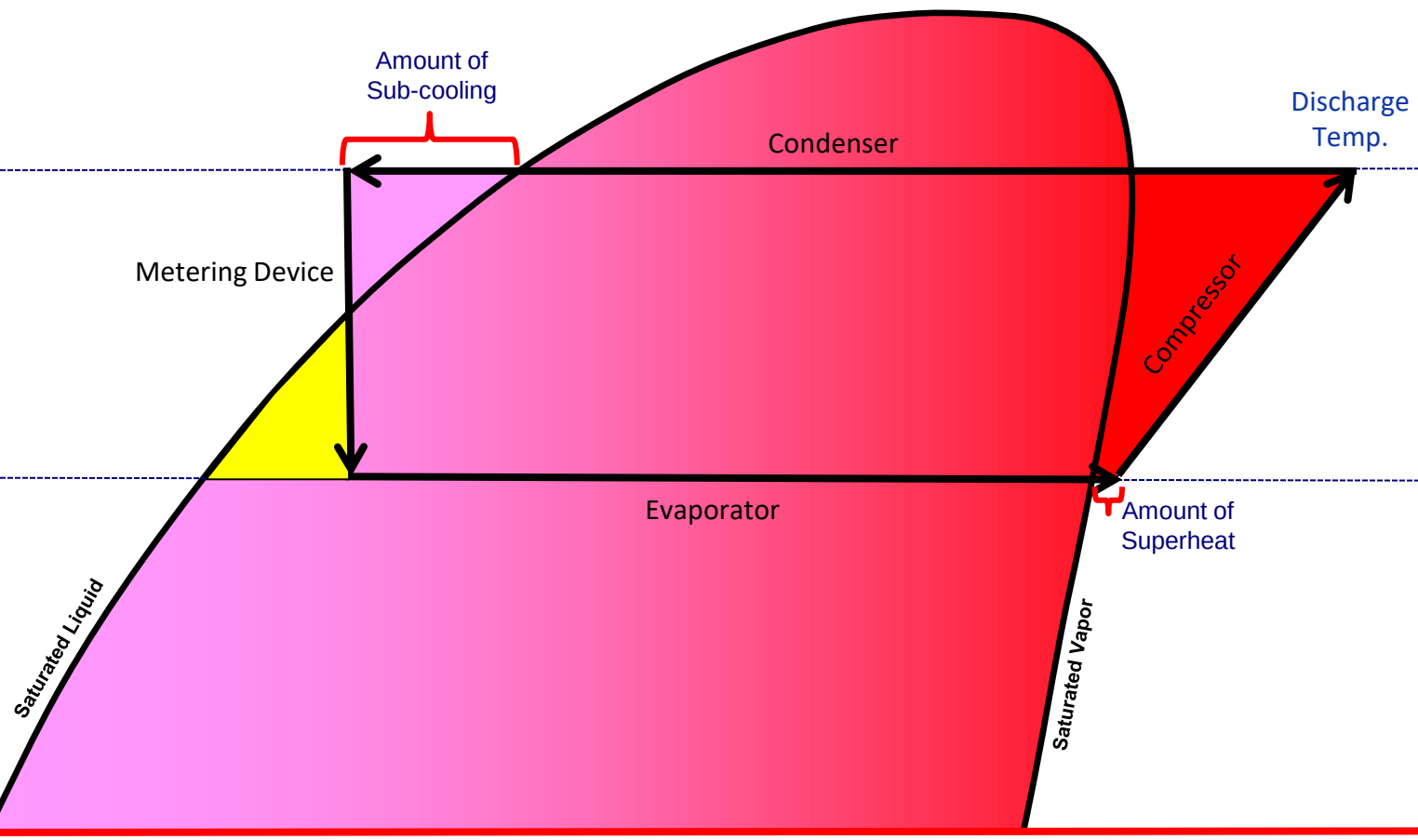
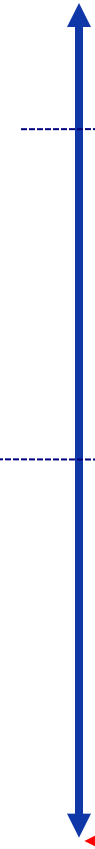
Pressure
PSIG

High Side
411 PSIA

40°

Low Side
133PSIA

Pressure
PSIA



Discharge
Temp.

Condenser

Metering Device

Evaporator

Compressor

Saturated Liquid

Saturated Vapor

Amount of
Superheat

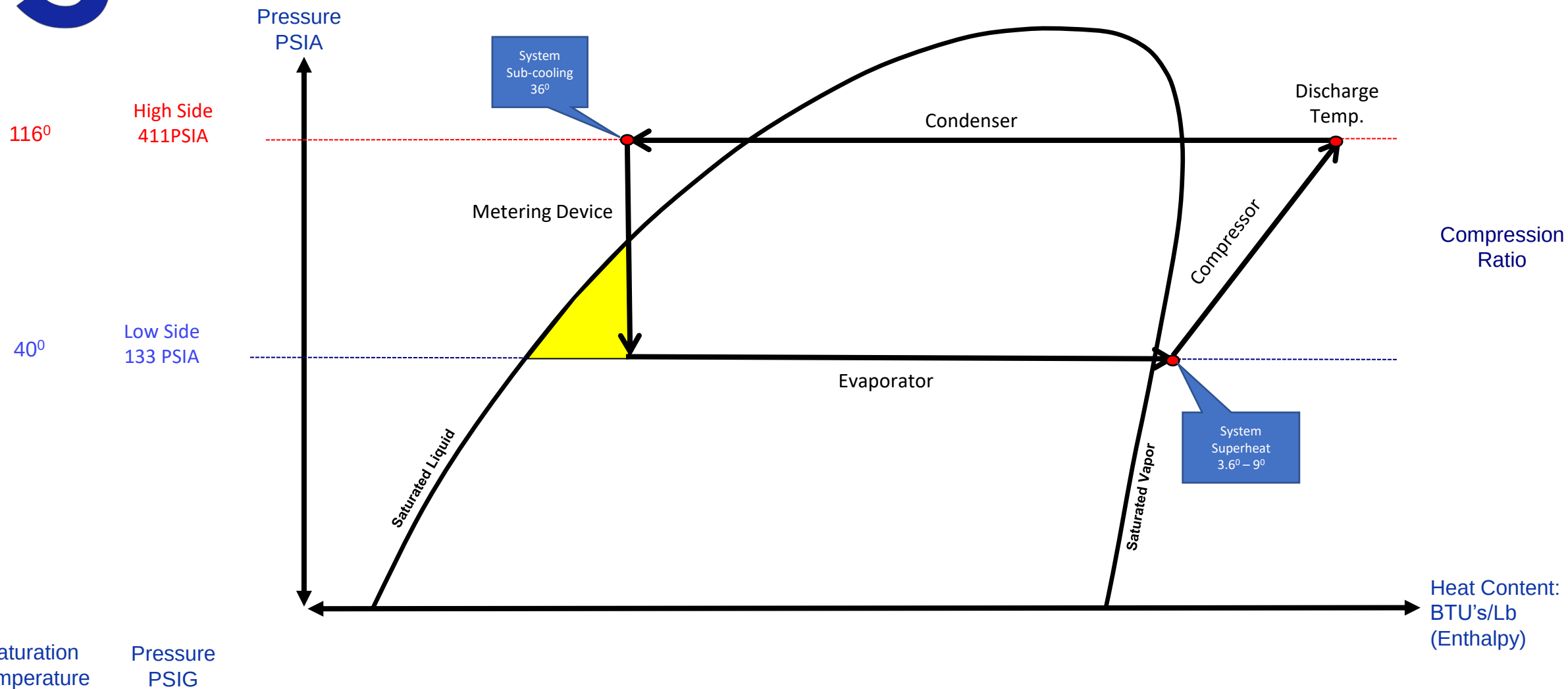
Amount of
Sub-cooling

Heat Content:
BTU's/Lb
(Enthalpy)



PE Diagram Basics

System Design





PE Diagram Basics

What are the good for?

Saturation
Temperature

116°

Pressure
PSIG

High Side
411 PSIA

Pressure
PSIA

Low Side
133PSIA

40°

Metering Device

Total Heat of
Rejection

Amount of
Sub-cooling

Condenser

Discharge
Temp.

Compression
Ratio

Evaporator

Compressor

Amount of
Superheat

Saturated Liquid

Saturated Vapor

Heat Content:
BTU's/Lb
(Enthalpy)

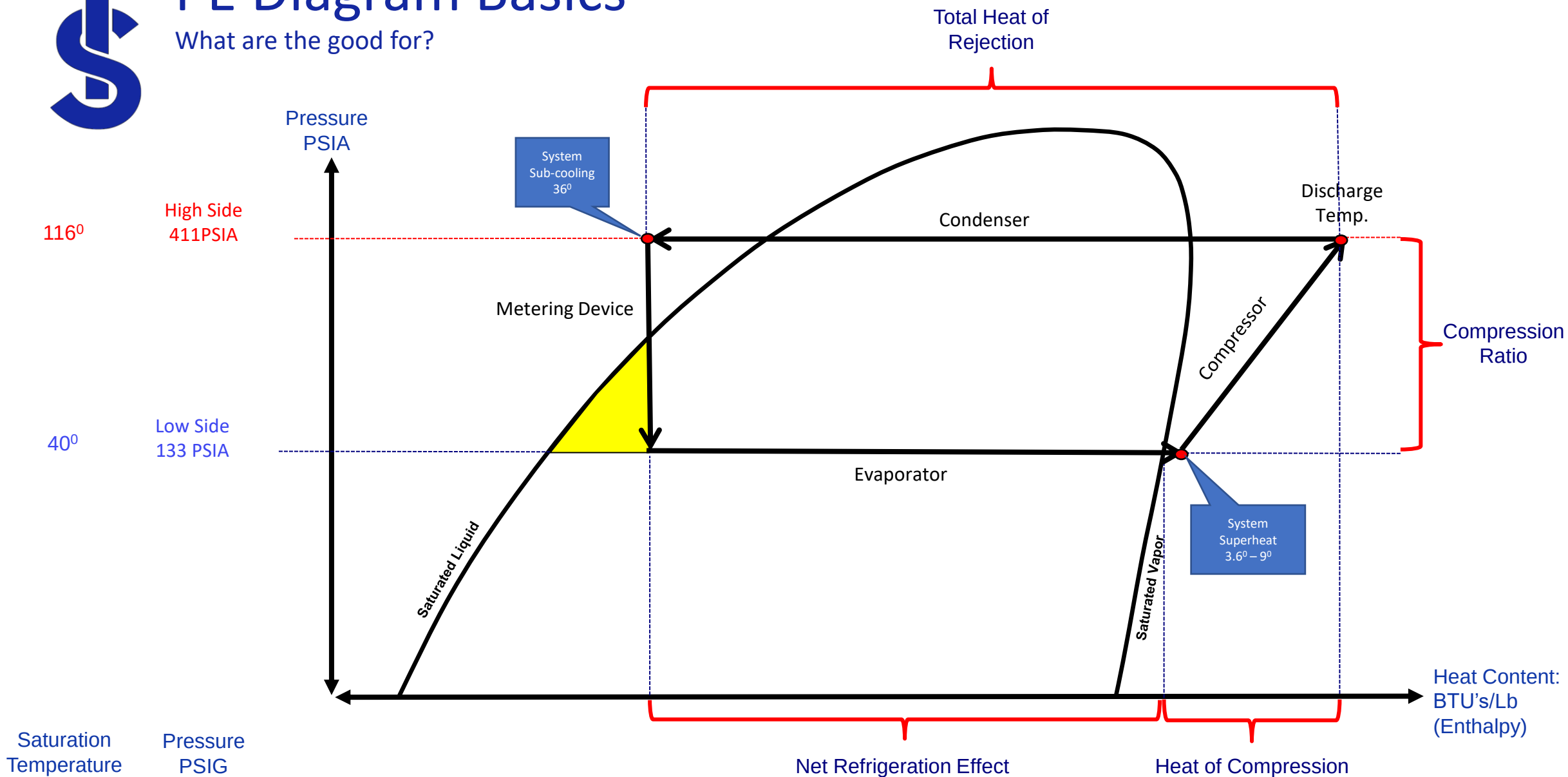
Net Refrigeration Effect

Heat of Compression



PE Diagram Basics

What are the good for?





Ductless EEV & Liquid Line

Saturation
Temperature

116°

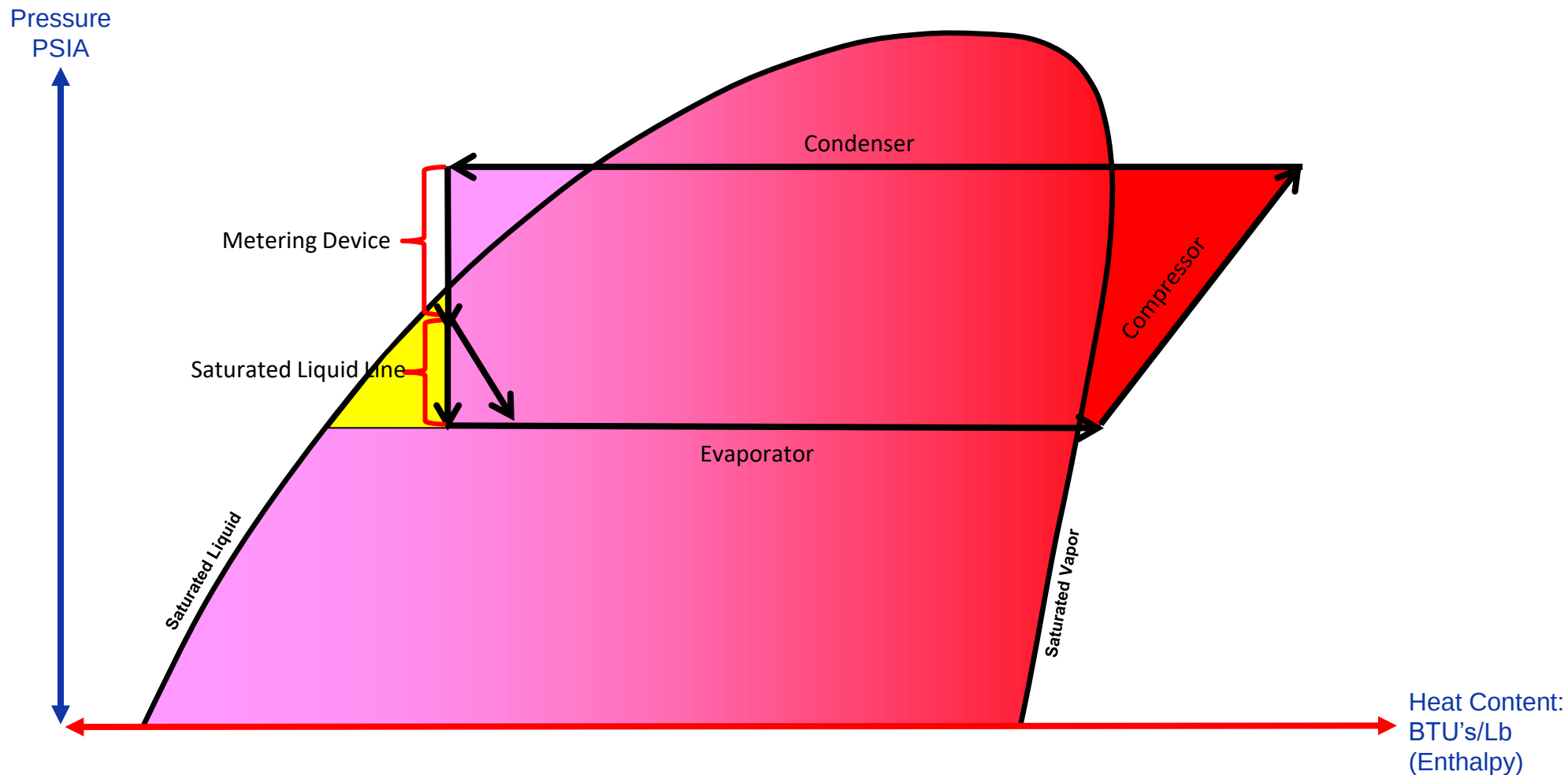
Pressure
PSIG

High Side
411 PSIA

40°

Low Side
133PSIA

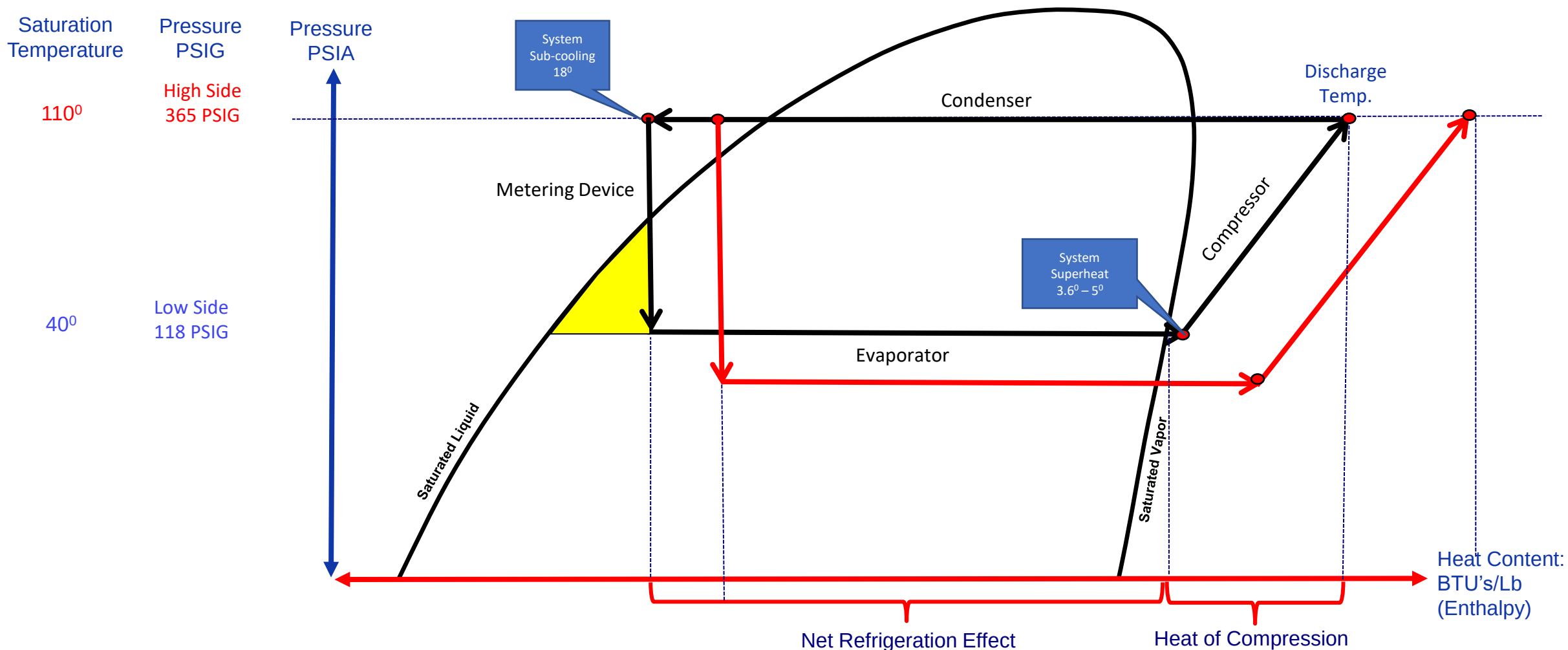
What if the piping is
run longer than
specification or
It travel through a hot
attic?

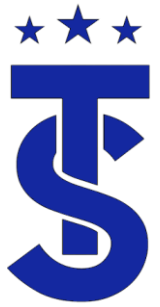




PE Diagram

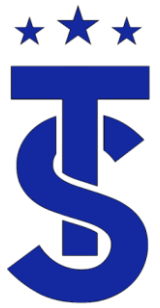
Why are they a useful training aide?
What happens when a system is short on gas?





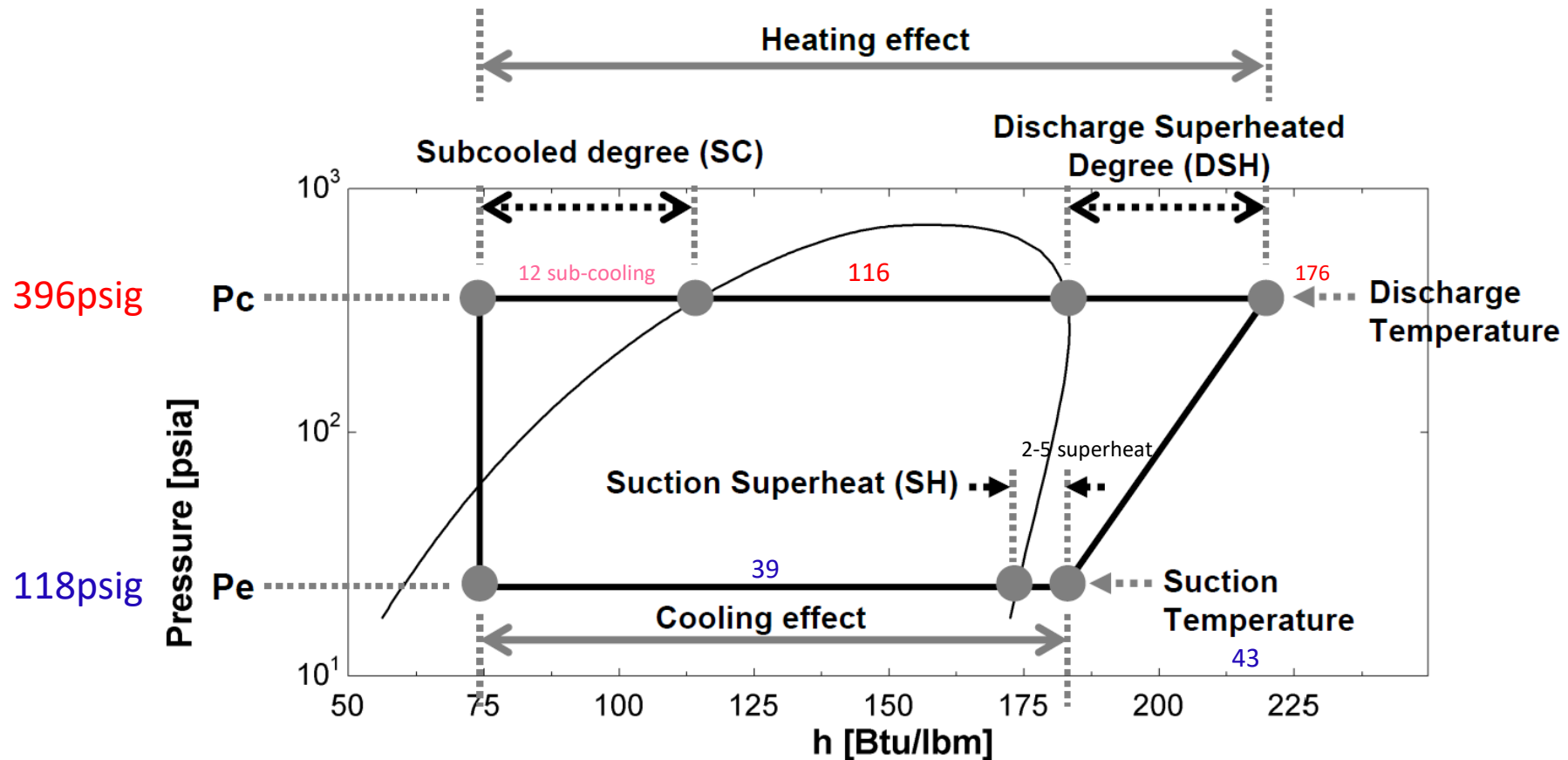
Let's put it to use

- Systems are designed to operate at a given set of conditions.
- At those conditions there will be a specific:
 - Evaporator temperature
 - Air flow across the evaporator
 - Condensing temperature
 - Space temperature
 - Evaporator super heat
 - Sub-cooling
 - Outdoor temperature
 - Refrigerant charge



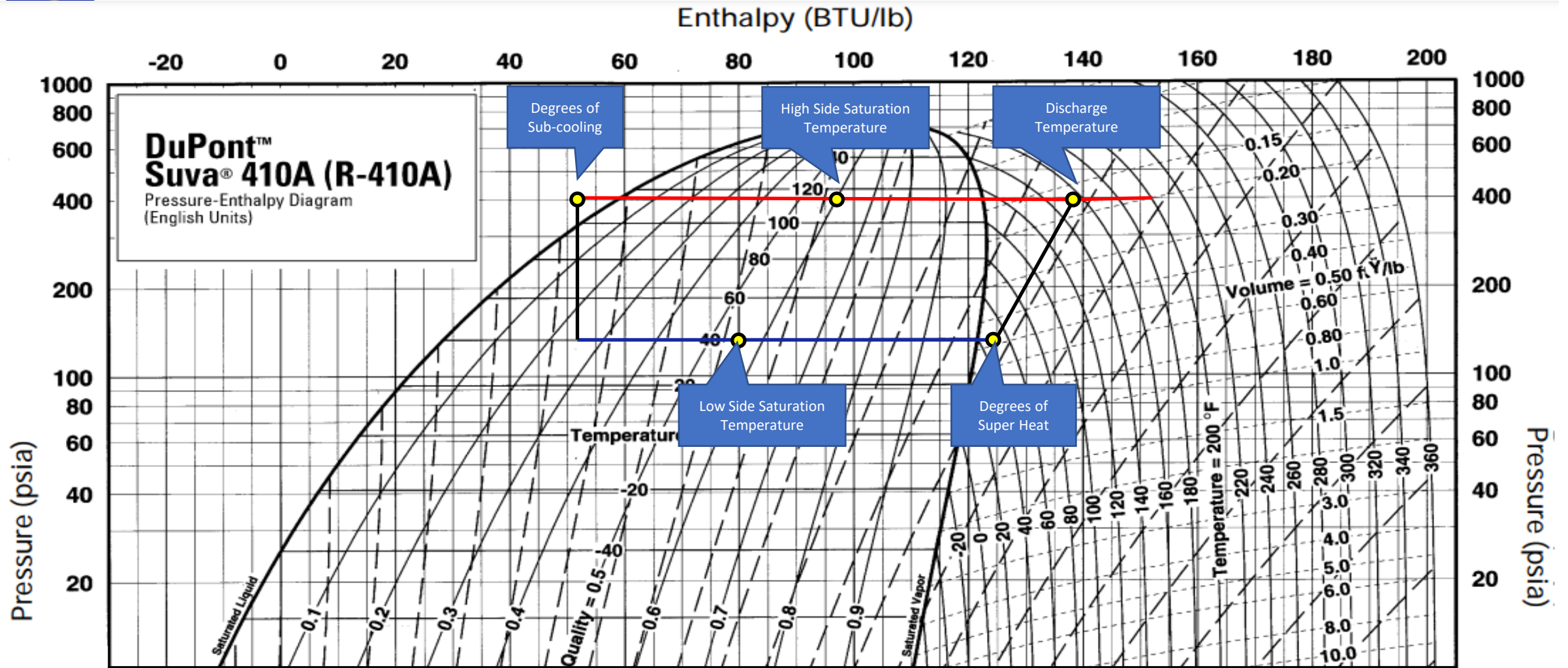
PE Diagrams

Used for system design





Cooling Cycle Design

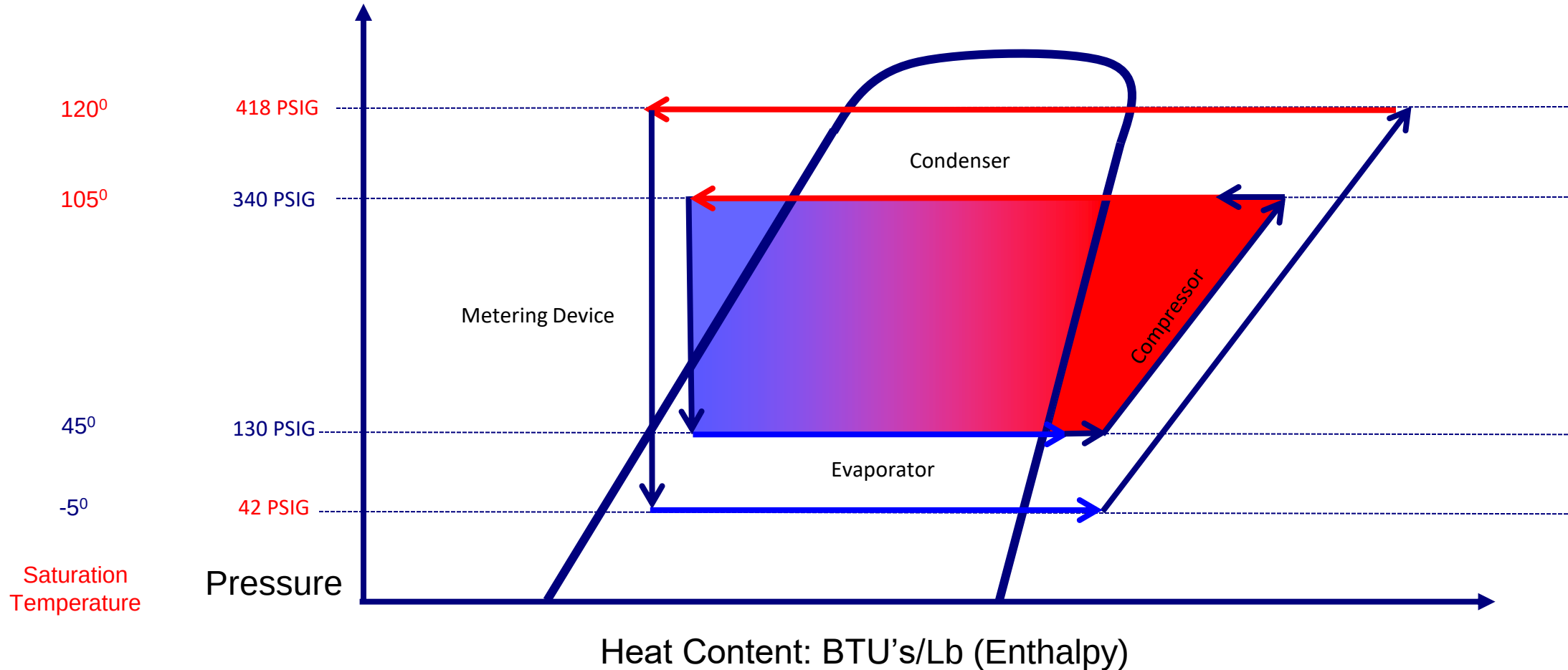


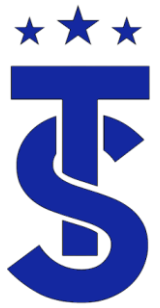




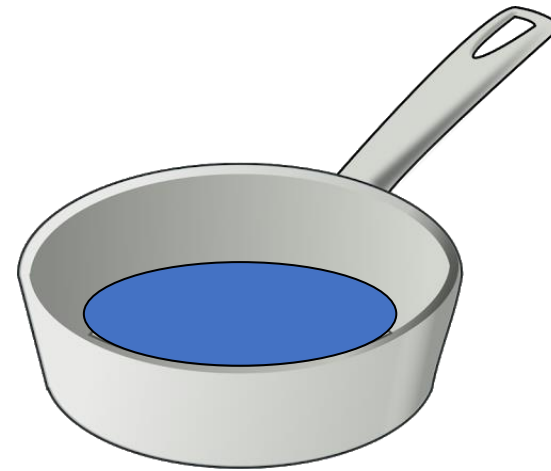
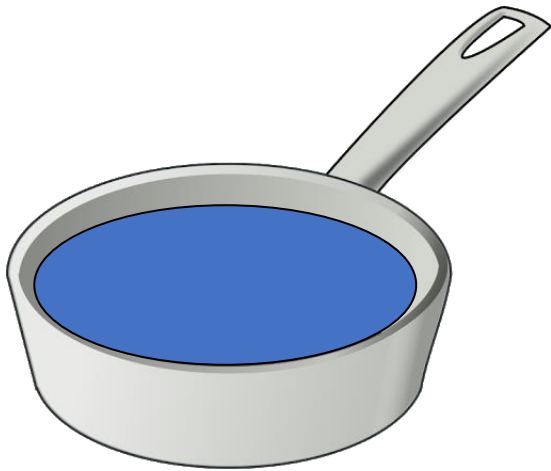
Cooling Cycle vs Heat Cycle

Which Cycle uses more energy?





Remember this?



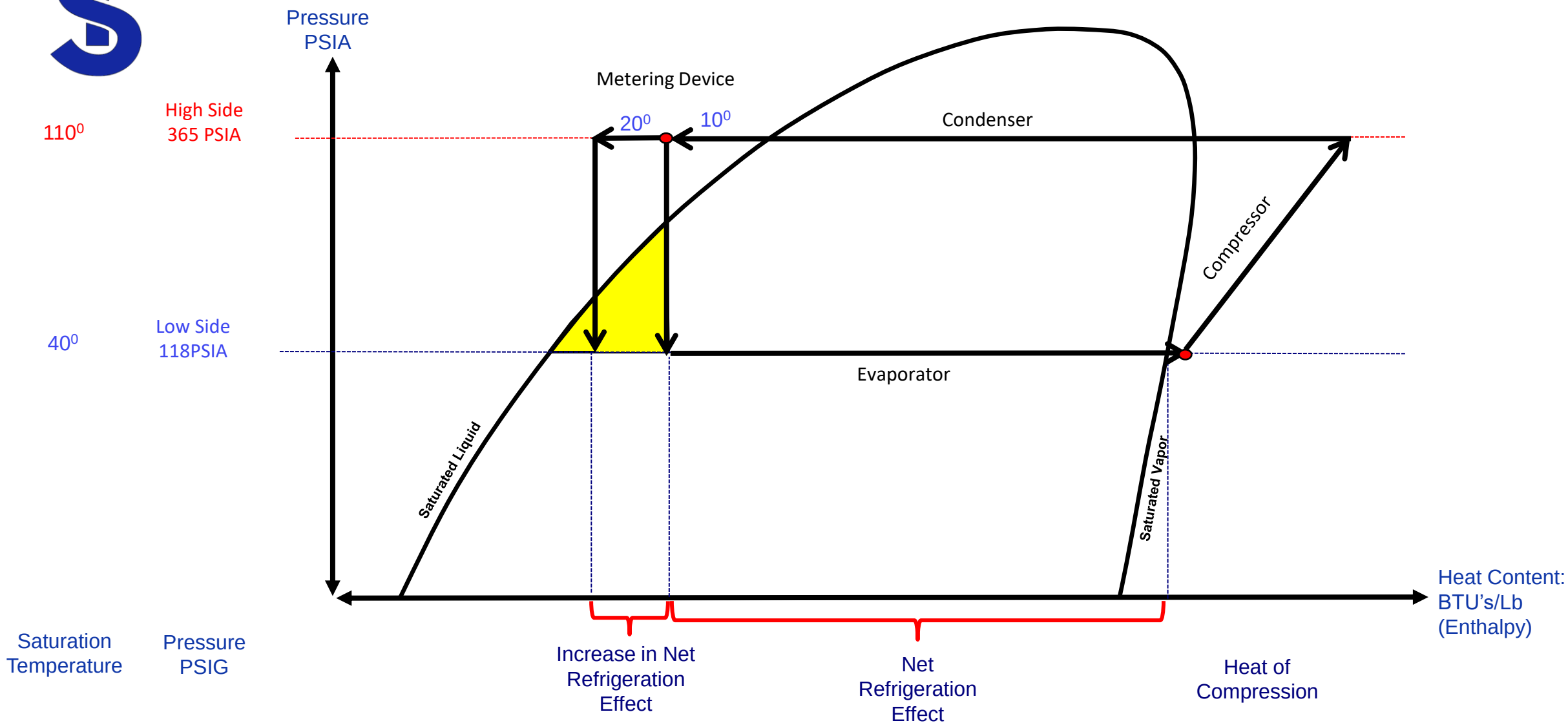
Which pot requires more heat to boil all the water?

Why?

The more liquid refrigerant in the evaporator the more heat that can be removed from the space
This is why high efficiency units have lower evaporator superheats and high subcooling?



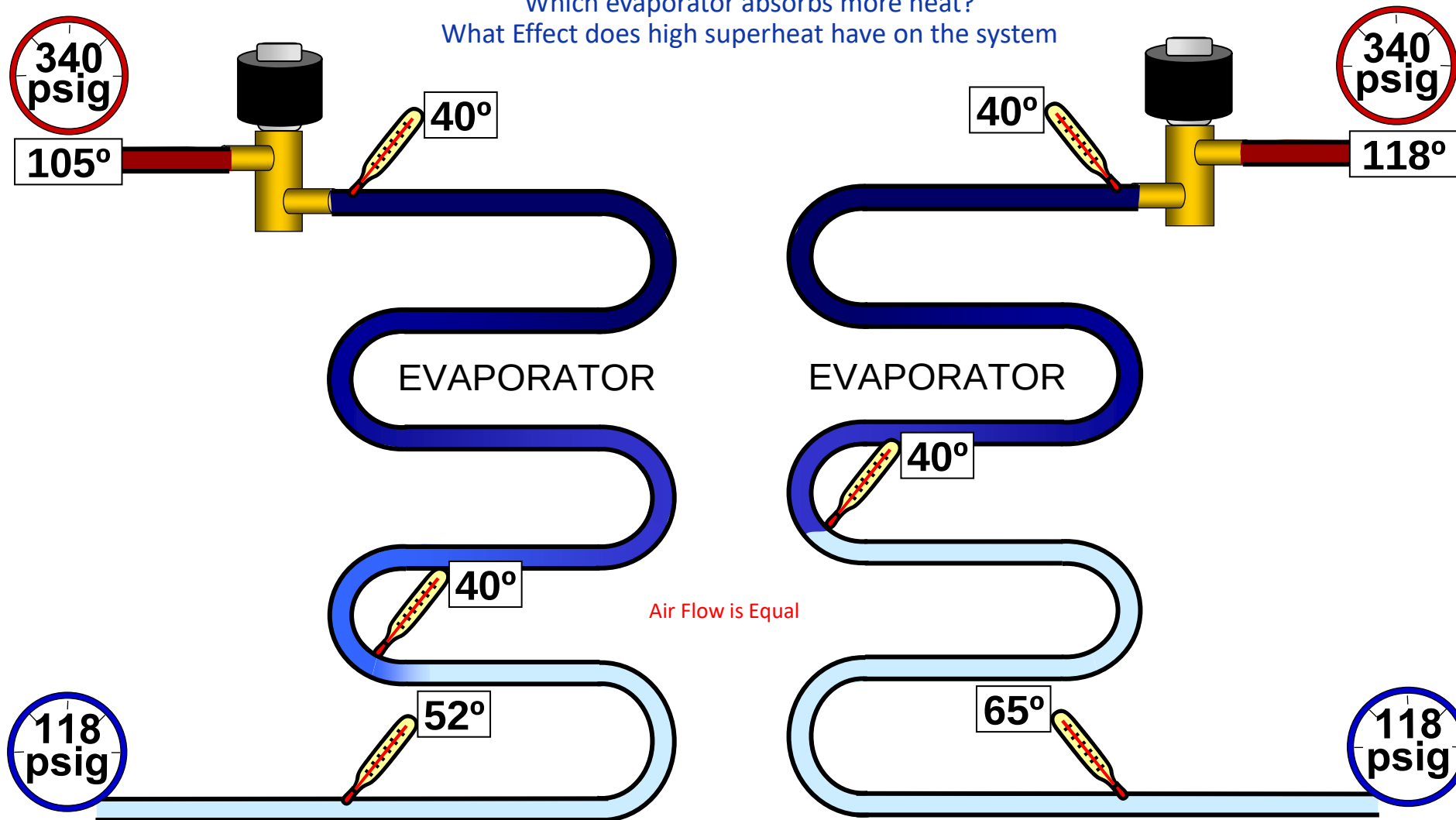
Example

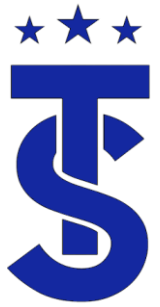




Remember this?

Which evaporator absorbs more heat?
What Effect does high superheat have on the system





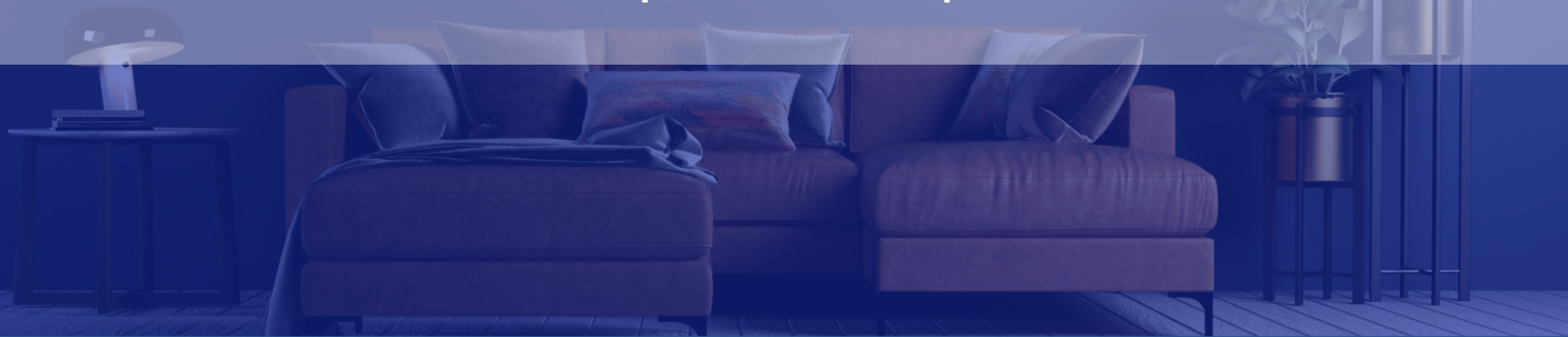
Review

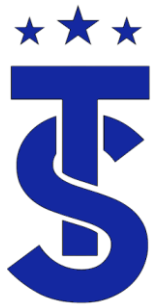
- Refrigeration- Air conditioning systems are designed using the law of thermodynamics.
- System are designed based on temperature and Not Pressure
- Refrigeration system operation is predictable and can charted on PE Diagrams.
- Manufacturer KNOW what to expect out of the system at given conditions as long as the CHARGE is CORRECT and the system is INSTALLED CORRECTLY



System Operation

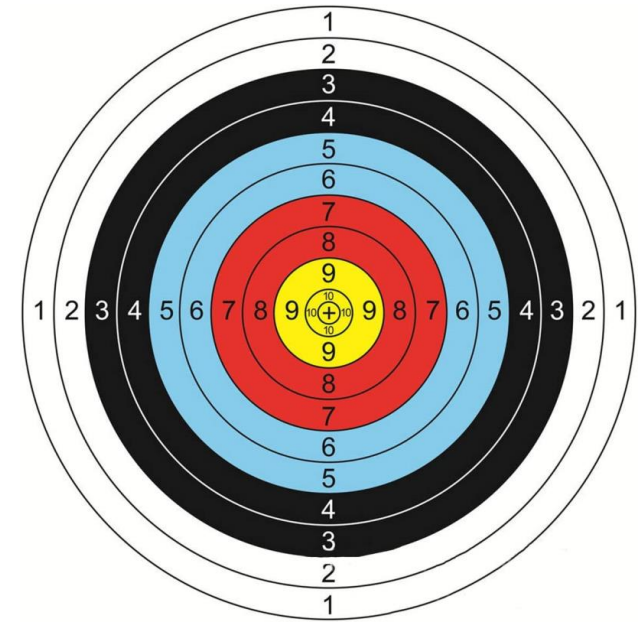
Interrelationship Between Components

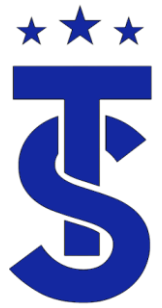




Putting It All Together

- The operating characteristics are determined during the testing process using PE diagrams.
- From these tables engineers can calculate the operating characteristics at any given condition for each mode.
- They use this information to establish targeted operating values of the system for the conditions and mode it is running at.
- The system components will adjust to varying conditions to maintain the “Targeted” operation.

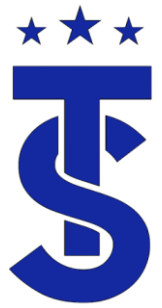




What to look for.

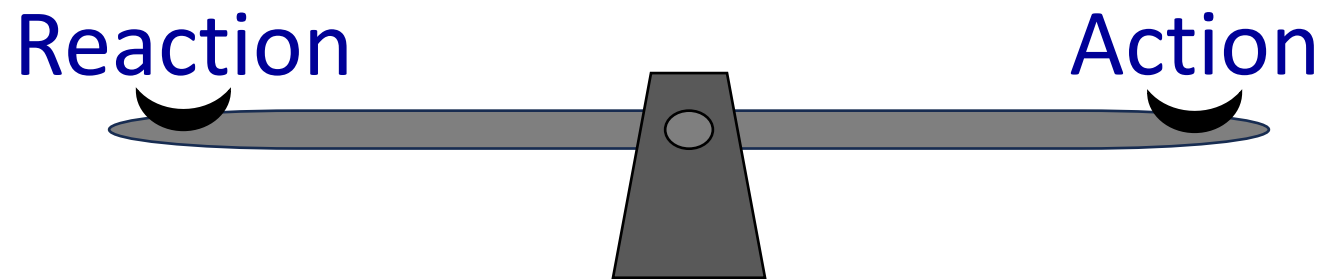
- Here is the caveat
 - The system are dynamic and constantly adjusting
 - We can not precisely identify these targets.
 - There for when troubleshooting we look for measurements to be with in a range, Not a specific value
- Sometimes it feels like you're hitting a moving target.
- We can use the controls sequence of operation to get an operating range for the heating and cooling cycles.





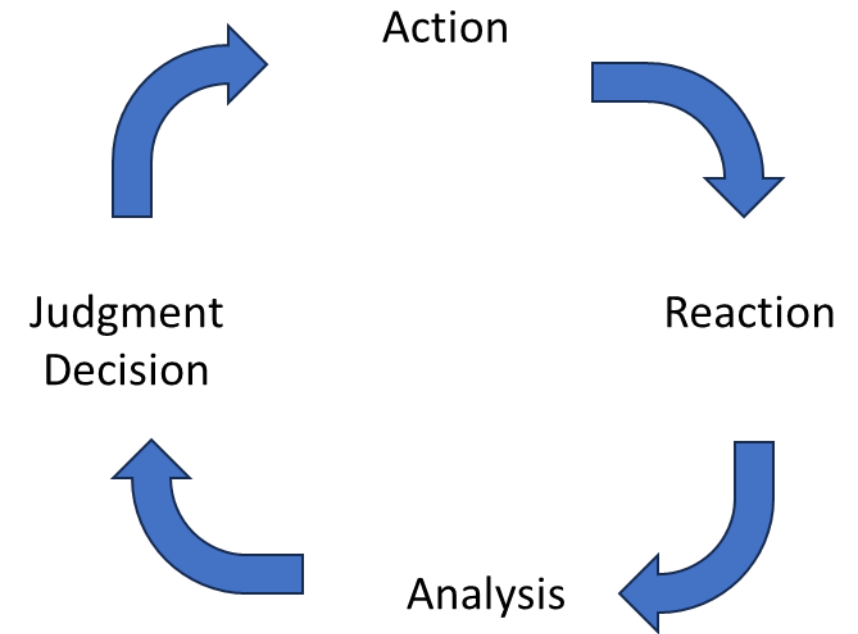
VRF System Operation

- The operation of a VRF system is a constant balancing act between changes in the system and the operation of components.
- For every change within the system there is a reaction by the components to maintain their target operation.
- One change within the system can cause a domino effect of component reactions.
- It's important to understand the cause and effects that take place within the VRF system to understand the system operation.



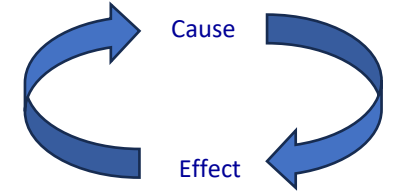
System Operation

- The operation of a Variable system is a constant balancing act between changes in the system and the operation of components.
- For every change within the system there is a reaction by the components to maintain their target operation.
- One change within the system can cause a domino effect of component reactions.
- It's important to understand the cause and effects that take place within the VRF system to understand the system operation.



Under Charge

- In the cooling mode the suction pressure will drop due to a shortage of gas in the system.
- The compressor will slow down to try to maintain target suction pressure.
- The IDU superheat will be high and the EEV's will be driven open more than normal.
- Evaporator super heat will be high
- The discharge temperature will be high
- There will be a reduced amount of Sub-Cooling.
- There will be a lack of cooling performance



Over Charge

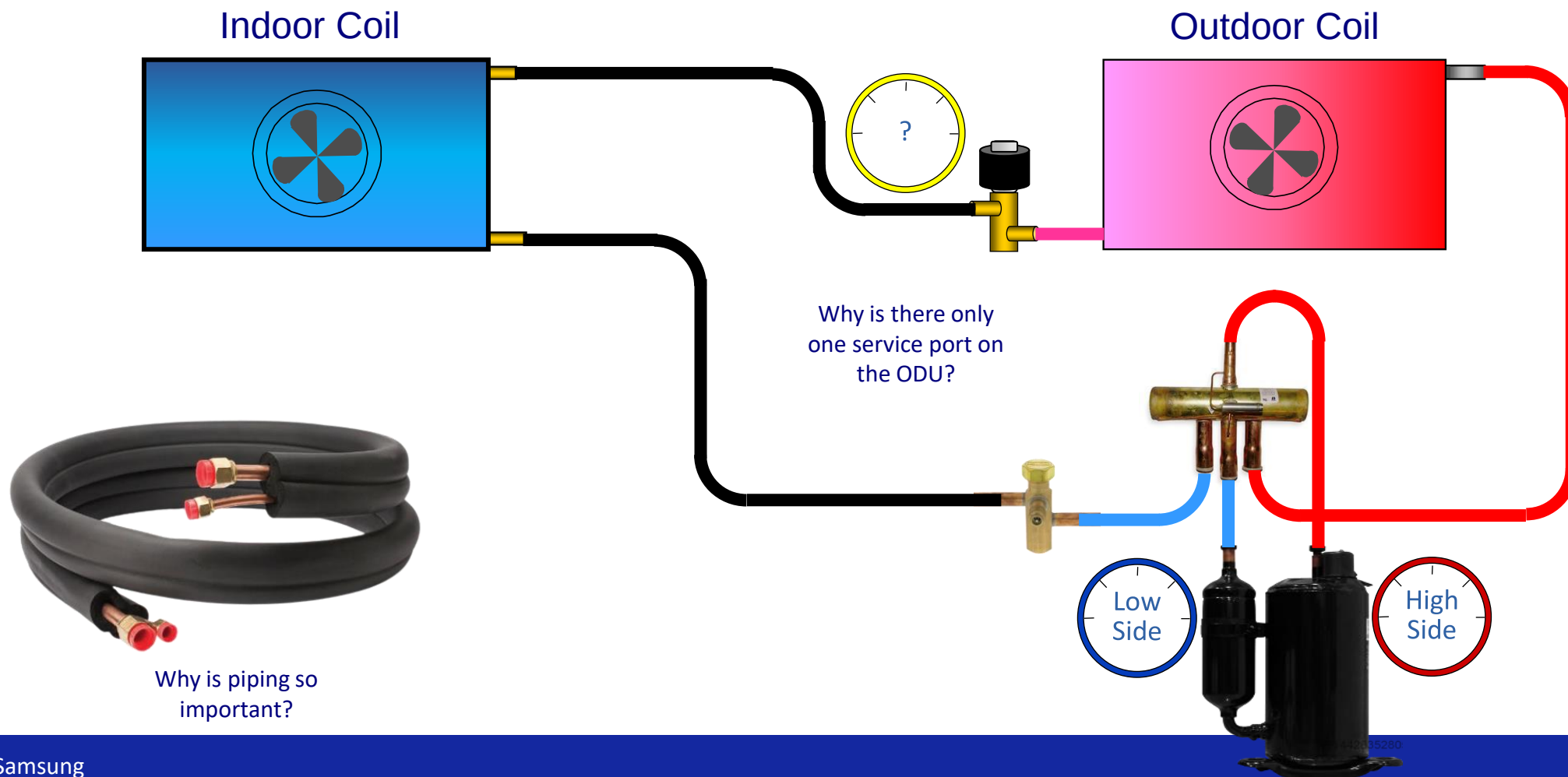
- In the heating mode the head pressure will be high due to a excess refrigerant in the system.
- The compressor will achieve target head pressure at lower than need Hz.
- The discharge temperature will be low
- The IDU superheat will be lower - normal and the EEV's will be driven closed to their lower limit.
- There will be a large amount of Sub-Cooling.
- There will be a lack of heating performance because of the reduced compressor speed.

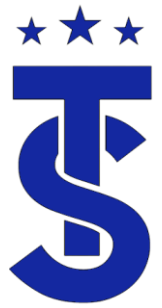




System Piping

What effect does piping have on the system?



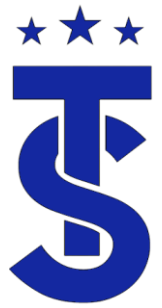


What is “Normal”

- When troubleshooting it is important know what each manufacturer’s normal or targets values of operation are.
- There are no more “rules of thumbs”.
- Make sure you ask the right questions and use the manufacturer’s information when troubleshooting.

There is none!!!!
It depends on what Outdoor unit the
indoor is connected to and the
outdoor air temperature





System Targets

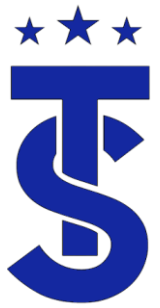
How does the unit operate in each mode?



Evaporator Temperature
Evaporator Super-Heat
Discharge Temperature

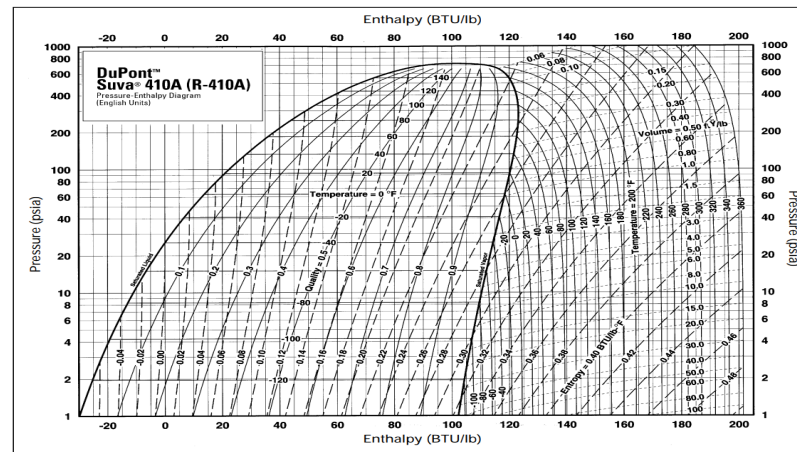
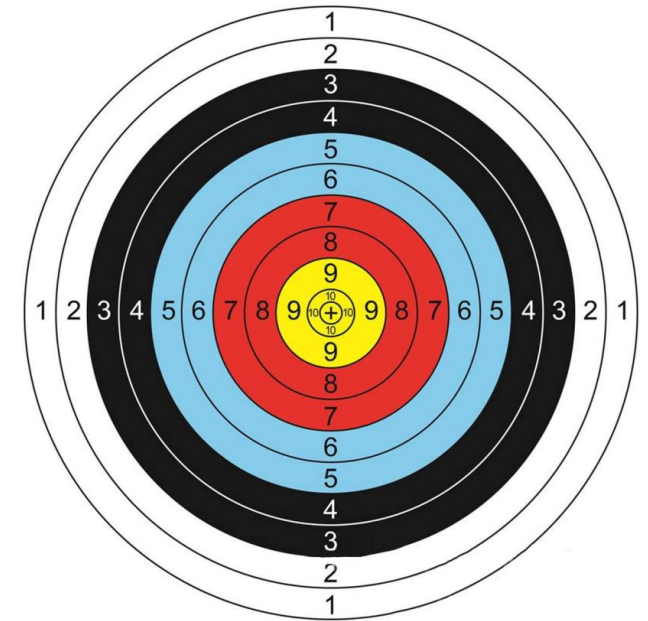


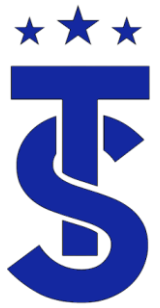
Condenser Temperature
Condenser Sub-Cooling
Discharge Temperature



Where do the targets come from?

- The systems are tested at various conditions.
- The operating characteristics are determined during the testing process.
- The system uses this information to determine how the system should operate under a given set of conditions.





Compressor Regulation



IN HEATING MODE, THE COMPRESSOR RUNS TO
MAINTAIN A HIGH SIDE SATURATION TEMPERATURE OF
120 DEGREES



IN COOLING MODE, THE COMPRESSOR MODULATES TO
MAINTAIN AN EVAPORATOR TEMPERATURE OF 40
DEGREES



THE ARE SEVERAL SAFETY CONDITIONS THAT WILL
OVERRIDE COMPRESSOR OPERATION
FLOOD BACK CONDITION
HIGH DISCHARGE TEMPERATURE



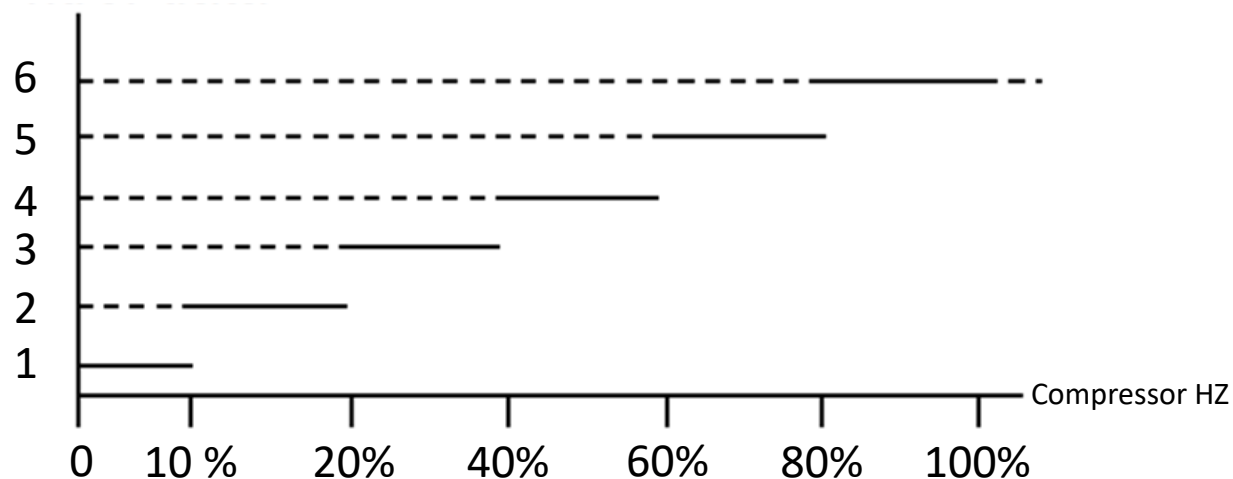
Target Operation

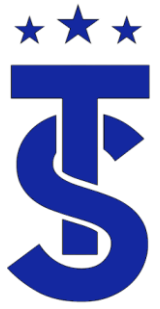
Target discharge temp = Function(Hz & Room temp & ambient temp)

- The target discharge temperature is set based on the demand of the system.
- The system looks at compressor Hz and outside air temperature to determine the target discharge temperature.
- Target Discharge temperature can be viewed on the CAC and FJM Products.
- Target discharge temperatures are different in the heating and cooling modes.

Target Discharge
Temperature Range

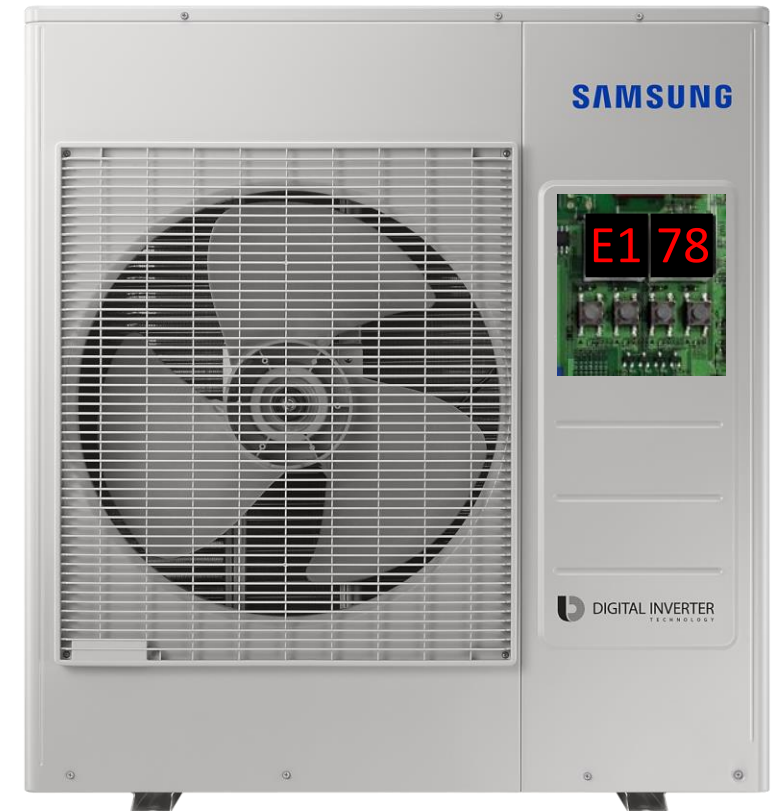
For Demonstration Purpose Only

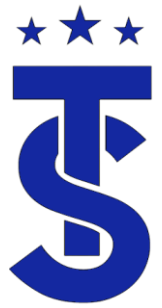




Target Discharge Temperature FJM

- The only way to see the target discharge temperature is to use the K 4 Button
- Push the K 4 button 14 times (E) to display the target discharge temperature
- Push the K 4 button 9 times (9) to display the discharge temperature
- Push the K 4 button 16 times (G) to see if the system in a protection mode

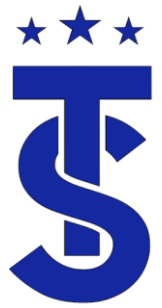




Viewing Target Information

K 4 Buttons CAC

Times Pushed	Displayed Information	Times Pushed	Displayed Information
1	Order Frequency	11	EEV
2	Current Frequency	12	Capacity Sum of Indoor Unit
3	Number of preset indoor units	13	Protective Control
4	Ambient Temperature	14	IPM Temperature
5	Compressor Discharge Temperature	15	Number of Connected Indoor Units
6	EVA Mid Thermistor	16	ESC EEV Steps (CAM)
7	Condenser Sensor	17	ESC In Sensor
8	Current	18	ESC Out Sensor
9	Outdoor Fan RPM	19	End View Mode
10	Target Discharge Temperature		



Viewing Target Information

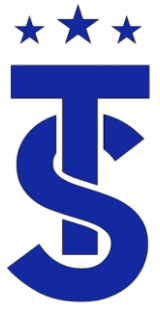
K 4 Buttons FJM

Times Pushed	Displayed Information	Times Pushed	Displayed Information
1	Present Compressor Frequency	9	Discharge Temperature
2	Target Compressor Frequency	10	OLP Temperature
3	EEV 0 Step	11	Condenser Temperature
4	EEV 1 Step	12	Outdoor Temperature
5	EEV 2 Step	13	Running Current
6	EEV 3 Step	14	Target Discharge Temperature
7	EEV 4 Step	15	Total capacity of Indoor units
8	Fan RPM (H-high, L-low, Blank-)off	16	Safety Control



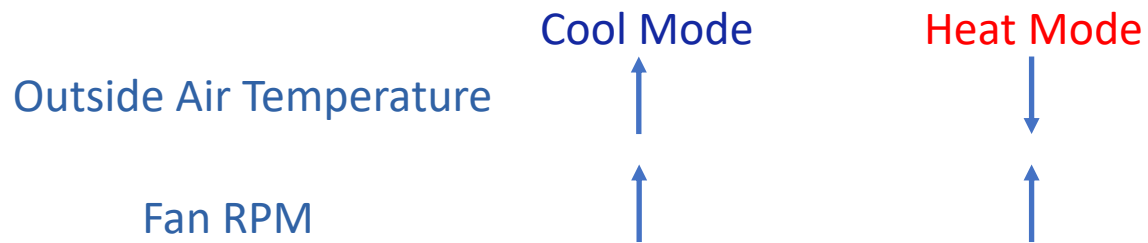
System Controls

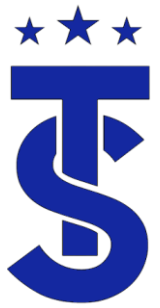
- The following slides are taken for the RLC basic installation class.
- They give us a look inside how the system is controlled.
- Using the information learned we can get an acceptable operating pressure range in the heating and cooling mode.
- Remember if we know the temperature of the refrigerant, we know the pressure!
- Our system provides a gas line service port.
- You can measure the low side pressure in the cooling mode and high side pressure in the heating mode.



Outdoor Fan Control

- On start up the fan speed is set based on outdoor air temperatures in the heating and cooling modes.
- After start up the fan speed is controlled by outside air temperature and compressor frequency.
- The settings are based on data obtained from the PE diagrams and testing to determine the best operation for the conditions.





Compressor capacity control

Cooling capacity

1. Fuzzy control

- Use Room temp. & target temp of running units
- Gap is big : Hz up
- Gap is small : Hz down

Heating capacity

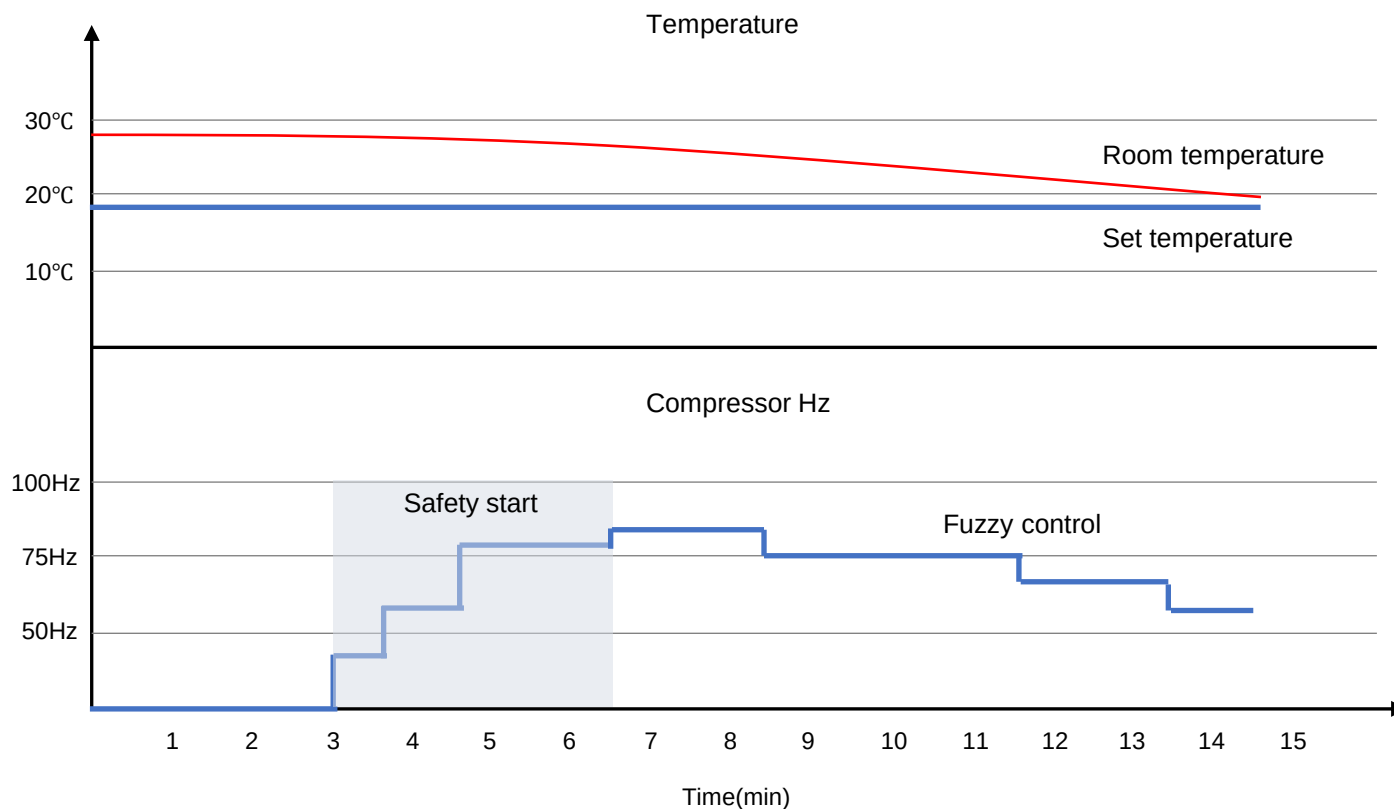
1. Target evaporator temperature control

- * Target : 47~52°C (117~126°F) (only 1 IDU on = target -5°C(9°F))
- * At this time we use calculated evaporator temperature as a target
- Calculate average evaporator temperature of running indoor units
- average temperature < target : Hz up
- average temperature > target : Hz down
- average temperature - target $\leq 1^{\circ}\text{C}(2^{\circ}\text{F})$: no change



Compressor control

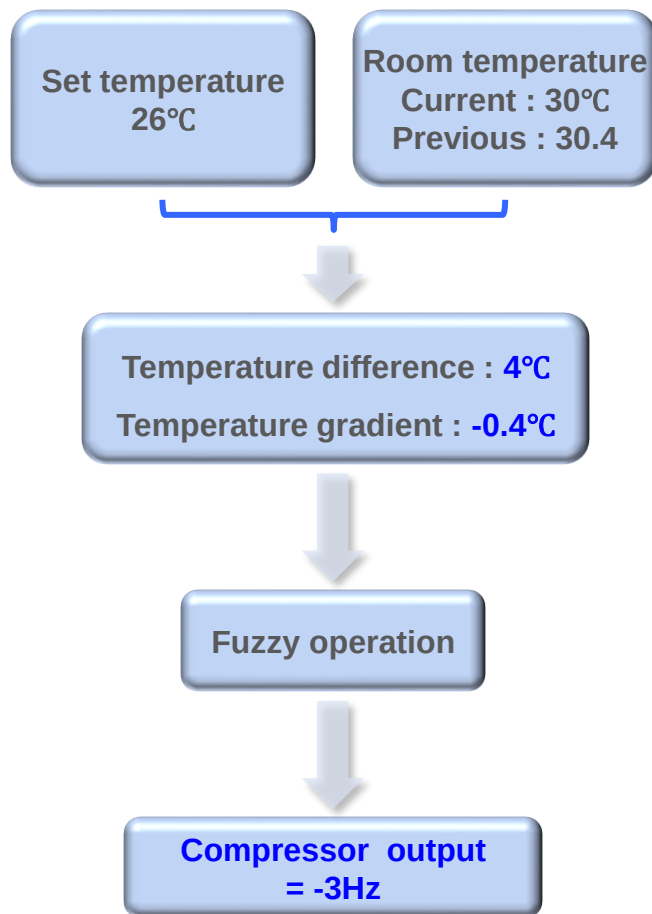
- Compressor Hz is determined by between **outdoor temperature** and ΔT .
(ΔT is difference of between the **room temperature** and **set temperature**.)
 - When the room temperature approaches the set temperature, compressor Hz reduces.
- Fuzzy control





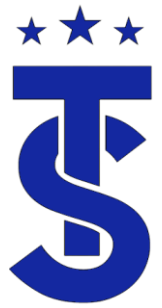
Fuzzy control

- Make rules based on knowledge of experts.



✓ Fuzzy table

		Temperature gradient(°C)				
		← -0.6	-0.5	-0.4	-0.3	-0.2 →
Temperature difference(°C)	↑					
	+2.0	-8	-8	-7	-3	0
	+2.5	-7	-8	-6	-2	0
	+3.0	-6	-6	-5	-1	1
	+3.5	-5	-6	-4	0	1
	+4.0	-4	-4	-3	1	1
	+4.5	-3	-4	-2	2	1
	+5.0	-2	-3	-1	3	2
		↓				



Operational Modes

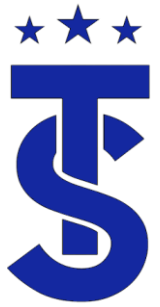
Freeze protection control

- Anti-freeze control prevents frost formation on the indoor unit evaporator coil.
 - Indoor coil temperature below 37°F compressor frequency reduced to 15Hz – 35Hz.
 - Indoor coil temperature drops below 31 °F, compressor is stopped.
 - Freeze protection control de-activated when indoor coil temperature rises to 45°F - 50°F.
 - If indoor coil temperature drops below 40°F, the compressor frequency will be held.

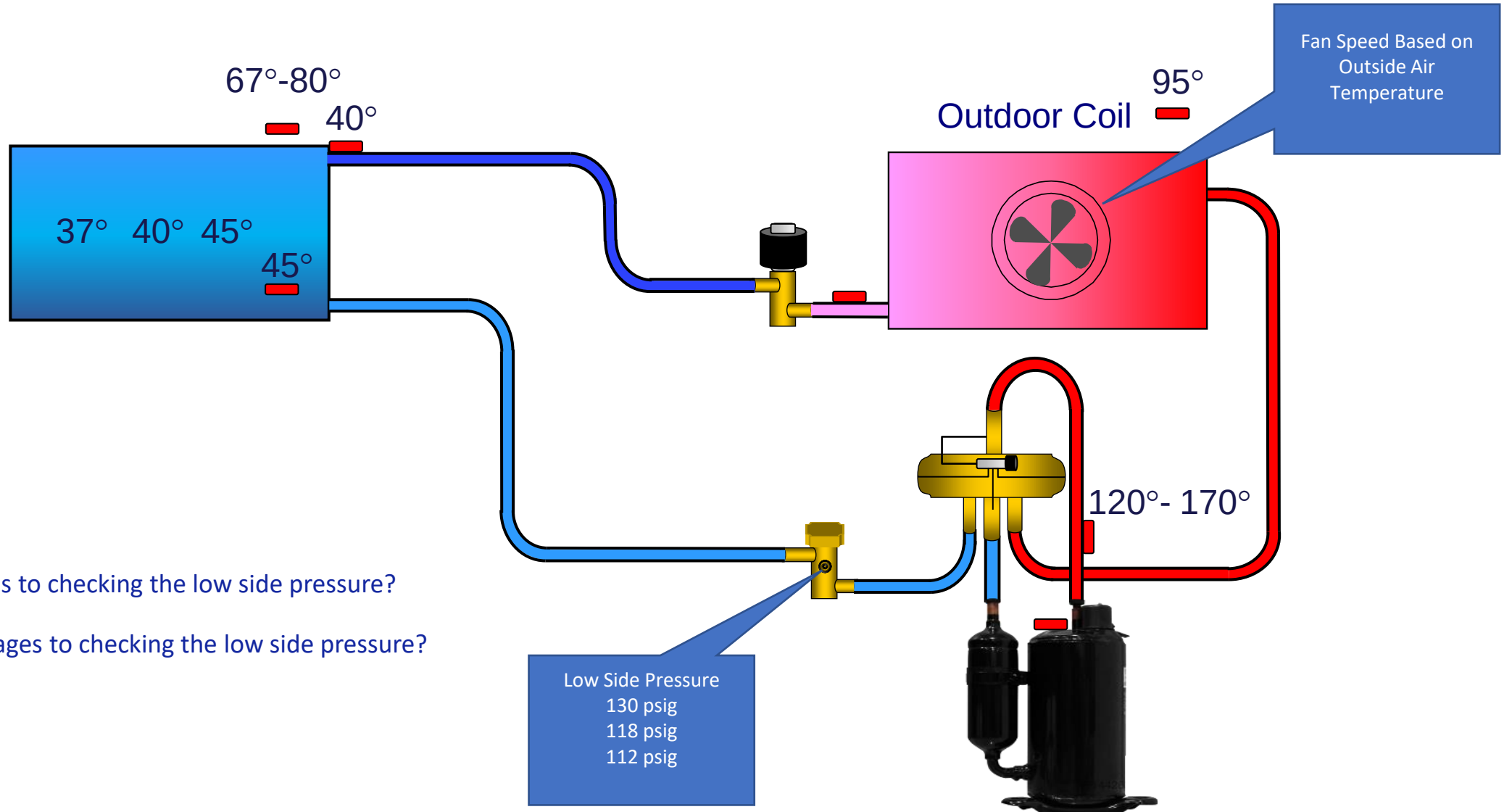
What Can we take away from this slide?

How id it useful in understanding the operation of the system?



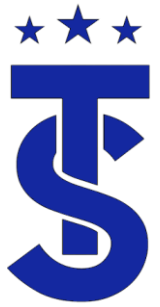


Cooling Cycle Operation



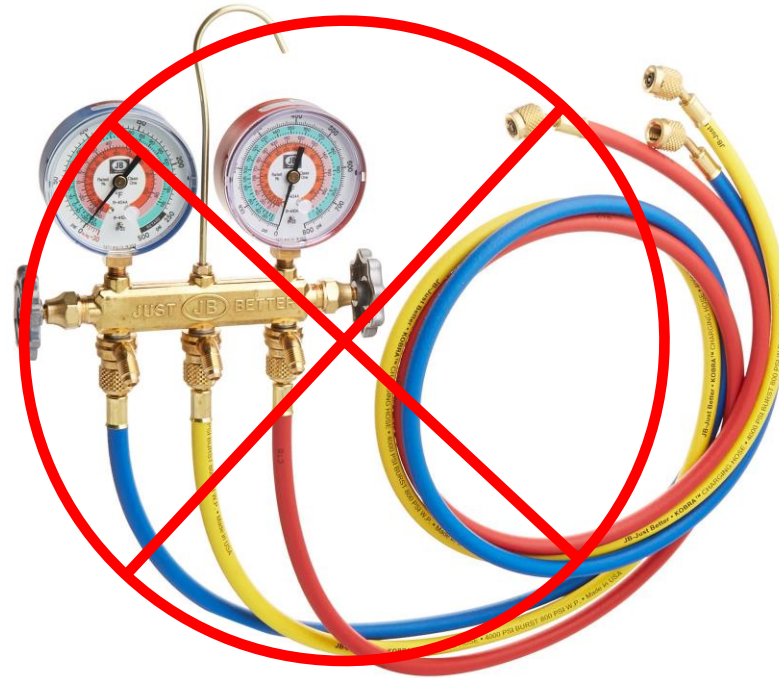
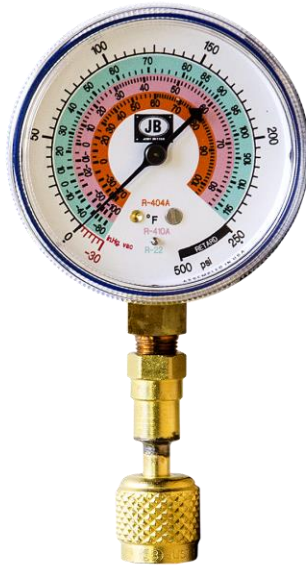
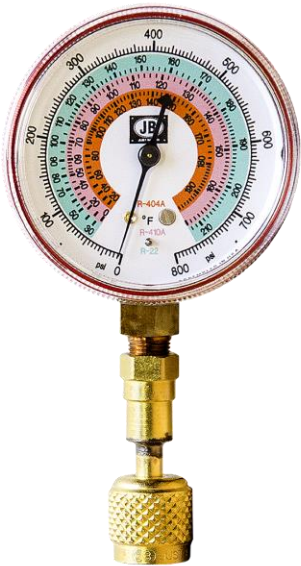
Are there any advantages to checking the low side pressure?

Are there any disadvantages to checking the low side pressure?



Measuring System Pressures

Yes – No ?



Why?



FJM Freeze Protection Cooling Mode

✓ Use lowest temperature among pipe in / evap in / evap mid or evap out / pipe out

- 1st Freezing Protection

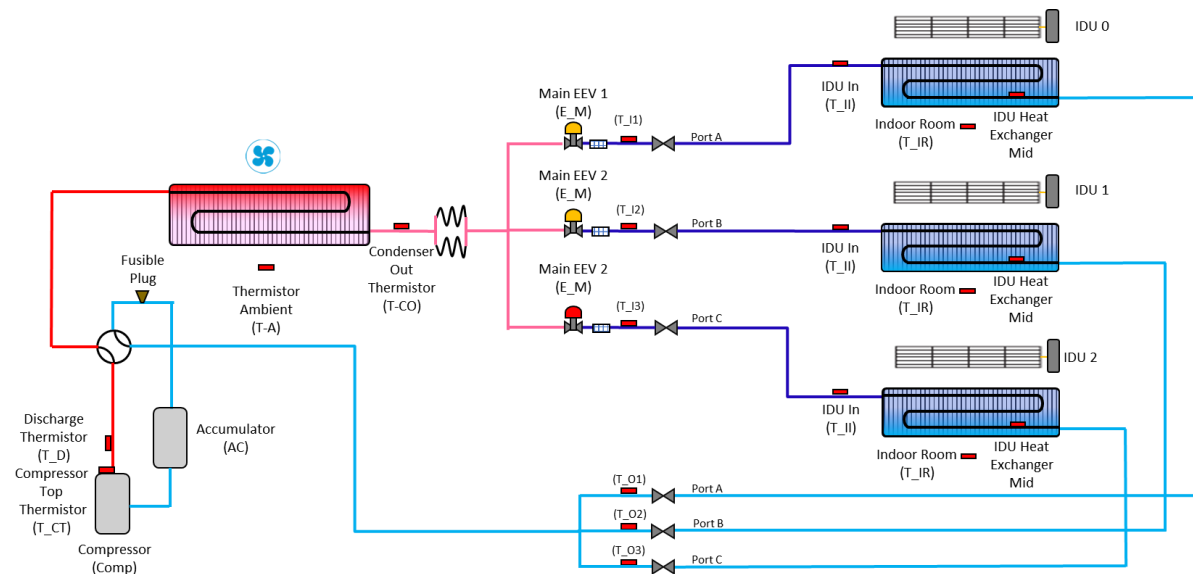
- Evaporator temperature $< 4^{\circ}\text{C}$ (39°F) for 1min
- Action
 - Hz down (Indoor unit's capacity code down)

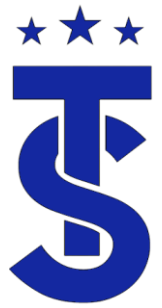
- 2nd Freezing Protection

- Cooling Mode
 - Evaporator temperature $< 2^{\circ}\text{C}$ (36°F)
 - Evaporator temperature $< 6^{\circ}\text{C}$ (43°F) for 9mins
- Action
 - Hz down (Indoor unit's capacity code = 0)
 - EEV close fully
 - Error display E401

- Release condition for 1st & 2nd protection & E401

- Evaporator temperature $\geq 6^{\circ}\text{C}$ (43°F) (Use evap. in/mid/out for release)
- Action
 - Release freezing protection
 - * Hz will be increased 1hz/min for 30mins

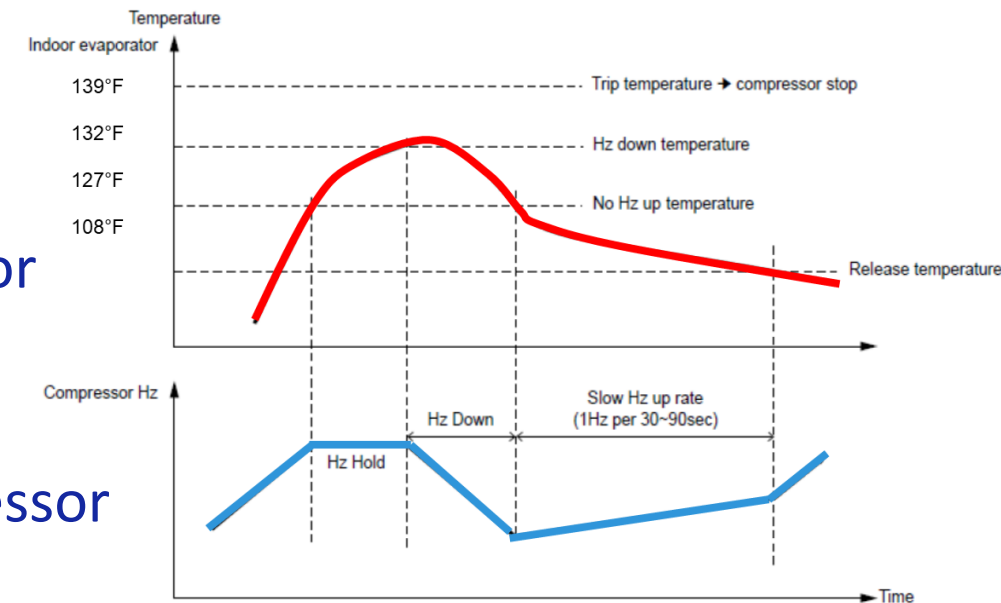




Operational Modes

OLP (overload protection)

- Initiated in heat mode if the indoor coil temperature reaches 127°F.
 - Compressor frequency held.
 - If the coil temp to rises to 132°F the compressor frequency is reduced.
 - Rate of reduction determined by fuzzy logic
 - When indoor coil temp drops to 108°F. compressor will ramp up but at a reduced rate.
 - 1hz every 30-90 seconds

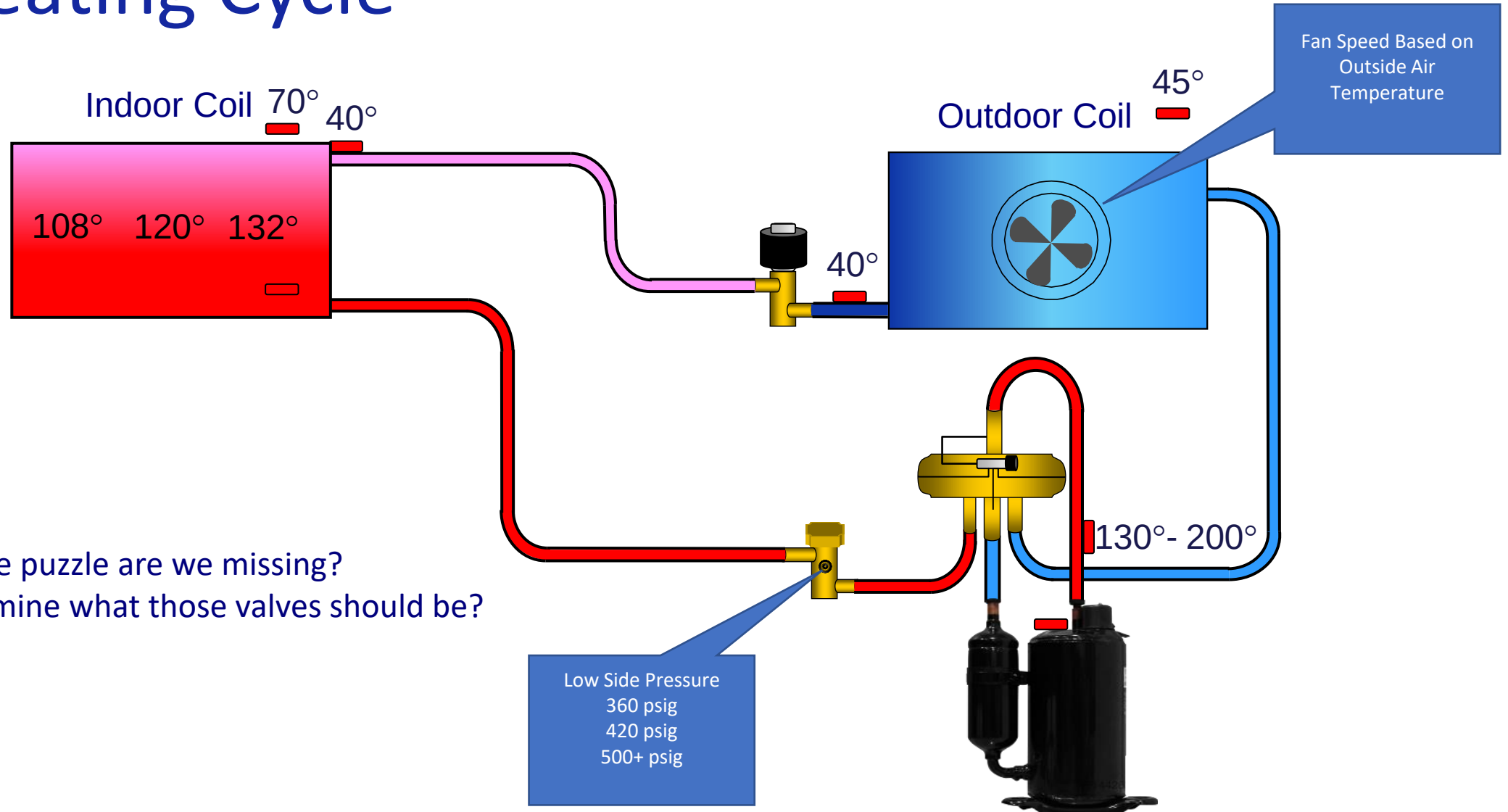


What Can we take away from this slide?

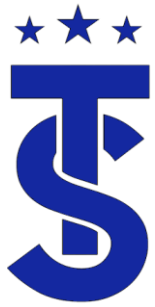
How id it useful in understanding the operation of the system?



Ductless Heat Pump Heating Cycle



What Pieces of the puzzle are we missing?
How do we determine what those valves should be?



Protection Mode



CAC
K 4 13 Times

D or G	0: Cooling 1: Heating	Protective control	Frequency status	
		0: No Protective control 1: Freezing 2: Non-stop defrosting 3: Over-load 4: Discharge 5: Total electric current	0: Normal 1: Hold 2: Down 3: Up_limit 4: Down_limit	

Use this Function to see if the system operation is being restricted by a safety function.

This feature is under utilized!



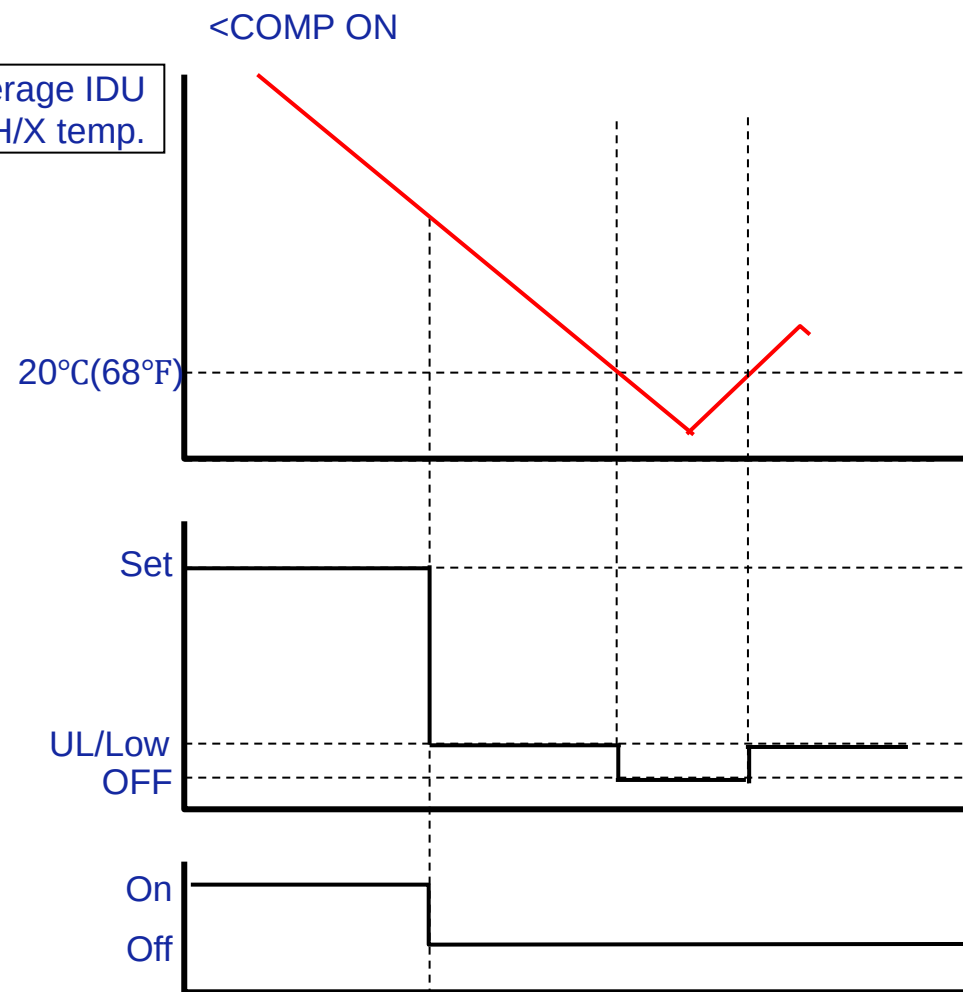
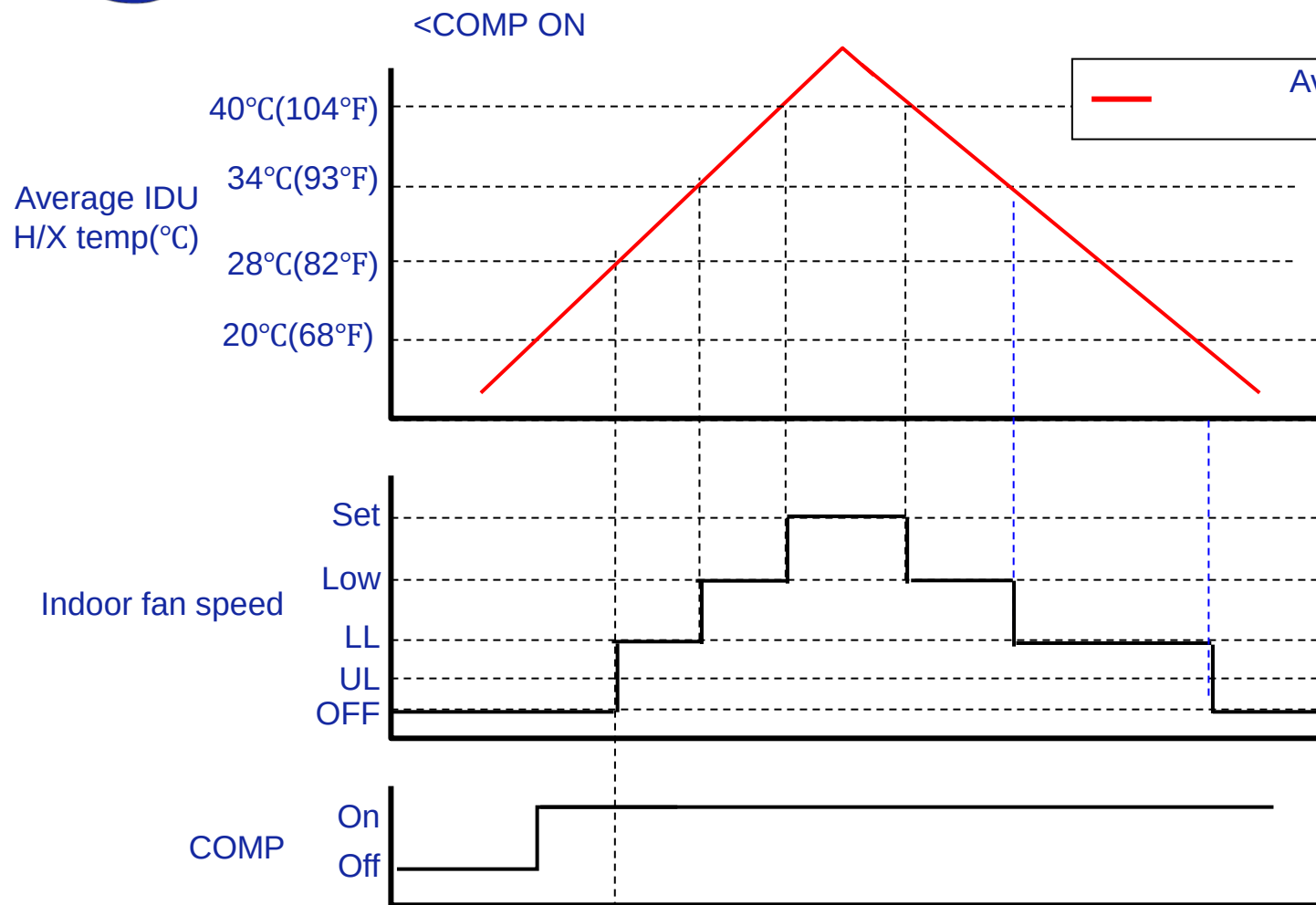
FJM
K 4 16 Times

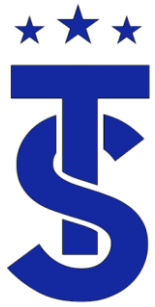


Cold air prevention

✓ To prevent cold draft when heating starting → warm up the evaporator then blow out the air

* Control value may be various by each model(Average IDU H/X temp. = average of (evap in /evap out))

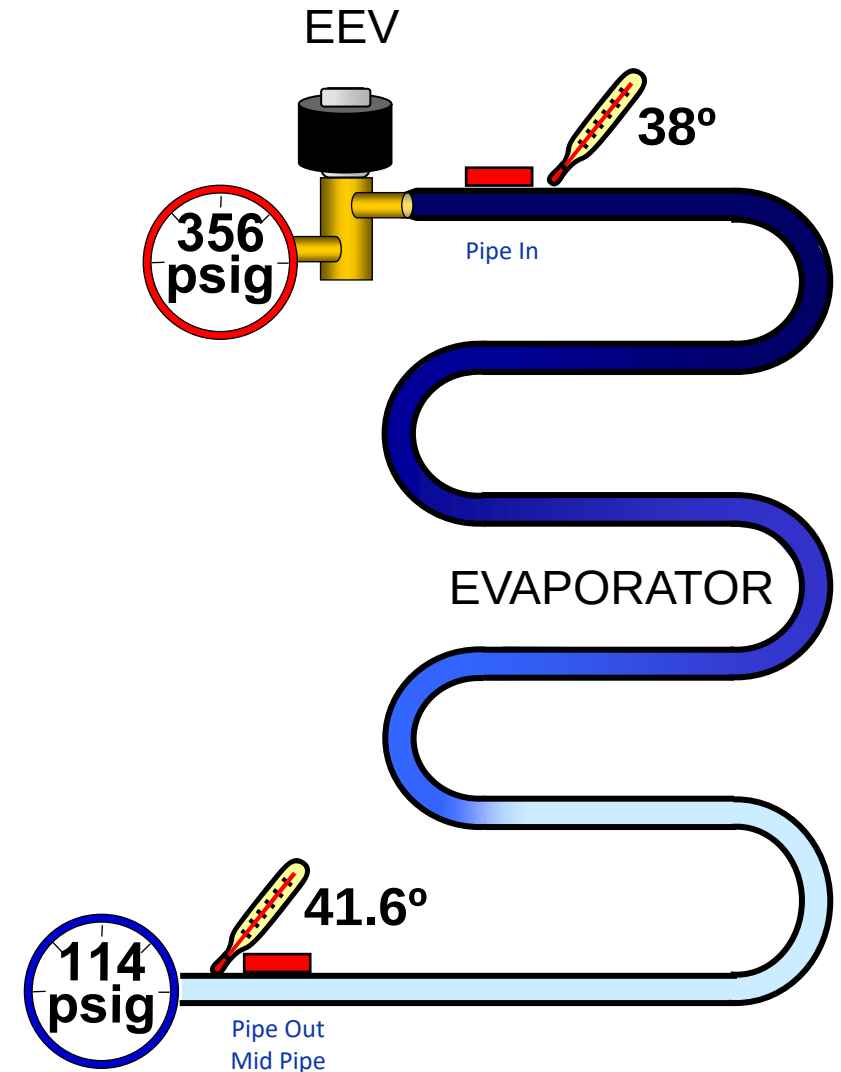




EEV Operation

Cooling Mode (Typical)

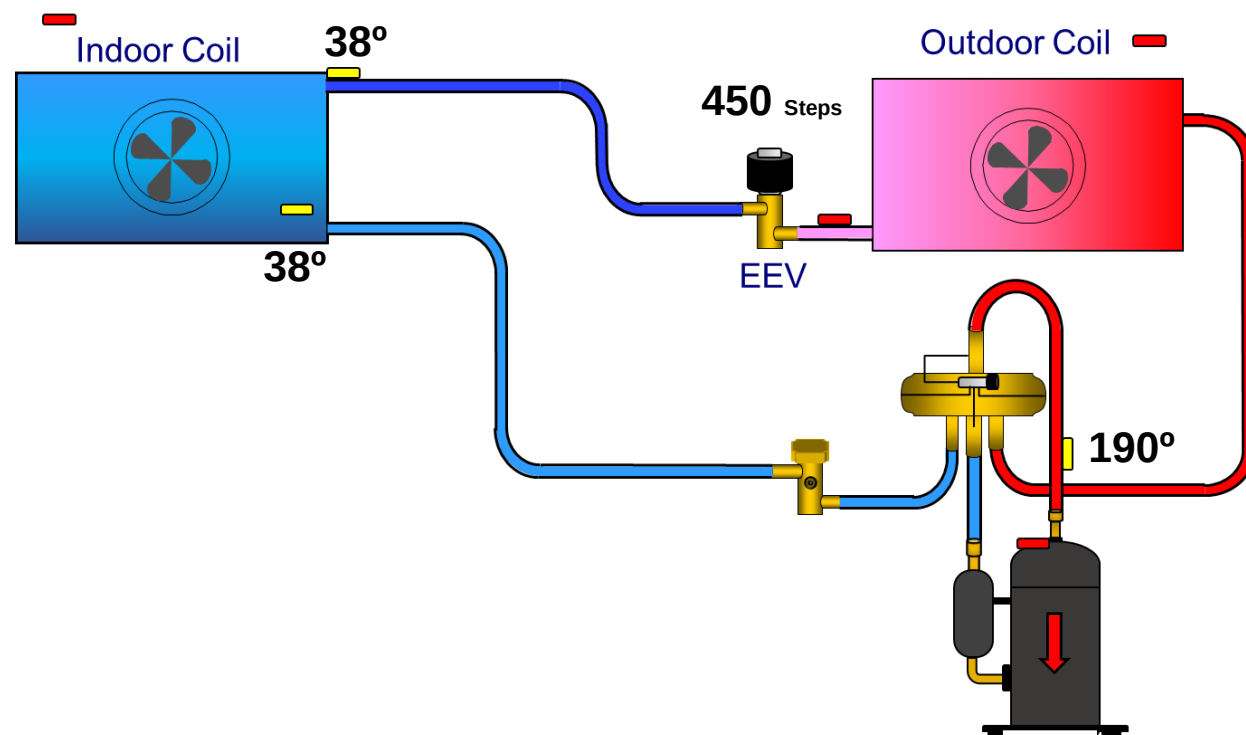
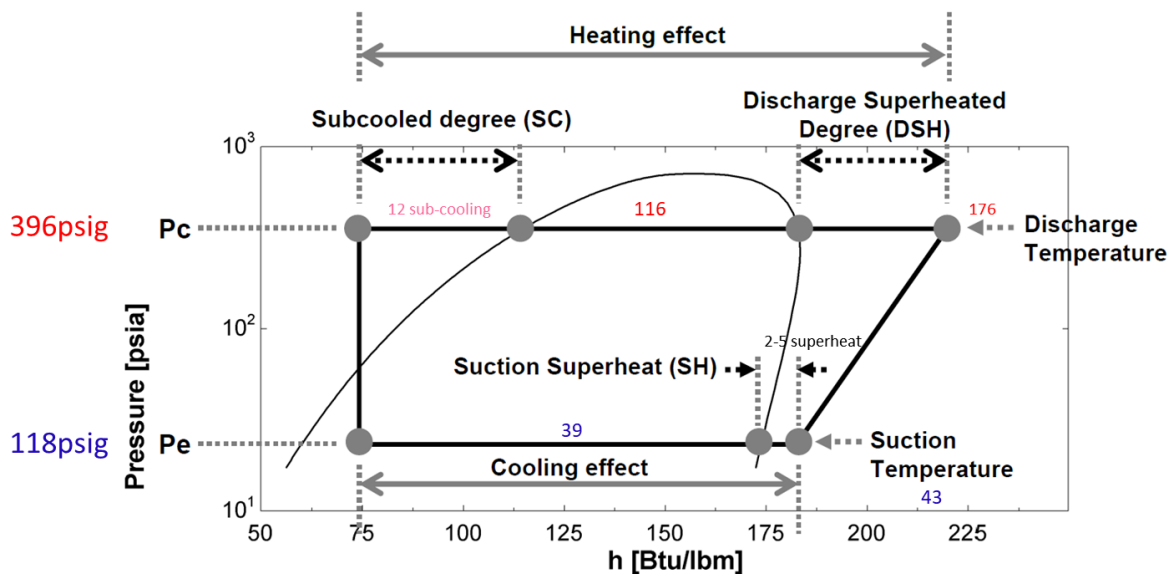
- The EEV in the cooling mode will modulate to maintain a 3.6-degree differential between the pipe in (Low Side Saturation Temperature) and pipe (Suction Pipe Temperature).
- If load increases the valve will open to maintain that differential.
- If the load drops, the valve will close to maintain that differential.
- What are some thing that cause the load to change?





1:1 EEV Operation

- If Evaporator superheat is correct discharge temperature is correct.
- Maintain Evaporator Superheat
- Maintain Discharge Temperature





FJM EEV control

Cooling

1. Target discharge control
2. Evap. super heat control(target 1~4°C(2~7°F))

SH high = EEV open / SH low = EEV close

* Indoor unit stop = 0step

Heating

1. Target discharge control
 2. Evaporator in temperature control
 - Current temperature < average : EEV open
 - Current temperature > average : EEV close
- Refrigerant balance between each units

* Stop unit : 80 or more.(prevent ref. stagnation)

Note . When new indoor unit is turned on EEV step will go back to initial step.

This is to secure system capacity & reliability

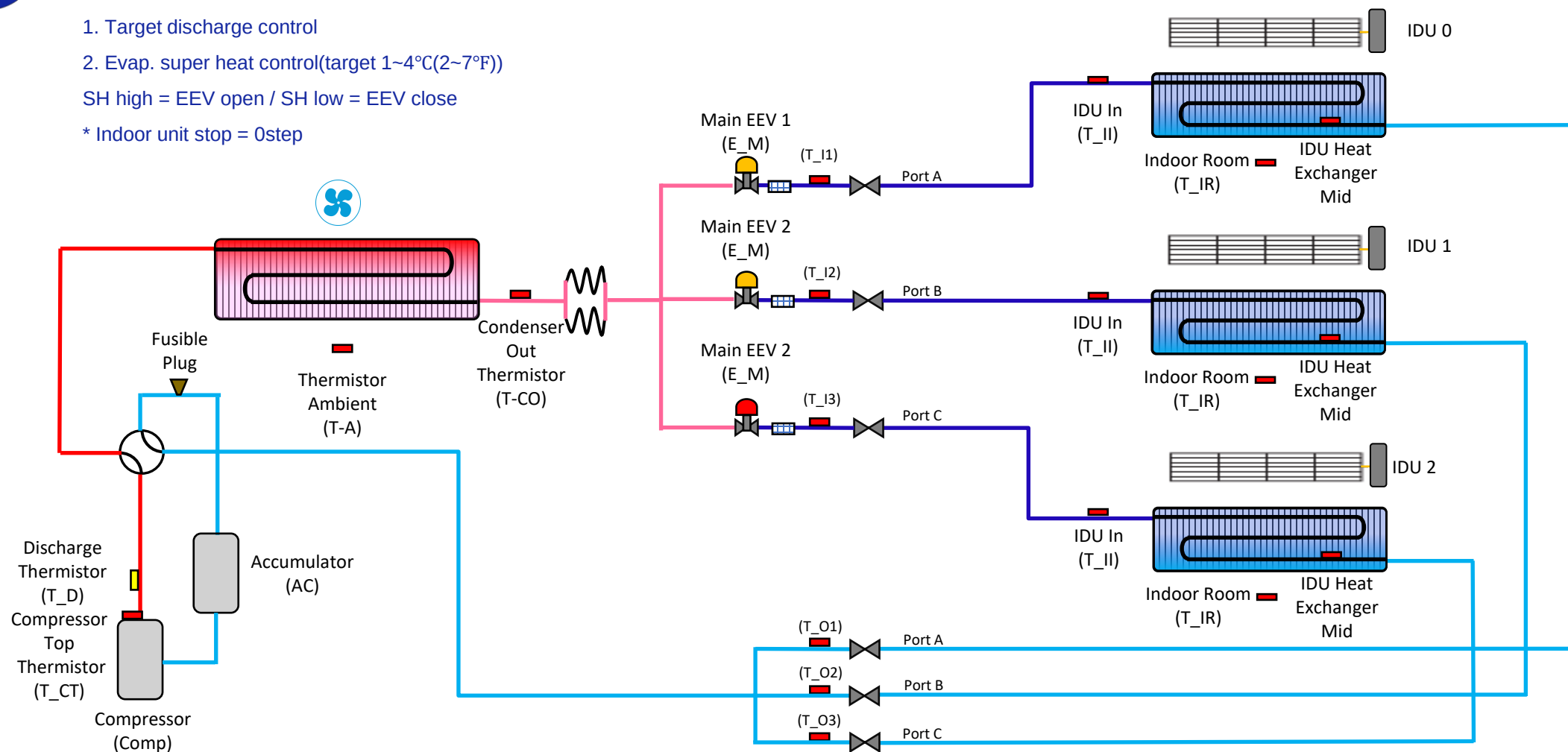
If eeV step is maintain(when new IDU on) → ref. flow more to certain IDU → required more time for balancing

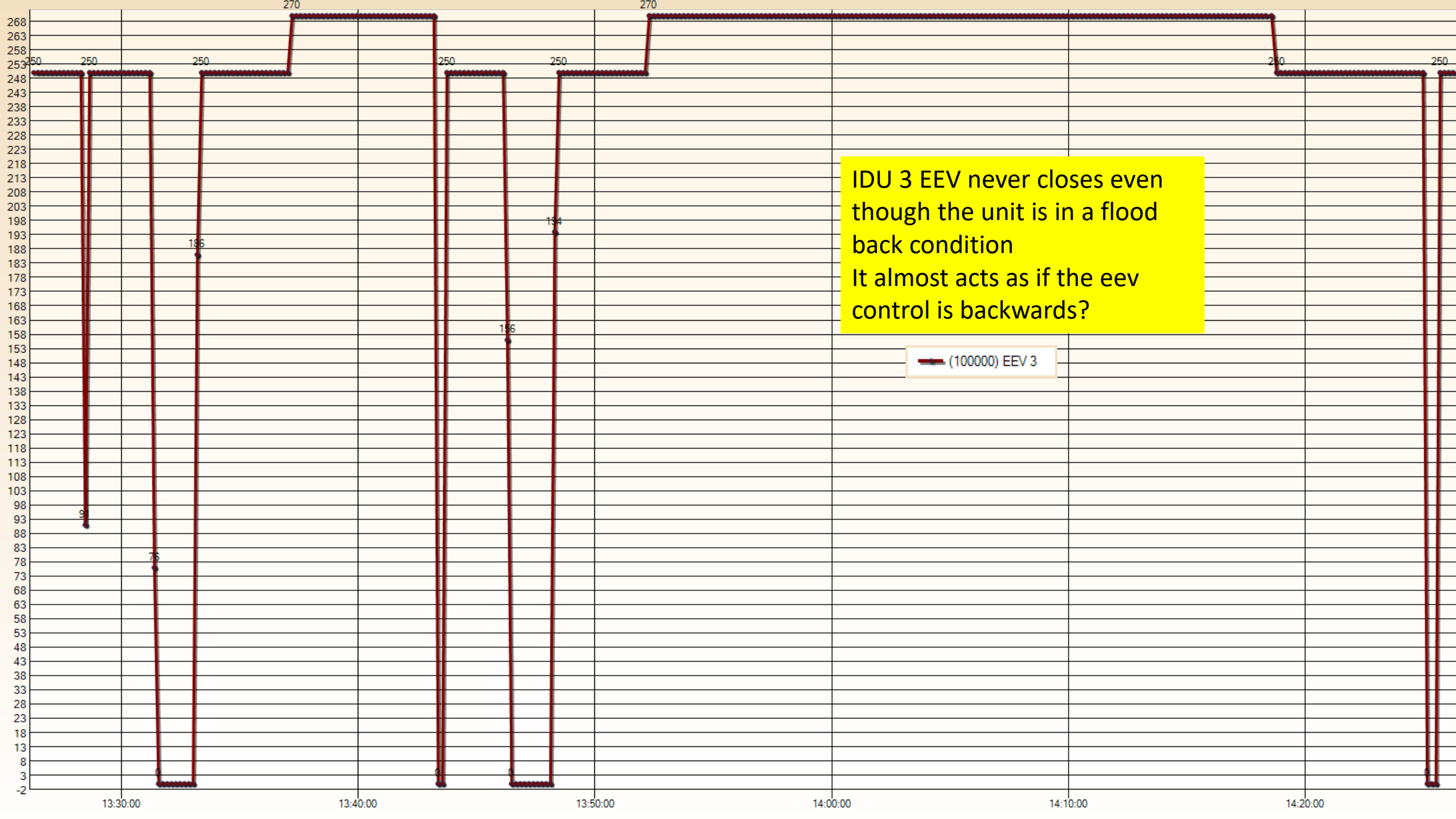
Target discharge temp = Function(Hz & Room temp & ambient temp)

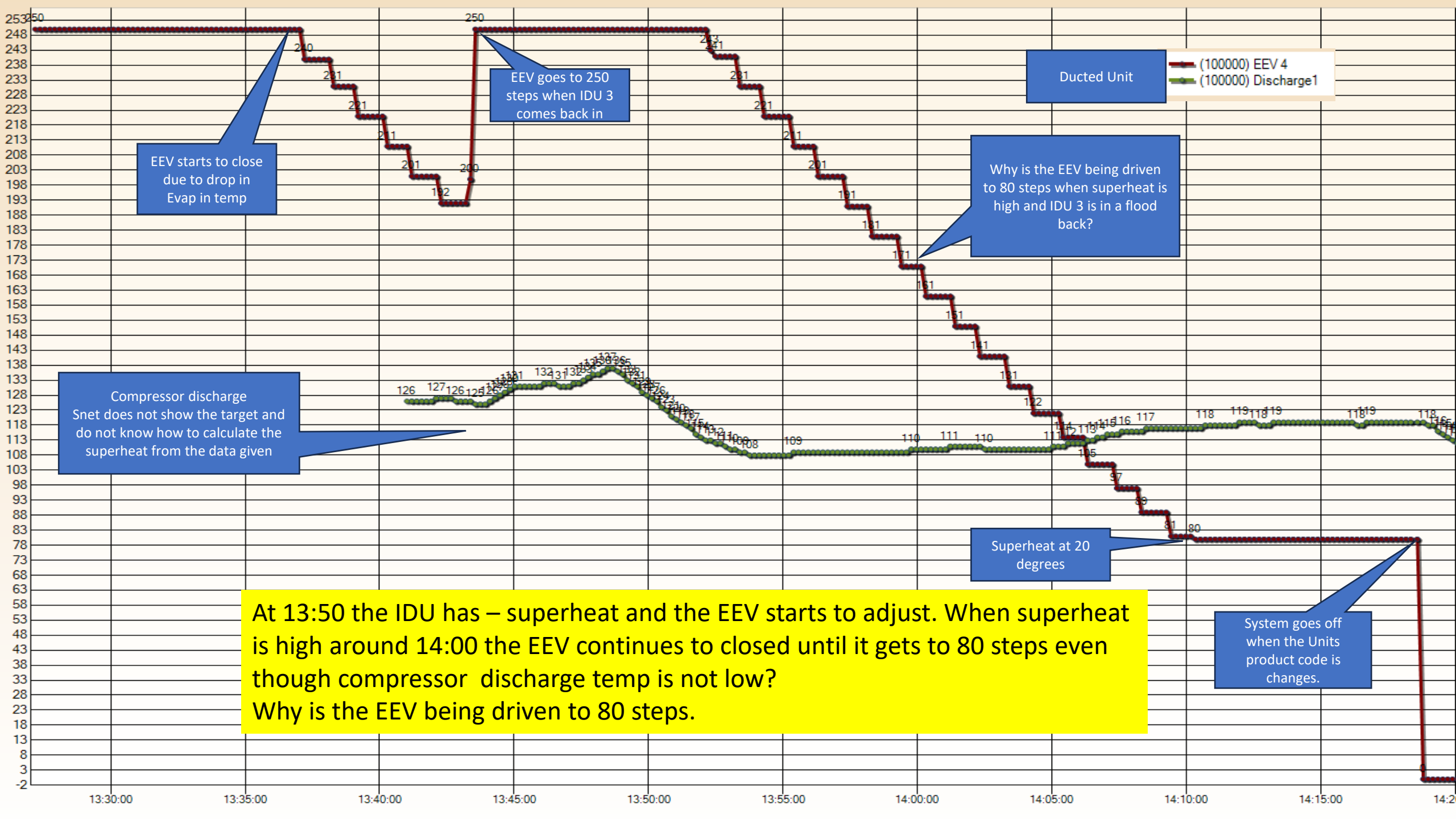


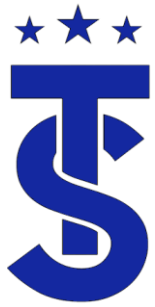
FJM EEV Control Cooling

1. Target discharge control
 2. Evap. super heat control(target 1~4°C(2~7°F))
- SH high = EEV open / SH low = EEV close
- * Indoor unit stop = 0step





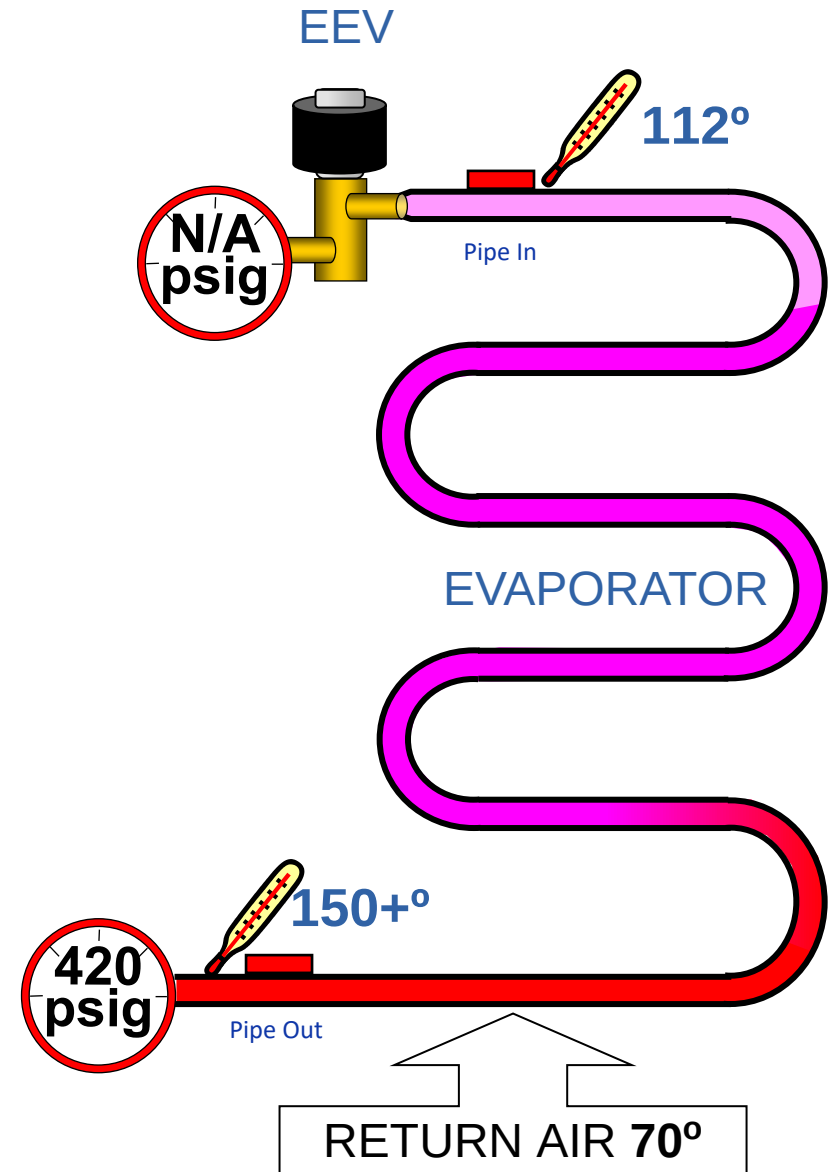




EEV Operation

Heating Mode

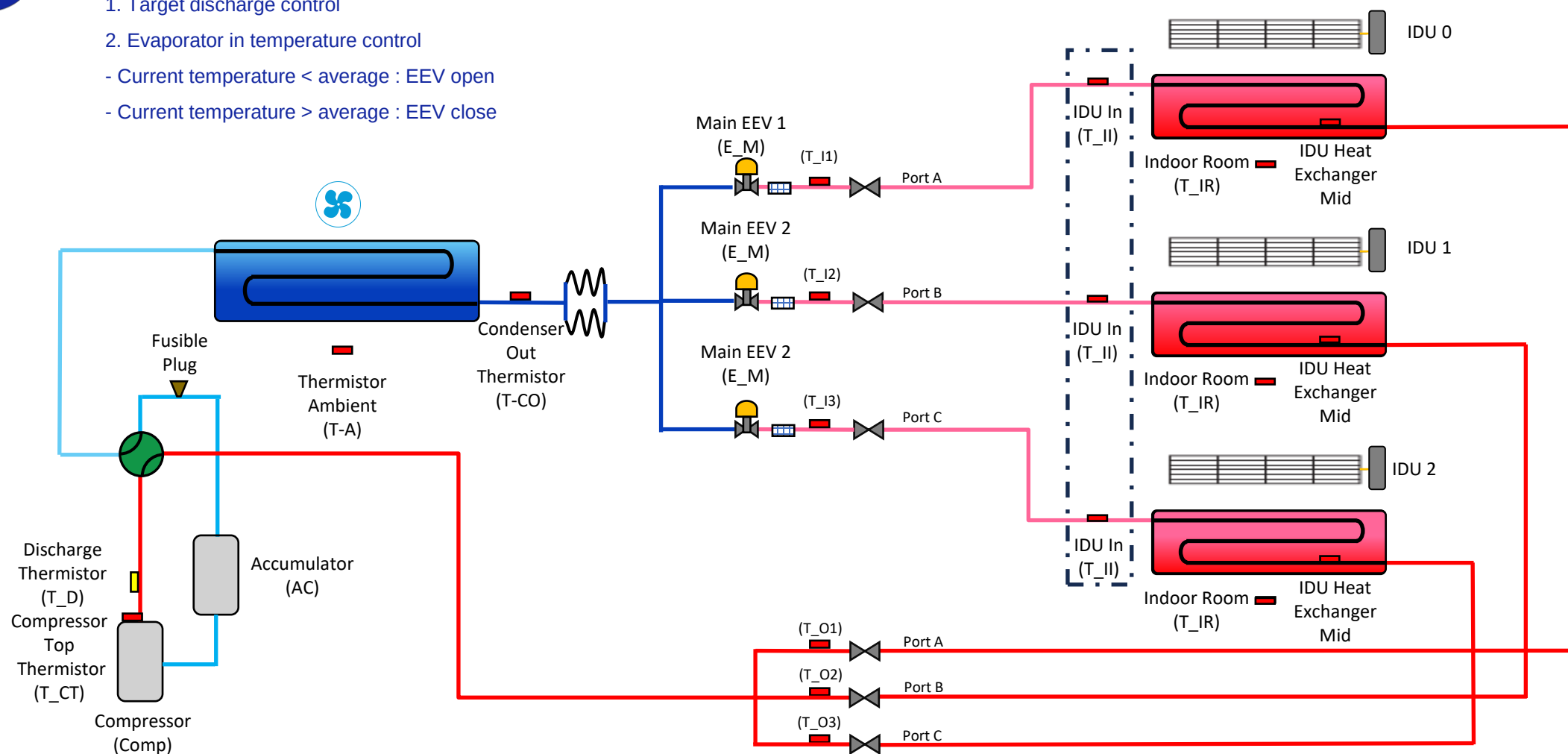
- The EEV in the heating mode will modulate to maintain a 9 degree differential between the pipe in (Liquid Temperature) and pipe (High Side Saturation Temperature)
- If the load increases the valve will open to maintain that differential.
- If the load drops, the valve will close to maintain that differential.
- What effect does low space temperatures have on the operation of the system?



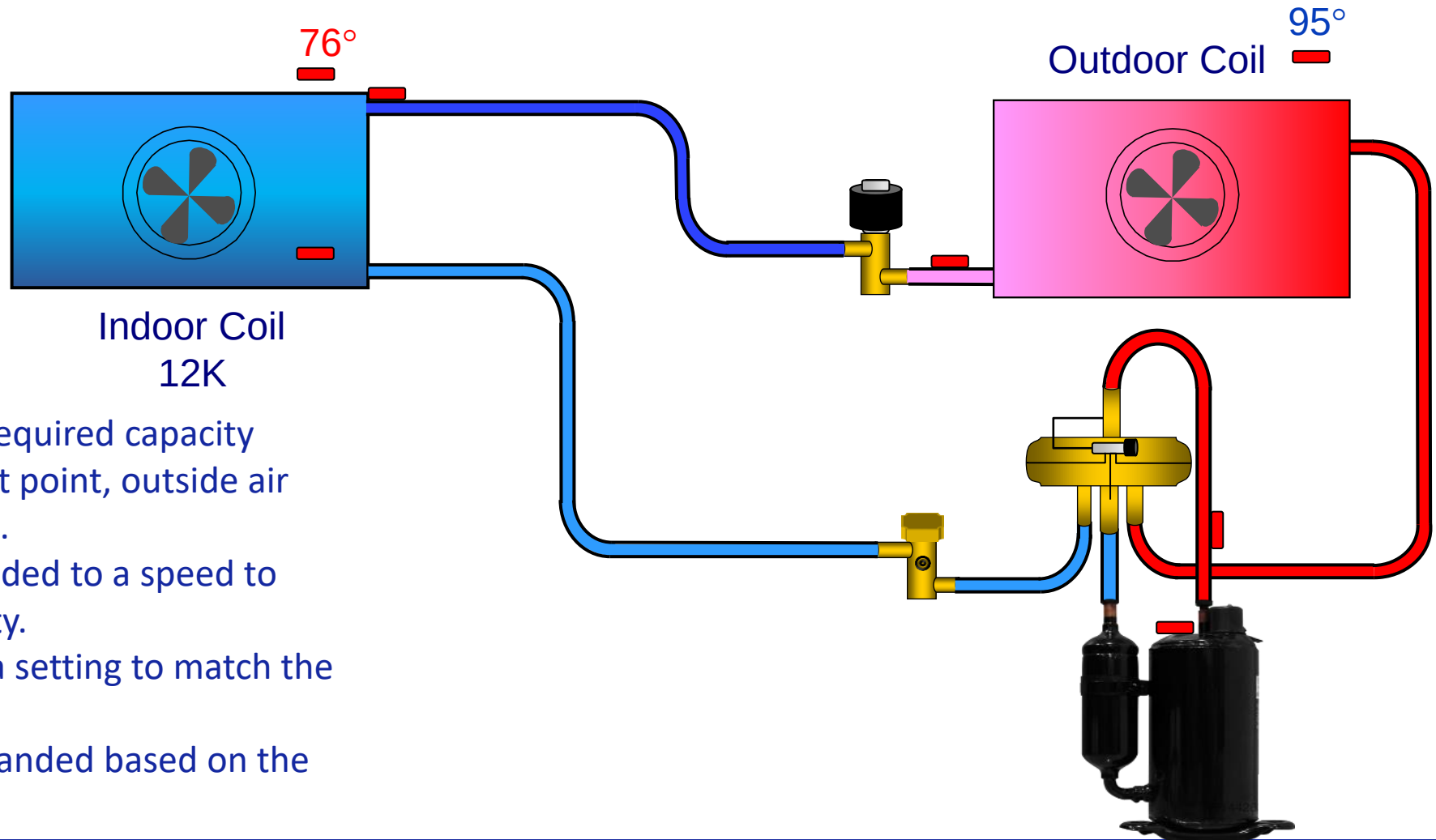


FJM EEV Control Heating

1. Target discharge control
 2. Evaporator in temperature control
- Current temperature < average : EEV open
 - Current temperature > average : EEV close



★★★ Review Start up Calculations



1. The system calculates the required capacity based on deviation from set point, outside air temperature and fan speed.
2. The compressor is commanded to a speed to delivery the desired capacity.
3. The EEV is commanded to a setting to match the compressor capacity.
4. Outside fan speed is commanded based on the outside air temperature.



Air Flow Frequency Relationship



$ T_s - T_i \leq 1.8^\circ\text{F}$	$1.8^\circ\text{F} < T_s - T_i \leq 5.4^\circ\text{F}$	$5.4^\circ\text{F} < T_s - T_i $
Low	Mid	High

The fan speed is adjusted based on the difference between the set temperature(T_s) and indoor temperature (T_i)



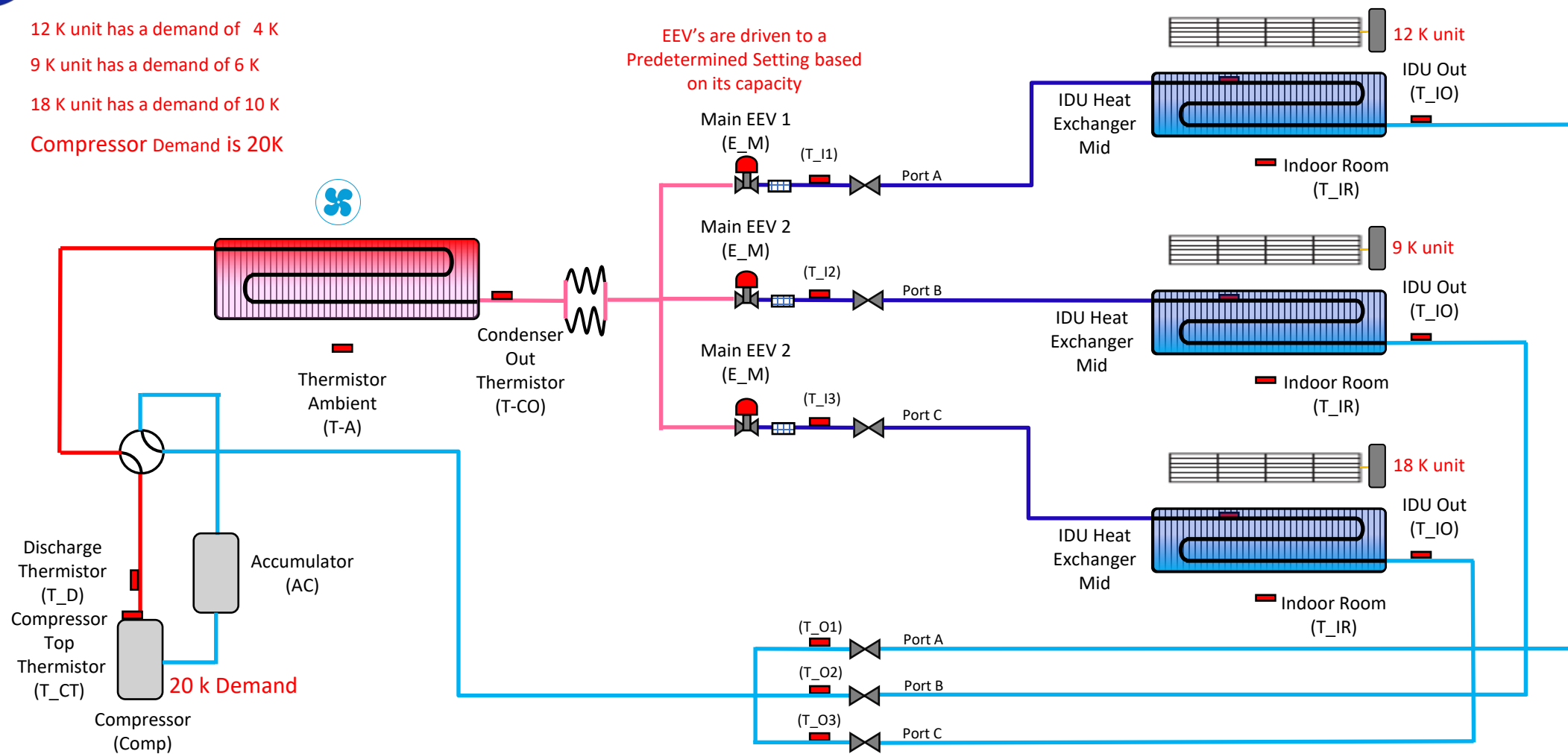
FJM Start Up Calculations

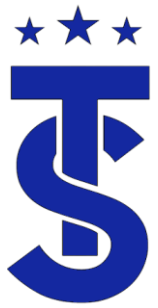
12 K unit has a demand of 4 K

9 K unit has a demand of 6 K

18 K unit has a demand of 10 K

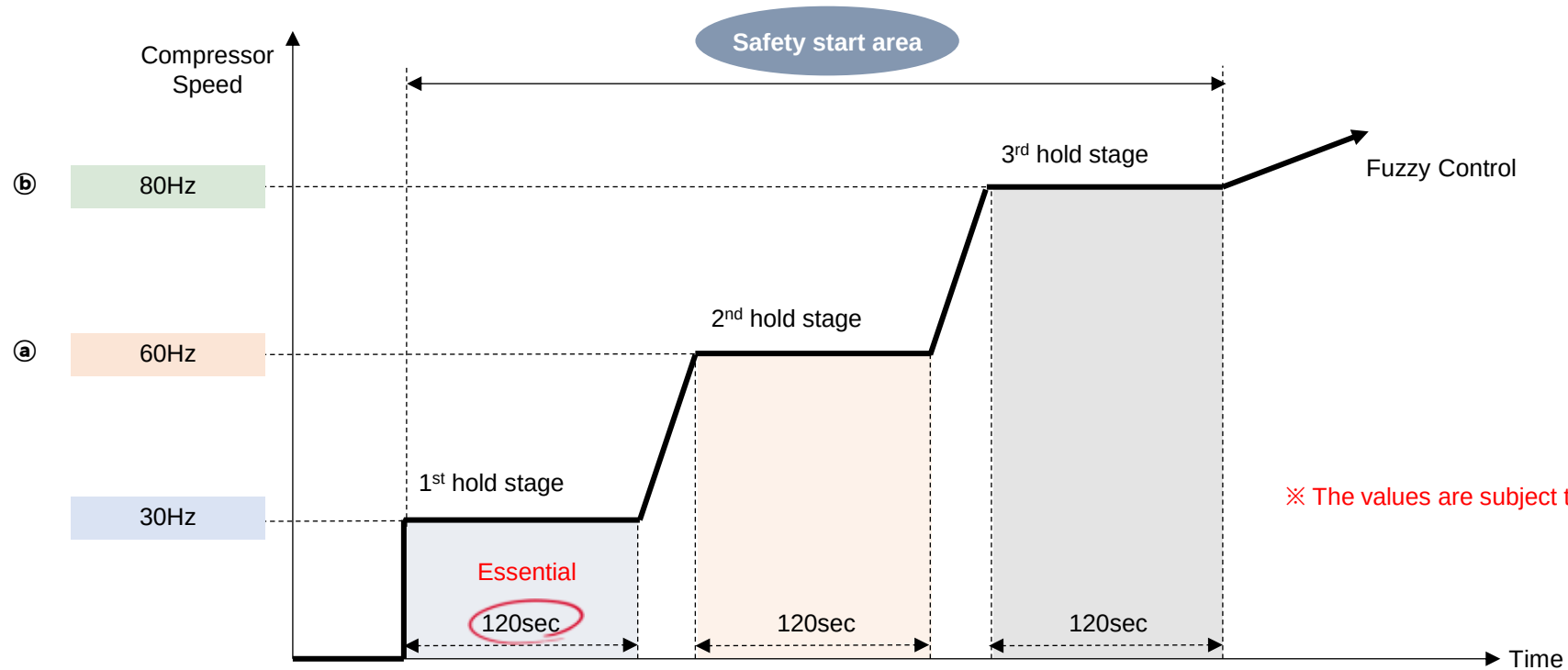
Compressor Demand is 20K

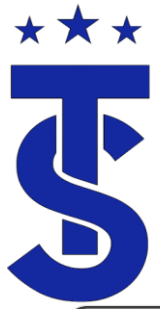




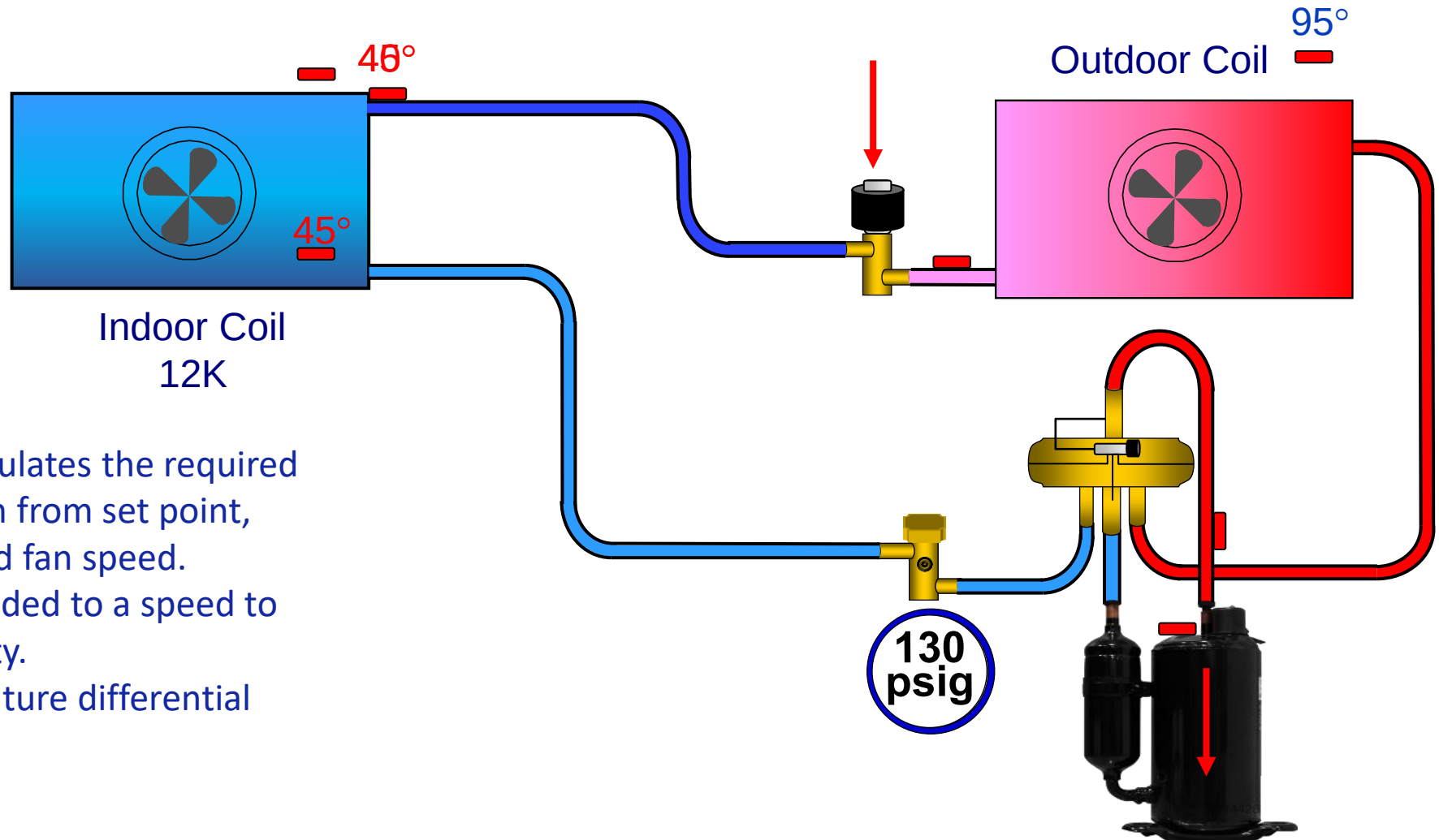
System operation

- Compressor Control
 - At the beginning of the operation,
 - Prevent rapid change of load to protect the system.
 - Prevent excessive discharge of oil in compressor.
- If the target Hz is lower than ① or ②, 2nd or 3rd hold stage will be skipped

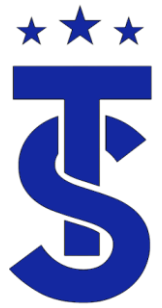




Steady State Operation



1. The system continually calculates the required capacity based on deviation from set point, outside air temperature and fan speed.
2. The compressor is commanded to a speed to delivery the desired capacity.
3. The EEV maintains temperature differential

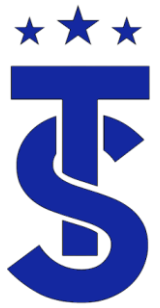


Steady State Control

Discharge Temperature Control

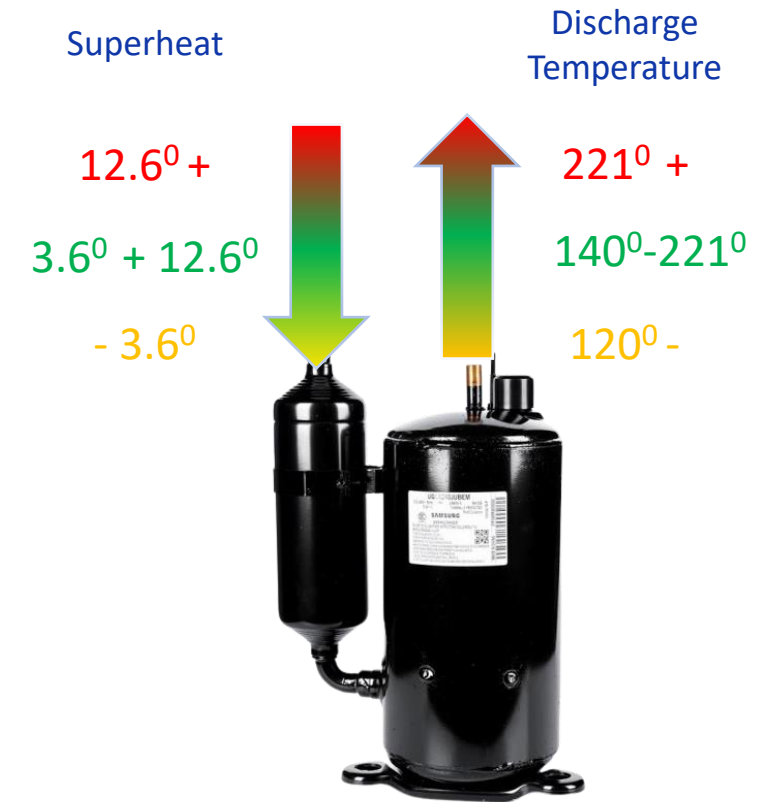
- The system monitors the discharge temperature.
- The discharge temperature is determined by the system based on the operating conditions. (taken from the PE diagrams)
- Discharge temperature is directly related to the system superheat.
- There is an acceptable range for discharge temperature based on PE diagrams and testing.
- If the discharge stays within that range the system is happy.





Discharge Temperature Range

- Normal 140° - 221°
- $221^{\circ} +$ May be low on Charge
- Below 140° system may be overcharged
- Discharge Superheat $54^{\circ} +$ during trial heat operation

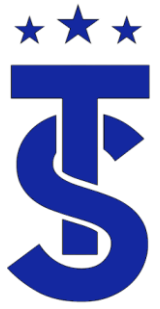




Component operation

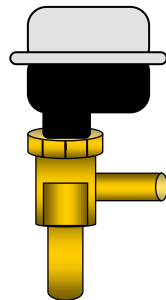
Cause and Effect

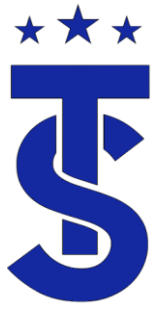




Interrelationship between components

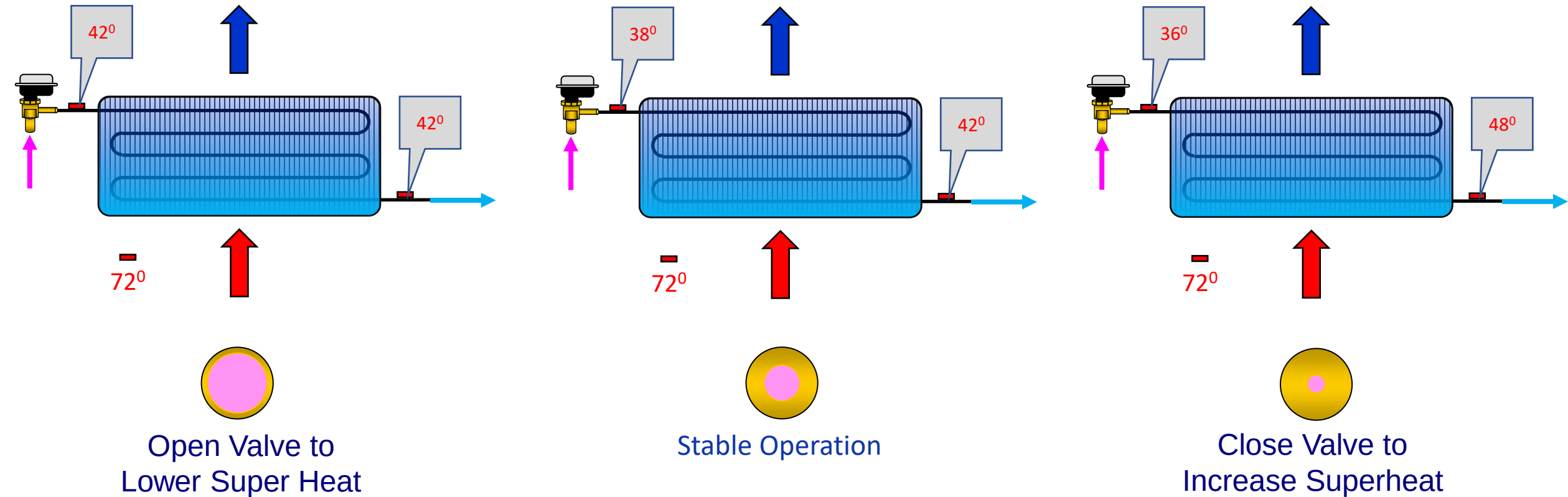
- Once the targeted operation is set you need to know how the components within the system operate to maintain those targets.
- You also need to know how the adjustment of 1 component effect that others in the system.

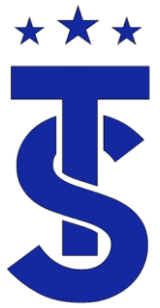




LEV opening Effects on Super Heat

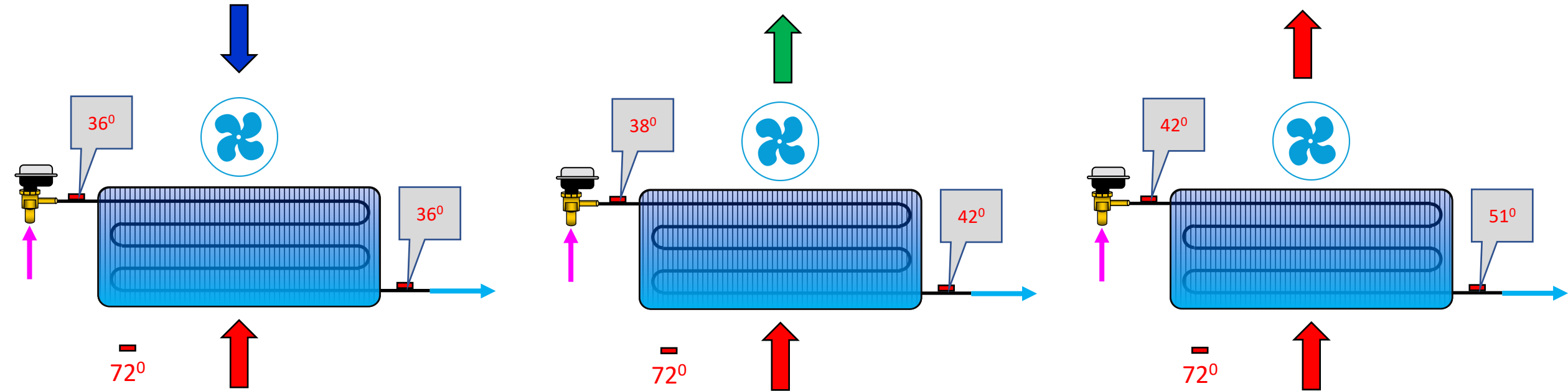
(With no other changes in the system)





Air flow effects on superheat heat

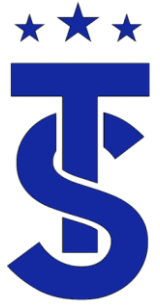
(With no other changes in the system)



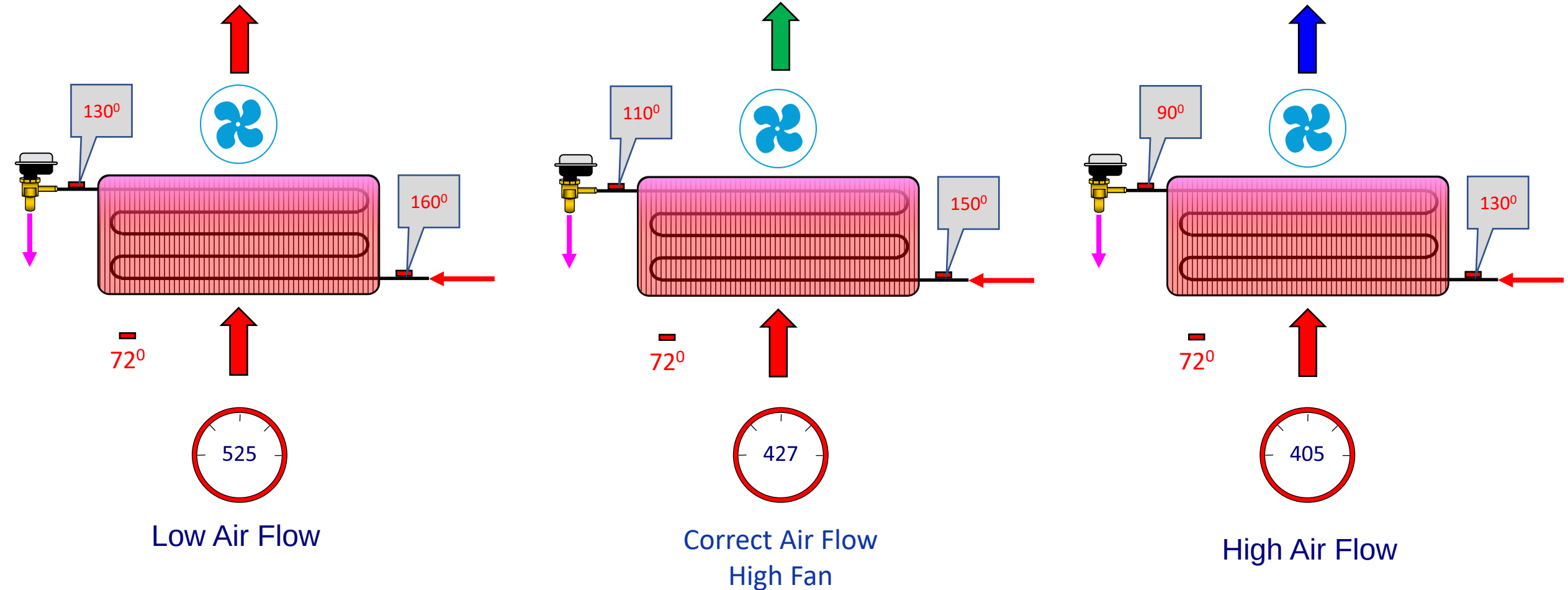
Fan cycles to low speed

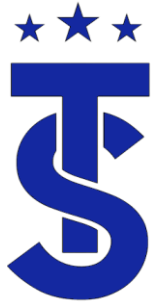
Stable Operation
Medium Fan

Fan cycles to high

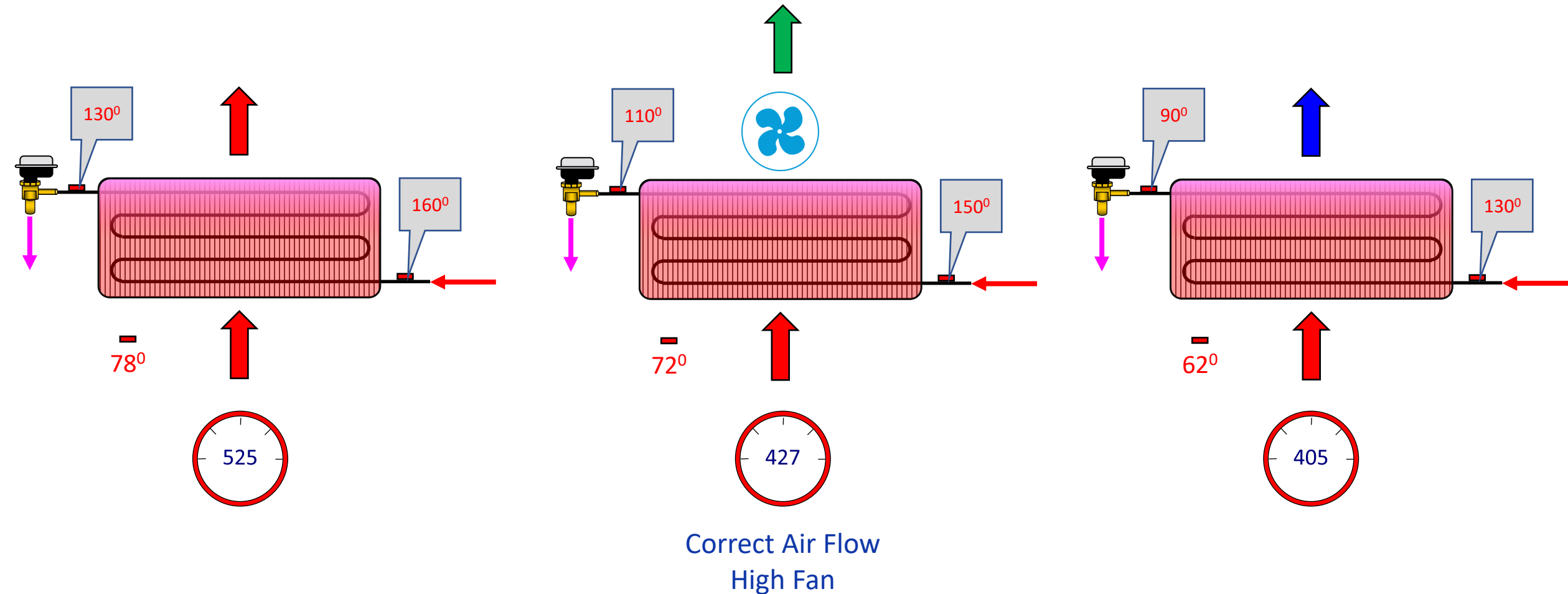


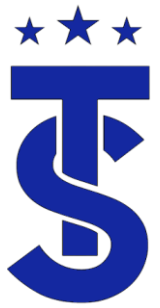
Air flow effects on heat PSI





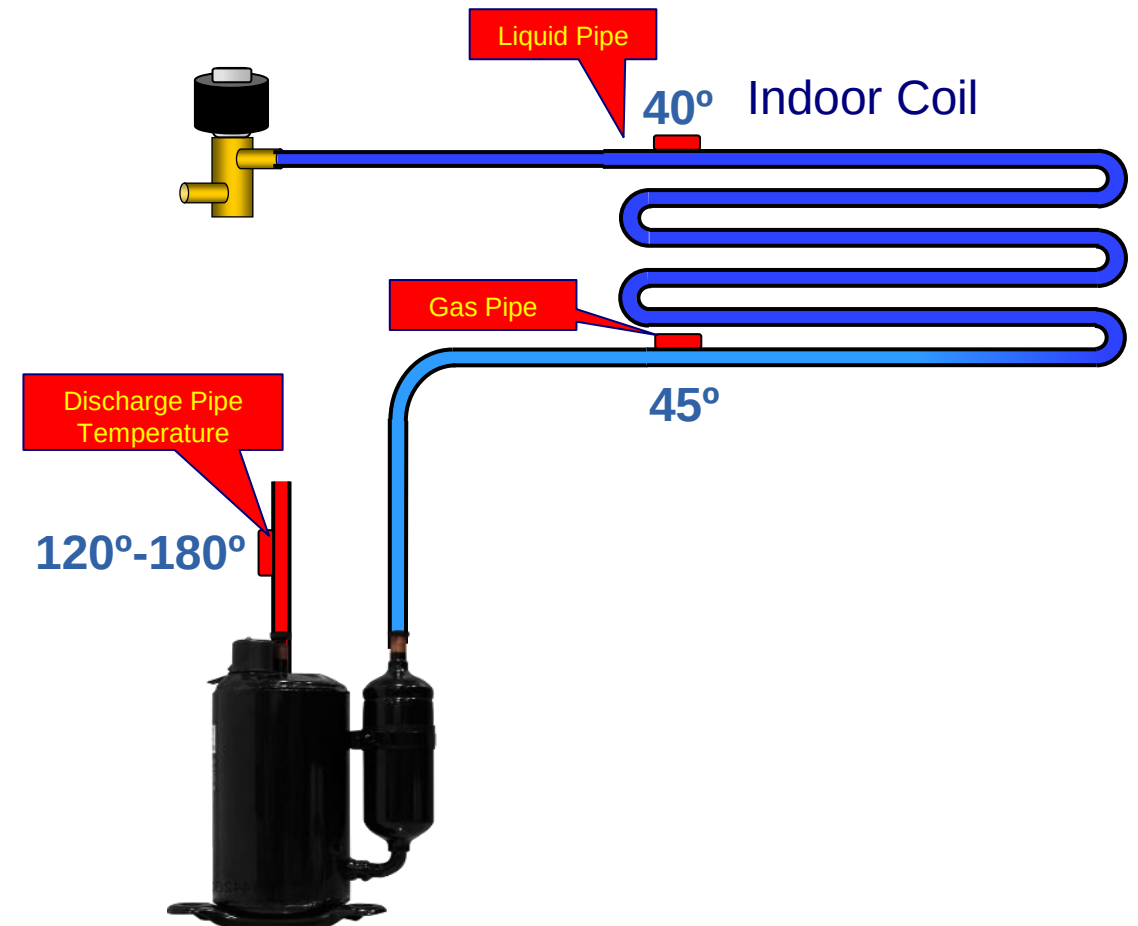
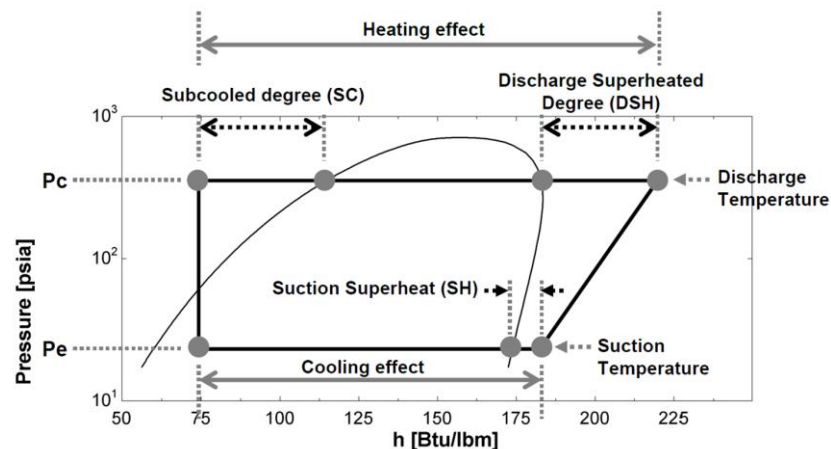
Space Temperature effects on heat PSI

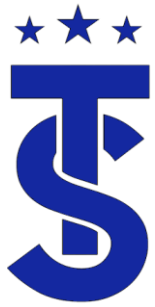




EEV – Discharge Temperature Control

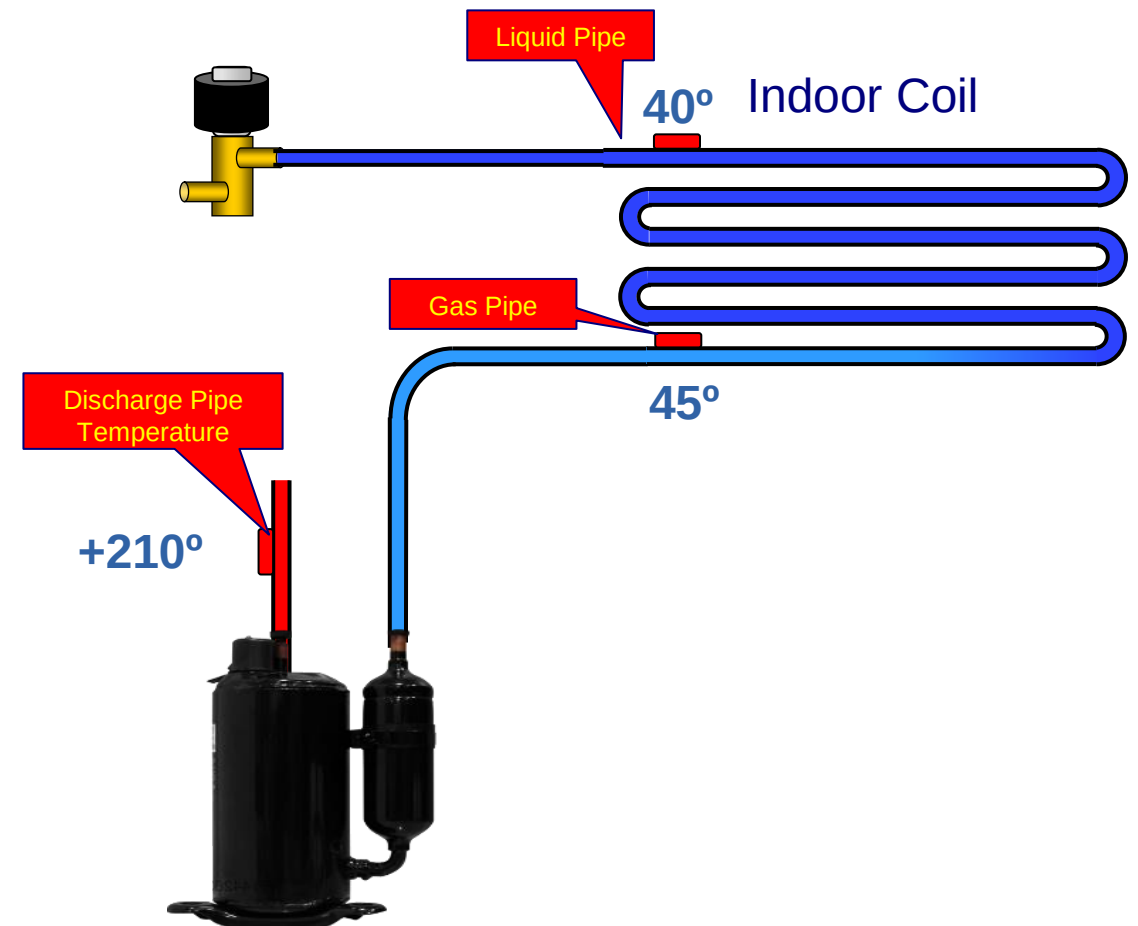
- The EEV maintains the proper differential across the evaporator in so maintaining discharge temperature
- If superheat goes up the discharge temperature goes up.





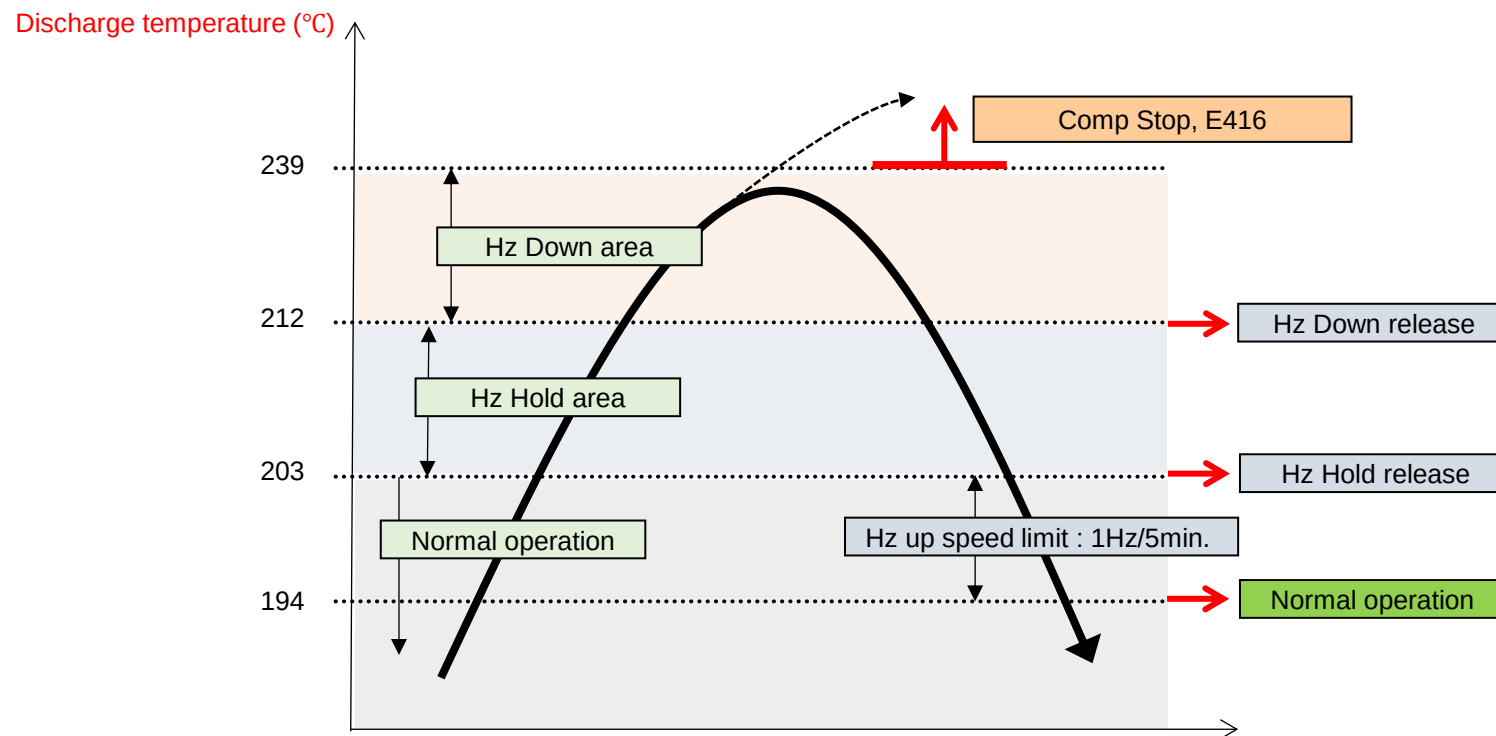
Discharge Temperature Limits

- If the discharge temperature is low and the system wants to increase the discharge temperature, then it will slow down the compressor allowing the refrigerant to sit in the evaporator longer to absorb more heat (more contact time with the coil)
- If the system wants to lower the discharge temperature it will increase the speed of the compressor (less contact time with the coil)
- If the discharge temperature continues to rise the and gets to an unsafe temperature the compressor will slow down.
- If the discharge temperature can not be reduced the compressor shuts off.





Discharge temperature protection control



※ The values are subject to change according to the model.

- Prevent abnormal discharge temperature to protect the compressor & system.



The Compressor Speed Vs Compression Ratio

(With no other changes in the system)

Normal Compression Ratio



CR goes down



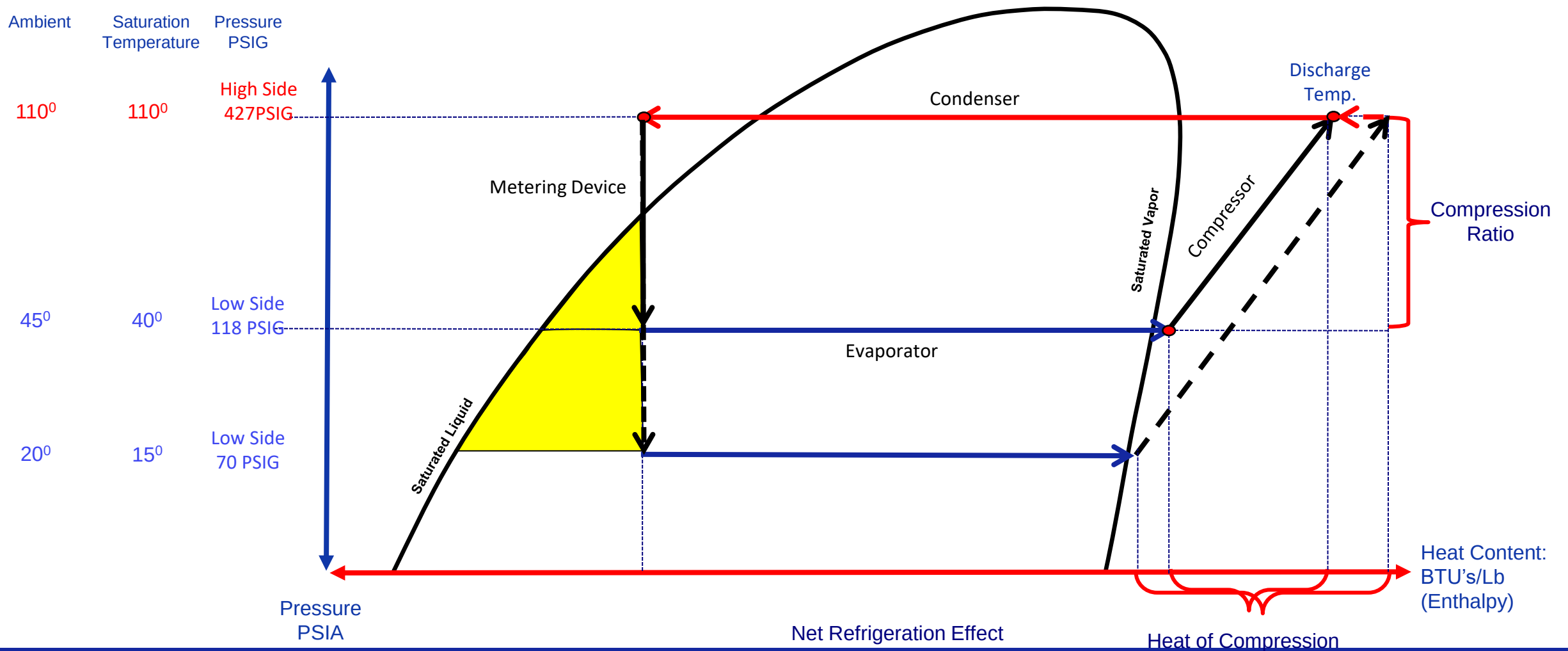
CR goes up





Increase in Compression Ratio Heating Operation

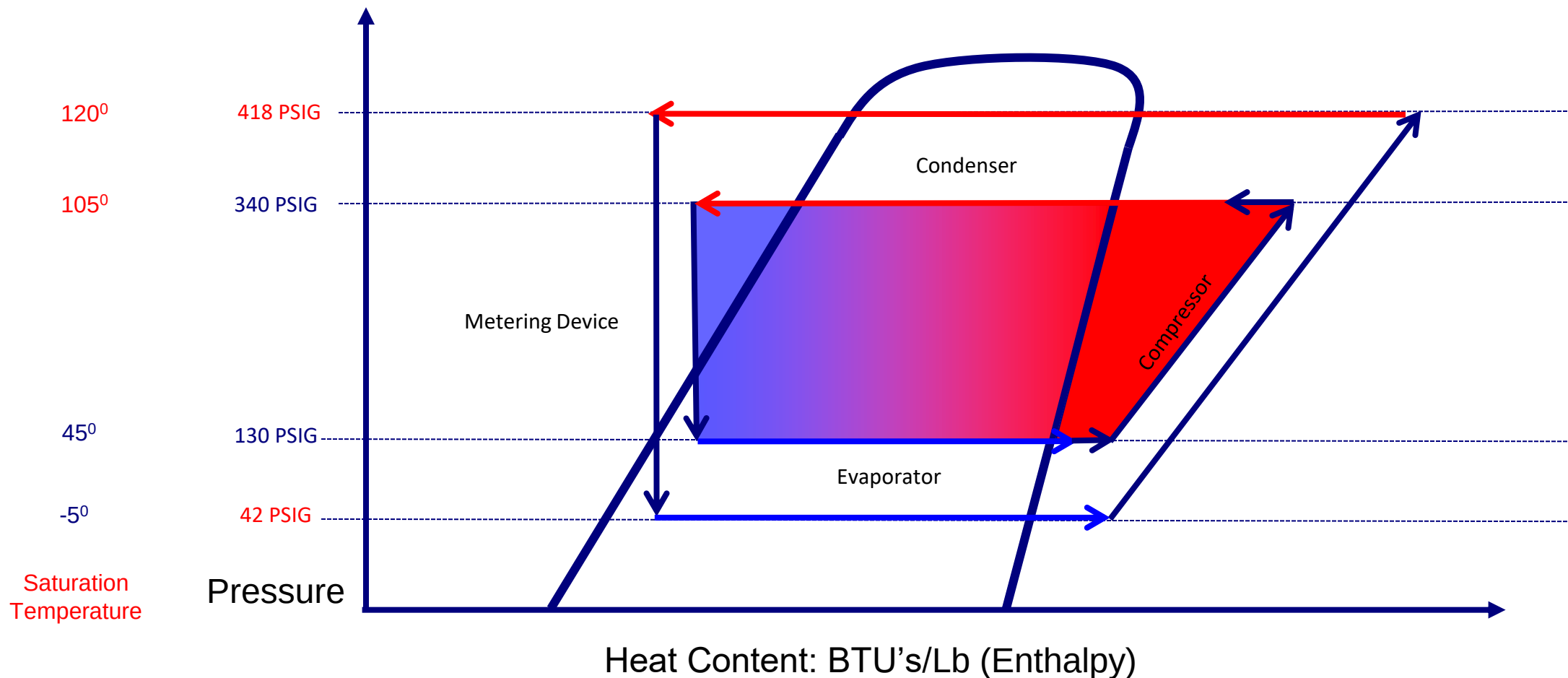
How can we increase the capacity of the evaporator at low ambient conditions





Remember This

Cooling Cycle vs Heat Cycle
Which Runs Hotter





LEV Opening Vs Discharge Temperature Relationships

(With no other changes in the system)

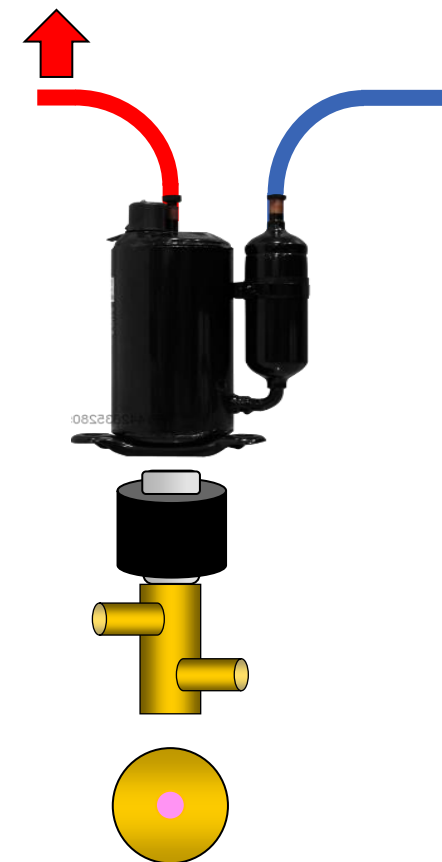


Open Valve to Lower Temp

Target Temp



Stable Operation



Close Valve to Increase Temp



Discharge Temperature Protections Vs Compressor Frequency

(With no other changes in the system)

Flood Back Condition



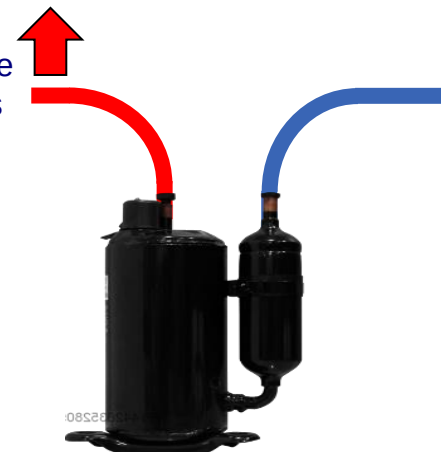
Compressor Slow Down

Target Temp

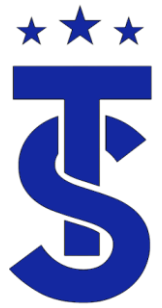


Stable Operation

High Discharge Temperatures



Compressor Slows Down

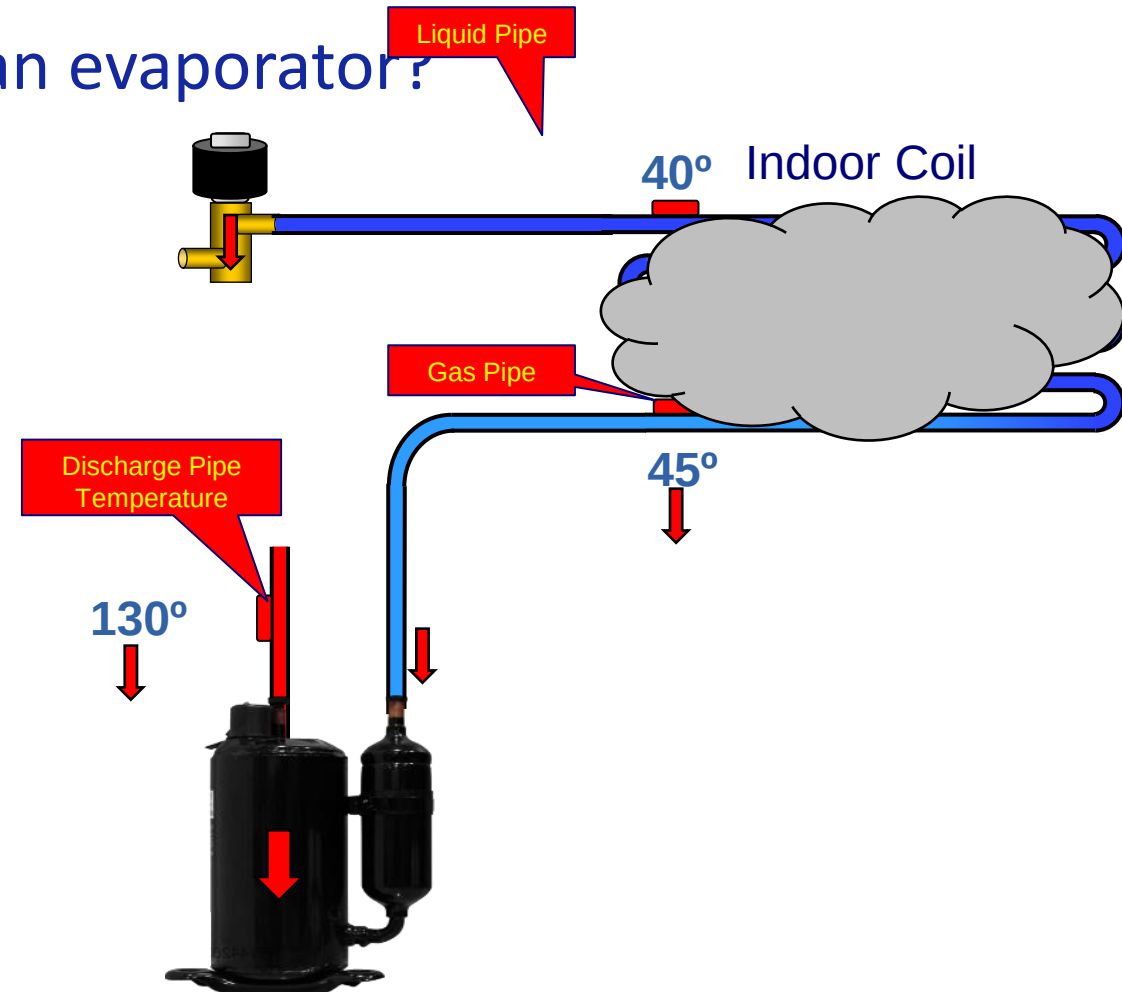


Discharge Temperature Control

Example: Dirty Coil

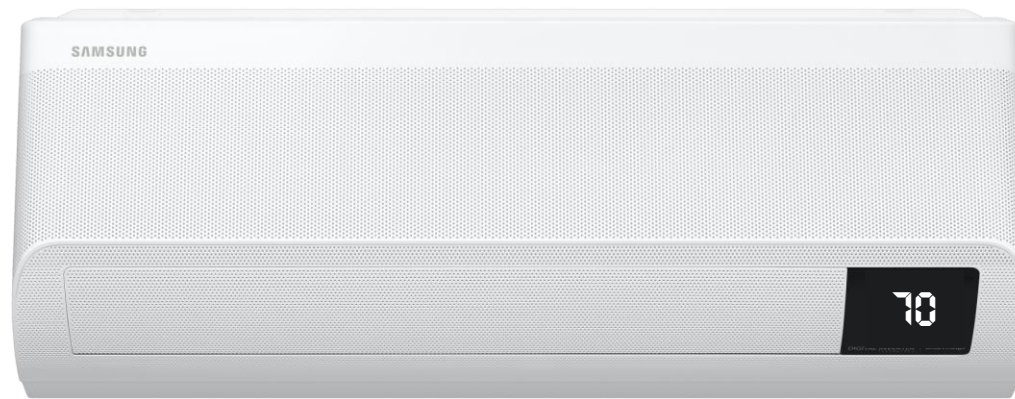
- What affect does a dirty coil have on an evaporator?

- ☐ A dirty coil causes low super heat.
- ☐ The EEV closes to increase super heat but is can not.
- ☐ The coils is flooding back to the compressor.
- ☐ Discharge temperature drops.
- ☐ Compressor slows down.



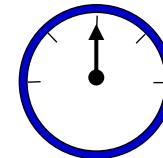
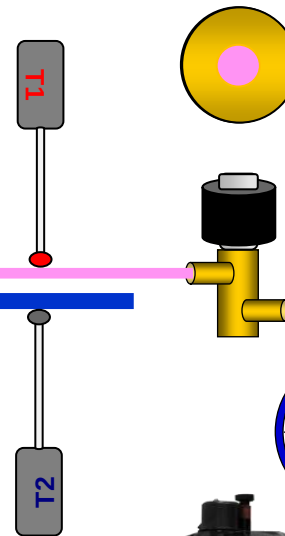


Putting it all together System Operation



High Fan Speed

Valve Opening



Target
Suction



Hz. Medium





System Testing

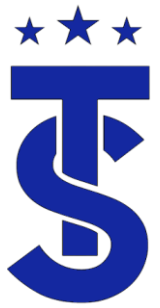


IDU Capacity



- The IDU capacity is directly related to the air flow setting, refrigerant temperature and space conditions.
- If the air flow setting is low the output capacity is low.
- If the air flow setting is high the output capacity will be high.
- On start up the system looks at air flow setting and deviation from set point and calculates the desired BTU output.
- Heating Capacities are higher than cooling capacities. Why?

Performance	Nominal Capacity ¹	Cooling / Heating (Btu/h)	12,000 / 12,000
	Capacity Range	Cooling (Btu/h)	3,000 - 14,500
		Heating (Btu/h)	2,600 - 20,000
	SEER / EER		21.5 / 12.05
	COP	Nominal Heating	3.66
	HSPF		12.0
	AHRI Reference Number		207349354

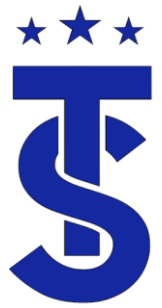


Calculating the IDU Heat Output

A+B=C

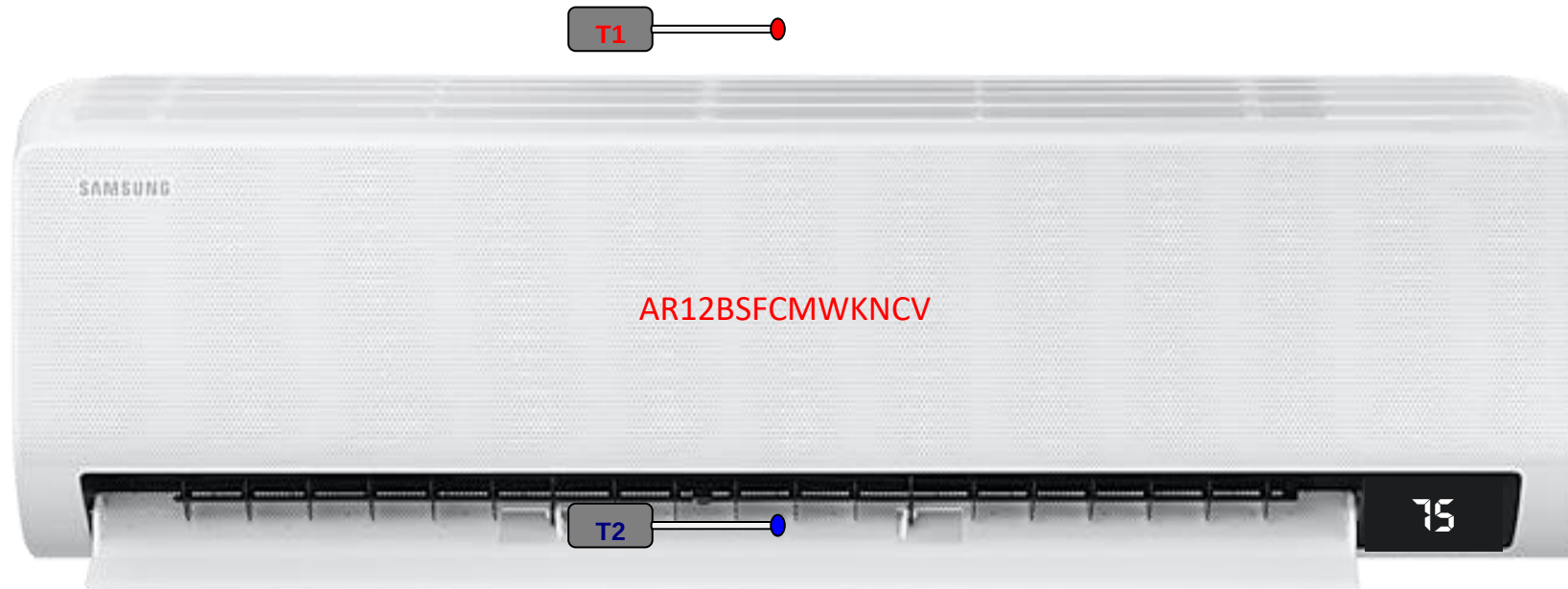
■ Heating BTUs = CFM x Δt x (Sensible Heat Factor) 1.08

- CFM = the fan CFM plotted on the fan tables, or measured at several points in the system
- Δt = the dry bulb temperature change through the system
- 1.08 = the heating BTU multiplier at sea level, or .075-lbs. of air per CFM x .24 (the specific heat of air) x 60 (minutes in an hour.) This factor will vary at higher altitudes and temperatures



Indoor Unit Testing

Verifying the IDU Capacity



Test the system on high fan speed



Dry Bulb temperatures are required
for heating calculations.
Dry and Wet-bulb temperatures are
required for cooling capacities



- Finding the indoor CFM for a ductless unit is quite simple.
- If the coil and fan wheel is clean.
- And there are no issues with the fan motor the CFM for the unit can be found using the specification sheet.
- Using the Mobile App. Find the CFM for High in heating mode.

SAMSUNG

SR12BSCFMWKXCV / AR12BSCFMWKXCV

Page 1 of 4

Samsung "WindFree™ 2.0e", wall mounted evaporator, split system

Job Name _____
 Purchaser _____
 Submitted to _____
 Unit Designation _____

Location _____
 Engineer _____
 Reference ☐ Approval ☐ Construction ☐
 Schedule # _____

Specifications

Model	US Code	Indoor Unit	RNS12CMB
		Outdoor Unit	RXS12CMB
Model Number	Indoor Unit	AR12BSFCMWKXCV	
	Outdoor Unit	AR12BSFCMWKXCV	
Performance	Nominal Capacity ¹	Cooling / Heating (Btu/h)	12,000 / 12,000
	Capacity Range	Cooling (Btu/h)	3,000 - 14,500
		Heating (Btu/h)	2,600 - 20,000
	SEER / EER		21.5 / 12.05
	COP	Nominal Heating	3.66
Power	HSPF		12.0
	AHRI Reference Number		207349354
	Voltage	Ø / V / Hz	1 / 208-230 / 60
	Working Voltage Range (VAC)		176 - 254
	Operating Current (Min./Std./Max.)	Cooling (A)	1.2 / 4.8 / 6.2
Dimensions		Heating (A)	1.1 / 4.6 / 9.3
	Max. Breaker	Amps	20
	Min. Circuit Ampacity (A)		12.5
	W X H X D (in.)	Indoor Unit	32 5/16 X 11 3/4 X 8 7/16
		Outdoor Unit	31 1/8 X 21 9/16 X 11 1/4
Sound Pressure Level	Indoor Unit	Low / High (dB)	20 / 38
	Outdoor Unit	High (dB)	46
Operating Temperatures	Outdoor	Cooling	14° ~ 115°F (-10° ~ 46.1°C)
		Heating	-5° ~ 75°F (-20.5° ~ 23.9°C)
	Indoor	Cooling	61° ~ 90°F (16° ~ 32°C)
		Heating	81°F (27°C) or less
	Indoor & Outdoor	High side (flare)	1/4"
Pipe Connections		Low side (flare)	3/8"
	Maximum / Minimum Line Set Length (ft.)		66 / 9.8
	Maximum Vertical Separation (ft.)		49
	Type		R410A
	Control Method		Electronic Expansion Valve
Refrigerant	Factory Charge	oz.	35.3
	Charged for		25 feet
	Additional Refrigerant		0.16 oz. / ft. over 25 ft.
Compressor	Manufacturer		Samsung
	Type		BLDC Rotary
Evaporator Fan	FLA		9.2
	Type		BLDC motor with cross-flow fan
	Air Volume (ILMH/Turbo)	Cooling (CFM)	300 / 335 / 371 / 385
		Heating (CFM)	335 / 371 / 403 / 420
	Consumption	Watts	27 X 1
Condenser Fan	FLA		0.12
	Motor		BLDC motor with axial fan (1)
	Output	Watts	40
	FLA		0.18
	Air Volume	CFM (max.)	1,589
Safety	Certifications		ETL (UL 1995)
	Devices		PCB fuses, indoor unit terminal block thermal fuse, current transformer, over-voltage protection, crankcase heating, temperature limit protection logic, compressor overload sensing

¹ Certified in accordance with the AHRI Unitary Small Air-Source Heat Pumps (USHP) Certification Program which is based on the latest edition of AHRI Standard 210/240.

*The WindFree™ unit delivers an air current that is under 0.15 m/s while in WindFree™ mode. Air velocity that is below 0.15 m/s is considered "still air" as defined by ASHRAE (American Society of Heating, Refrigerating, and Air-Conditioning Engineers).

Samsung HVAC maintains a policy of ongoing development, specifications are subject to change without notice. Refer to www.AHRIdirectory.org for current reference numbers.

(actual equipment appearance may vary)

General Information

- The indoor unit shall feature WindFree™ mode. In cooling mode, as room temperature nears set temperature, the unit will close its louvers and will disperse air into the space through thousands of micro-holes on the front of the indoor unit preventing cold air drafts on occupants.
- The indoor unit shall have Wi-Fi capability as standard
- Outdoor unit shall provide 208/230V power to indoor unit via 14 AWG X 3 interconnect power cable

Construction

- Indoor unit chassis shall be UL94 V0 with a galvanized steel mounting bracket
- The indoor unit shall have easy-access to wire, pipe, and drain connections via access panel on the bottom of the unit for simple installation and service
- The outdoor unit shall be galvanized steel with a baked on powder coated finish for durability

Heat Exchanger

- The heat exchangers shall be mechanically bonded fin to copper tube

Refrigerant System

- The compressor shall be hermetically sealed, inverter controlled, BLDC Rotary
- Refrigerant flow shall be controlled by an electronic expansion valve at the outdoor unit

Indoor Fan

- The indoor fan shall be a single, antibacterial cross-flow type
- Three fan speed settings and auto setting
- Automatic (motorized) vertical swing (up/down) and horizontal swing (left/right) louvers

Controls

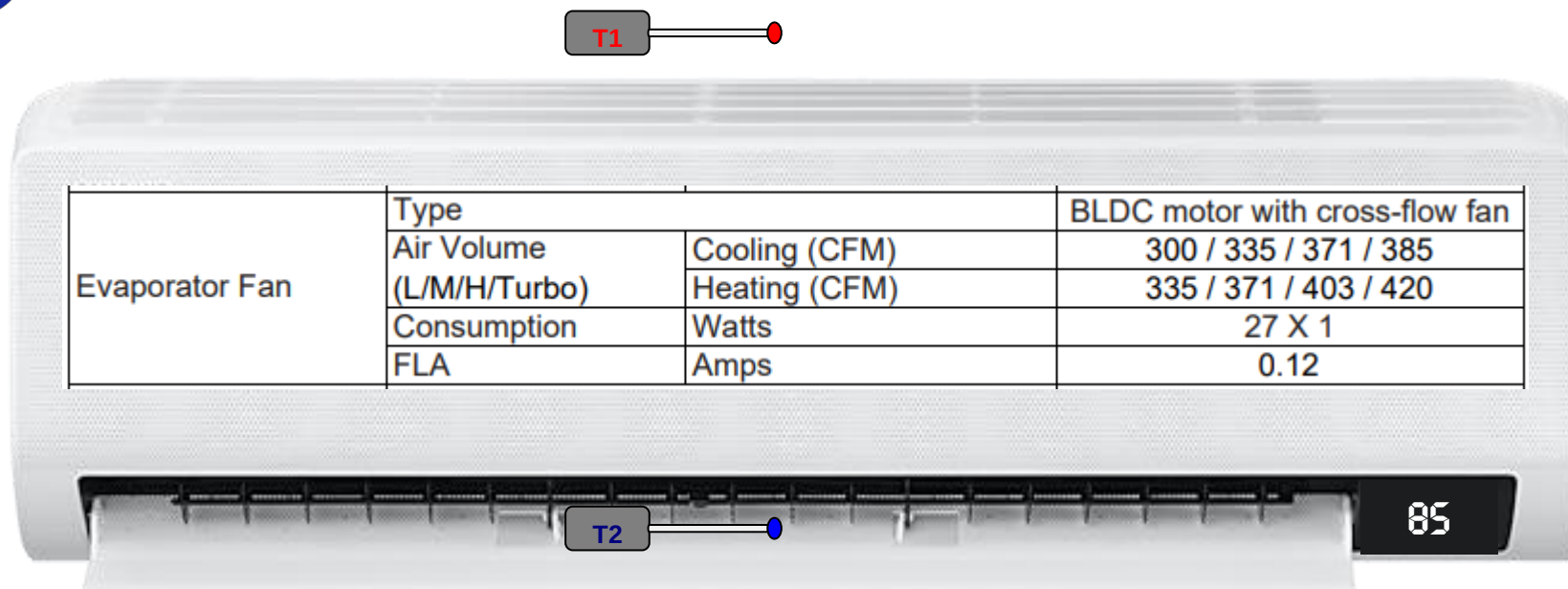
- The system shall have a built in Wi-Fi adapter as standard to allow control and monitoring using the Samsung SmartThings app (Android, iOS)
- Dual set temperature support when connected to MWR-WG00UN Advanced Wired Controller and supported central control options.
- The indoor unit shall have a simple connection for overflow detection devices or any other normally closed contact for simple unit shutdown
- The indoor unit shall support a wireless controller, holder, and batteries
- Wired controller options available
- Samsung central control compatible (MM-R10UN accessory required)
- Interconnect control wire between outdoor and indoor unit shall be 16AWG X 2

Convenience

- System energy consumption can be viewed using the Samsung SmartThings mobile app or on the indoor unit display using the included wireless controller
- AI (artificial intelligence) Auto Mode technology monitors factors such as indoor temperature, outdoor temperature, set temperature, and operating time to learn the patterns within your home to automatically adjust system operation to maximize occupant comfort and efficiency (Wi-Fi connection required)
- Eco Mode to reduce energy consumption during low demand operation
- Smart install mode - startup system diagnostics operation to ensure system readiness during initial operation
- Auto restart
- Auto Clean Function
- 7-se



Calculate the Indoor Unit Heating Capacity



High Speep 403 CFM





Heat Calculation Example

■ Heating BTUs = CFM x Δt x (Sensible Heat Factor) 1.08

□ CFM = 403

□ Δt = 33

• $403 \times 33 \times 1.08 = 14,363 \text{ BTU}$

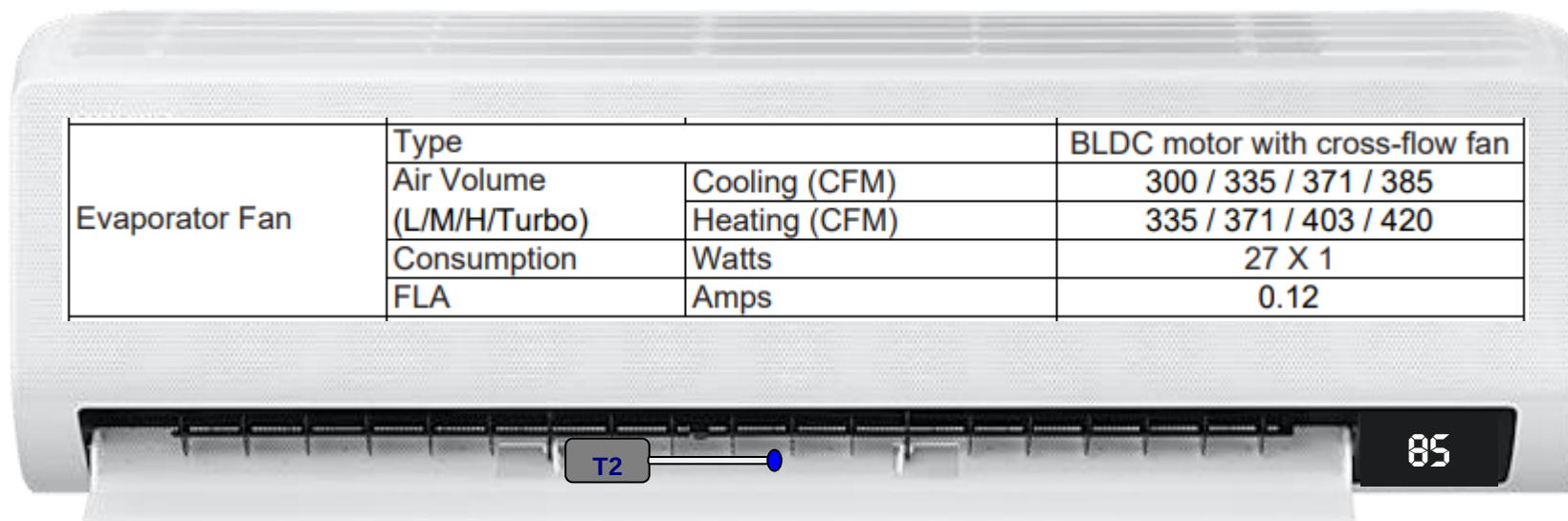
• Is this good or bad?

Performance	Nominal Capacity ¹	Cooling / Heating (Btu/h)	12,000 / 12,000
	Capacity Range	Cooling (Btu/h)	3,000 - 14,500
		Heating (Btu/h)	2,600 - 20,000
	SEER / EER		21.5 / 12.05
	COP	Nominal Heating	3.66
	HSPF		12.0
	AHRI Reference Number		207349354



Calculate the Indoor Unit Cooling Capacity

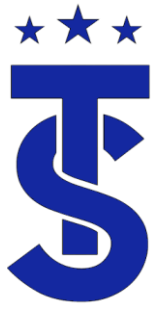
T1 —● Supply Air: 55 BD – 51.3 WB



Return Air: 75 BD – 65.5 WB



High Speed 385 CFM



Cooling Capacity Calculations

- Cooling BTUs = CFM x Δh x 4.5
- CFM = The fan CFM plotted on the fan tables, or measured at several points in the system
- Δh = the enthalpy change through the system (we get this number by converting wet bulb readings.)
- 4.5 = the cooling BTU multiplier at sea level, or .075-lbs. of air per CFM x 60 (minutes in an hour.) This factor will also change at higher altitudes. BTU multipliers will also change due to extreme air temperatures



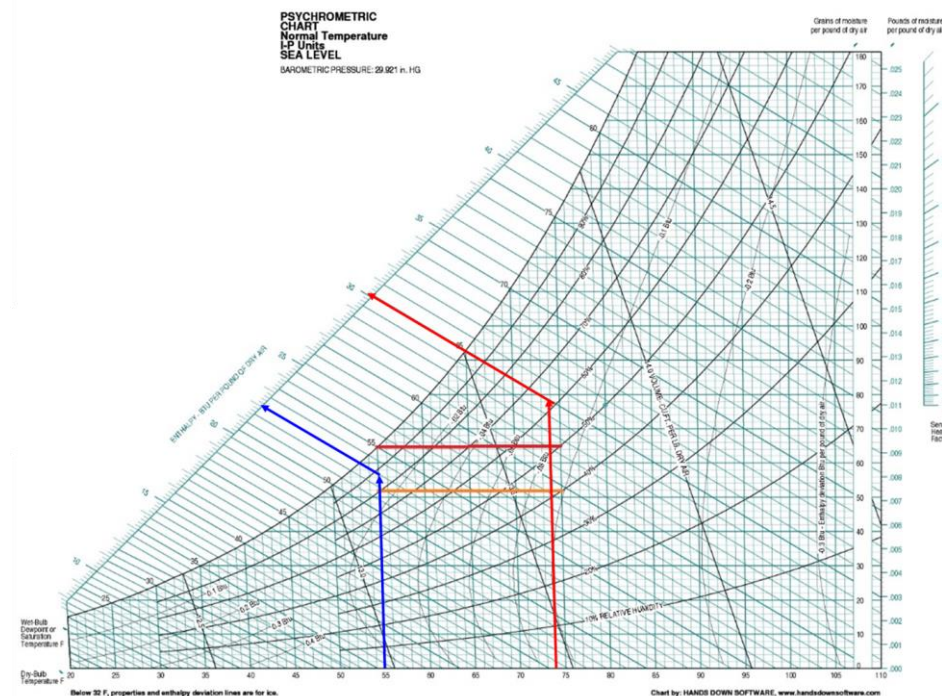
Calculate Enthalpy Differential

- Using a psychrometric calculator determine the enthalpy of the return and supply air then calculate the difference
 - Return Air: 75 BD – 65.5 WB – Enthalpy 29.2
 - Supply Air: 55 BD – 51.3 WB – Enthalpy 22.2
 - Enthalpy 7

Free online Psychrometric Calculator [Print](#)

☒ IP ☐ SI altitude ft b press in.Hg

T (°F) dry bulb	rH (%) rel.humidity	TDP (°F) dew point	W (GPP) abs. humidity	h (btu/lb) enthalpy	Tw (°F) wet bulb	sp. vol (cft/lb)		
75	55	61.2	80	29.2	65.5	13.78	Calculate	Remove
55	90	48.9	51	22.2	51.3	13.14	Calculate	Remove





Cooling Capacity Formula

- $BTUH = CFM \times 4.5 \times \text{Delta Enthalpy}$
- As an **example**, measurements inside the main ducts of the air handler:
 - Before the evaporator (return): DB: 75° F, WB: 65.5° F, Enthalpy = 29.2 BTU/lb.
 - After evaporator (supply): DB: 55° F, WB: 51.3° F, Enthalpy = 22.2 BTU/lb.
- **System Enthalpy:** 29.2 - 22.2 = 7. BTU/lb.
- **System CFM = 385CFM**
- **BTUH = 385 x 4.5 x 8**
- **BTUH = 12,126 BTUH**
- Is this good or bad?

Evaporator Fan	Type		BLDC motor with cross-flow fan
	Air Volume (L/M/H/Turbo)	Cooling (CFM)	300 / 335 / 371 / 385
		Heating (CFM)	335 / 371 / 403 / 420
	Consumption	Watts	27 X 1
	FLA	Amps	0.12

Performance	Nominal Capacity ¹	Cooling / Heating (Btu/h)	12,000 / 12,000
	Capacity Range	Cooling (Btu/h)	3,000 - 14,500
		Heating (Btu/h)	2,600 - 20,000
	SEER / EER		21.5 / 12.05
	COP	Nominal Heating	3.66
	HSPF		12.0
	AHRI Reference Number		207349354





Job site Sheets

Samsung FJM Job Site Sheet

System Information

Outdoor Unit	Model #	AJ036BXS4CH/AA	Serial #	Mode	Cool
Indoor Units	Circuit A	Circuit B	Circuit C	Circuit D	Circuit E
Location					
Model #					
Serial #					
Type of Controller					

Outdoor Unit:

Are there any faults being displayed at the outdoor unit? (Y or N) Yes ☐ No ☐
If yes list the fault being displayed: **E 460**

ODU Electrical

Supply Voltage L1-L2: _____
Zone A Voltages: L1 - L2 _____ F1 - F2 _____
Zone B Voltages: L1 - L2 _____ F1 - F2 _____
Zone C Voltages: L1 - L2 _____ F1 - F2 _____
Zone D Voltages: L1 - L2 _____ F1 - F2 _____
Zone E Voltages: L1 - L2 _____ F1 - F2 _____

IDU Piping

Zone A Pipe Sizes _____ Length: _____
Zone B Pipe Sizes _____ Length: _____
Zone C Pipe Sizes _____ Length: _____
Zone D Pipe Sizes _____ Length: _____
Zone E Pipe Sizes _____ Length: _____

ODU Information (Using the K4 Button).

What is the Target Discharge Temperature: _____ Push the K4 Button 14 Times (Reading is in Celsius).
Is the system in Safety Control. Push the K4 Button 16 Times and Record the LED Display Reading: _____

What is the size and type of wire used to connect the indoor units? _____

Is there a disconnect at the indoor units? (Y or N) No ☐ Yes ☐

Are there wire splices or junction box connections in the indoor unit wiring circuit? (Y or N) _____

Is there a condensate pump installed? (Y or N) No ☐ Yes ☐

What is the Gas Pressure at the Service Port of the OU? _____

What is the ambient temperature? _____

Customer Complaint: _____

Samsung FJM Job Site Sheet

Indoor Units:

Diagnostic Instructions:

Run all the units in heat or cool with the fan speed set to high. Set the controller as high or low as possible for the mode you are using. Wait 20 minutes then record the following information.

Are there any faults being displayed on the indoor unit? (Y or N)
If yes list the fault being displayed and the indoor unit address:

Indoor Unit A

Mode: Heat Set point: _____ Room Temperature: _____ Fan speed setting: _____ ESP of Ducted Unit: _____
Voltage at IDU: L1-L2 _____ F1-F3 _____ F3-F4 _____
ODU Pipe In Temperature DB°F _____
IDU Pipe In Temperature DB°F _____
IDU Pipe Out Temperature DB°F _____
ODU Pipe Out Temperature DB°F _____
Degrees of Superheat or Subcooling at IDU: _____
Return Air Temperature DB°F _____ Supply Air Temperature DB°F _____ Measured at the IDU

Indoor Unit B

Mode: Off Set point: _____ Room Temperature: _____ Fan speed setting: _____ ESP of Ducted Unit: _____
Voltage at IDU: L1-L2 _____ F1-F3 _____ F3-F4 _____
ODU Pipe In Temperature DB°F _____
IDU Pipe In Temperature DB°F _____
IDU Pipe Out Temperature DB°F _____
ODU Pipe Out Temperature DB°F _____
Degrees of Superheat or Subcooling at IDU: _____
Return Air Temperature DB°F _____ Supply Air Temperature DB°F _____ Measured at the IDU

Indoor Unit C

Mode: Cooling - Heating Set point: _____ Room Temperature: _____ Fan speed setting: _____
Voltage at IDU: L1-L2 _____ F1-F3 _____ F3-F4 _____
ODU Pipe In Temperature DB°F _____
IDU Pipe In Temperature DB°F _____
IDU Pipe Out Temperature DB°F _____
ODU Pipe Out Temperature DB°F _____
Degrees of Superheat or Subcooling at IDU: _____
Return Air Temperature DB°F _____ Supply Air Temperature DB°F _____ Measured at the IDU

Indoor Unit D

Mode: Cooling - Heating Set point: _____ Room Temperature: _____ Fan speed setting: _____
Voltage at IDU: L1-L2 _____ F1-F3 _____ F3-F4 _____
ODU Pipe In Temperature DB°F _____
IDU Pipe In Temperature DB°F _____
IDU Pipe Out Temperature DB°F _____
ODU Pipe Out Temperature DB°F _____
Degrees of Superheat or Subcooling at IDU: _____
Return Air Temperature DB°F _____ Supply Air Temperature DB°F _____ Measured at the IDU

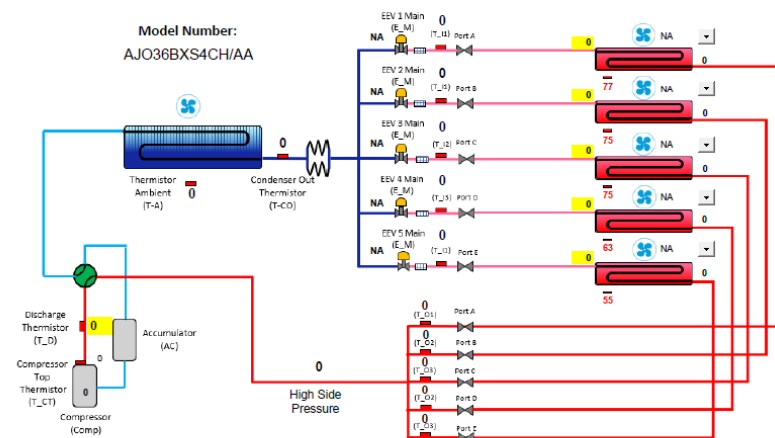
Indoor Unit E

Mode: Cooling - Heating Set point: _____ Room Temperature: _____ Fan speed setting: _____
Voltage at IDU: L1-L2 _____ F1-F3 _____ F3-F4 _____
ODU Pipe In Temperature DB°F _____
IDU Pipe In Temperature DB°F _____
IDU Pipe Out Temperature DB°F _____
ODU Pipe Out Temperature DB°F _____
Degrees of Superheat or Subcooling at IDU: _____
Return Air Temperature DB°F _____ Supply Air Temperature DB°F _____ Measured at the IDU

SAMSUNG

FJM Operation Heating Mode

Model Number:
AJ036BXS4CH/AA



Comments:

K 4 Button Pushes	Information displayed
1	Compressor Frequency
2	Target Frequency
3	EEV 0 Steps
4	EEV 1 steps
5	EEV 2 Steps
6	EEV 3 steps
7	EEV 4 Steps
8	Fan RPM
9	Discharge Temperature
10	OLP Temperature
11	Condenser Out Temperature
12	Outdoor Temperature
13	Running Current
14	Target Discharge Temperature
15	Total Capacity of indoor units
16	Safety Control

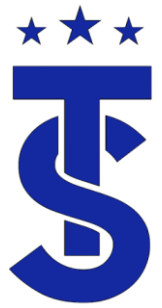
Indoor Capacity Set Point

A 9k
B 7k
C 15k
D 15k

15k

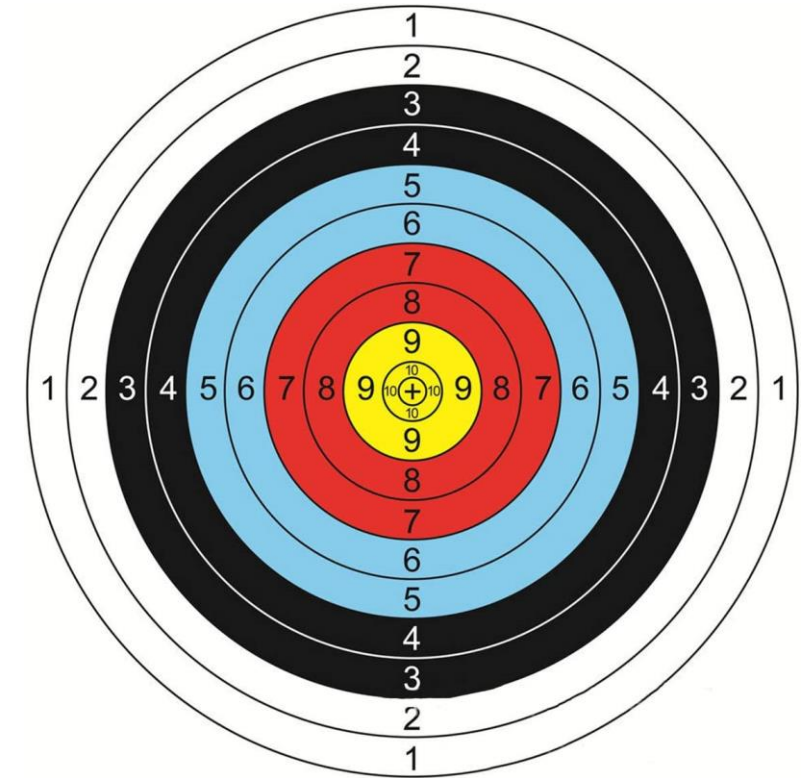
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Increasing your accuracy for troubleshooting.

- You must have a good understanding of refrigeration basics.
- You must understand the cause and effect of the system components.
- Look at the system over an extended run time.
- The more information you gather will increase your troubleshooting accuracy.
- Using the right tools and diagnostic information is the key to success.



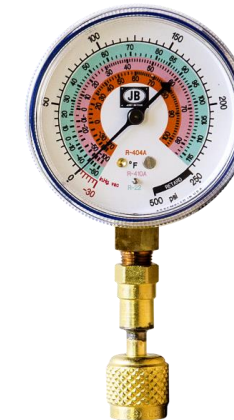
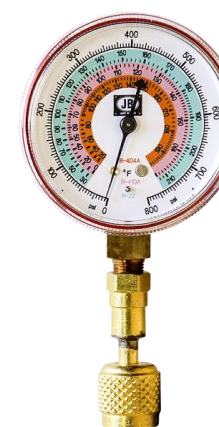
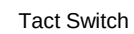
How can you increase your accuracy in making the correct diagnosis?.

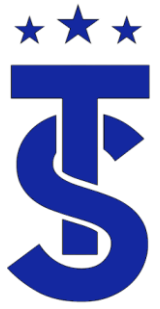


The Importance of Multiple Data

The System is Experiencing Low, Low Side Pressure in Cooling. What's the problem?

Air Conditioning Troubleshooting Chart				
System Problem	INDICATORS			
	Discharge Pressure	Suction Pressure	Superheat	Sub-Cooling
Over Charge	High	High	Low	High
Undercharge	Low	Low	High	Low
Liquid Line Restriction (in outdoor)	High	Low	High	High
Liquid Line Restriction (at indoor)	Low	Low	High	High
Low Evaporator Air Flow	Low	Low	Low	Low
Dirty Condenser	High	High	Low	Low
Low Outdoor Temperature	Low	Low	High	High
Inefficient Compressor	Low	High	High	High
Metering Device Not Opening	Low	Low	High	High
Un-insulated Sensing Bulb	High	High	Low	Low
Low Load	Low	Low	Low	Low

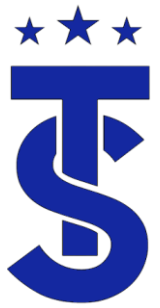




Gathering operational Data

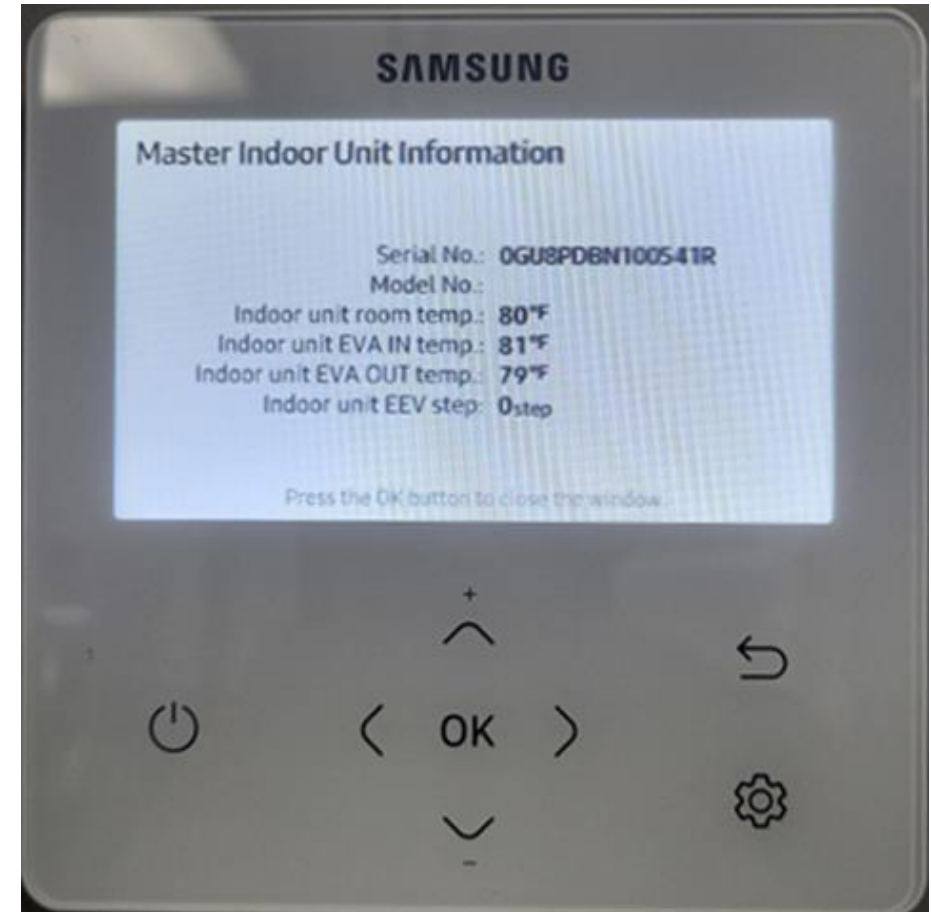
- The K 4 Button can be used to gather ODU operating data
- The advanced controller can be used to gather indoor operating data under Master Indoor Unit information

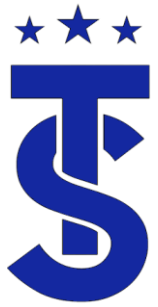




Service feature of the advanced controller

- Enter service mode PW 0202
- Select indoor unit option
- Select Connection information
- Select Masters Indoor Unit Information.





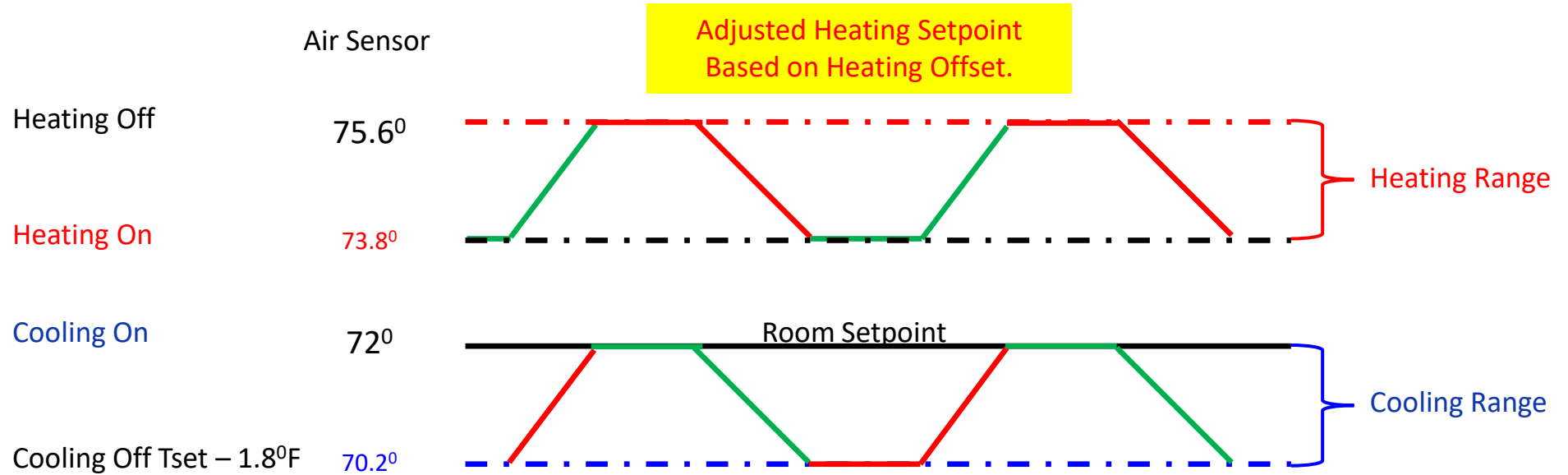
Gathering operational Data

Times Pushed	Displayed Information	Times Pushed	Displayed Information
1	Present Compressor Frequency	9	Discharge Temperature
2	Target Compressor Frequency	10	OLP Temperature
3	EEV 0 Step	11	Condenser Temperature
4	EEV 1 Step	12	Outdoor Temperature
5	EEV 2 Step	13	Running Current
6	EEV 3 Step	14	Target Discharge Temperature
7	EEV 4 Step	15	Total capacity of Indoor units
8	Fan RPM (H-high, L-low, Blank-)off	16	Safety Control





Thermal On & Off



Cooling mode

Thermo on: Room temp. $\geq$ Set temp

Thermo off: Room temp. $\leq$ Set temp. - 1.8°F

Heating Mode

Thermo on: Room temp. $\leq$ Set temp. - 1.8°F + ΔT^*

Thermo off: Room temp. $\geq$ Set temp. + ΔT



Factory default (product option code SEG6)

4way type : $\Delta T^* = 9^\circ\text{F}$

others : $\Delta T^* = 3.6^\circ\text{F}$



Segment 6 of the 01-Product Code.

- Here are examples of the heat offset settings for different units
- The chart that explains the heat offset and the heating off calculations will vary depending on the type of equipment.

Address Δ	Model	RMC	MCU ADDRESS	MCU PORT	Location	Product Option	Installation Option	Installation Option2
0	RAC	02	-	-	Monkey Room	[0]11A35-[1]7C0F7-[2]71A20-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000
1	RAC	00	-	-	Living Room	[0]11A35-[1]7C217-[2]72323-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000
2	RAC	01	-	-	Master Bedroom	[0]11A35-[1]7C0F7-[2]71A20-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000

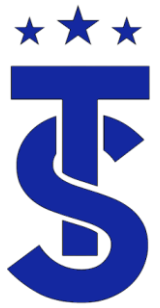
Indoor Unit Installation Data								
Indoor Unit Installation Data								
Address Δ	Model	RMC	MCU ADDRESS	MCU PORT	Location	Product Option	Installation Option	Installation Option2
0	Duct	00	-	-	⦿⦿⦿⦿⦿⦿ ⦿⦿⦿⦿⦿⦿	[0]1E26C-[1]05020-[2]76470-[3]70005	[0]20610-[1]00100-[2]00000-[3]00000	[0]5FFFF-[1]FFFFF-[2]FFFFF-[3]FFFFF



Adjusting The Heat Offset Using Segment 6 of The 01 Product Code.

- The heat offset can be disabled using a couple of methods
- Changing segment 4 of the 02-option code to a setting that uses an external sensor should remove the heat offset.
- If that does not work, you can change segment 6 of the 01 product code.
- When using segment 6 of the 01-product code you will need to verify IDU type, the default setting and use the appropriate chart.

Address ▲	Model	RMC	MCU ADDRESS	MCU PORT	Location	Product Option	Installation Option	Installation Option2
0	RAC	02	-	-	Monkey Room	[0]11A35-[1]7C0F7-[2]71A20-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000
1	RAC	00	-	-	Living Room	[0]11A35-[1]7C217-[2]72323-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000
2	RAC	01	-	-	Master Bedroom	[0]11A35-[1]7C0F7-[2]71A20-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000



Segment 4 of the 02-installation code

- Selecting a setting that “uses” the external room temperature sensor “should” remove the heat off set because the system thinks is using a room temperature sensor for sensing.
- If this setting does not work change segment 6 of the 01-product code.

Value	External Room Temp Sensor	Minimizing fan operation when Thermal-OFF	
		Cooling Thermal-OFF	Heating Thermal-OFF
0 (Default)	Disuse	Fan operation: ON	EVA-IN temp ≥ 68°F: Ultra Low EVA-IN temp < 68°F: OFF
1	Use	Fan operation: ON	EVA-IN temp ≥ 68°F: Ultra Low EVA-IN temp < 68°F: OFF
2	Disuse	Fan operation: ON	EVA-IN temp ≥ 68°F: Ultra Low for 20 seconds every 5 minutes EVA-IN temp < 68°F: OFF
3	Use	Fan operation: ON	EVA-IN temp ≥ 68°F: Ultra Low for 20 seconds every 5 minutes EVA-IN Temp < 68°F: OFF
4	Disuse	Fan operation: OFF	EVA-IN Temp ≥ 68°F: Ultra Low EVA-IN Temp < 68°F: OFF
5	Use	Fan operation: OFF	EVA-IN Temp ≥ 68°F: Ultra Low EVA-IN Temp < 68°F: OFF
6	Disuse	Fan operation: OFF	EVA-IN Temp ≥ 68°F: Low Low for 20 seconds every 5 minutes EVA-IN Temp < 68°F: OFF
7	Use	Fan operation: OFF	EVA-IN Temp ≥ 68°F: Low Low for 20 seconds every 5 minutes EVA-IN Temp < 68°F: OFF
8	Disuse	Fan operation: ON at Ultra Low speed	EVA-IN Temp ≥ 68°F: Low Low EVA-IN Temp < 68°F: OFF
9	Use	Fan operation: ON at Ultra Low speed	EVA-IN Temp ≥ 68°F: Low Low EVA-IN Temp < 68°F: OFF
A	Disuse	Fan operation: ON at Ultra Low speed	EVA-IN Temp ≥ 68°F: Ultra Low for 20 seconds every 5 minutes EVA-IN Temp < 68°F: OFF
B	Use	Fan operation: ON at Ultra Low speed	EVA-IN Temp ≥ 68°F: Ultra Low for 20 seconds every 5 minutes EVA-IN Temp < 68°F: OFF



Segment 6 of the 01-Product Code.

- Here are examples of the heat offset settings for different units
- The chart that explains the heat offset and the heating off calculations will vary depending on the type of equipment.

Address Δ	Model	RMC	MCU ADDRESS	MCU PORT	Location	Product Option	Installation Option	Installation Option2
0	RAC	02	-	-	Monkey Room	[0]11A35-[1]7C0F7-[2]71A20-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000
1	RAC	00	-	-	Living Room	[0]11A35-[1]7C217-[2]72323-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000
2	RAC	01	-	-	Master Bedroom	[0]11A35-[1]7C0F7-[2]71A20-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000

Indoor Unit Installation Data								
Indoor Unit Installation Data								
Address Δ	Model	RMC	MCU ADDRESS	MCU PORT	Location	Product Option	Installation Option	Installation Option2
0	Duct	00	-	-	⬢⬢⬢⬢⬢⬢⬢⬢ ⬢⬢⬢⬢⬢⬢⬢⬢	[0]1E26C-[1]05020-[2]76470-[3]70005	[0]20610-[1]00100-[2]00000-[3]00000	[0]5FFFF-[1]FFFFF-[2]FFFFF-[3]FFFFF



Hot air rise and in some cases the temperature at the unit will be higher than living space.
Because of this there is an automatic Heat Compensation offset built into the systems.

Segment 6 of the 01product code - Heat Compensation Values = 5

OFF: 79.1°

ON: 75.4°

RAC IDU 1:1 or with a FJM
Heat Mode Control Algorithm:

Thermo on:

Room temp. ≤ Set temp. + ΔT

Thermo off:

Room temp. ≥ Set temp. + 3.6°F + ΔT

Example:

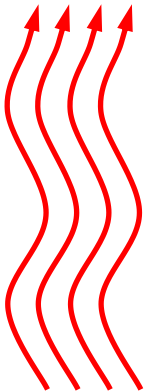
Heating mode Thermo on:

Room temp. ≤ 70 + 5.4 = 75.4

Thermo off:

Room temp. ≥ 70 + 3.6 + 5.4 = 79.1

ΔT = 5.4°F



Set Point
70°



01 SEG6	ΔT °C	Note
0	-2 / 0	Heating ΔT / Cooling ΔT
1	-1 / 0	
2	0 / 0	
3	1 / 0	
4	2 / 0	
5	3 / 0	
6	4 / 0	
7	5 / 0	
8	3 / 1	
9	3 / 2	
A	3 / 3	
B	3 / -1	

Segment 6 option chart from an older presentation



Hot air rise and in some cases the temperature at the unit will be higher than living space.
Because of this there is an automatic Heat Compensation offset built into the systems.

Segment 6 of the 01product code - Heat Compensation Values = 5

OFF: 79.1°

ON: 75.4°

RAC IDU 1:1 or with a FJM

Heat Mode Control Algorithm:

Thermo on:

Room temp. $\leq$ Set temp. + ΔT

Thermo off:

Room temp. $\geq$ Set temp. + 3.6°F + ΔT

Example:

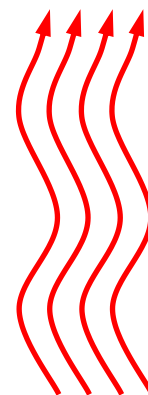
Heating mode Thermo on:

Room temp. $\leq 70 + 5.4 = 75.4$

Thermo off:

Room temp. $\geq 70 + 3.6 + 5.4 = 79.1$

$\Delta T = 5.4^{\circ}\text{F}$



Set Point
70°



Product option code(01) SEG6		
Value	ΔT	ΔX
0	-2 °C / -3.6 °F	0 °C
1	-1 °C / -1.8 °F	
2	0 °C / 0 °F	
3	1 °C / 1.8 °F	
4	2 °C / 3.6 °F	
5	3 °C / 5.4 °F	1 °C / 1.8 °F
6	4 °C / 7.2 °F	
7	5 °C / 9 °F	
8	-2 °C / -3.6 °F	
9	-1 °C / -1.8 °F	
A	0 °C / 0 °F	
B	1 °C / 1.8 °F	
C	2 °C / 3.6 °F	
D	3 °C / 5.4 °F	
E	4 °C / 7.2 °F	
F	5 °C / 9 °F	

Segment 6 option chart from an newer presentation



Hot air rise and in some cases the temperature at the unit will be higher than living space.
Because of this there is an automatic Heat Compensation offset built into the systems.

Segment 6 of the 01product code - Heat Compensation Values = c

OFF: 75.4⁰

ON: 71.8⁰

Non RAC IDU

Heat Mode Control Algorithm:

Thermo on:

$$\text{Room temp.} \leq (\text{Set temp.} - 1.8) + \Delta T^{\circ}\text{C}$$

Thermo off:

$$\text{Room temp.} \geq \text{Set temp.} + \Delta X^{\circ}\text{C} + \Delta T^{\circ}\text{C}$$

Example:

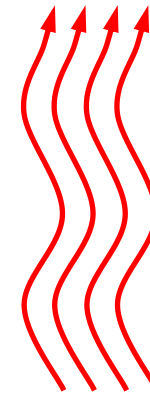
Heating mode Thermo on:

$$\text{Room temp.} \leq (70 - 1.8) + 3.6 = 71.8$$

Thermo off:

$$\text{Room temp.} \geq 70. + 1.8 + 3.6 = 75.4$$

$$\Delta T = 3.6^{\circ}\text{F}$$



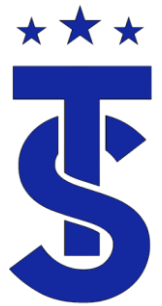
Set Point
70⁰



Default setting Segment 6 of the 01product code for a MPAH is C

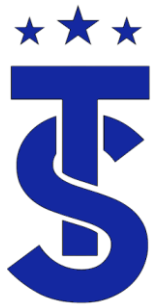
Product option code(01) SEG6		
Value	ΔT	ΔX
0	-2 °C / -3.6 °F	0 °C
1	-1 °C / -1.8 °F	
2	0 °C / 0°F	
3	1 °C / 1.8 °F	
4	2 °C / 3.6 °F	
5	3 °C / 5.4 °F	
6	4 °C / 7.2°F	
7	5 °C / 9°F	1 °C / 1.8 °F
8	-2 °C / -3.6 °F	
9	-1 °C / -1.8 °F	
A	0 °C / 0°F	
B	1 °C / 1.8 °F	
C	2 °C / 3.6 °F	
D	3 °C / 5.4 °F	
E	4 °C / 7.2°F	
F	5 °C / 9°F	

($\Delta X^{\circ}\text{C}$: Heating Thermo off
Hysteresis, $\Delta T^{\circ}\text{C}$ = Heating
compensation temperature)



Handheld Controller Setting Option Codes

1. Remove the batteries and wait for the screen to go blank. While holding the temperature up and down buttons, replace the batteries.
2. Use the FAN DOWN button to adjust the segment on the left and the FAN UP button to adjust the segment on the right to enter code into the wireless controller as shown in the above images.
3. Press MODE to advance to the next page (noted by "Auto", "Cool", and "Dry").
4. Once the numbers shown above are entered into the controller, press the power button several times while pointing remote at the indoor unit.
5. All 6 segments must be entered to prevent undesired programming and operation.
6. Remove batteries. After the screen goes blank, replace batteries and cover to return to normal operation.
7. Cycle power to system.



Understanding the single option code

Position 1 is used to tell the system which type of code is being programmed

D is for a single setting of a code

Position 2 is used to tell the system which code to change.

1 is for the 01 Product code

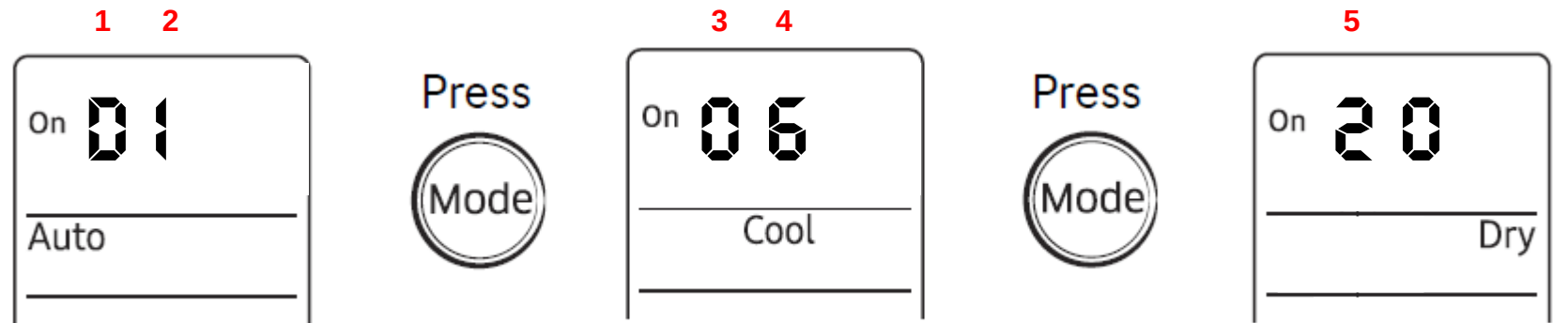
2 is for the 02 Installation code

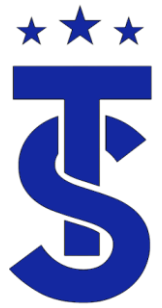
Position 3 & 4 are used to identify which Segment of the code is being changed .

Segments are 2-digit numbers. Single digit numbers have a 0 in front of them

Position 5 is used to tell the system which option of the segment to change

* In this example segment 6 of the 01-Product code is being set to 2

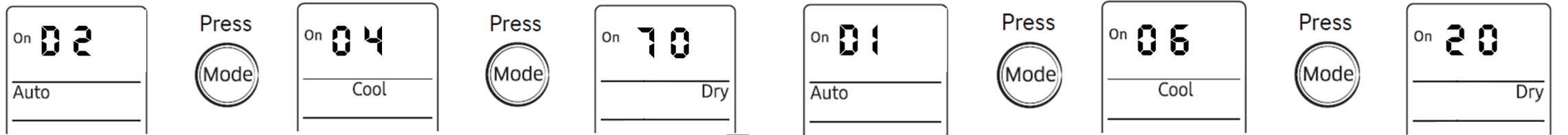




Single option setting with the handheld controller

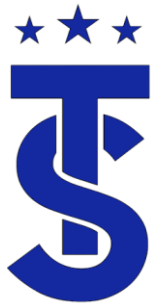
1. Here are the single option setting for:

1. Segment 4 of the 02 product code
2. Segment 6 of the 01 product code.



Setting Segment 4 of the 02-installation code to 2

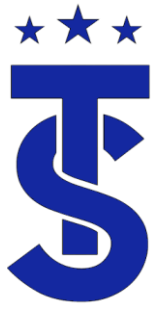
Setting Segment 6 of the 01product code to 2



Air Flow

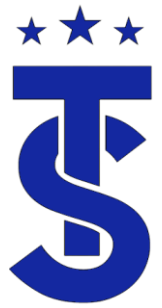
Installation and Start Up

- In this section we will review the processes required to set up and test air flow on ducted units.
- Calculate the CFM of a MPAH using the ESP, fan setting and fan performance.
- This information is general in nature.
- Always refer to the specific installation manual of the unit you are working on.
- Manuals can be found at www.samsunghvac.com



What is ESP?

- External Static Pressure is the measurement of all the resistance in the duct system that the fan must work against.
- Examples are filters, grills, A/C coils and the ductwork.
- It is the sum of the suction pressure (negative) and discharge pressure (positive) created by the equipment blower.
- External Static pressure is measured using a manometer and is expressed in inches of water column (i.e., #” WC)

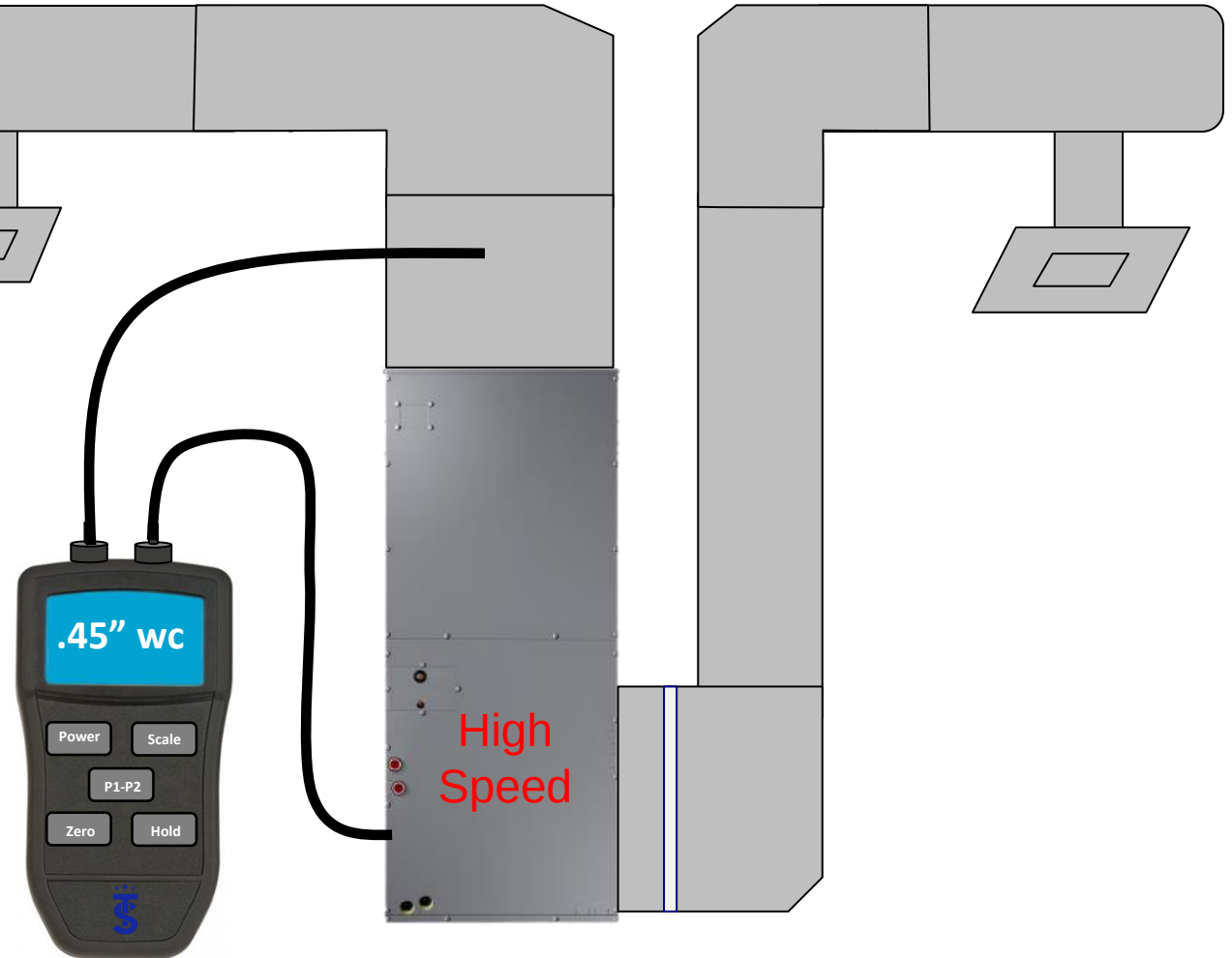


Air Flow Measuring ESP

Manometers are used to test the ESP of a System.

ESP is the difference between the suction and discharge of the fan.

ESP is used to determine the operation and CFM of a fan.



Air Flow

Ducted Units

- Ducted units can contain either a BLDC motor or a constant torque motor.
- Each unit require field setting to achieve proper operation.
- Units are rated for a given CFM at a maximum External Static Pressure (ESP) on high speed.
- The CFM and ESP are needed to correctly size the duct work.
- If the ESP exceeds the manufacturers recommendations the system cannot move the correct air flow.



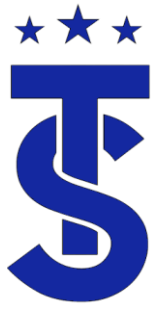
Contact Time Air Flow

- There are no more rules of thumb when it comes to air flow.
- The air flow has to be correct for the system you're working on.
- Too much air flow is a bad as not enough.
- It is recommended that the ESP of any ducted unit be tested during start up.
- Do you know what the system ESP was tested at?

Specifications			
Performance	Nominal Capacity (Btu/h)	Cooling	24,000
		Heating	27,000
Power	Voltage	ø / V / Hz	1 / 208-230 / 60
	Nominal Input Current*	Cooling (A)	1.12
	MCA*	Amps	0.9
	MOCP*	Amps	10
Fan	Type	Double-inlet, forward curve, centrifugal	
	Motor	Type	Constant-torque (ECM)
		HP	1/3
		Output (W)	290
Airflow	CFM @ 0.4" ESP (UL)	L / M / H	450 / 560 / 706
External Static Pressure	Standard	"WC	0.4
	Min. / Max.	"WC	0.1 / 0.7

Always verify the proper air flow for each unit





Indoor Unit Rating at High Speed

- In order to get the correct capacity out of the indoor unit the fan must be on high speed.
- The coil and filter must be clean.
- How does the speed of the fan effect contact time?



CFM: L-293/ M-321/ H-364



CFM: L-300/ M-353/ H-424

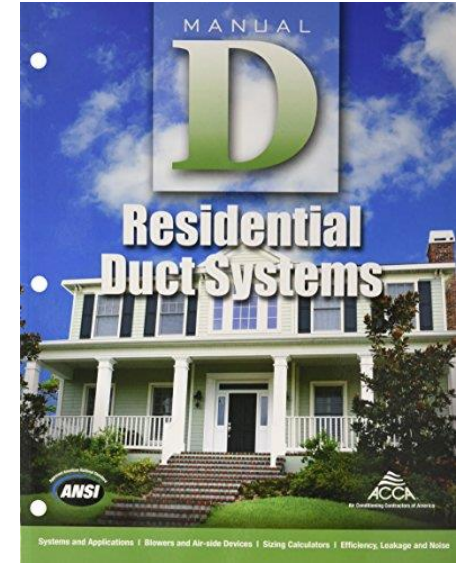


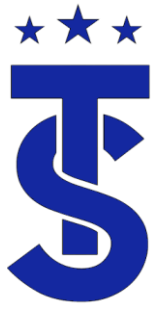
CFM: L-459/ M-494/ H-547.

Each unit has a capacity of 12,000 Btu

★ ★ ★ T Air Flow Duct Design

- It is up to the contractor to design a duct system that will provide the proper amount of air to the indoor unit (CFM), moving at the correct speed across the coil (velocity in FPM) and distributed correctly to the zones as need.
- If it is a retrofit application the duct work should be checked to verify that it will work with the new system.
- Measuring the ESP of the unit and the Velocity of the air can tell if the duct system is not designed correctly.





Understanding ESP and Velocities

- There is a balance between ESP, component pressure drops and Velocity.
- This must be considered when applying and servicing a ducted system.



Air Flow

- Air flow requirement can be found in the specification sheets.
- The CFM requirement is also given in the specification sheet.
- Make no assumptions regarding air flow.
- Do not apply rule of thumb to duct design or application
- System capacity and performance is directly related to air flow



SUBMITTAL

AC048KNZDCH/AA / AC048BXADCH/AA (CNH48ZDK / CXH48ADB)

Page 1 of 4

Samsung Multi-Position Air Handler, Single Zone Duct, Split System

Job Name
 Purchaser
 Submitted to
 Unit Designation

Location
 Engineer
 Reference ☐ Approval ☐ Construction ☐
 Schedule #

Specifications			
Model	Indoor Unit Model Number (US Code)	AC048KNZDCH/AA (CNH48ZDK)	
	Outdoor Unit Model Number (US Code)	AC048BXADCH/AA (CXH48ADB)	
Performance ¹	Nominal Capacity	Cooling / Heating (Btu/h)	48,000 / 53,000
	Capacity Range	Cooling (Btu/h)	12,000 – 52,000
		Heating (Btu/h)	11,000 – 63,000
	SEER / EER	18.2 / 9.2	
	COP (nominal heating)	3.00	
	HSPF	9.8	
	AHRI Certification Number	207598072	
Power	Voltage	ø / V / Hz	1 / 208-230 / 60
	Working Voltage Range (VAC)	176 - 254 (max. 3% deviation from each)	
	Operating Current (min. / std. / max.)	Cooling (A)	5.1 / 23.4 / 29.4
		Heating (A)	4.4 / 23.2 / 32.0
	Max. Breaker Amps	40	
	Min. Circuit Ampacity (A)	32.8	
Dimensions	W X H X D (in.)	Indoor Unit	24 1/2 X 59 3/4 X 21 3/4
		Outdoor Unit	37 X 47 11/16 X 13
	Duct Connections (W X H)	Supply (in.)	14 1/4 X 22
		Return (ID, in.)	24 1/2 X 21 3/4
	Weight (lbs.)	Indoor Unit	163.14
		Outdoor Unit	191.8
Sound Pressure Level	Indoor Unit dB(A)	L / M / H	36 / 39 / 42
	Outdoor Unit dB(A)	Cooling / Heating (high)	53 / 55
Operating Temperatures	Outdoor	Cooling	23 ~ 122°F (-5 ~ 50°C)
		Heating	0 ~ 122°F (-18 ~ 50°C) w/ baffle
	Indoor	Cooling	-4 ~ 75°F (-20 ~ 24°C)
		Heating	61 ~ 90°F (16 ~ 32°C)
Pipe Connections	Indoor & Outdoor	High side (brazed / flare)	3/8"
		Low side (brazed / flare)	5/8"
	Maximum (ft.)	240	
	Maximum Vertical Separation (ft.)	98.4	
	Condensate Connection (with included adapter)	FPT 3/4" PVC	
Refrigerant	Type	R410A	
	Control Method	Electronic Expansion Valve	
	Factory Charge	7.50	
	Charged for Additional Refrigerant	24.6 ft	
		0.11 oz/ft over 24.6 ft	
Compressor	Manufacturer	Samsung	
	Type	Inverter Driven, Twin BLDC Rotary	
	RLA	20.9	
Evaporator Fan	Type	Double-Inlet, forward curve, centrifugal (with ECM motor)	
	Air Volume	CFM (L/MH)	1,130 / 1,271 / 1,413 (at standard ESP)
	Output (W) / FLA (A)	410 W / 1.66 A	
	Static Pressure	Standard (\"WC)	0.28
	Min. / Max. (\"WC)	0.01 - 0.98	
Condenser Fan	Motor	BLDC With Axial Type Fan (2)	
	FLA / Watts / CFM (max.)	1.25 A / 125 W / 3,531 CFM	
Safety	Certifications	ETL (UL 1995)	
	Devices	PCB fuses, indoor unit terminal block thermal fuse, current transformer, over-voltage protection, crankcase heating, temperature limit protection logic, compressor overload sensing	

¹ Certified in accordance with the AHRI Unitary Small Air-Source Heat Pumps (USHP) Certification Program which is based on the latest edition of AHRI Standard 210/240.

Samsung HVAC maintains a policy of ongoing development; specifications are subject to change without notice. Refer to www.AHRI.org for current reference numbers.




General Information

- The outdoor unit shall supply power to indoor unit via 14 AWG X 3 power wire
- High-voltage terminal block temperature sensor to disable unit in the event of power connection overheating
- Auto-restart after power loss
- Soft-start compressor minimizing current inrush
- All heat exchangers shall be mechanically bonded aluminum fin to copper tube
- The condensing unit heat exchanger salt spray test method: ISO-9227; the heat exchanger showed no unusual rust or corrosion development to 3,000 hours.

Option settings

- The outdoor unit shall have snow accumulation prevention option setting to prevent snow drifting against an idle outdoor unit.
- Night-time Quiet Mode: reduction of operational sound during (automatic or manual activation).
- System can be set up as heating/cooling, cooling only, or heating only via outdoor unit option setting.
- Maximum Current Control configurable from 50% - 100% via outdoor unit, wired controller, or central controller

Indoor Fan

- Indoor fan is a double inlet, forward curve, centrifugal type
- Three fan speed settings and auto setting
- Field configurable for downflow operation with optional conversion kit

Construction

- Outdoor unit shall be galvanized steel with a baked-on powder coated finish for durability
- Indoor Unit: Insulated, galvanized steel.

Controls

- Control wiring shall be 2 X 16 AWG
- No additional interface modules/adapters are required when connecting to Samsung central control options.
- The unit shall be operated via a wireless or wired remote control with DDC type signal
- Dual set temperature support when connected to MWR-WG00UN Advanced Wired Controller or central control options.
- Wired or wireless controllers must be purchased separately

Refrigerant System

- The compressor shall be hermetically sealed, inverter-controlled BLDC rotary type.
- Refrigerant flow shall be controlled by an electronic expansion valve at outdoor unit

Warranty

10 Years compressor, 10 years parts, 1 year limited labor when registered




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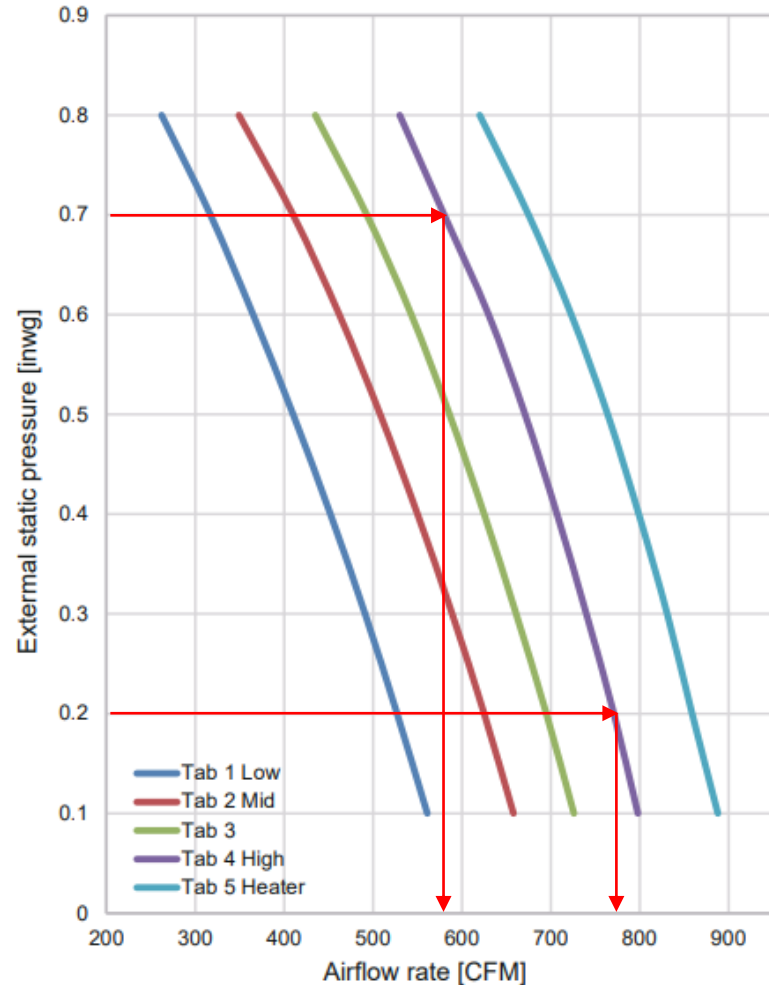
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SAMSUNG



Air Handler CFM

AC024KNZDCH/AA



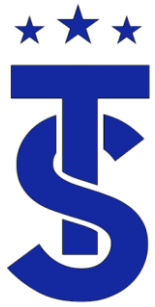
What If the Measured
ESP is .7"

Reduced Air Flow will
decrease the capacity of
the system

WHY?

SAMSUNG		SUBMITTAL AC024KNZDCH/AA / AC024BXADCH/AA (CNH24ZDK / CXH24ADB) Page 1 of 4		
Samsung Multi-Position Air Handler, Single Zone Duct, Split System				
Job Name		Location		
Purchaser		Engineer		
Submitted to		Reference	☐ Approval ☐ Construction ☐	
Unit Designation		Schedule #		
Specifications				
Model	Indoor Unit Model Number (US Code)	AC024KNZDCH/AA (CNH24ZDK)		
	Outdoor Unit Model Number (US Code)	AC024BXADCH/AA (CXH24ADB)		
Performance ¹	Nominal Capacity	Cooling / Heating (Btu/h)	24,000 / 27,000	
	Capacity Range	Cooling (Btu/h)	8,000 - 36,000	
		Heating (Btu/h)	7,000 - 35,000	
	SEER / EER		18.7 / 11.0	
	CCP (nominal heating)		2.50	
Power	HSPF		10.9	
	AHRU Certification Number		207598068	
	Voltage		1 / 208-232 / 160	
	Working Voltage Range (VAC)		176 - 264 (max. 10% deviation from each)	
	Operating Current (min. / std. / max.)	Cooling (A)	4.3 / 8.9 / 14.8	
Dimensions		Heating (A)	3.5 / 12.8 / 24.0	
	Max. Breaker		Amps	30
	Min. Circuit Ampacity (A)			24.1
	W X H X D (in.)	Indoor Unit		17 1/2 X 43 X 21
		Outdoor Unit		37 X 39 5/16 X 13
Sound Pressure Level	Pipe Connections	Supply (in.)	12 1/4 X 18 1/2	
		Return (in.)	17 1/2 X 21	
	Weight (lbs.)	Indoor Unit	98.1	
		Outdoor Unit	118.4	
		Indoor Unit (slp.)	1.18 / 14	
Operating Temperatures		Outdoor Unit (slp.)	35 / 38 / 41	
		Cooling / Heating (high)	50 / 52	
		Cooling	23 ~ 122°F (-5 ~ 50°C)	
		Heating	0 ~ 122°F (-18 ~ 50°C) air baffle	
		Heating	-13 ~ 75°F (-25 ~ 24°C)	
Pipe Connections		Cooling	61 ~ 80°F (16 ~ 32°C)	
		Heating	1 ~ 80°F (27°C)	
	Indoor & Outdoor	High side (brake / flare)	1/4"	
		Low side (brake / flare)	3/8"	
		Maximum Vertical Separation (ft.)	98.4	
Refrigerant	Condensate Connection (with included adapter)		PPH 3/4" PVC	
	Type		4410A	
	Control Method		Electronic Expansion Valve	
	Factory Charge (lbs.)		5.73	
	Charged for Additional Refrigerant		24.6 lb.	
Compressor		0.11 each over 24.6 lb.		
	Manufacturer		Samsung	
	Type		Inverter Driven, Twin BLDC Rotary	
	RLA		15.9	
	Amps			
Evaporator Fan	Type		Double-inlet, forward curve, centrifugal (with ECM motor)	
	Air Volume	CFM (L/M/H)	547/ 636 / 759 (at standard ESP)	
	Output (W) / FLA (A)		290 W / .72 A	
	Static Pressure	Standard ("WC)	0.20	
		Min. / Max. ("WC)	0.01 - 0.79	
Condenser Fan	Motor		BLDC With Auto Type Fan (1)	
	FLA / Watts / CFM (max.)		1.25 A / 125 W / 2.684 CFM	
	Certifications		ETL (UL 1995)	
	Safety		Device: PCB leak, indoor unit terminal block thermal fuse, current transformer, over-voltage protection, over-temperature heating, temperature limit protection fuse, compressor overload sensing	
			which is based on the latest edition of AHRI Standard 210/240.	
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Evaporator Fan	Type		Double-inlet, forward curve, centrifugal (with ECM motor)
	Air Volume	CFM (L/M/H)	547/ 636 / 759 (at standard ESP)
	Output (W) / FLA (A)		290 W / .72 A
	Static Pressure	Standard ("WC)	0.20
		Min. / Max. ("WC)	0.01 - 0.79



Air Flow Requirements

Ducted Units

- Ducted units need to supply the correct amount of air to the coil for the proper operation.
- The CFM and ESP are needed to correctly size the duct work.
- Air flow and fan requirements can be found in the unit's submittal sheets.



Airflow	CFM	L/M/H	258 / 314 / 399
External Static Pressure	Standard	"WC	0.18
	Min / Max.	"WC	0.10 / 0.59

AJ012BNHDCH/AA



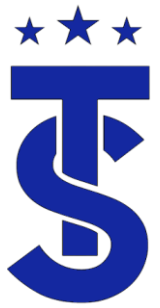
Airflow	CFM	L/M/H	244 / 286 / 374
External Static Pressure	Standard	"WC	0.4
	Min / Max.	"WC	0.10 / 0.50

AC012BNZDCH/AA

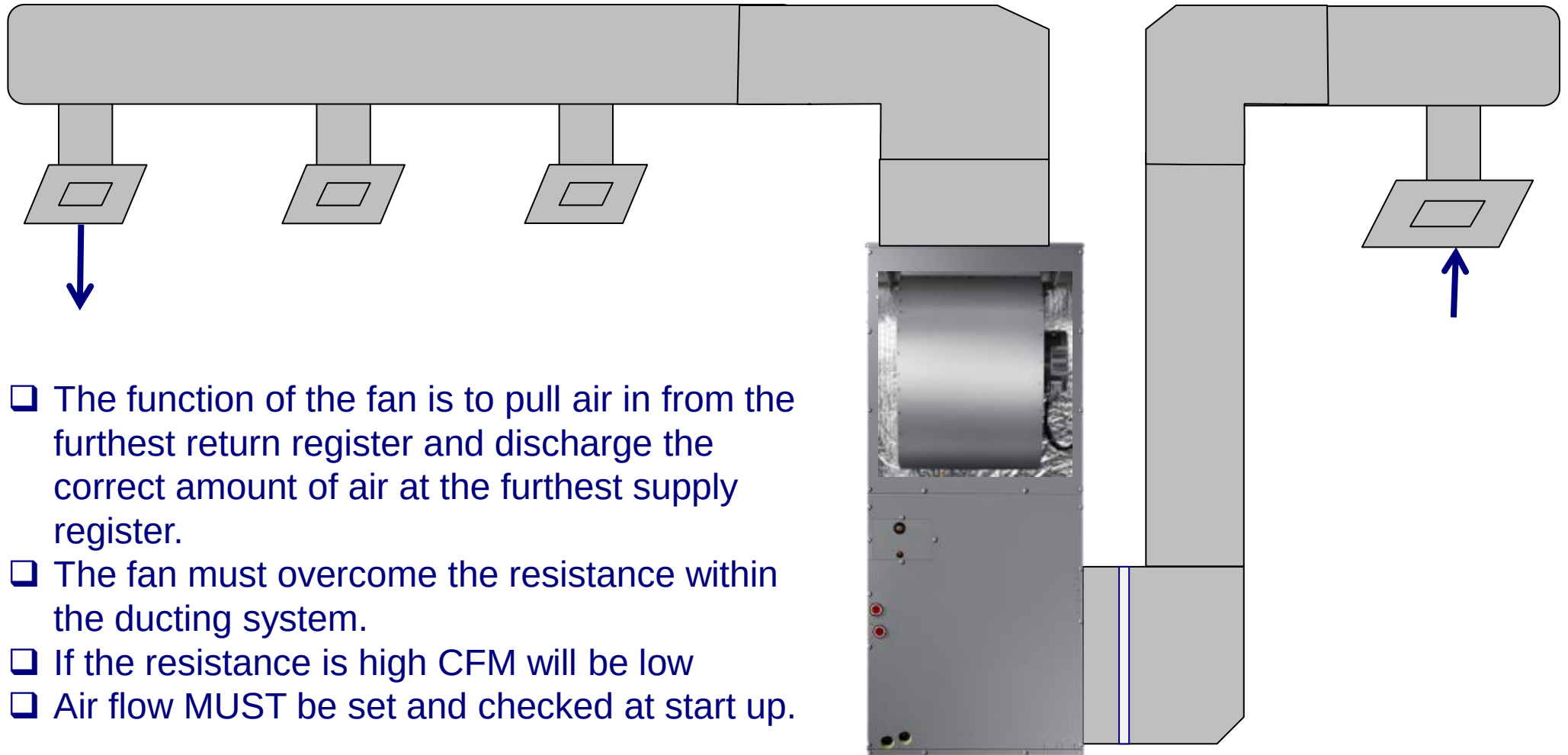


Evaporator Fan	Type	BLDC (1) With Sirocco Fan (2)	
	Air Volume	CFM (L/M/H)	300 / 353 / 399
	Output	Watts	84
	External Static Pressure	Standard ("WC)	0.1
		Min. / Max. ("WC)	0.0 / 0.24

AC012BNLDCH/AA



What's the Purpose of the fan ?



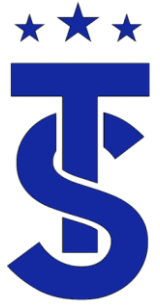
- ❑ The function of the fan is to pull air in from the furthest return register and discharge the correct amount of air at the furthest supply register.
- ❑ The fan must overcome the resistance within the ducting system.
- ❑ If the resistance is high CFM will be low
- ❑ Air flow **MUST** be set and checked at start up.

Air Flow

Types of Motors

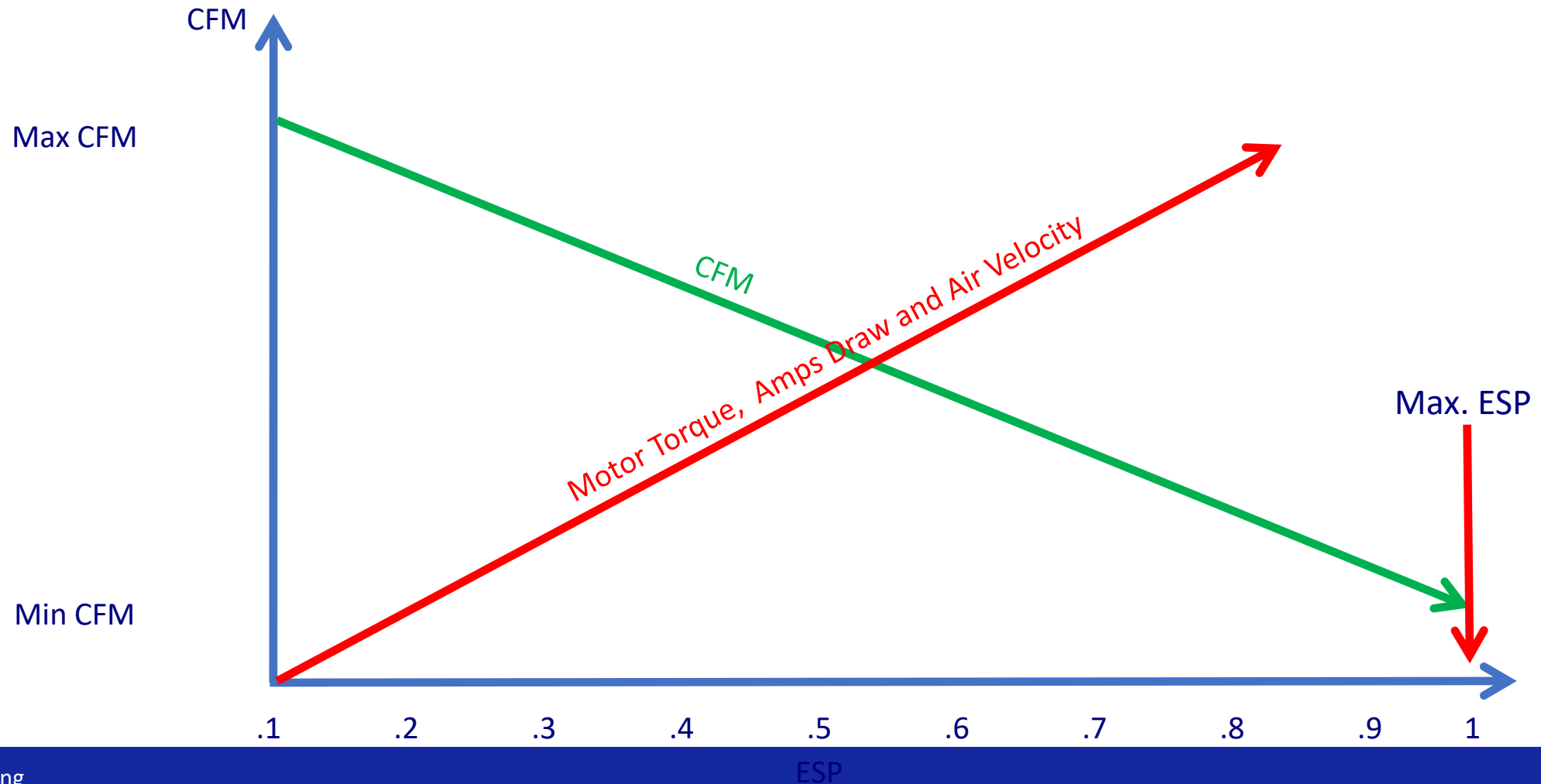
- BLDC Motor
 - Requires field programming to achieve the proper performance.
- Constant torque setup
 - Require field adjustment to set the ESP of system.

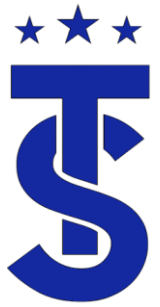




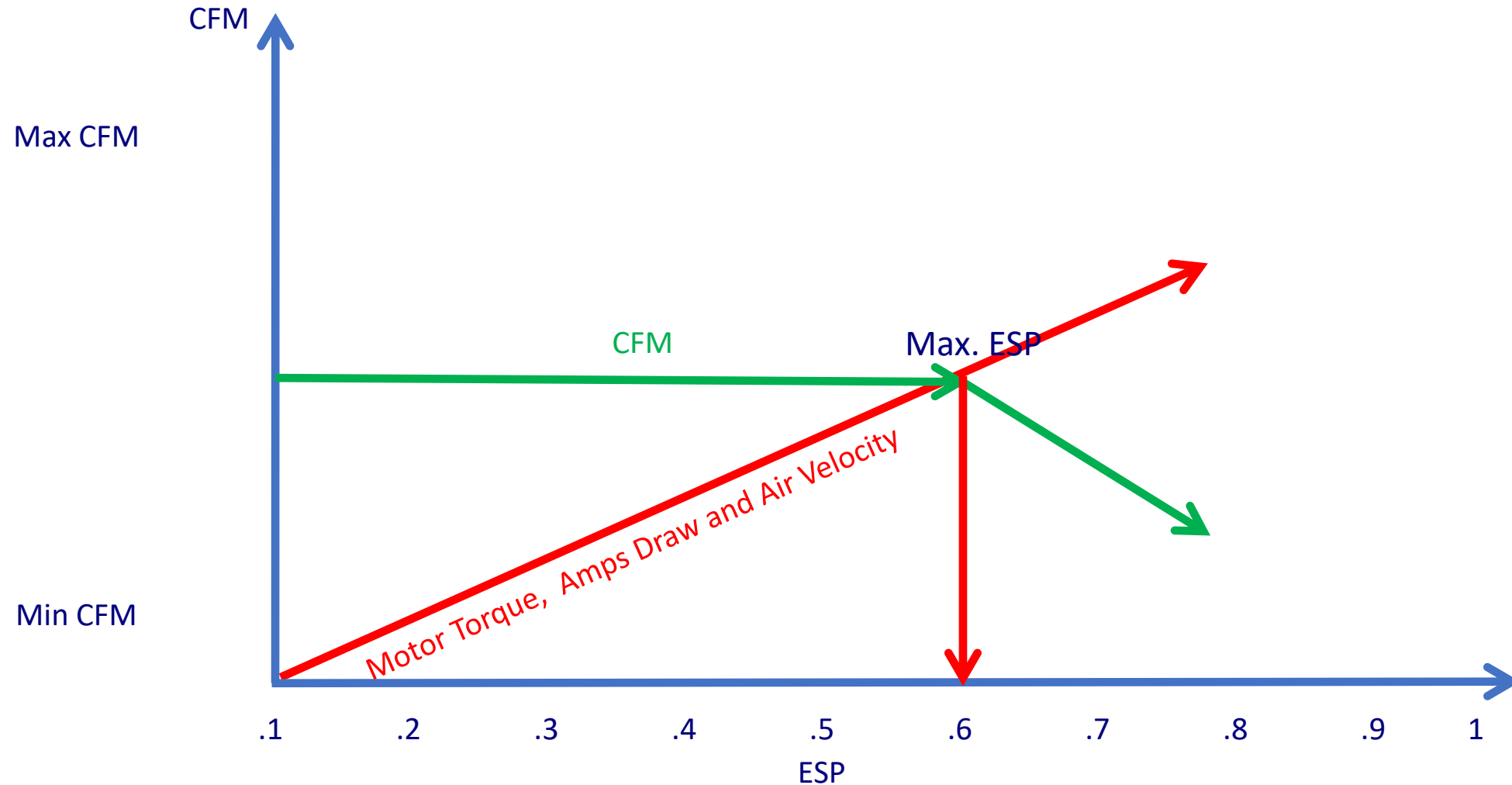
CFM to ESP Chart

Constant Torque Motor





CFM to ESP Chart BLDC Motor

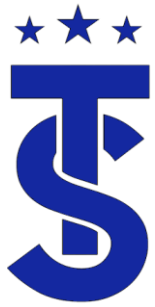


Air Flow

Motor Set Up

- BLDC set up
 - Automatic Air Volume setting
 - Product Code option setting
 - Fine tuning adjustment
- Constant torque setup
 - Uses fan speed taps to adjust air flow





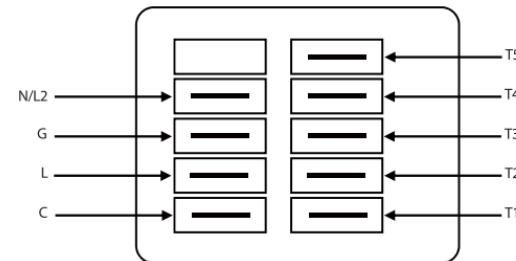
Constant Torque Motor

Terminal Designation



Terminal	Connection	Default speed tap settings					
		AC018KNZDCH/ AA	AC024KNZDCH/ AA	AC030KNZDCH/ AA	AC036/042KNZDCH/ AA	AC048KNZDCH/ AA	AC054KNZDCH/ AA
C	Speed tap common - 24 VAC common						
L	Supply voltage - 240 VAC Line 1						
G	Ground connection						
N/L2	Supply voltage - 240 VAC Line 2						
T1	Low speed tap - 24 VAC input		"Low" speed	"Low" speed	"Low" speed	"Low" speed	"Low" speed
T2	Medium-low speed tap - 24 VAC input	"Low" speed	"Mid" speed	"Mid" speed		"Mid" speed	"Mid" speed
T3	Medium speed tap - 24 VAC input	"Mid" speed		"High" speed			
T4	Medium-high speed tap - 24 VAC input	"High" speed "	"High" speed		High speed for electric heat	"High" speed	High speed for electric heat
T5	High speed tap - 24 VAC input	High speed for electric heat	High speed for electric heat	High speed for electric heat	"High" speed	High speed for electric heat	"High" speed

Motor control/voltage taps



Always refer to the specific unit's motor data!



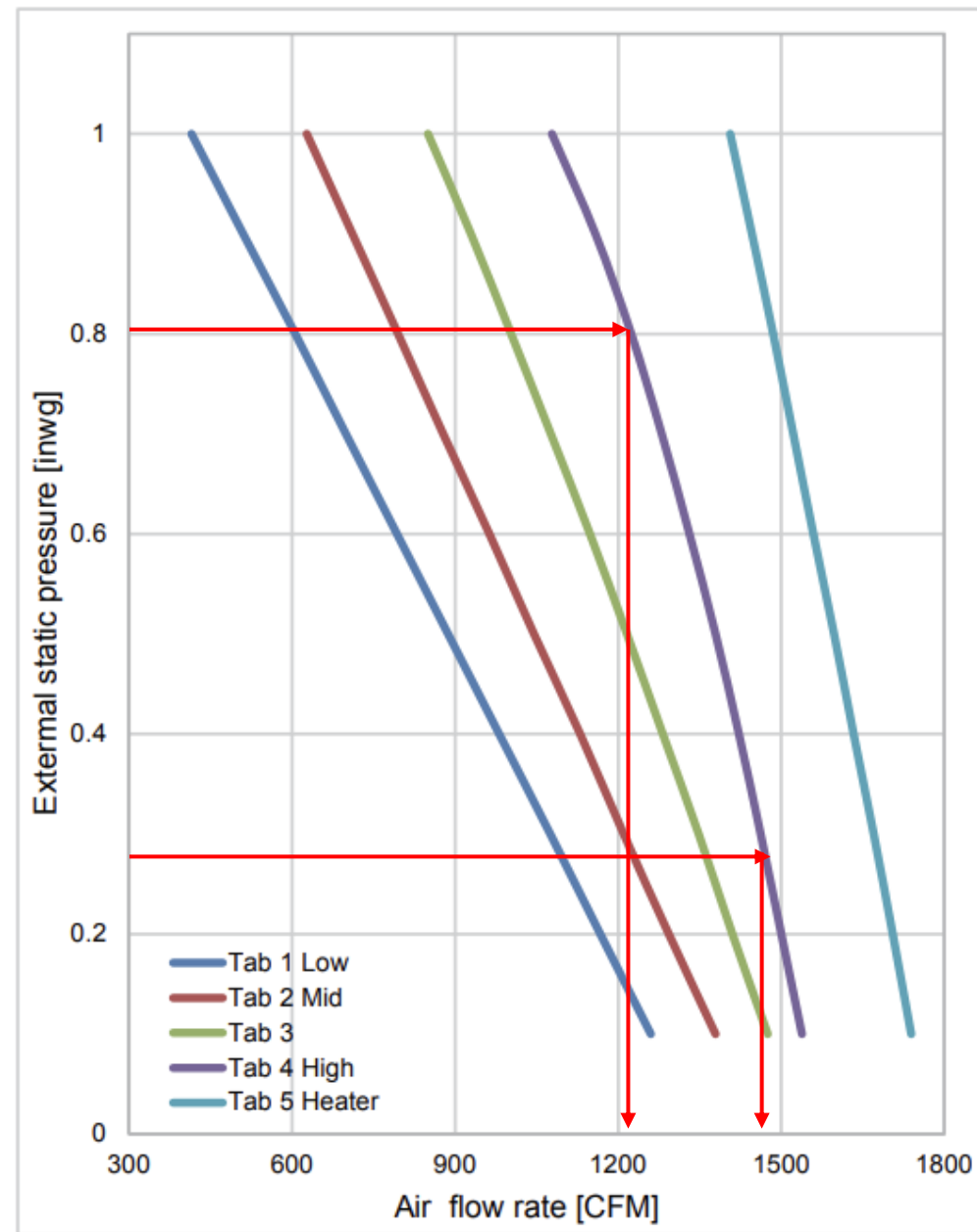
Fan Curves



- The air handler was rated with the fan on high speed with an external static pressure of .28"wc and 1,413 CFM
- If the ESP of the blower goes up CFM goes down.
- IF CFM goes down the delivered capacity goes down

Evaporator Fan	Type		Double-inlet, forward curve, centrifugal (with ECM motor)
	Air Volume	CFM (L/M/H)	1,130 / 1,271 / 1,413 (at standard ESP)
	Output (W) / FLA (A)		410 W / 1.66 A
	Static Pressure	Standard ("WC)	0.28
		Min. / Max. ("WC)	0.01 - 0.98

6) AC048KNZDCH/AA



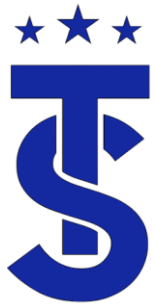


Calculating CFM:

Fan Performance charts



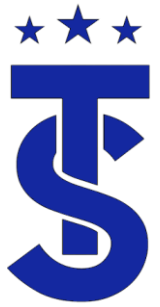
Motor Tap	P	(inwg)	CFMM	otor Tap	P	(inwg)C	FM
1		0.1	1260	4		0.1	1538
		0.2	1167			0.2	1500
		0.3	1075			0.3	1462
		0.4	981			0.4	1422
		0.5	887			0.5	1379
		0.6	793			0.6	1331
		0.7	699			0.7	1280
		0.8	606			0.8	1224
		0.9	510			0.9	1158
		1	415			1	1078
2		0.1	1379	5		0.1	1739
		0.2	1293			0.2	1705
		0.3	1210			0.3	1670
		0.4	1130			0.4	1633
		0.5	1045			0.5	1597
		0.6	963			0.6	1559
		0.7	878			0.7	1522
		0.8	795			0.8	1485
		0.9	711			0.9	1446
		1	627			1	1406
3		0.1	1475	= Default Setting AC048KNZDCH/AA HP : 3/4 Default motor taps: High / Mid /Low = 4/2/1			
		0.2	1410				
		0.3	1348				
		0.4	1282				
		0.5	1216				
		0.6	1148				
		0.7	1076				
		0.8	1003				
		0.9	929				
		1	850				



BLDC ESP Setup

Automatic Air-Volume Feature

- This feature is activated using the remote controller.
- Refer to the installation manual for the controller you are using for direction on to initiate this process.
- The system will cycle the fan to determine the best setting for the application.
- You will receive a pass or fail notification.



Initiating Automatic Air Volume

MWR-WG00UN Wired Controller

- Press the up and down buttons at the same time for more than 3 seconds.
- The password entry screen appears.
 - Enter the password, “0202,” and then press the ok button.
- The settings screen for installation/service mode appears.
- Using the up and down buttons, change the settings to **Automatic Air Volume**.
- Press the right button to move to the next setting.
- Press the down button until **Operation** is displayed
- Press up and down button until **Run** is displayed
- After the process is over verify that the screen displays **Passed**
- Press OK
- Press the Back button to move to the Home screen.
- If the system failed the test, the ESP would need to be set using the product code option.



BLDC ESP Setup

Product Codes

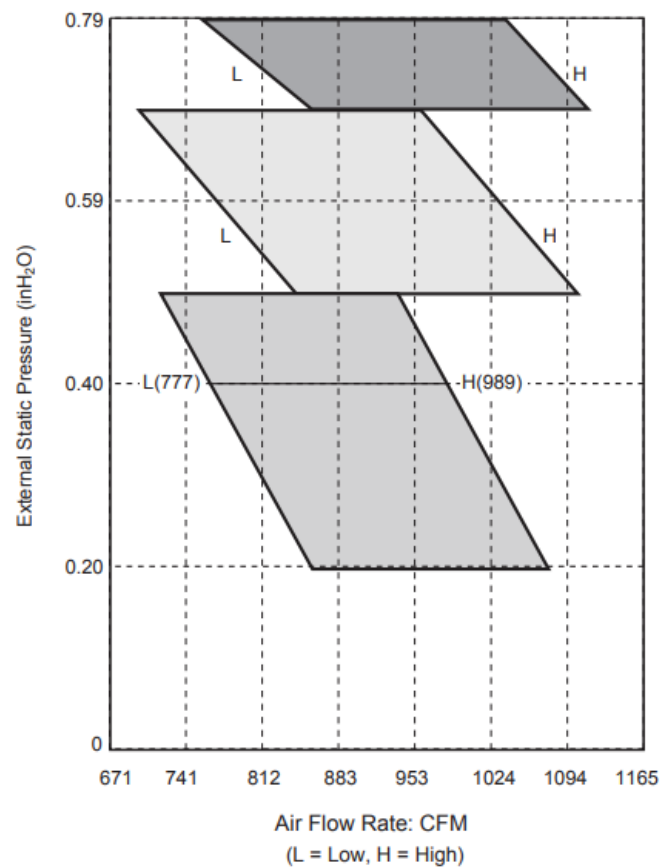
- Product Codes can be used to set the ESP of the fan.
- After the ESP is set run the fan and measure the ESP.
- If the ESP is not within range adjust the ESP and test again.

Model	AJ009BNHDCH	AJ012BNHDCH
Static Pressure (mmAq)	Option Code for Indoor Unit	
$2.5 \leq P \leq 5$	01B3FC-1C50D3-271A23-370000	01B3FC-1C5404-272328-370000
$5 < P \leq 7.5$	01B3FC-1C5466-271A23-370000	01B3FC-1C5477-272328-370000
$7.5 < P \leq 10$	01B3FC-1C54D9-271A23-370000	01B3FC-1C54EA-272328-370000
$10 < P \leq 12.5$	01B3FC-1C582C-271A23-370000	01B3FC-1C583D-272328-370000
$12.5 < P \leq 15$	01B3FC-1C5970-271A23-370000	01B3FC-1C5981-272328-370000



Has the system been set up properly?

Fan Performance Data



External Static pressure (InH ₂ O / mmAq)	Option code
0.2 / 5	010054-1355C6-206E6E-331110
0.4 / 10	010054-1355C6-206E6E-331110
0.6 / 15	010054-13596E-206E6E-331110
0.8 / 20	010054-135AE4-206E6E-331110



AM036FNHDCH/AA

Airflow	CFM (UL)	H/M/L	989 / 883 / 777
	Total CFM Range ²		706 - 1,116
External Static Pressure	Standard	"WC	0.39
	Min. / Max.	"WC	0.20 - 0.79



Product Option Code

Adjusting Static Pressure

0" - 0.04" ESP



Model number: AM012MNMDCH/AA

Description: DVM S, Duct S - 12000 Btu/h,
0" - 0.04" ESP (DEFAULT)

Option code: 010054-1E5072-202323-331102

Static pressure (inches): 0" - 0.04" (DEFAULT)

0.12" - 0.20" ESP



Model number: AM012MNMDCH/AA

Description: DVM S, Duct S - 12000 Btu/h, 0.12" - 0.20" ESP

Option code: 010054-1E5449-202323-331102

Static pressure (inches): 0.12" - 0.20"

0.5" - 0.60" ESP



Model number: AM012MNMDCH/AA

Description: DVM S, Duct S - 12000 Btu/h, 0.5" - 0.60" ESP

Option code: 010054-1E59CA-202323-331102

Static pressure (inches): 0.5" - 0.60"



Product Option Code

Example

Model number: AM024MNHDCH

Description: DVM S, Duct S - 24,000 Btu/h,

Option code: 010054-1E54BB-204646-331115

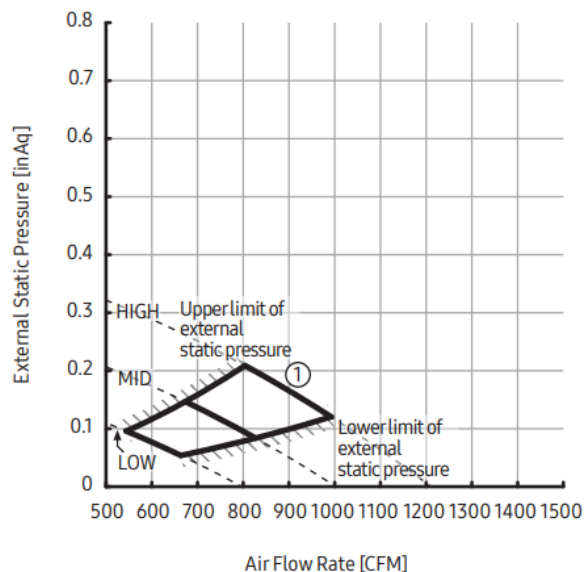
Static pressure (inches): 0.12" - 0.20"

7) AM024MNHDCH

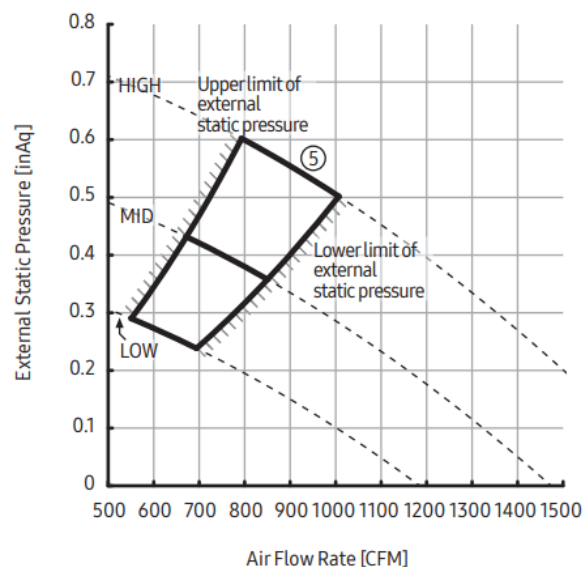


7) AM024MNHDCH

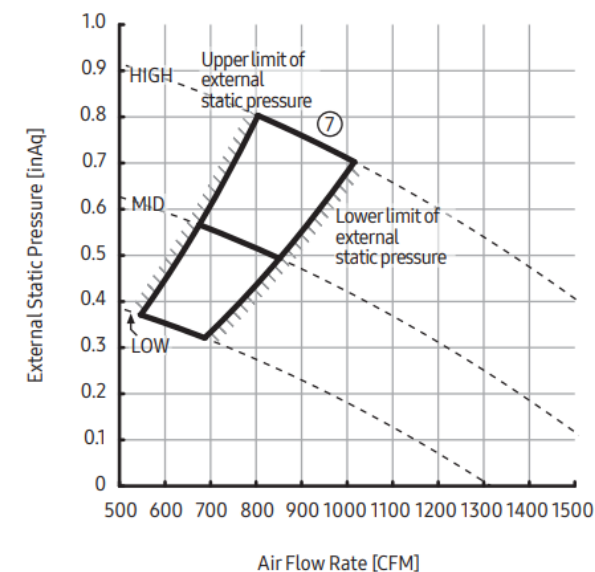
①	External Static Pressure(inAq)	Option Code
	0.12≤ SP ≤0.20	010054-1E54BB-204646-331115



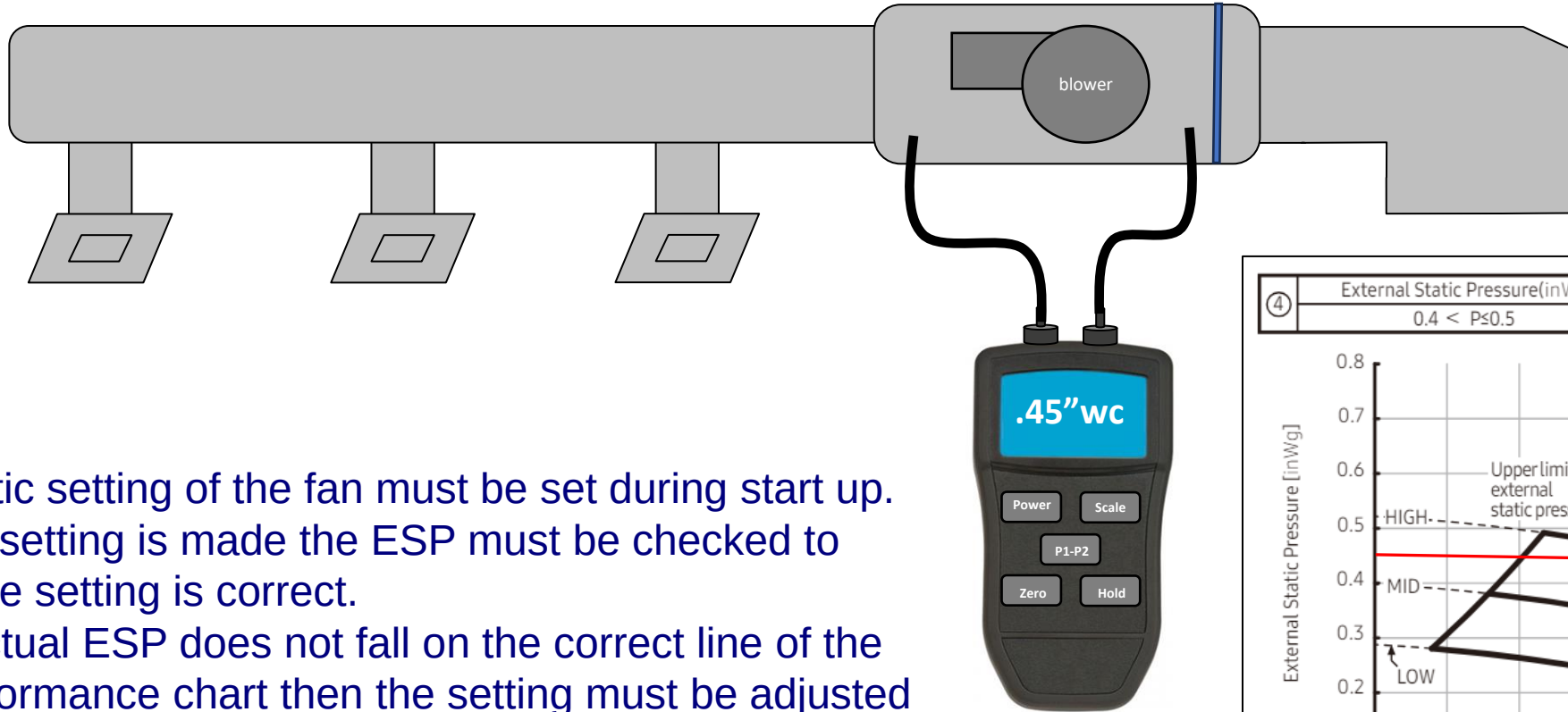
⑤	External Static Pressure(inAq)	Option Code
	0.50< SP ≤0.60	010054-1E5D1A-204646-331115



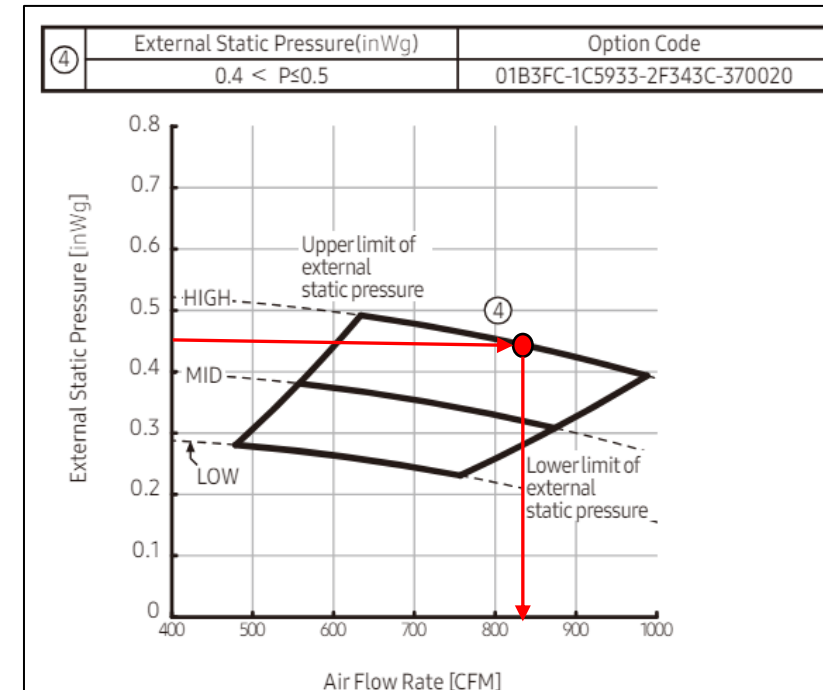
⑦	External Static Pressure(inAq)	Option Code
	0.69< SP ≤0.79	010054-1E5EA0-204646-331115



Testing



- ❑ The static setting of the fan must be set during start up.
- ❑ Once a setting is made the ESP must be checked to verify the setting is correct.
- ❑ If the actual ESP does not fall on the correct line of the fan performance chart then the setting must be adjusted and tested until it does



S-NET pro 2 - DVM S Water

HomeTrend GraphReplayAdd-OnHelp

8/4/2022 8:49:59 AM

Duration 00:11:57

8/4/2022 9:39:10 AM

Current Time '4/2022 9:01 AM

16xFastSlow

Player

Indoor Unit Installation Data														
Address	Model	RMC	MCU ADDRES	MCU PORT	Location	Product Option	Installation Option	Installation Option2	Main Micom	Error History1	Error History2	Error History3	Error History4	Error History5
0	Slim 1Way	00	7	A	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
1	Slim 1Way	01	6	D	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
2	Slim 1Way	03	6	A	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
3	Duct	03	7	B	-	[0]10054-[1]E54BB-[2]04646-[3]31115	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
4	Duct	00	6	C	-	[0]10054-[1]E54BB-[2]04646-[3]31115	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
5	Slim 1Way	05	7	E	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
6	Duct	06	7	F	-	[0]10054-[1]E5438-[2]03535-[3]31104	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
7	Slim 1Way	07	6	B	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
8	Slim 1Way	02	7	C	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000

AM024MNHDC/AA

Fan options		ESP	Sound Pressure (dBA)		
		inAq	H	M	L
Default	010054-1E54BB-204646-331115	0.2	39	36	30
Option	010054-1E581F-204646-331115	0.3	41	36	31
	010054-1E5973-204646-331115	0.4	43	38	33
	010054-1E59C6-204646-331115	0.5	45	40	34
	010054-1E5D1A-204646-331115	0.6	47	42	36
	010054-1E5D6D-204646-331115	0.7	48	43	37
	010054-1E5EAO-204646-331115	0.8	48	44	39

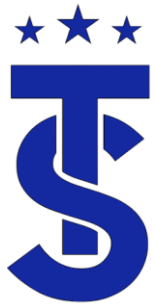
This customer is complaining about the temperatures in these 2 zones.
Units are AM024MNHDC/AA
System had been tested and balanced
This is a commercial system with long duct runs
What does the product and installations codes tell us.

Outdoor Unit DataOutdoor Unit Installation DataOutdoor Unit Cycle DiagramIndoor Unit DataIndoor Unit Installation DataControl for Unoccupied and Entering RoomMCU Unit Data

Version 1.13.0Unit - Temp.: FPower : BtuPressure : psi9/15/20224:59 PMCOM-1

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SAMSUNG



VRF Coder Reverse Look Up

Samsung HVAC America - VRF Coder

Indoor Option Setting

DVMS

RAC / CAC / FJM

UCK

Option Codes

DVM S & DVM 3

RAC / CAC / FJM

Indoor Addressing

DVM S & NASA RAC / FJM

FJM (A3***JN**CH/AA)

Outdoor

VRF

Chiller

DVMS System

CAC Setting

Mini Split Refrigerant

Control

Wired Indoor Motor Power Supply

Wireless

Misc.

Error Code

Option Codes [DVM S DVM 3]

Option codes are 24 segment programming codes programmed at the factory. These codes can be adjusted depending on the unit type to change operation characteristics and other unit variables.

Search by indoor unit model number or unit description Search by option code (reverse lookup)

Search by option code (used to determine indoor unit model based on known option code)

Enter option code (select or type) 010054-1E54B8-204646-331115

Model Number AM024MHHDC/AA

Unit Description DVM S, Duct S - 24000 Btu/h, 0.12" - 0.20" ESP (DEFAULT)

Static Pressure (in. WG) 0.12" - 0.20" (DEFAULT)

MWR-WE13N / MWR-WE11N / MWR-WE10N / MWR-WE10 - Wired controller MWR-SH10N / MWR-SH11UN Simple Touch Wired Controller MWR-SH00N - Wired controller Wireless controller Service software (SNET Pro 2) MWR-WH00 / MWR-WE00

Main Menu Sub Menu Data Bits

4 2 8888 88

ESC Mode Erase Store Set In

Hold SET and ESC for 5 seconds to enter service settings mode (actual button location may vary from the image above depending on wired controller model number).

Press the up arrow button, select Main Menu 4.

Press the right arrow button to advance.

Press the up arrow button to select sub menu 2.

Press the right arrow to advance.

Enter the code above into the data bit section using the up, down, and right arrow buttons.

Press the right arrow button to advance.

Page 1

4₂ 0100 54

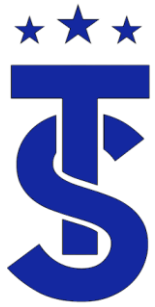
Page 2

4₂ 1E54 66

Page 3

4₂ 2046 46

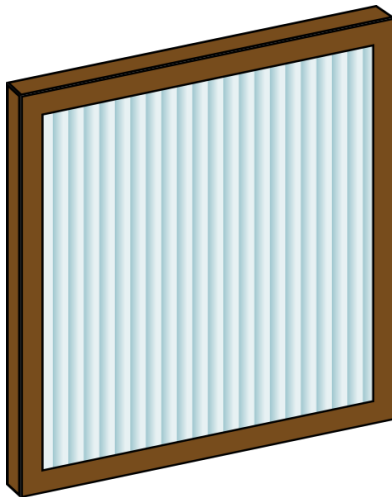
Page 4



Filter Selection

Know what's in your system!

Air Flow: 1000 CFM
300 FPM



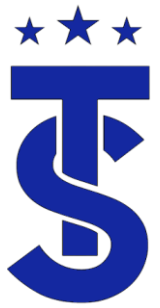
16 x 25 Air Filters @ 1,000 CFM

290	430	575	720	970*	*
0.09	0.15	0.22	0.31	0.46	nc

Air Flow: 1400 CFM
500 FPM



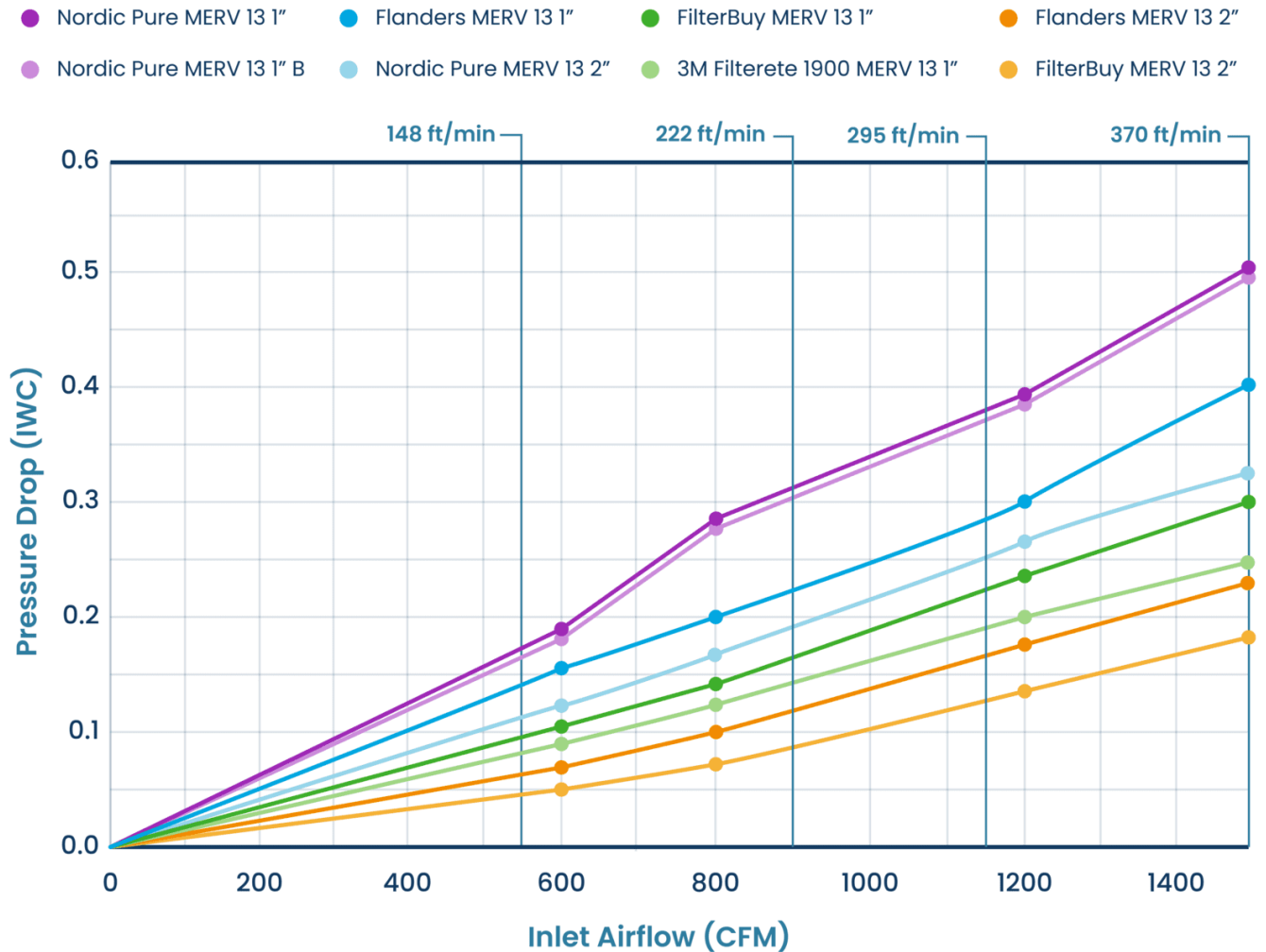
Resistance : .2" wc



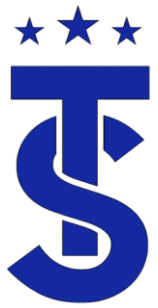
All filters are not created equally

- Know the characteristics of the filter you're selecting for a job.
- Misapplying a filter can cause performance issues with the system

MERV 13 Pressure Drop Measurements

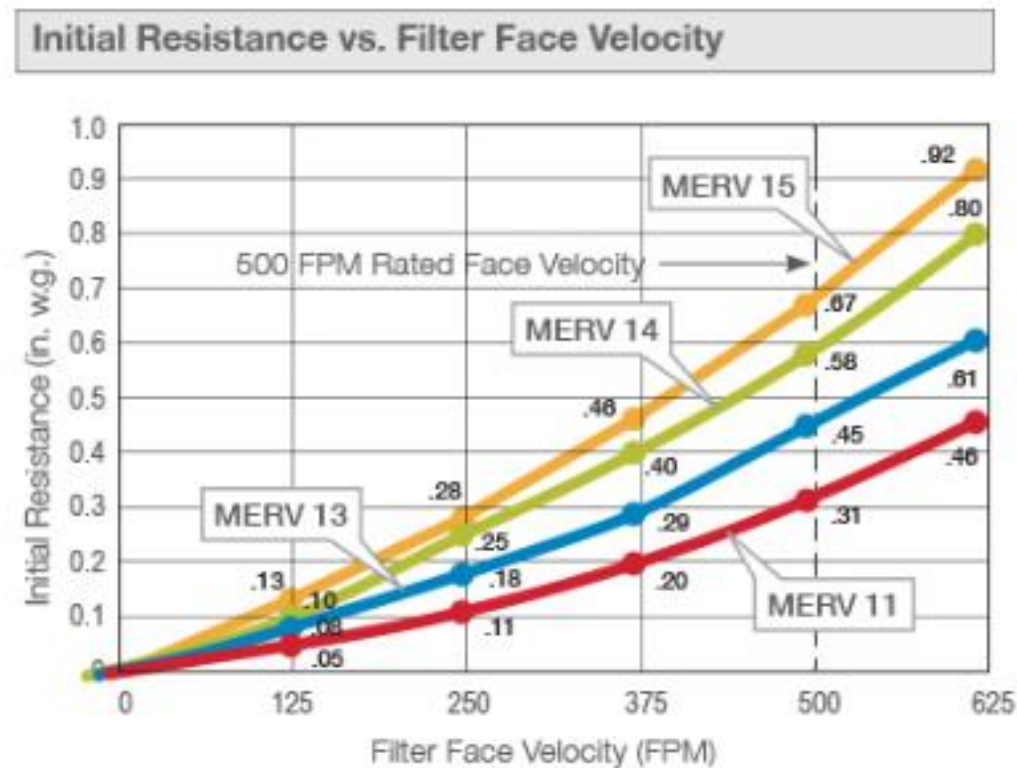


*Testing was conducted by the **California Energy Commission**. MERV Filters tested included various models from 3M, FilterBuy, Flanders, Nordic Pure, Honeywell, & US Home Filter.



Filter Selection

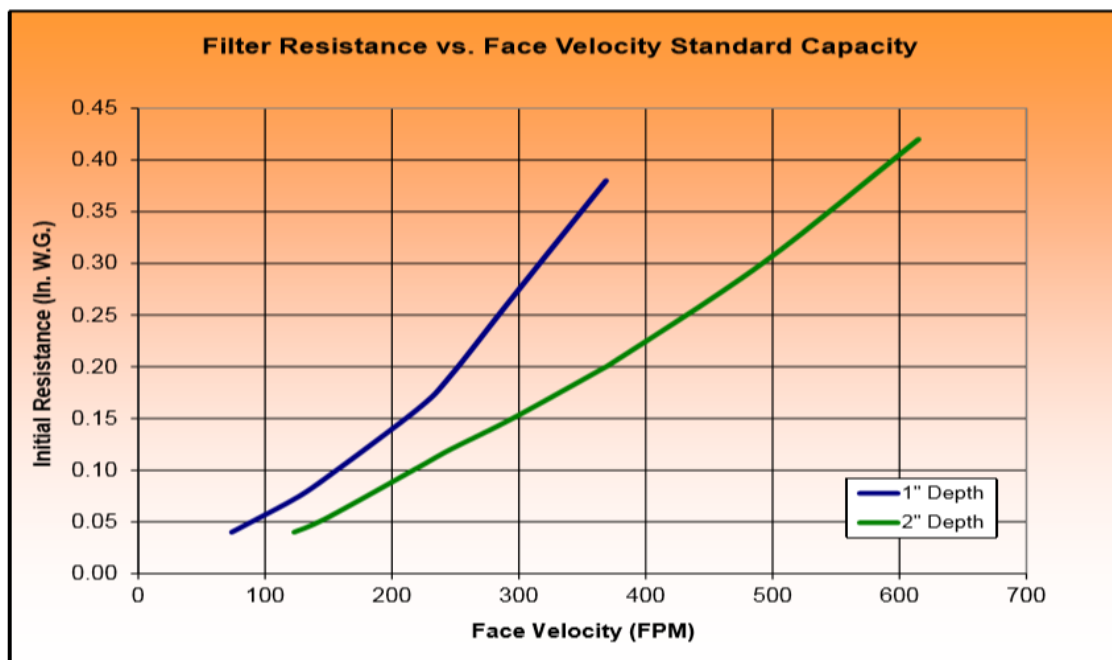
- If the homeowner purchases a filter with a higher MERV rating, can it effect the system performance?



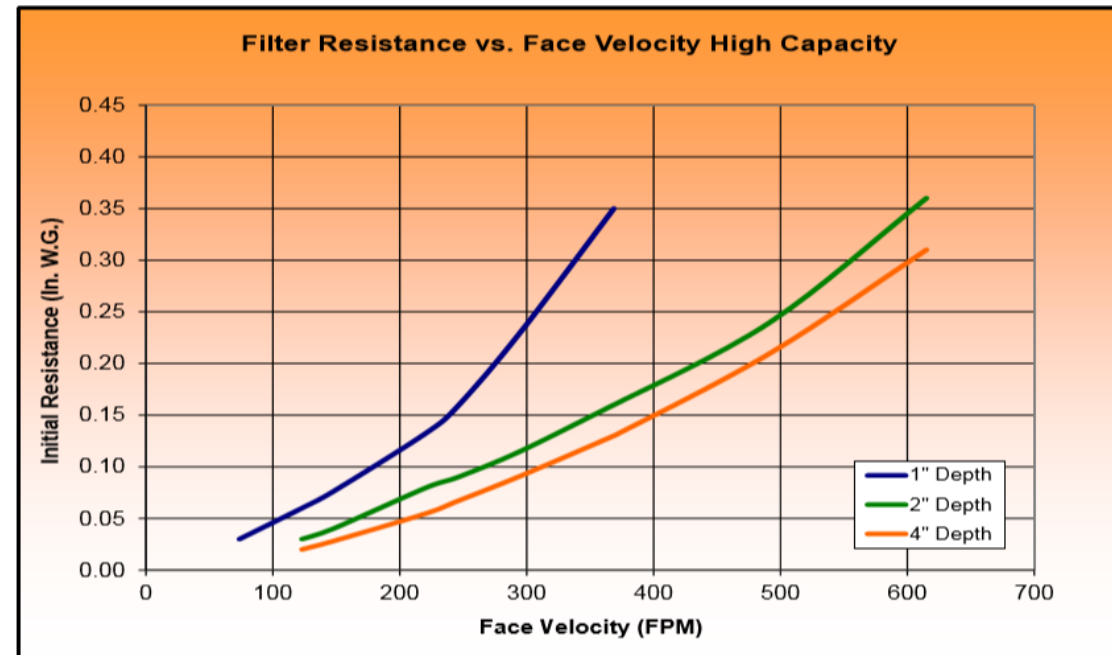


Filter Pressure Drop vs Face Velocity

- Understanding the velocity and pressure drop relationship.

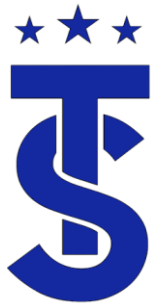


Standard-Capacity (KP)



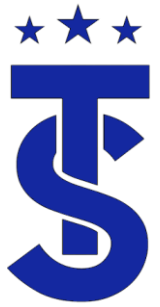
High-Capacity (KPH)

Purolator Key Pleat Merv 8 Filter Performance



Air Flow Review

- Air flow is the most common overlooked issue when it comes to ducted units.
- Having the correct air flow is crucial for system performance.
- Each unit has it's design CFM
- Ducts must be designed to deliver the proper CFM
- If the measured ESP exceeds the nameplate the delivered CFM will be reduced.
- If you know the CFM you can determine the performance of any unit.



Verify the operation of this system

AM036TNZDCH/AA

- On start up the system was put on high fan in the heating mode.
- The measured temperature rise in heat was 30 degrees.
- The measured TD in cooling mode was 20 degrees
- System passed the air balance test.
- The end user complains about lack of heating from time to time.
- What are some thing to look at

AM036TNZDCH

HP: 1/2

Default motor taps:
high=4, mid=3, low=1



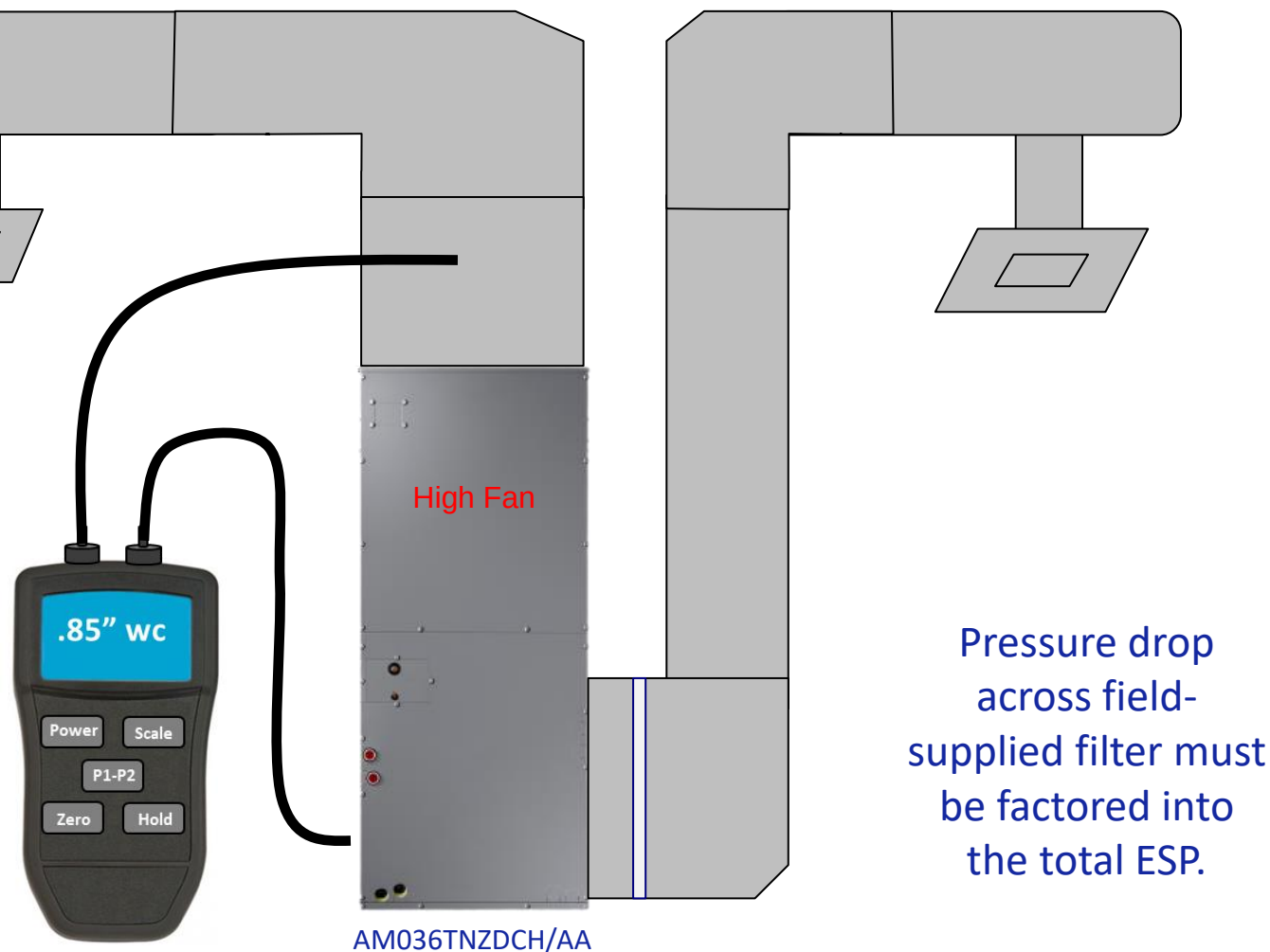
Specifications

Performance	Nominal Capacity (Btu/h)	Cooling	36,000
		Heating	40,000
Airflow	CFM @ 0.4" ESP (UL)	L / M / H	758 / 924 / 1,053
External Static Pressure	Standard	"WC	0.4
	Min. / Max.	"WC	0.1 / 1.0



Does the system provide the correct airflow?

- Set the fan to High Speed
- Measure the External Static pressure.
measured ESP was 1.16"wc?
- Use the fan performance chart to determine the CFM





Calculating CFM Example

- ❑ Set the fan to High Speed
- ❑ Fan tap in high is #4 (default)
- ❑ Measured External Static pressure is .85" w.c.
- ❑ The delivered CFM is approximately 889CFM
- ❑ Is this enough air flow (rated airflow is 1,053CFM)
- ❑ What can you do now that you know the CFM?
- ❑ You can determine the delivered capacity of the indoor unit in heating and cooling modes.

Motor Tap	ESP (inch)	CFM	RPM
2	0.1	989	670
	0.2	943	711
	0.25	940	737
	0.3	911	762
	0.4	878	810
	0.5	829	851
	0.6	791	881
	0.7	752	933
	0.8	682	1,010
	0.9	635	1,074
1	1	568	1,122
	0.1	888	615
	0.2	850	671
	0.25	826	691
	0.3	800	718
	0.4	762	771
	0.5	714	808
	0.6	663	879
	0.7	605	947
	0.8	524	1,016
	0.9	487	1,059
	1	431	1,105

Motor Tap	ESP (inch)	CFM	RPM
4	0.1	1,165	748
	0.2	1,121	786
	0.25	1,110	809
	0.3	1,089	832
	0.4	1,053	877
	0.5	1,021	903
	0.6	982	946
	0.7	941	976
	0.8	906	1,016
	0.9	872	1,042
3	1	816	1,108
	0.1	1,039	684
	0.2	992	743
	0.25	980	752
	0.3	952	780
	0.4	921	828
	0.5	884	860
	0.6	841	903
	0.7	809	929
	0.8	767	984
	0.9	715	1,049
	1	643	1,132

Motor Tap	ESP (inch)	CFM	RPM
5	0.1	1,318	826
	0.2	1,281	864
	0.25	1,270	876
	0.3	1,246	896
	0.4	1,207	938
	0.5	1,183	978
	0.6	1,152	997
	0.7	1,120	1,032
	0.8	1,077	1,061
	0.9	1,043	1,097
	1	994	1,126

AM036TNZDCH

HP: 1/2

Default motor taps:
high=4, mid=3, low=1



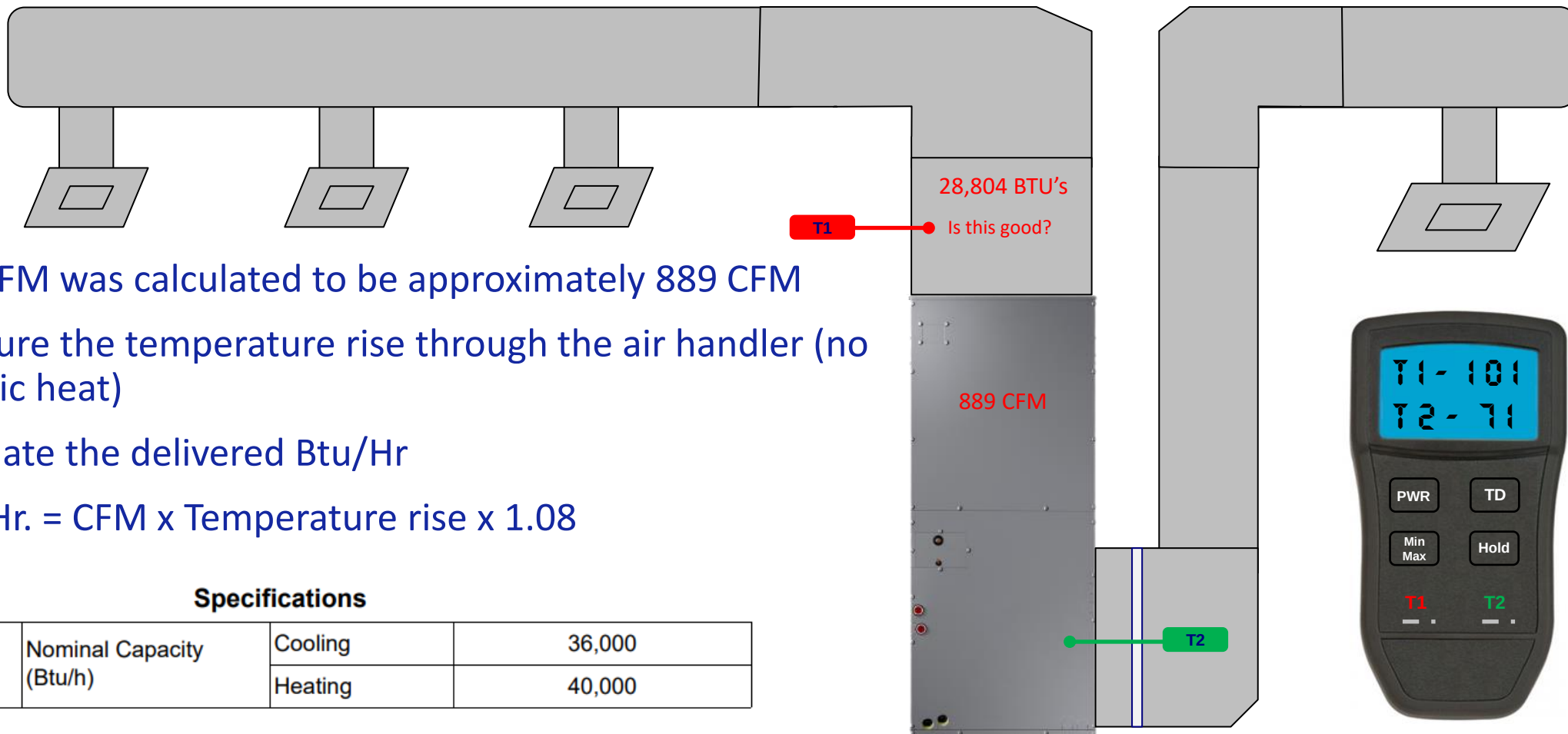


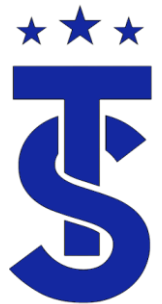
Calculate delivered BTUH

- The CFM was calculated to be approximately 889 CFM
- Measure the temperature rise through the air handler (no electric heat)
- Calculate the delivered Btu/Hr
- $\text{BTU/Hr.} = \text{CFM} \times \text{Temperature rise} \times 1.08$

Specifications

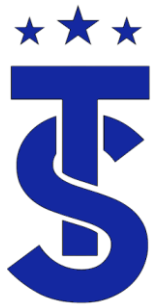
Performance	Nominal Capacity (Btu/h)	Cooling	36,000
		Heating	40,000





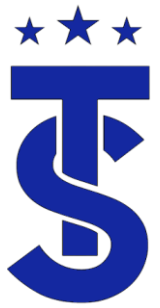
Conclusion

- When the system was tested the temperature rise was 30 degrees.
- It was stated that this is a “normal” temperature rise for a heat pump.
- If the temperature rise is correct, it must be moving the correct air flow.
- The balance report stated the CFM delivery was 1,053 CFM
- Should temperature rise be used as the only means of checking system performance???
- What is the correct temperature rise for this unit to deliver 40,000 Btu's of heat at the rated air flow?



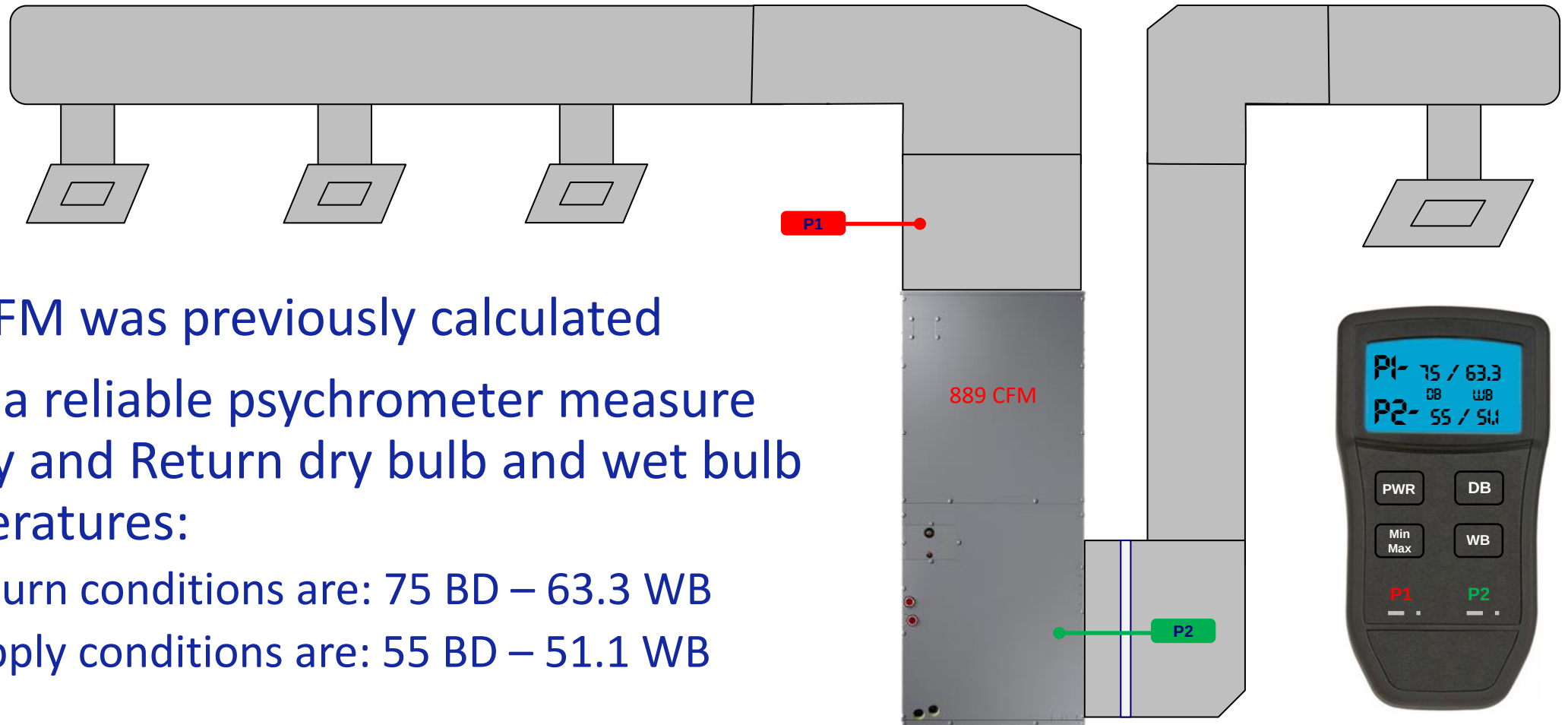
Cooling capacity in the cooling mode

- Cooling BTUs = CFM x Δh x 4.5
 - CFM = The fan CFM plotted on the fan tables, or measured at several points in the system
 - Δh = the enthalpy change through the system (we get this number by converting wet bulb readings.)
 - 4.5 = the cooling BTU multiplier at sea level, or .075-lbs. of air per CFM x 60 (minutes in an hour.) This factor will also change at higher altitudes. BTU multipliers will also change due to extreme air temperatures.



Calculate Cooling Capacity

- 889 CFM was previously calculated
- Using a reliable psychrometer measure Supply and Return dry bulb and wet bulb temperatures:
 - Return conditions are: 75 BD – 63.3 WB
 - Supply conditions are: 55 BD – 51.1 WB





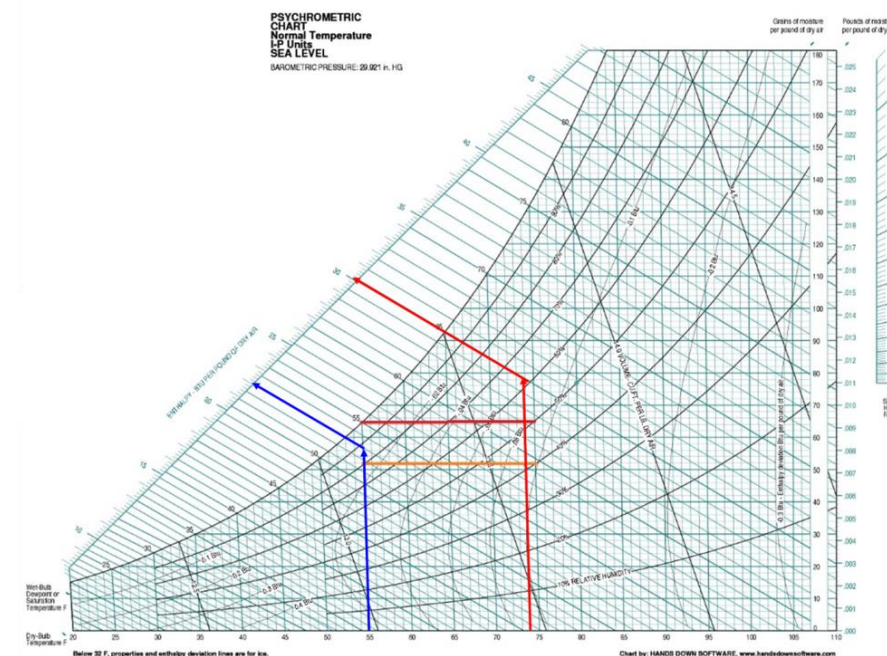
Calculate Enthalpy Differential

- Using a psychrometric calculator determine the enthalpy of the return and supply air then calculate the difference
 - Return Air: 75 BD – 63.3 WB – Enthalpy 29.2
 - Supply Air: 55 BD – 51.1 WB – Enthalpy 21.1
 - Enthalpy 8.1

Free online Psychrometric Calculator [Print](#)

☒ IP ☐ SI altitude ft b press in.Hg

T (°F) dry bulb	rH (%) rel.humidity	TDP (°F) dew point	W (GPP) abs. humidity	h (btu/lb) enthalpy	Tw (°F) wet bulb	sp. vol (cft/lb)		
75	55	57.7	71	29.2	63.3	13.78	Calculate	Remove
53.1	90	50.4	54	21.1	51.1	13.14	Calculate	Remove





Calculating Cooling Capacity

- Cooling BTUs = CFM x Δh x 4.5
 - CFM = 889
 - Δh = 8.1
- Cooling BTUs = 889 x 8.1 x 4.5
 - Cooling Capacity = 32,404 BTUH

AM036TNZDCH

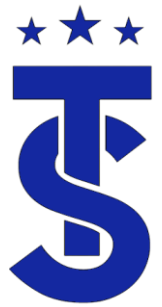
HP: 1/2

Default motor taps:
high=4, mid=3, low=1



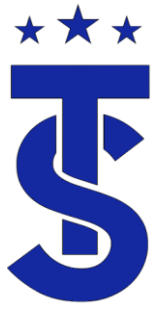
Specifications

Performance	Nominal Capacity (Btu/h)	Cooling	36,000
		Heating	40,000
Airflow	CFM @ 0.4" ESP (UL)	L / M / H	758 / 924 / 1,053
External Static Pressure	Standard	"WC	0.4
	Min. / Max.	"WC	0.1 / 1.0



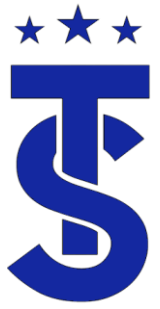
Review

- Air flow has a huge effect on the system performance
- Most contractors still size duct for the cooling CFM and not the Heating CFM.
- When the unit is tested, and air flow is set at the factory it is usually set for low ESP. This means that adjustments **MUST** be made during installation.
- A 30-degree temperature rise in the heating mode doesn't always give you the full story.

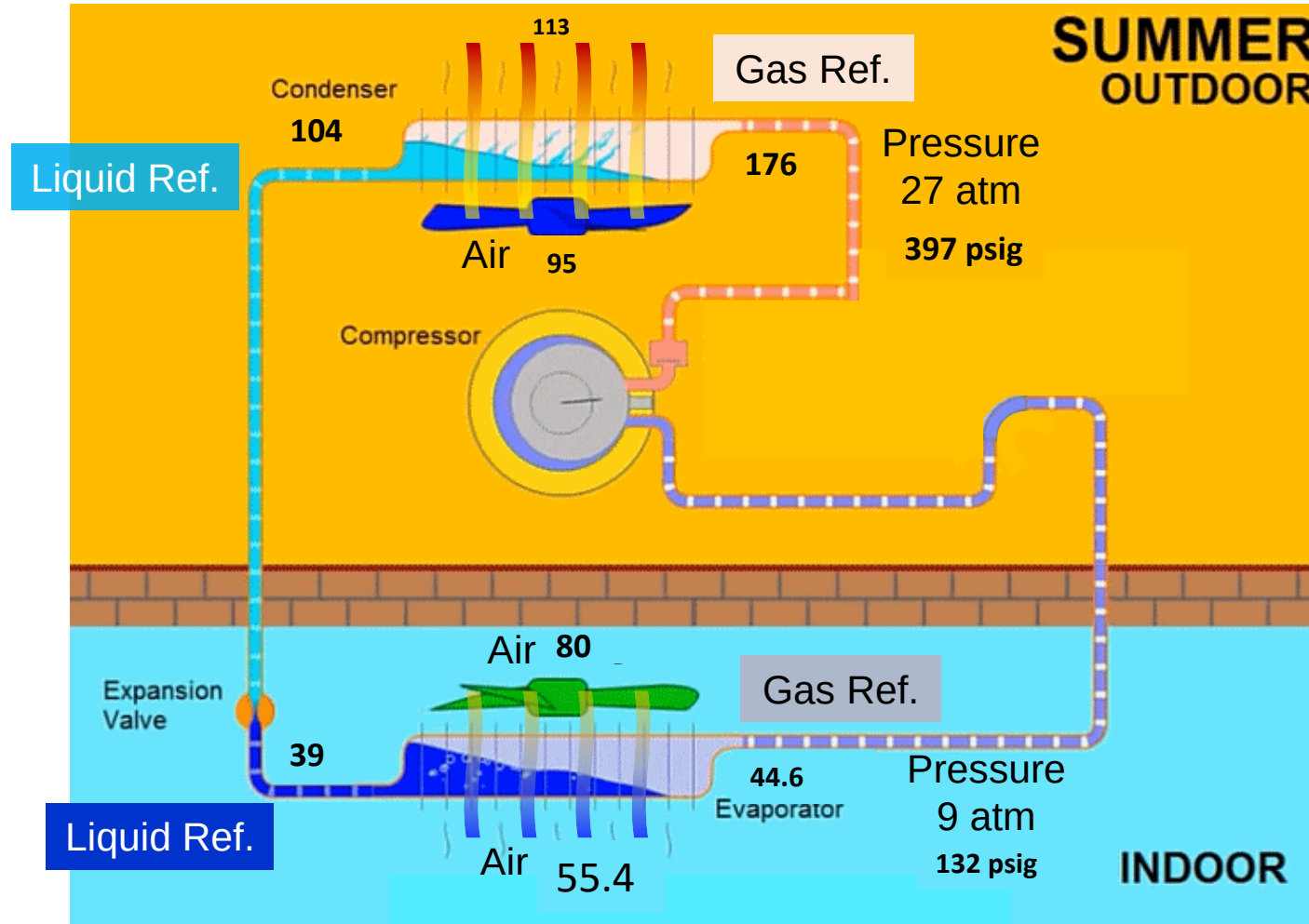


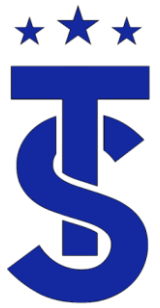
Reference Materials

- Information has been used from multiple Samsung training programs.
- Some example of the materials used on show on the following slides.

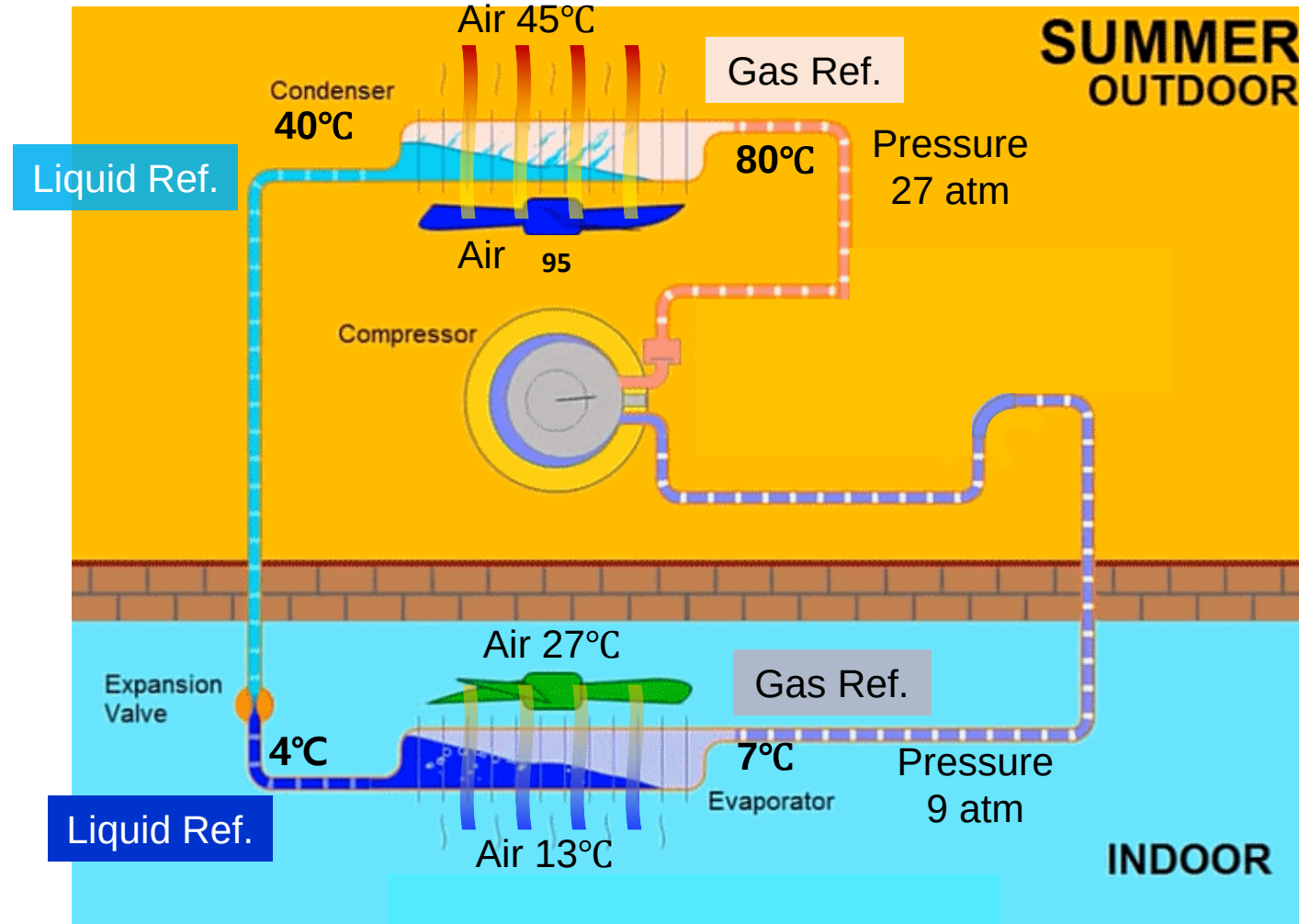


Basis of determining target operation conversion





Basis of determining target operation

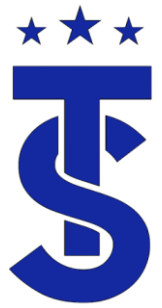




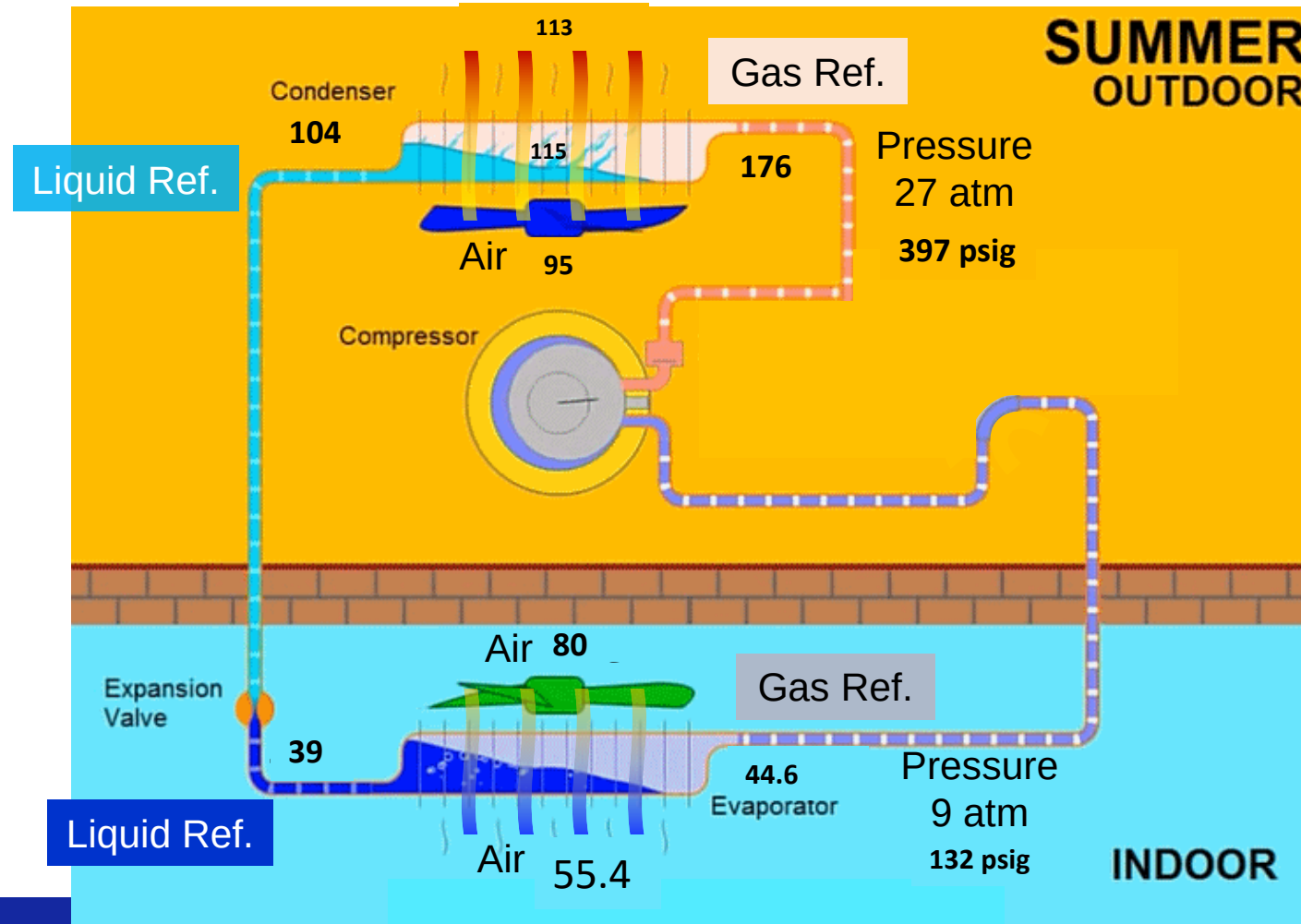


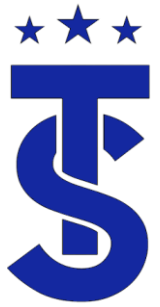
Refrigerant Troubleshooting



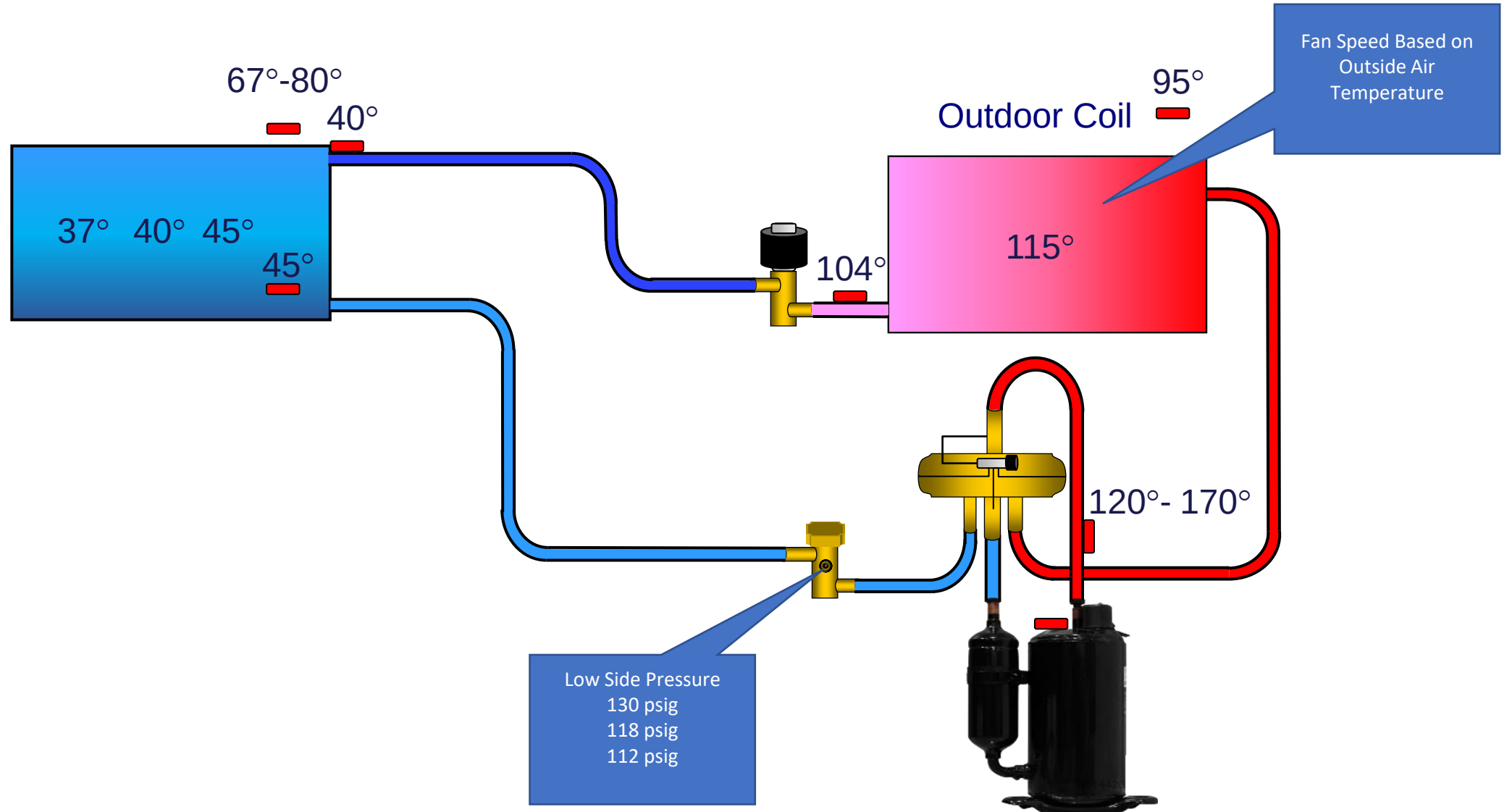


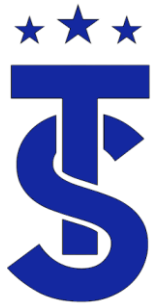
Basis of determining target operation conversion



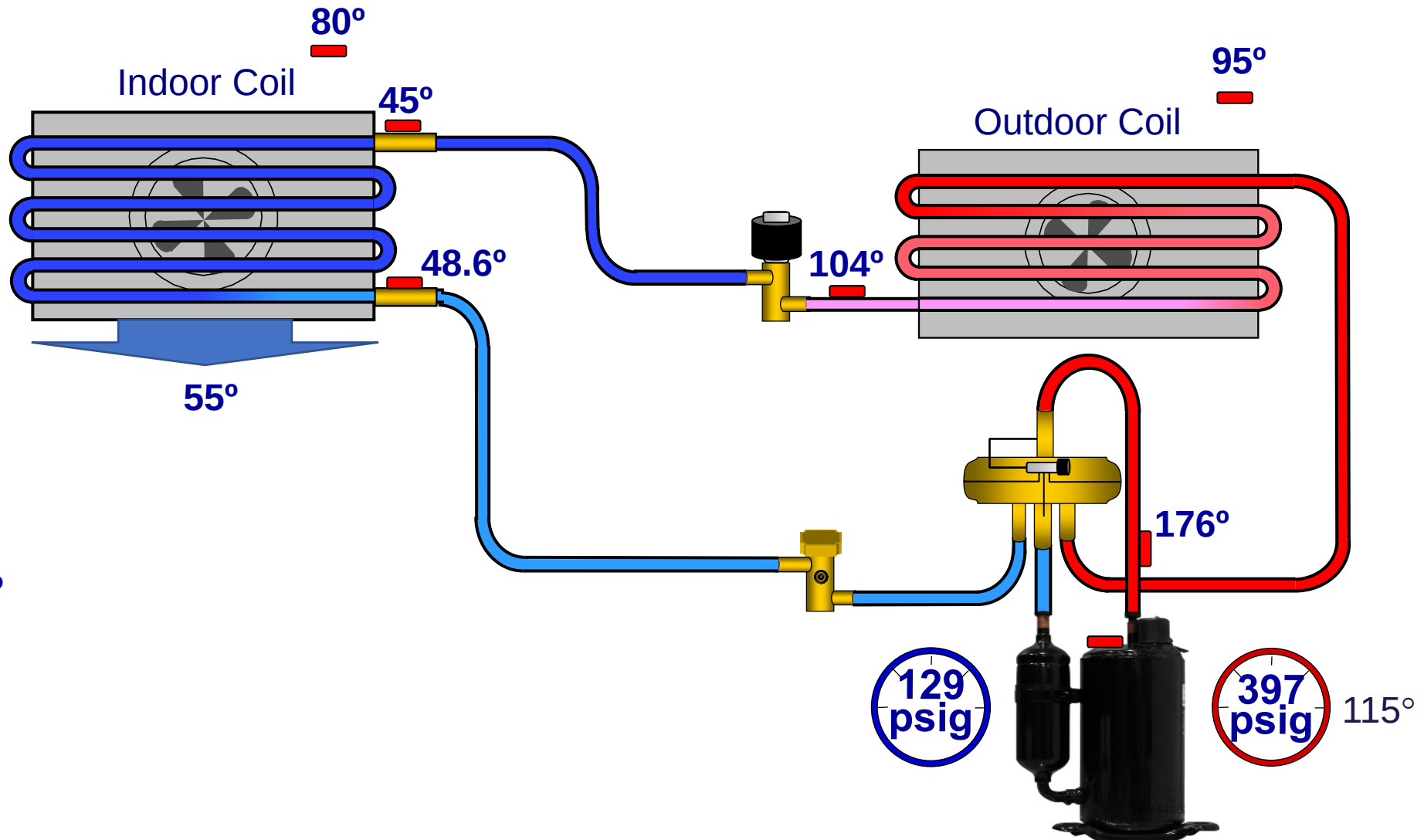


Cooling Cycle Ranges

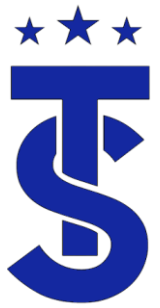




System Operation Standards

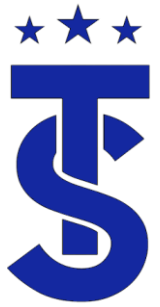


SH = 3.6°
SC = 12°
HS = 397 psig / 116°
LS = 129 psig / 45°

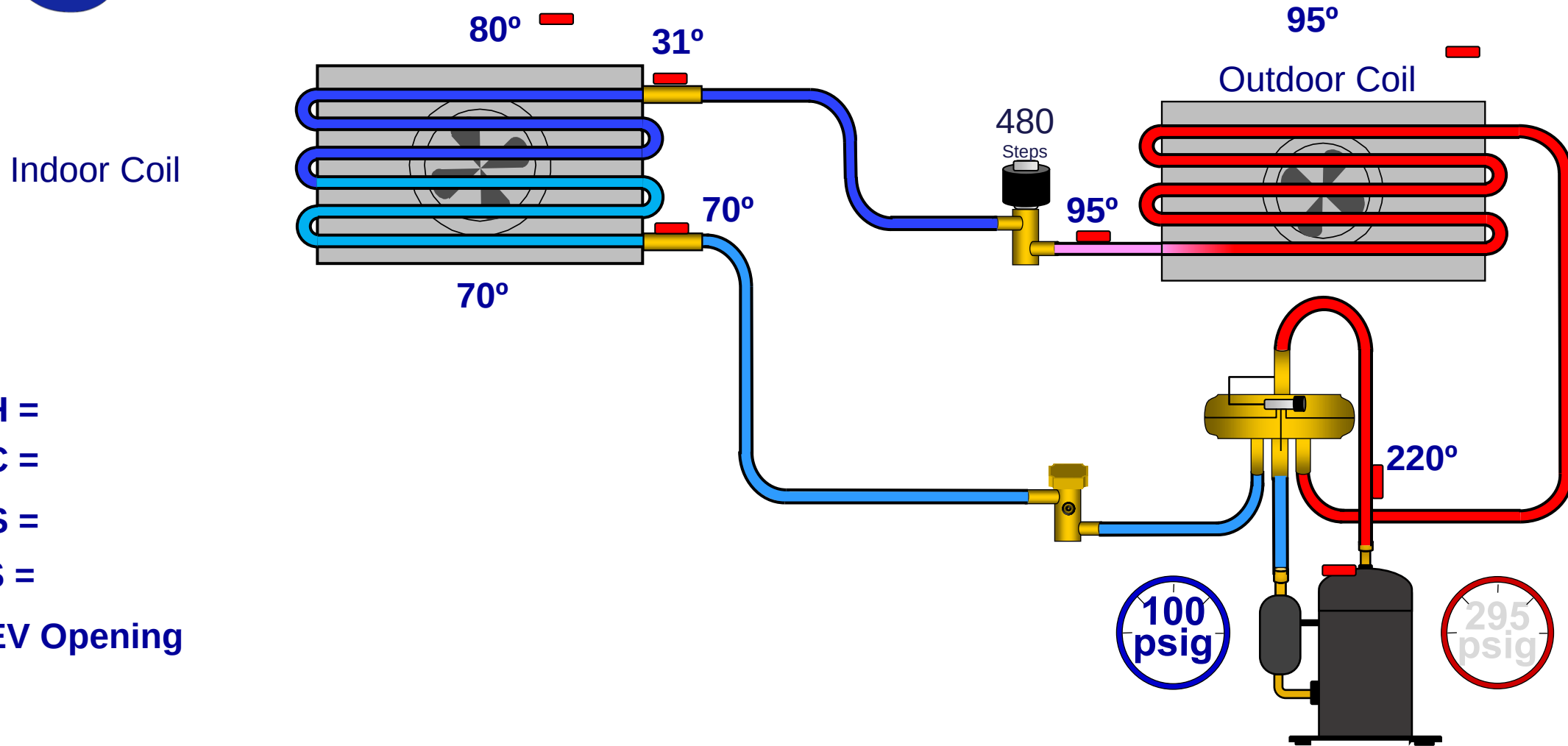


Diagnosing System Problems

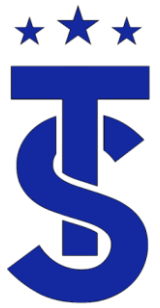
1. Undercharge of refrigerant
2. Overcharge of refrigerant
3. Dirty Condenser
4. Air in the system
5. Inefficient compressor
6. Restricted EEV
7. Restriction in the Gas line after the IDU
8. Restriction in the liquid line before the IDU
9. Dirty or iced evaporator, dirty filter, or low air flow



Diagnosis 1:



SH =
SC =
HS =
LS =
EEV Opening



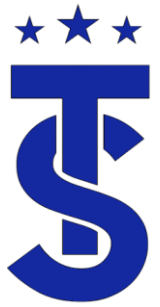
Undercharge of Refrigerant

Evaporator:

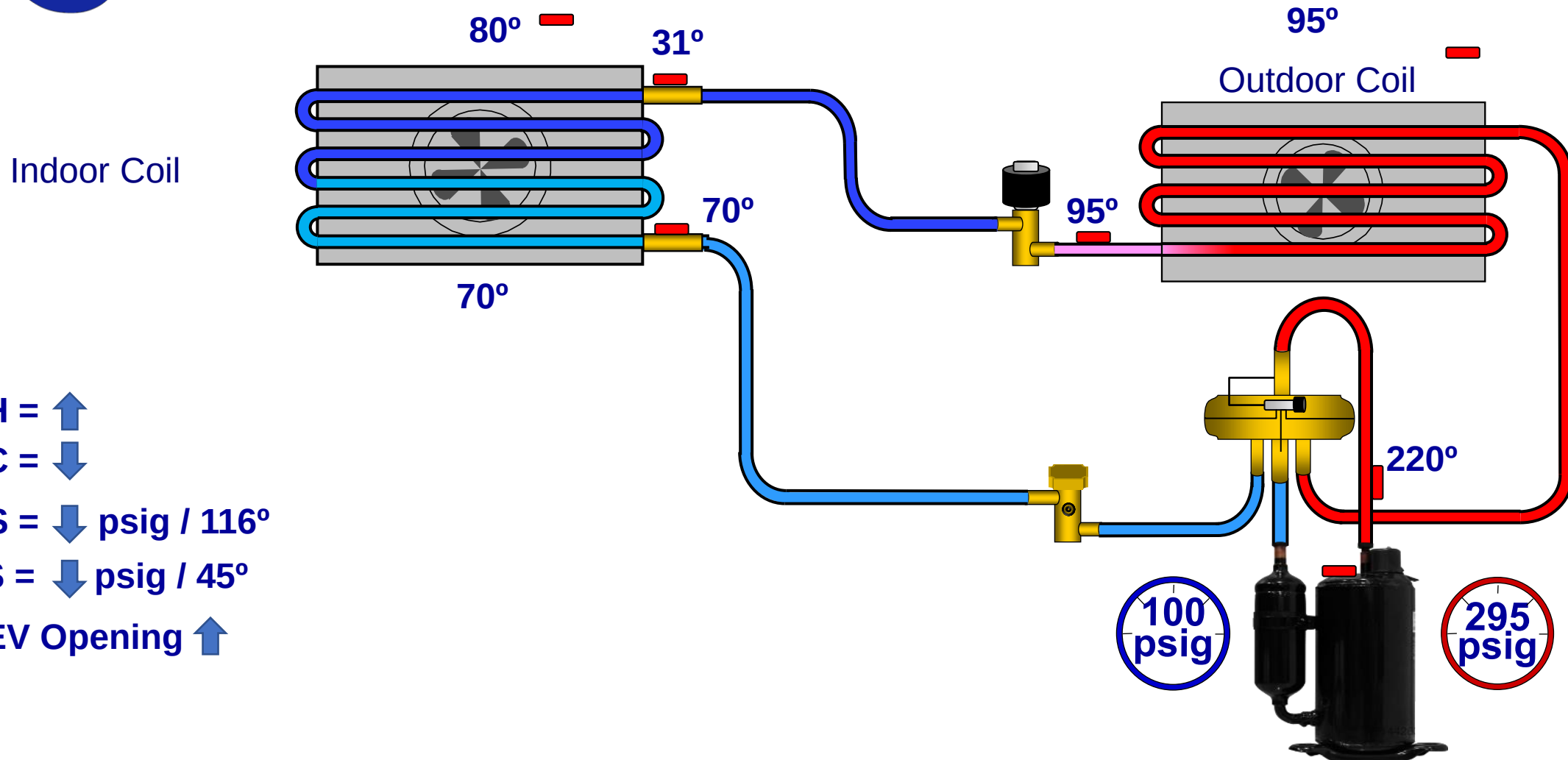
- Lower temperature
 - Lack of refrigerant, low pressure, low temperature
- High superheat
 - Limited refrigerant boiled off quickly

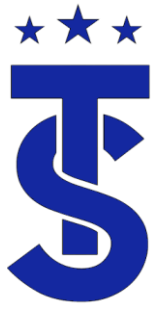
Condenser:

- Lower temperature
 - Little heat from the evaporator
- Low subcooling
 - Lack of refrigerant to subcool

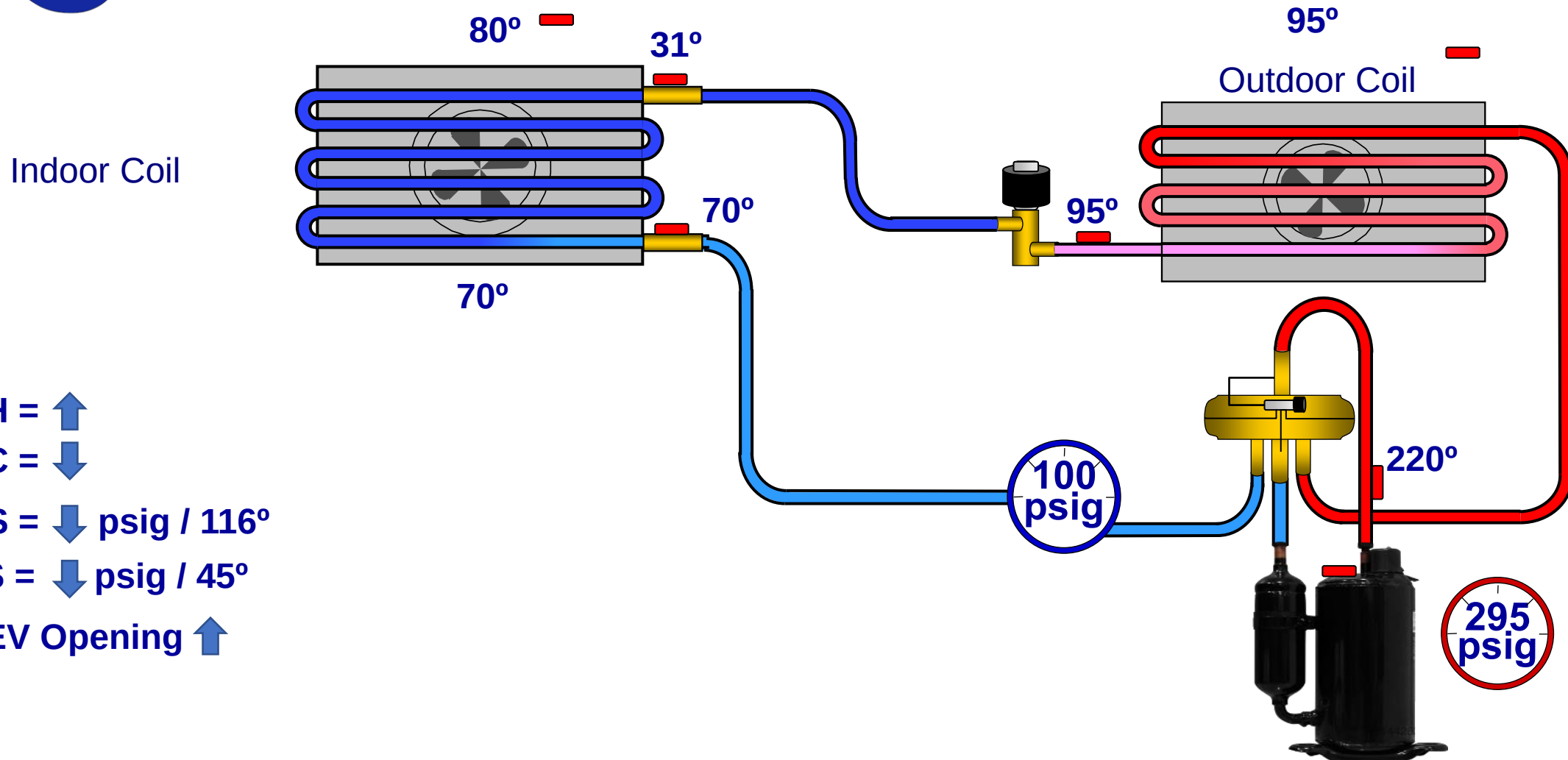


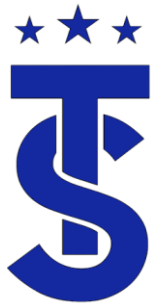
Low on Gas



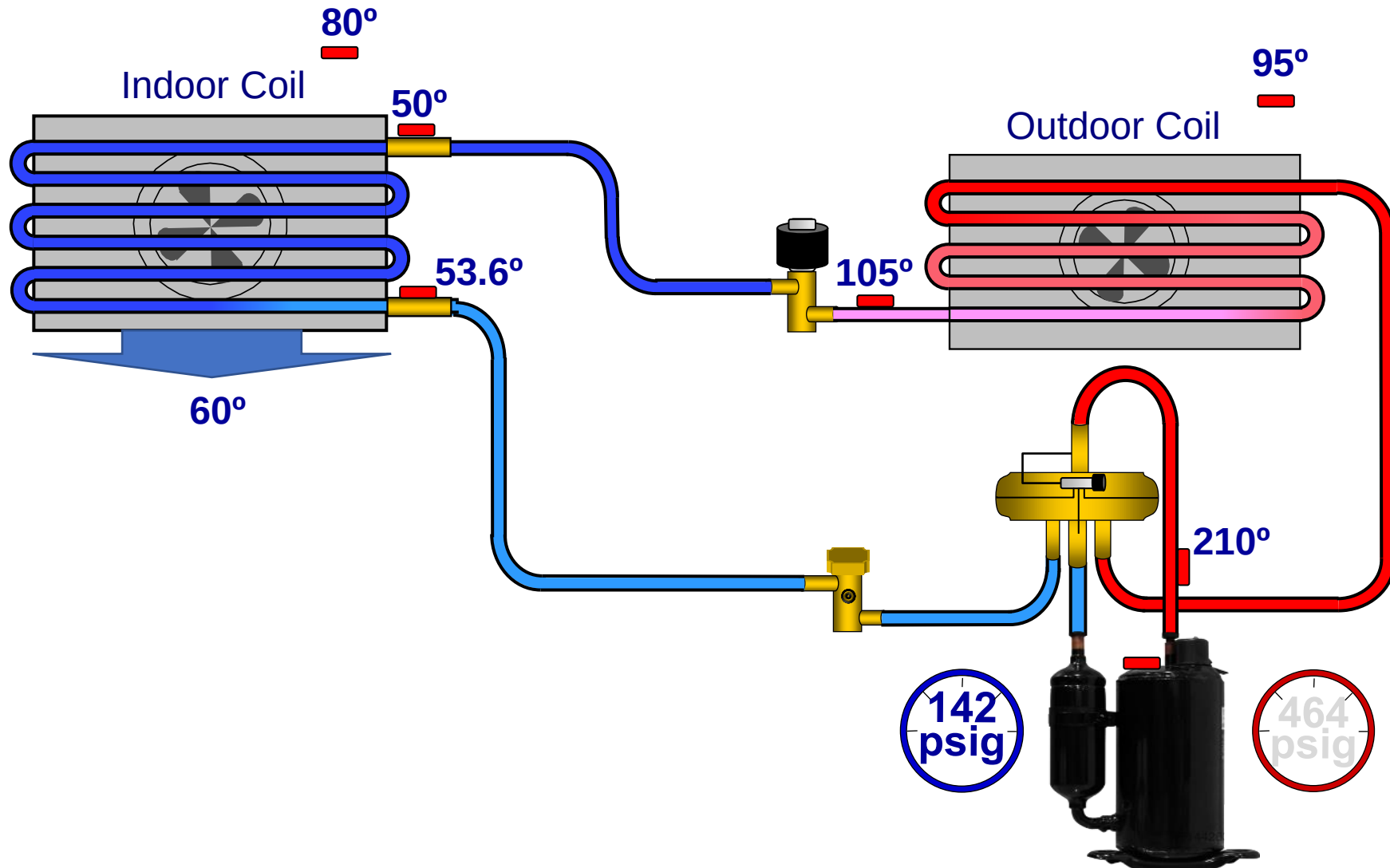


How Does the System Respond?

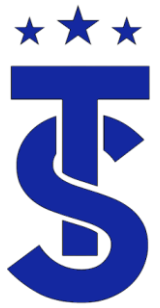




Diagnosis 2



SH =
SC =
HS =
LS =
EEV Opening



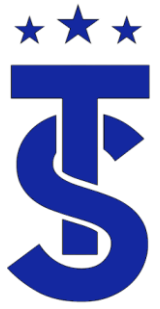
Overcharge of Refrigerant

Condenser:

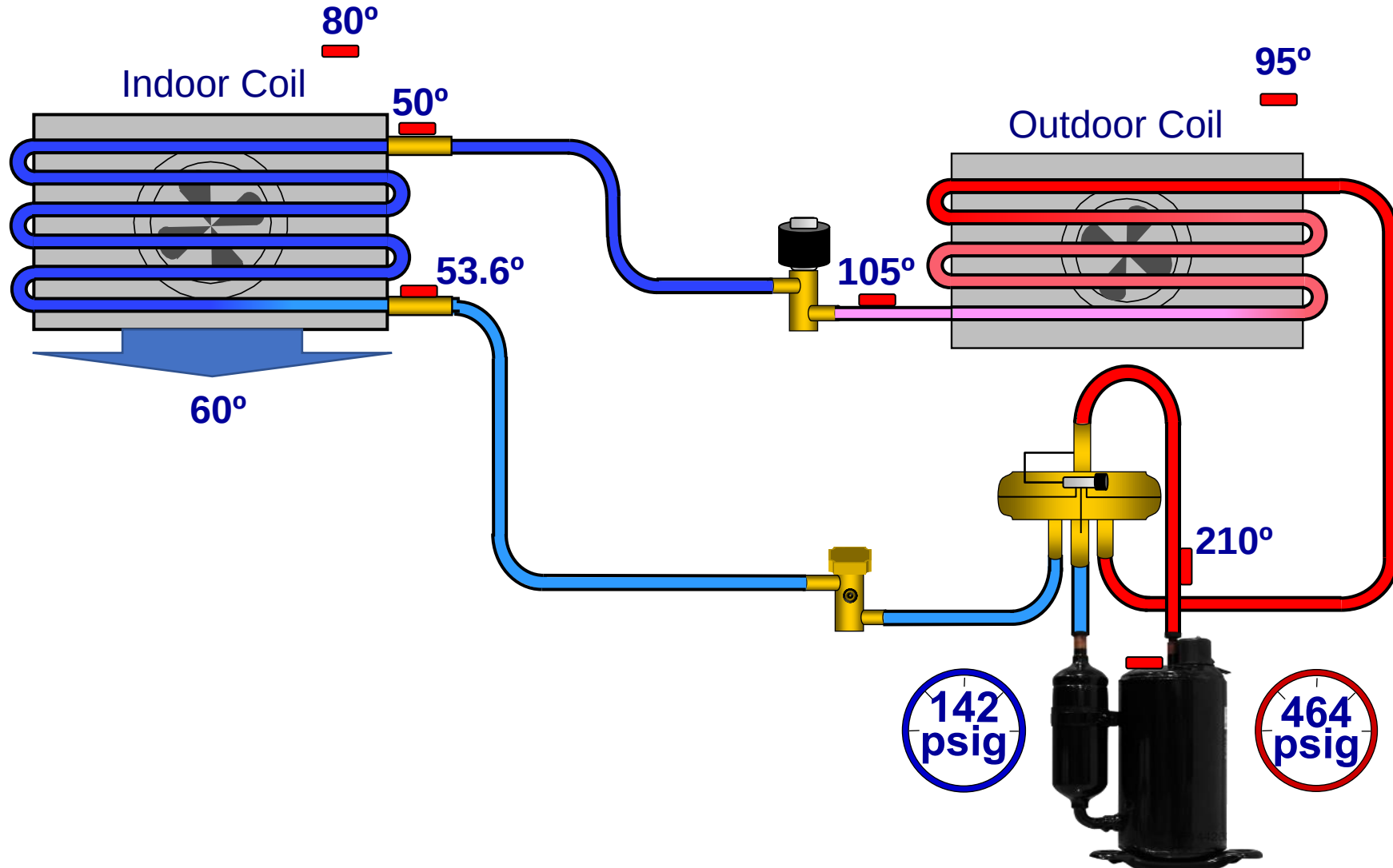
- High temperature
 - Excess refrigerant taking up condensing space
- High subcooling
 - More liquid for sensible cooling

Evaporator:

- Slightly higher temperature
 - High condensing pressure pushing liquid through valve
- Normal superheat
 - TEV tries to maintain suction line temperature



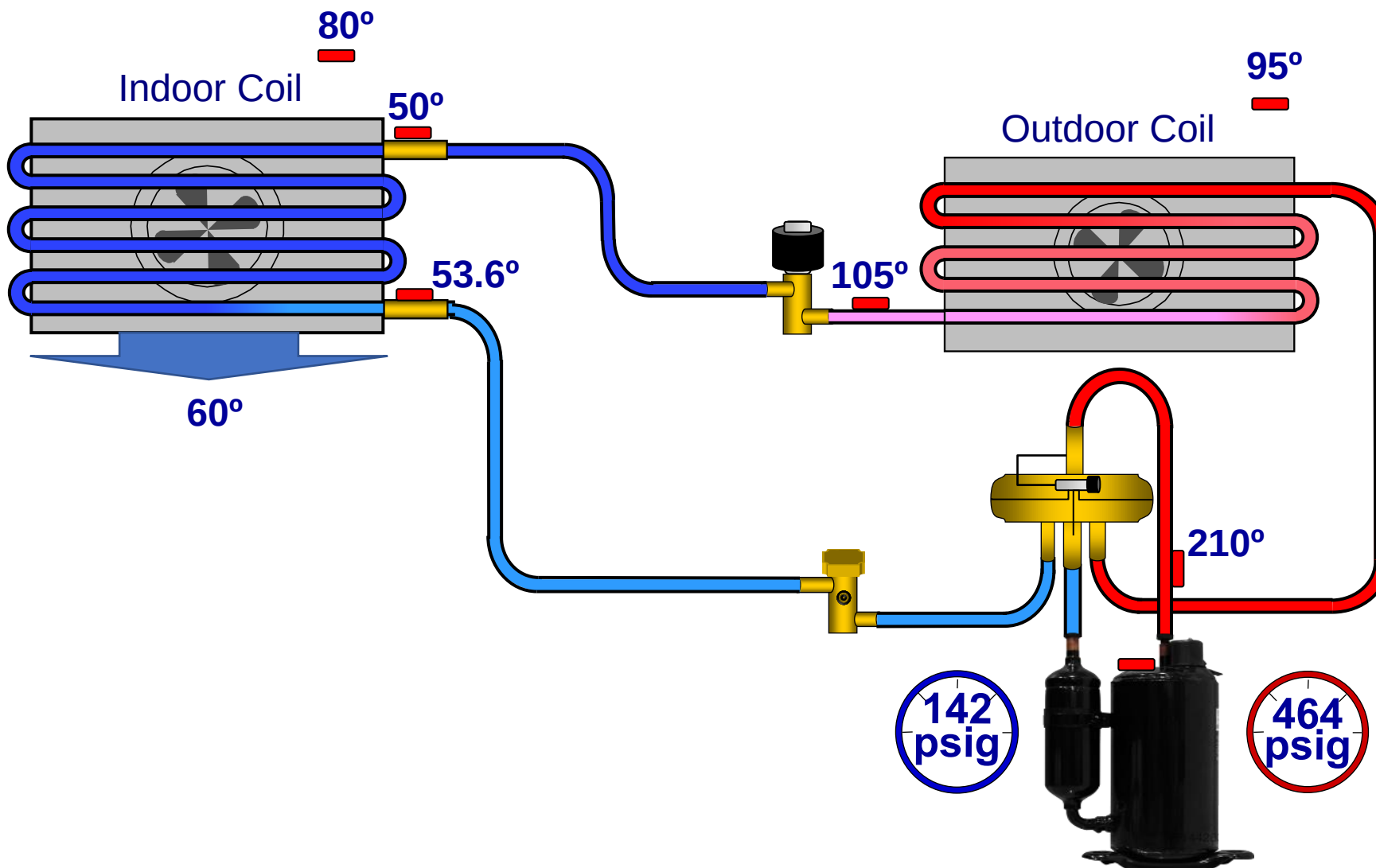
Overcharge



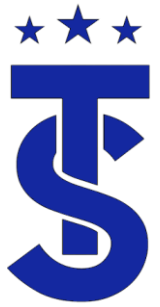
SH = 3.6
SC = ↑
HS = ↑ psig / 128°
LS = ↑ psig / 50°
EEV Opening ↓



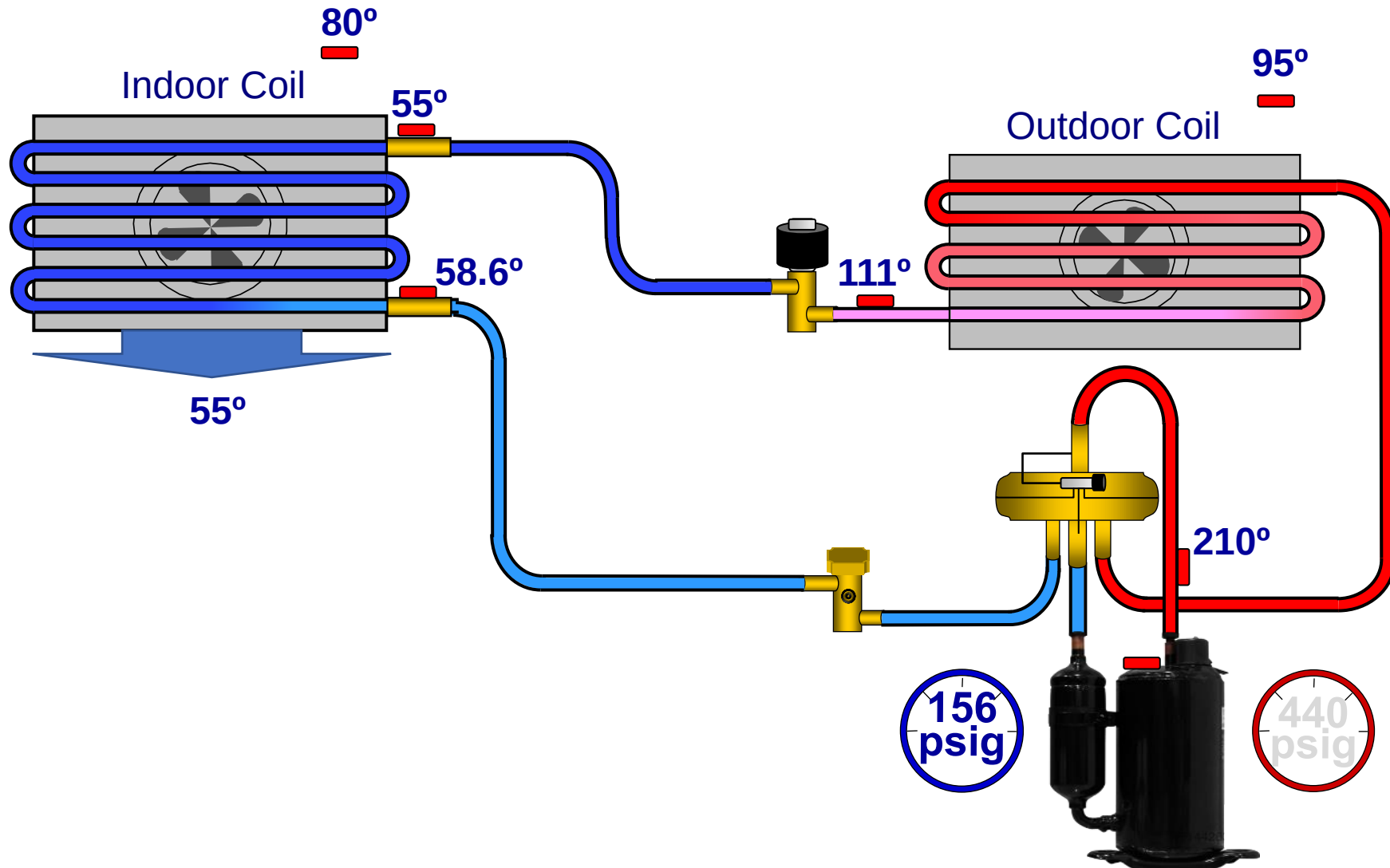
How Does the System Respond



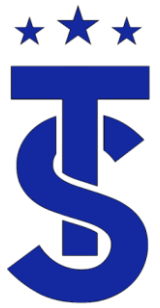
SH = 3.6
SC = ↑
HS = ↑ psig / 128°
LS = ↑ psig / 50°
EEV Opening ↓



Diagnosis 3



SH =
SC =
HS =
LS =
EEV Opening



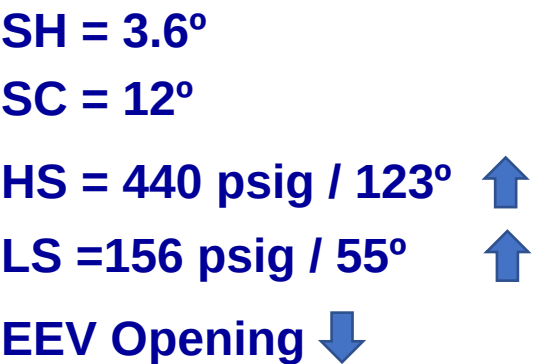
Dirty Condenser

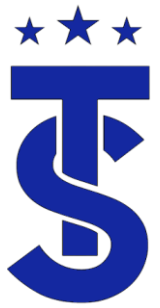
Condenser:

- Higher temperature
 - Heat cannot be rejected
- Normal subcooling

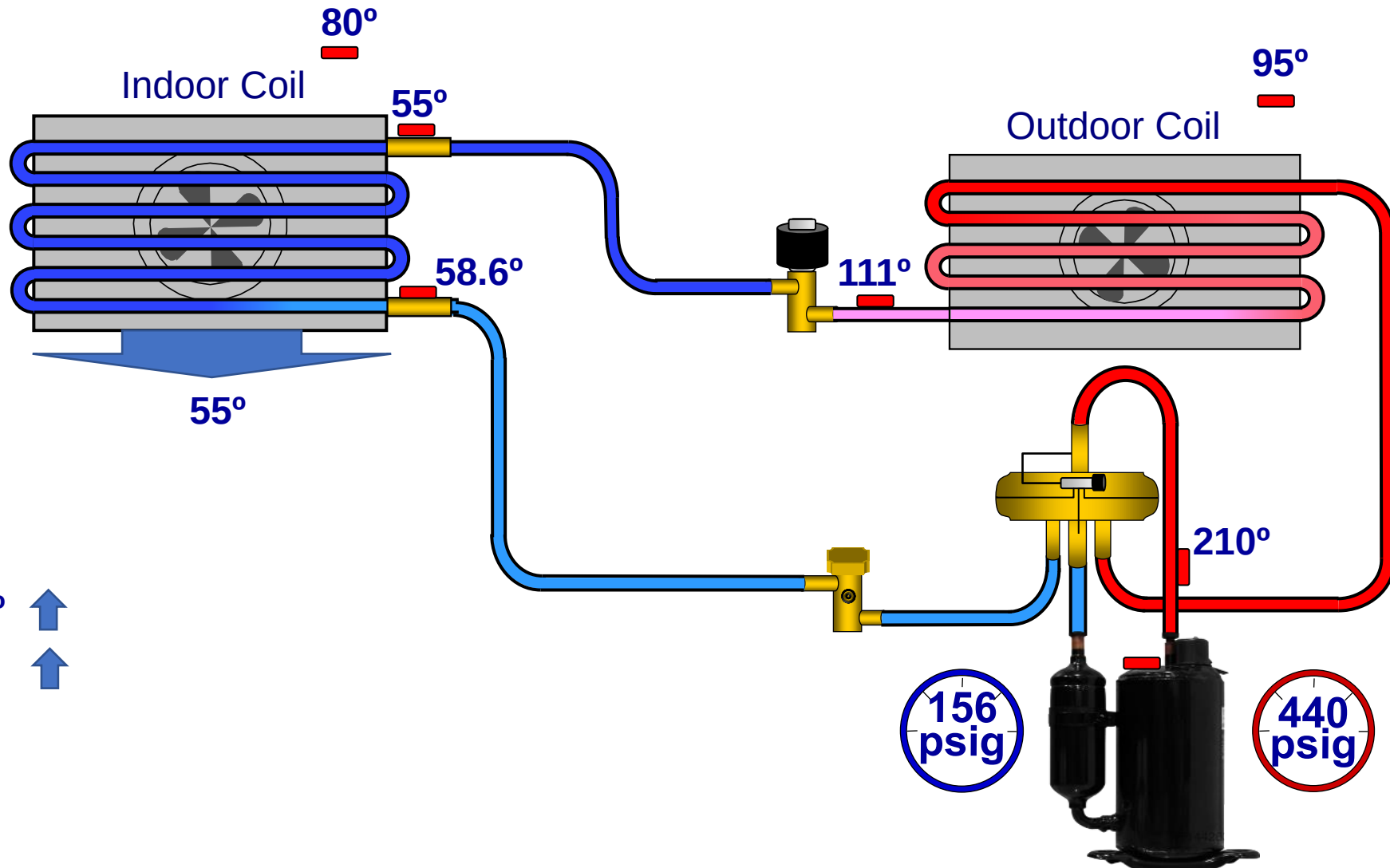
Evaporator:

- Slightly higher temperature
 - A little more refrigerant pushing through valve
- Normal superheat
 - TEV is trying to maintain superheat

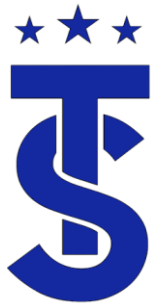




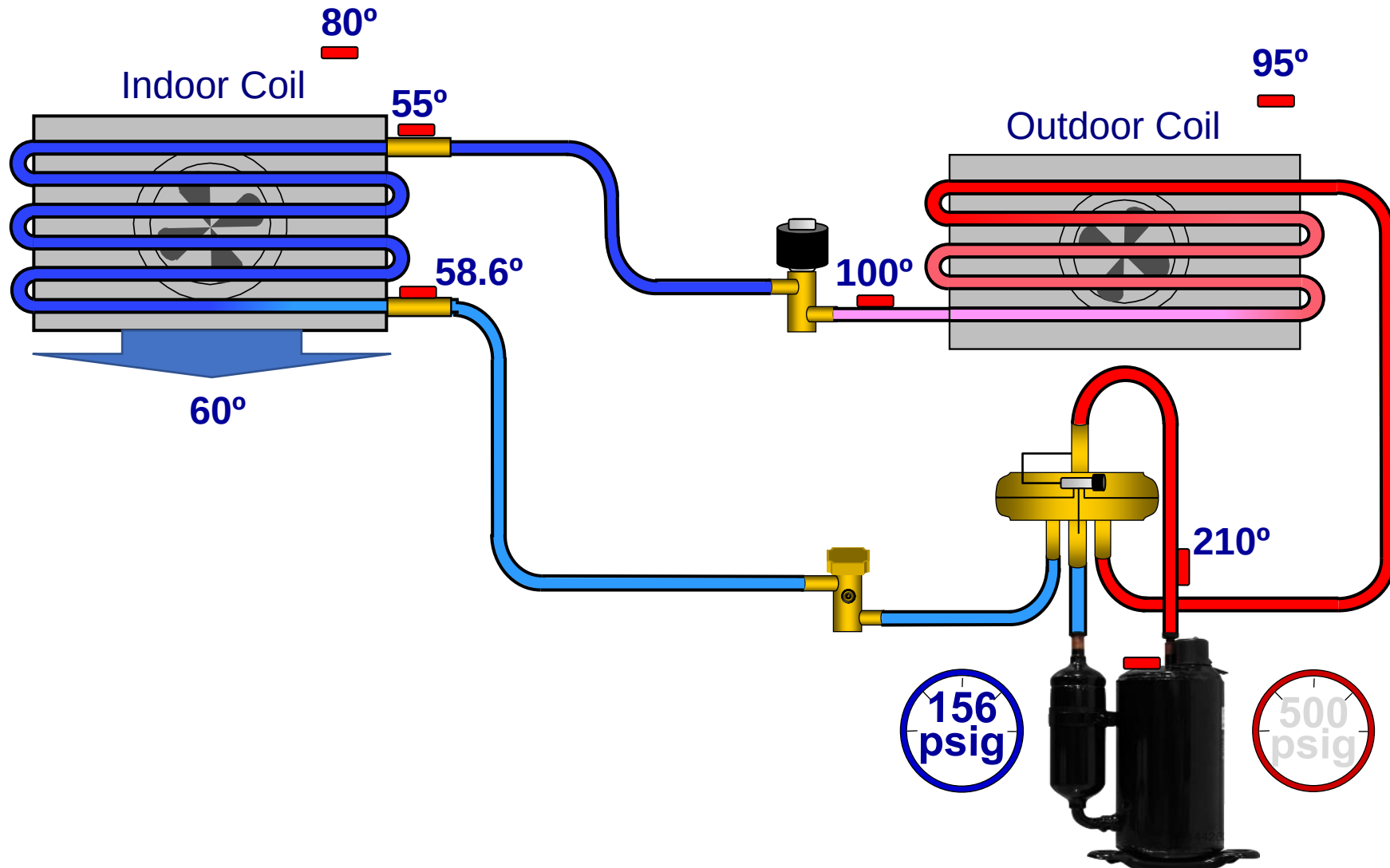
How Does the System Respond



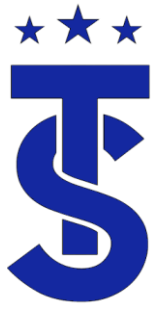
SH = 3.6°
SC = 12°
HS = 440 psig / 123° ↑
LS = 156 psig / 55° ↑
EEV Opening ↓



Diagnosis 4:



SH =
SC =
HS =
LS =
EEV Opening



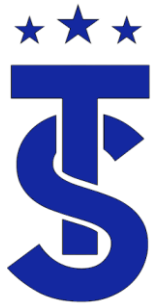
Air (non-condensables) in the system

Condenser:

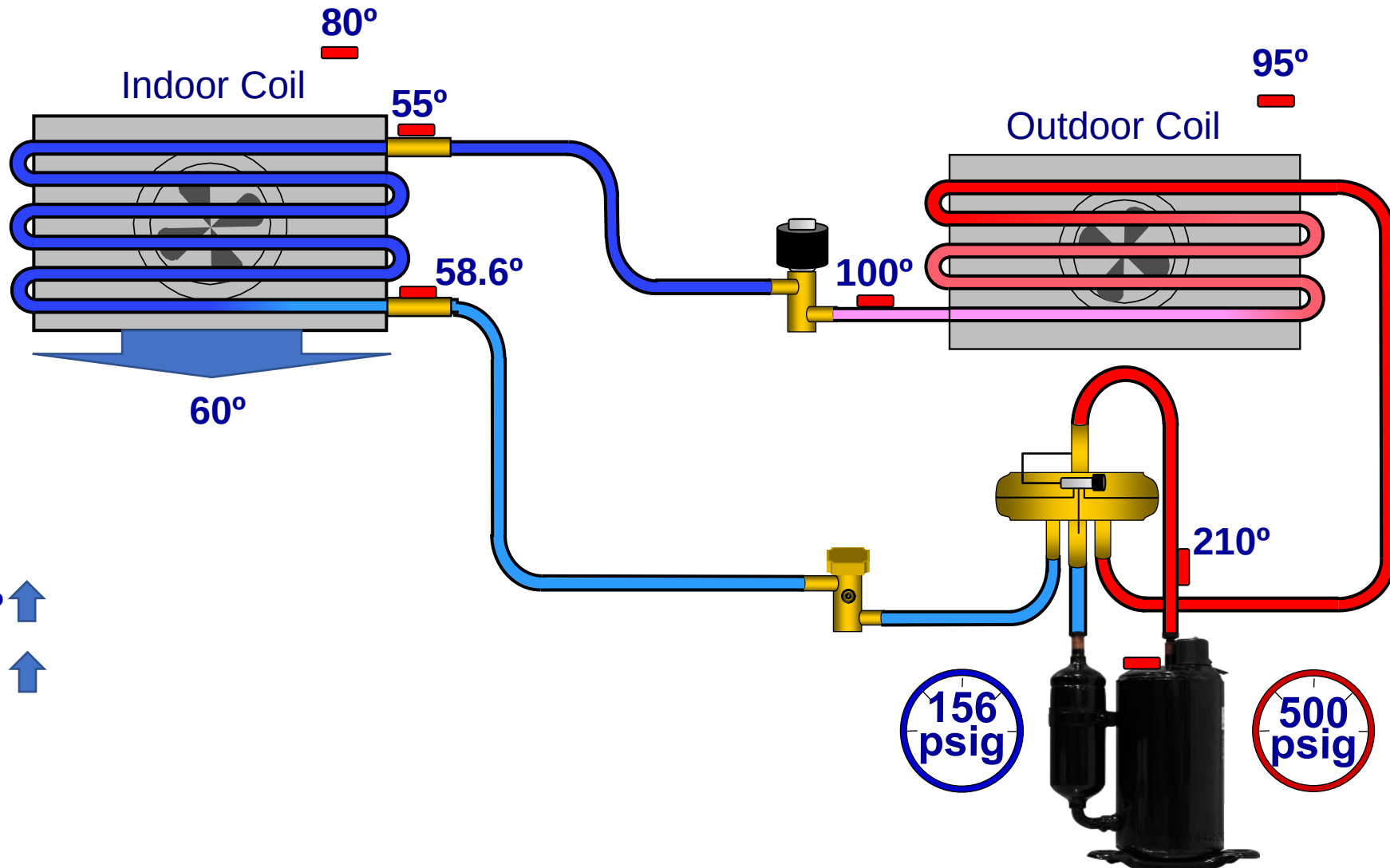
- High temperature
 - Air takes up space, less room for condensing
- Very high subcooling
 - Great difference between high condensing temperature and leaving liquid

Evaporator:

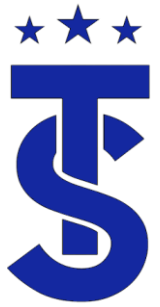
- Slightly higher temperature
 - High condensing pressure pushing through valve
- Normal superheat
 - TEV tries to maintain suction line temperature



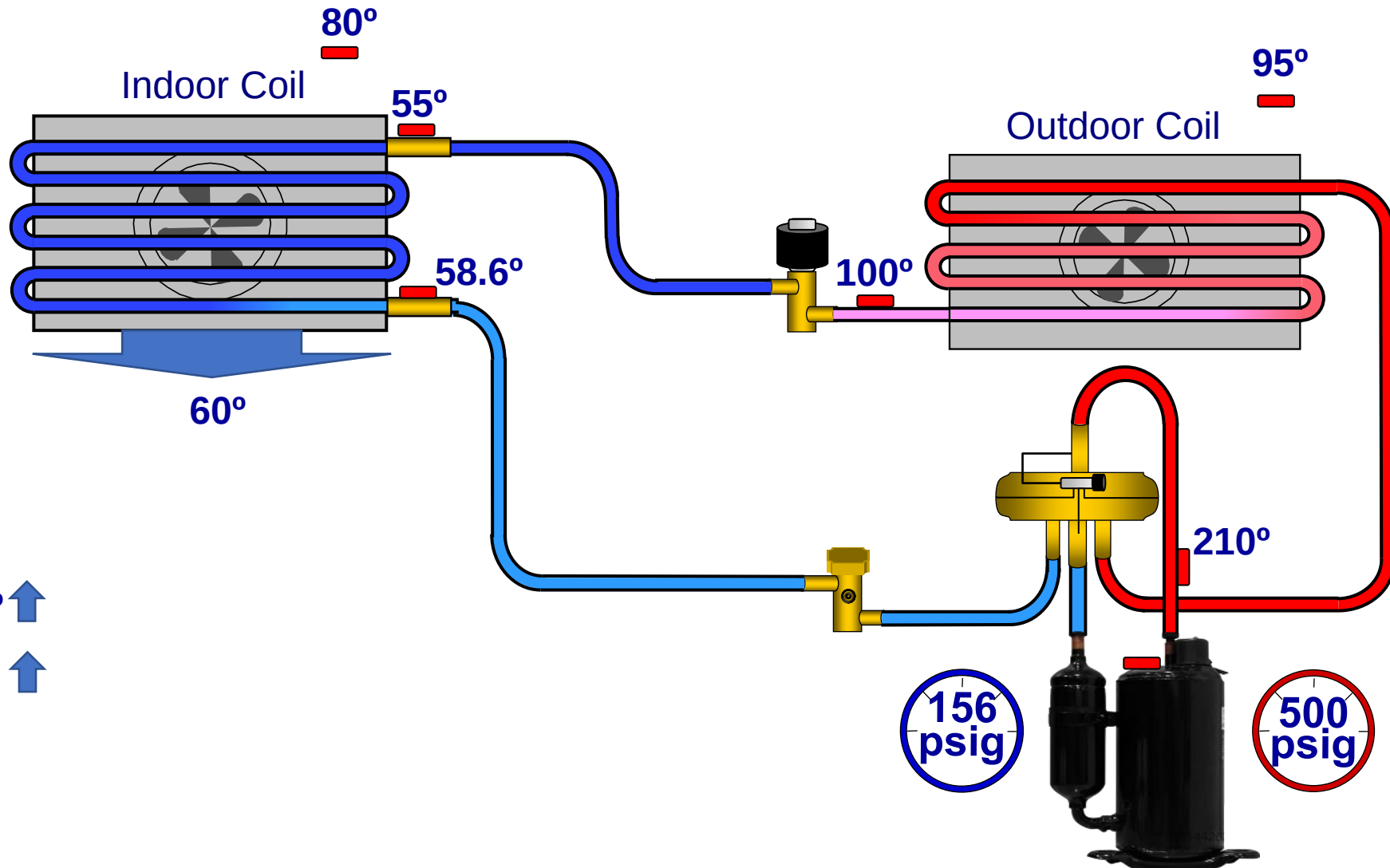
Non-Condensables



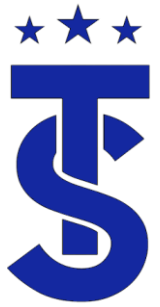
SH = 3.6
SC = ↑
HS = 500 psig / 134° ↑
LS = 156 psig / 55° ↑
EEV Opening ↓



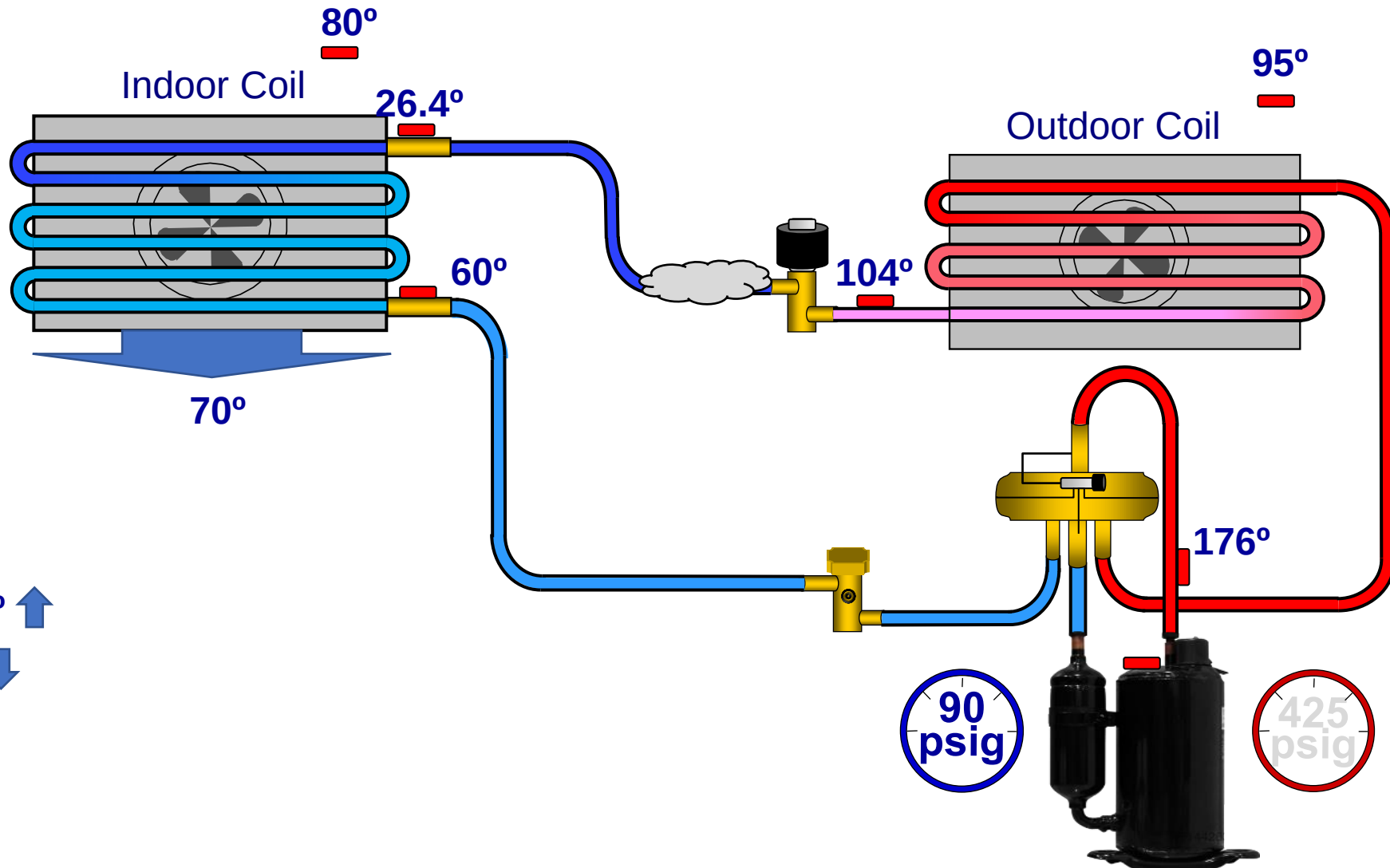
How Does the System Respond



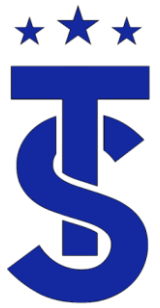
SH = 3.6
SC = ↑
HS = 500 psig / 134° ↑
LS = 156 psig / 55° ↑
EEV Opening ↓



Diagnosis 5:



SH = 26° ↑
SC = 17° ↑
HS = 425 psig / 121° ↑
LS = 90 psig / 26.4° ↓
EEV Opening ↑



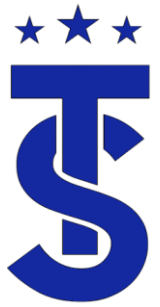
Partially Restricted EEV

Evaporator:

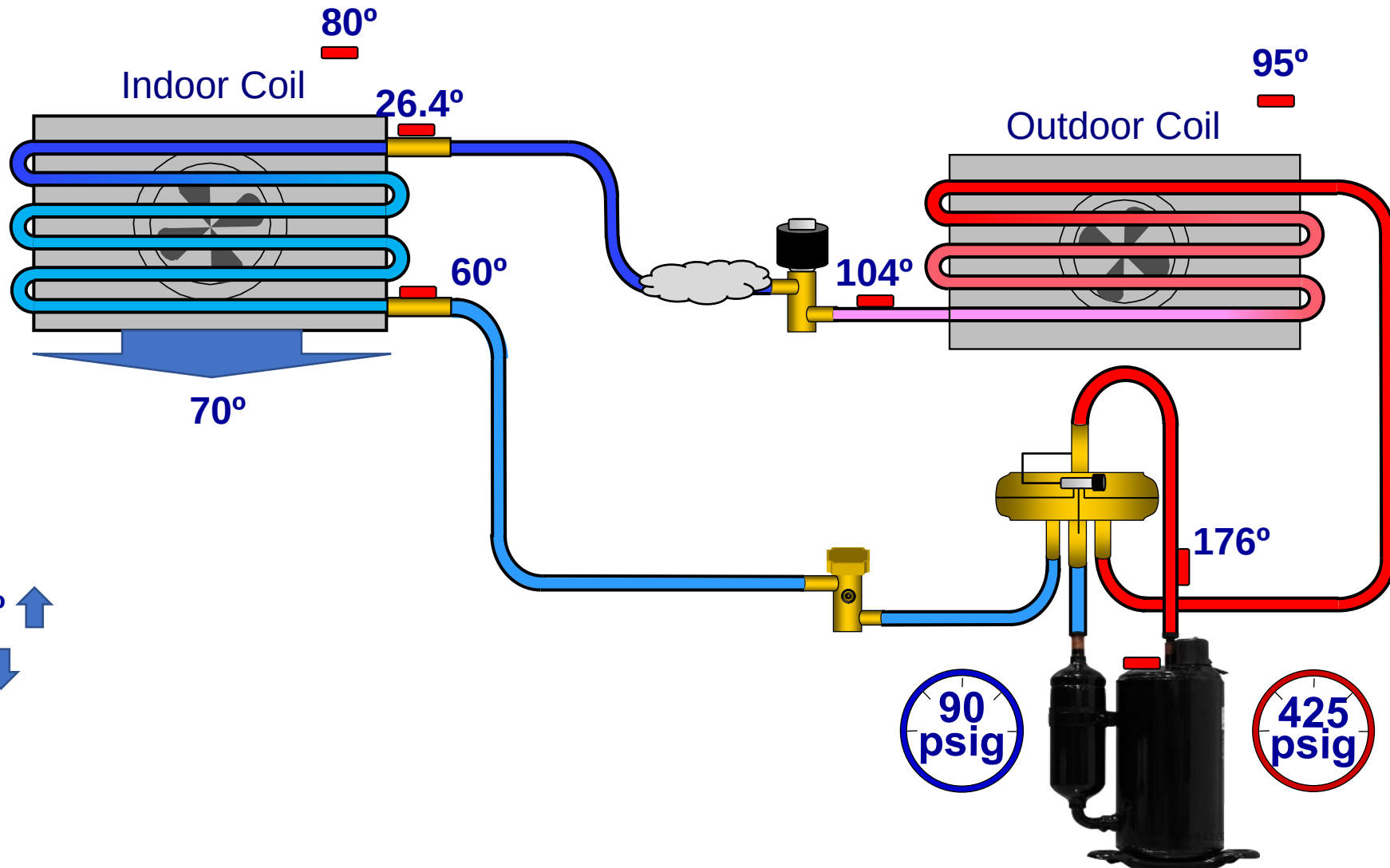
- Lower temperature
 - Lack of refrigerant, lower pressure, lower temperature
- High superheat
 - Limited refrigerant boiled off quickly

Condenser:

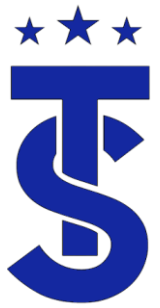
- Lower temperature
 - Little heat from the evaporator
- Normal subcooling to high
 - Refrigerant not used in evap is stored in the coil



Restricted EEV



SH = 26° ↑
SC = 17° ↑
HS = 425 psig / 121° ↑
LS = 90 psig / 26.4° ↓
EEV Opening ↑



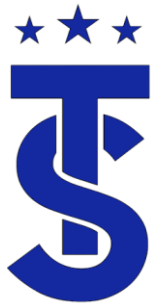
Dirty or iced evaporator, low air flow

Evaporator:

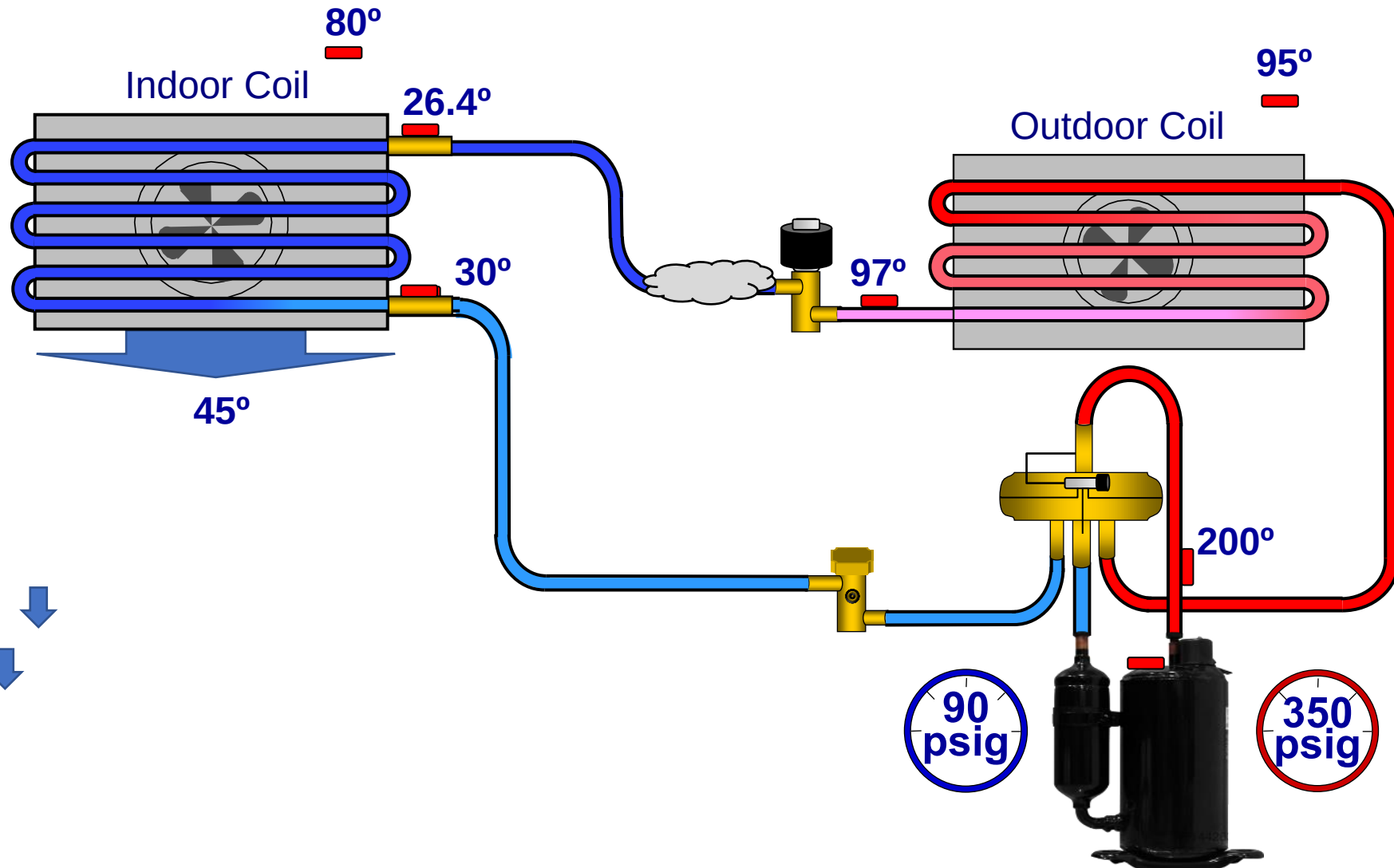
- Lower temperature
 - Refrigerant can't boil
- Low to normal superheat
 - TEV tries to maintain superheat

Condenser:

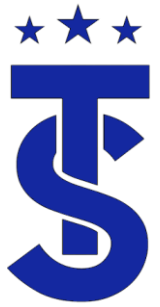
- Lower temperature
 - Little heat from the evaporator
- Normal subcooling
 - Adequate amount of refrigerant



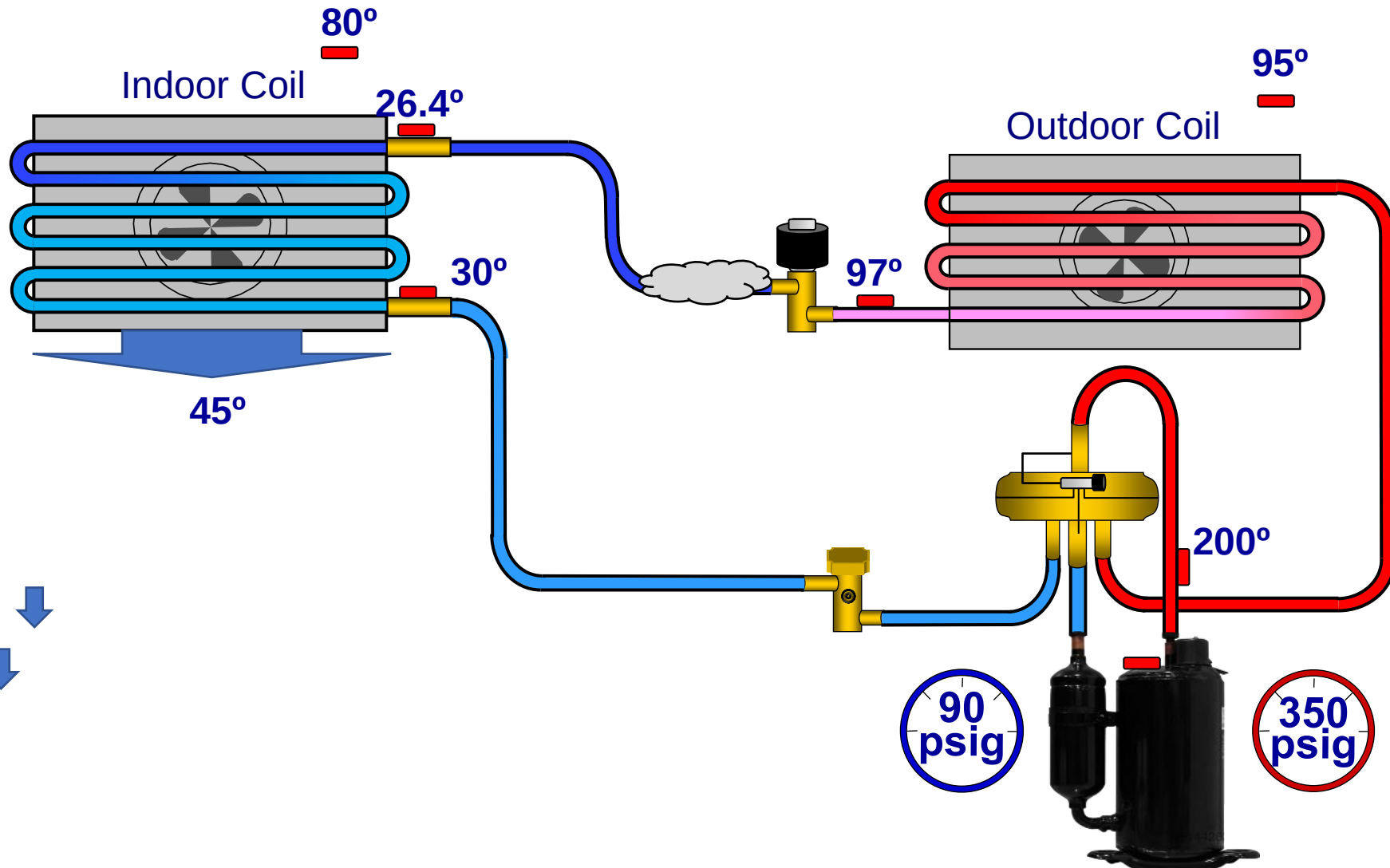
Low Air Flow



SH = 3.6° →
SC = 12° →
HS = 350 psig / 109° ↓
LS = 90 psig / 26.4° ↓
EEV Opening ↓



How Does the System Respond





THE END

